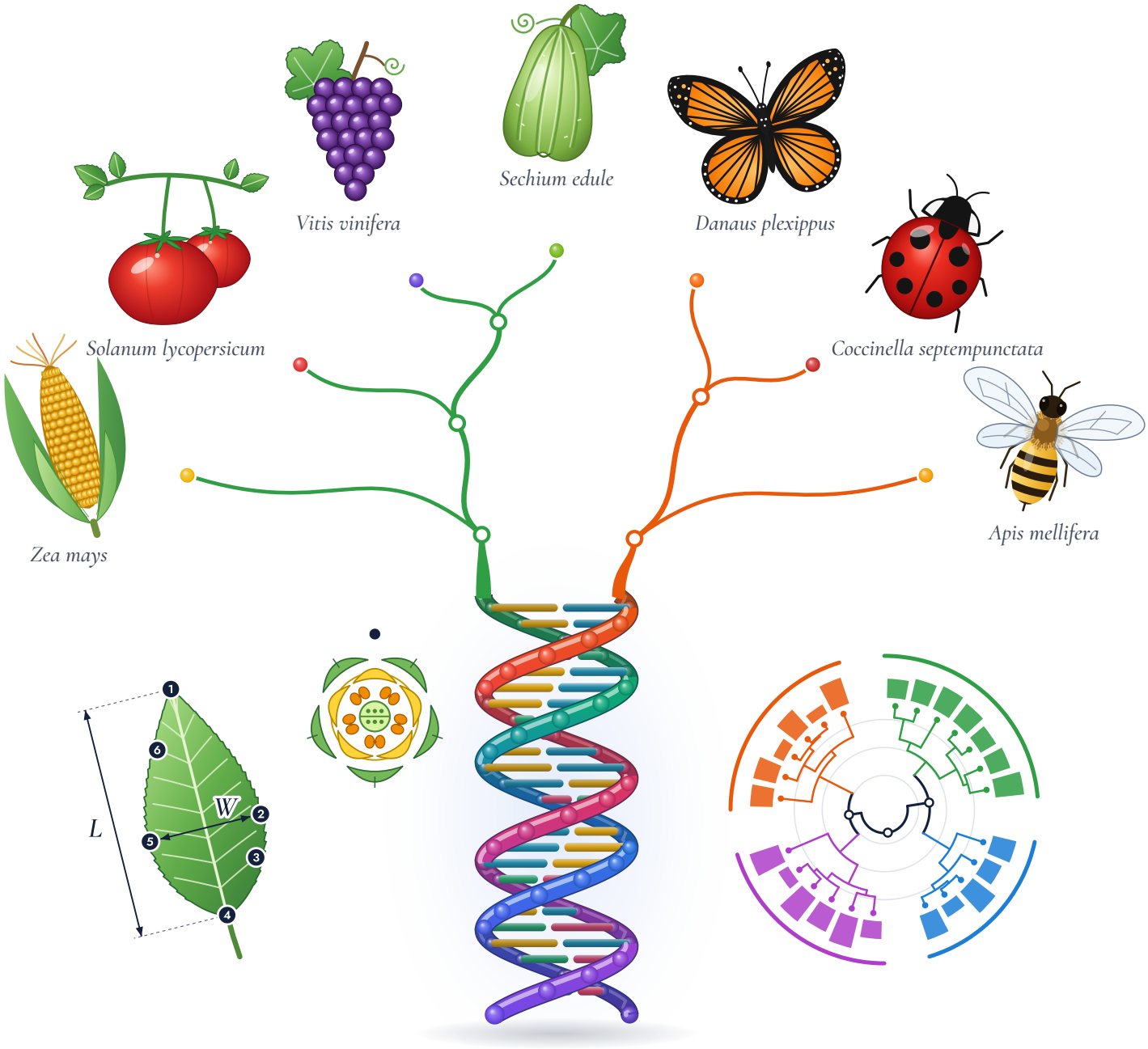
# PopGeneticsPro

## User's Manual
## *Manual de usuario*

POPULATION GENETICS · GENÉTICA DE POBLACIONES




Luis Ángel Barrera-Guzmán

VERSION 1.0 · VERSIÓN 1.0 · 2026

# PopGeneticsPro

*User's Manual · Population genetics*


| | |
|---|---|
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| | |
|---|---|
| SOFTWARE | PopGeneticsPro, version 1.0.1 (September 2026) |
| MANUAL | First English edition, 2026, translated from the first Spanish edition. It describes version 1.0.1 of the software. |



| | |
|---|---|
| SOFTWARE LICENCE | GNU General Public License, version 3 or later (GPL-3.0-or-later). |
| COPYRIGHT OF THE MANUAL | © 2026 Luis Ángel Barrera-Guzmán. All rights reserved. |


| | |
|---|---|
| INTERFACE LANGUAGE | English. In this manual, the names of buttons and menus are written exactly as they appear on screen, for example `Compute diversity`, followed by an explanation where needed. |

## HOW TO CITE THE SOFTWARE

Barrera-Guzmán, L. Á. (2026). *PopGeneticsPro: a browser-based platform for plant population genetics with codominant, dominant, sequence and morphological data* (version 1.0.1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.22740171

## HOW TO CITE THIS MANUAL

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The digital object identifier (DOI) 10.5281/zenodo.22740171 always resolves to the latest version of the software; each version also has its own DOI in the archive record (version 1.0.0, for example, has 10.5281/zenodo.22740172). Source code: https://github.com/luisangelbg/PopGeneticsPro · online version: https://luisangelbg.github.io/PopGeneticsPro/

> **PRIVACY**
>
> PopGeneticsPro runs entirely inside your browser. The files you open are read on your computer and **are never sent to any server**. You can analyse unpublished data, or data covered by confidentiality agreements, with no risk of them leaving your machine.

The illustrations, diagrams and screenshots in this manual are original and were generated with the software itself or with code written for the manual. The example datasets are described in section 1.7; the five-genera dataset is synthetic, and its taxon and fossil names are fictitious.

# Contents

FULL CONTENTS OF THE MANUAL


**P** How to read this manual iv

**1** Introduction to PopGeneticsPro 1

1.1 What is PopGeneticsPro? 1
1.2 Questions it answers 3
1.3 Five key ideas 4
1.4 Types of data it accepts 7
1.5 What you need and how to open it 8
1.6 A tour of the interface 9
1.7 The six included examples 11
1.8 Recommended workflow 13
1.9 How to read P values and intervals 14

**B1** Home, theory and simulator 16

2.1 The app's home page 16
2.2 The Wright–Fisher simulator 18
2.3 Five exercises with the simulator 22
2.4 Essential theory in the app 28
2.5 How to cite and references 31

**B2** Data and quality control 33

3.1 Input formats 34
3.2 The built-in data sheet 37
3.3 The template designer 39
3.4 Interpretation and the role of each column 44
3.5 Quality control 48
3.6 Which analyses your data allow 55
3.7 Step by step with the six examples 57

**B3** Allele frequencies and diversity 63

4.1 Theory in brief 63
4.2 The indices and what they mean 65
4.3 Decision rules 67
4.4 Block settings, tables and figures 68
4.5 Step by step: maize with microsatellites 70
4.6 Dominant data, haplotypes and morphology 76

**B4** Hardy–Weinberg and linkage disequilibrium 83

5.1 Hardy–Weinberg equilibrium 84
5.2 $F_{IS}$, inbreeding and null alleles 85
5.3 Linkage disequilibrium 88
5.4 Correction for multiple tests 89
5.5 Block settings, tables and figures 90
5.6 Step by step: maize and agave 92

**B5** Differentiation and AMOVA 99

6.1 Measuring differentiation 99
6.2 Hierarchical AMOVA 101
6.3 Pairwise matrices and gene flow 103
6.4 Decision rules 104
6.5 Block settings, tables and figures 106
6.6 Step by step: maize, agave and pine 107

B6 Distances, ordination and trees 116

7.1 Genetic distances 116
7.2 Principal coordinates (PCoA) 119
7.3 NJ and UPGMA trees with bootstrap 121
7.4 Trees with images of the OTUs 122
7.5 Mantel and isolation by distance 125
7.6 Block settings, tables and figures 127
7.7 Step by step: maize landraces with ISSR 127

B7 Bayesian clustering and assignment 132

8.1 The admixture model 132
8.2 How many clusters? ΔK, Ln P(D) and Puechmaille 134
8.3 DAPC 139
8.4 Assignment and migrants 142
8.5 Block settings, tables and figures 144
8.6 Step by step: agave with AFLP 145

B8 DNA sequences and molecular dating 148

9.1 Haplotype and nucleotide diversity 149
9.2 Neutrality tests and mismatch distribution 150
9.3 $N_{ST}$ versus $G_{ST}$ 152
9.4 Haplotype networks 153
9.5 Divergence times: fossils and rates 154
9.6 Block settings, tables and figures 156
9.7 Step by step: a pine and five genera 157

B9 Demography and spatial structure 167

10.1 Recent bottlenecks 168
10.2 Effective population size 171
10.3 Spatial genetic structure 172
10.4 Mating system 174
10.5 $P_{ST}$ versus $F_{ST}$ 175
10.6 Block settings, tables and figures 176
10.7 Step by step 177

B10 Figures and report 186

11.1 The figure editor 187
11.2 Exporting for publication 189
11.3 The automatic report and the ZIP package 191
11.4 Step by step 194

A Appendices 198

A File formats 198
B Summary of decision rules 206
C Glossary 224
D Troubleshooting 233
E References 239
F Third-party components 245

# How to read this manual

The manual follows the same order as the app: ten numbered blocks that run from left to right along the top bar. Each chapter explains one block: it opens with a little theory that is easy to digest, describes the indices the block computes and ends with a step-by-step example that uses the data already included in the software.

## A colour for each block

The colour of each block appears on its chapter opening, in the section numbers, in the theory boxes and in the table of contents, so you can tell at a glance which part of the app you are in.

| Block | Name in the app | What it does |
|---|---|---|
| 1 | Home | Home page, essential theory, simulator and how to cite |
| 2 | Data | File loading, data sheet, templates and quality control |
| 3 | Diversity | Allele frequencies and diversity indices |
| 4 | HWE & LD | Hardy–Weinberg equilibrium, inbreeding, null alleles and linkage |
| 5 | AMOVA | Differentiation among populations and analysis of molecular variance |
| 6 | Distance | Genetic distances, PCoA, trees and isolation by distance |
| 7 | Clustering | Bayesian clustering, DAPC and assignment of individuals |
| 8 | Sequences | DNA sequences, haplotype networks and molecular dating |
| 9 | Demography | Bottlenecks, effective population size, spatial structure and $P_{ST}$ |
| 10 | Report | Publication-ready figures and automatic report |

## Types of boxes

**NOTE**

Clarifications and practical tips that save time.

**IMPORTANT**

Points that can change your conclusions if they are overlooked.

**THEORY IN BRIEF**

The idea behind an analysis, without unnecessary formulae.

**EXAMPLE**

A real case worked with the data included in the software.

**DECISION RULE**

| | |
|---|---|
| **If** the value falls within this range… | **then** this is the recommended interpretation. |

These are guides to interpretation, not laws: the biological context of your species always takes precedence.

**900 dpi**

KEY FACT

A figure or fact worth remembering; for example, the maximum resolution at which figures are exported.

## Typographical conventions

| Written like this | Meaning |
|---|---|
| Compute diversity | A button, check box or menu in the app, with its text exactly as it appears on screen. |
| Edit figure › Style & typography | A path: open the first item, then the second. |
| `maize_ssr_populations.csv` | The name of a file, column or value, to be typed exactly as shown. |
| $H_e$, $F_{ST}$, $\Phi_{PT}$ | Indices and parameters; their meaning is explained the first time they appear. |
| ● ◐ — | Available · partly available or subject to conditions · not applicable. |

# 1

CHAPTER ONE

# Introduction to PopGeneticsPro

What the software is, which questions it answers, which data it accepts and how to work through it from start to finish. The chapter also covers the basic ideas of population genetics that you need to get the most out of the rest of the manual.


**1.1** What is PopGeneticsPro?

**1.2** Questions it answers

**1.3** Five key ideas

**1.4** Types of data it accepts

**1.5** What you need and how to open it

**1.6** A tour of the interface

**1.7** The six included examples

**1.8** Recommended workflow

**1.9** P values and intervals


## 1.1 What is PopGeneticsPro?

PopGeneticsPro is a platform for analysing the genetic variation of plant populations **from the field to the paper**: it takes the table that comes out of the laboratory, checks it, computes the classical and modern analyses of population genetics and delivers publication-ready figures and reports.

Everything happens inside the web browser. There is nothing to install, no programming knowledge is required and the data never leave the computer. The same steps work for microsatellites, AFLP or ISSR bands, chloroplast sequences and morphological characters, so you can compare different markers scored on the same material within a single workflow.

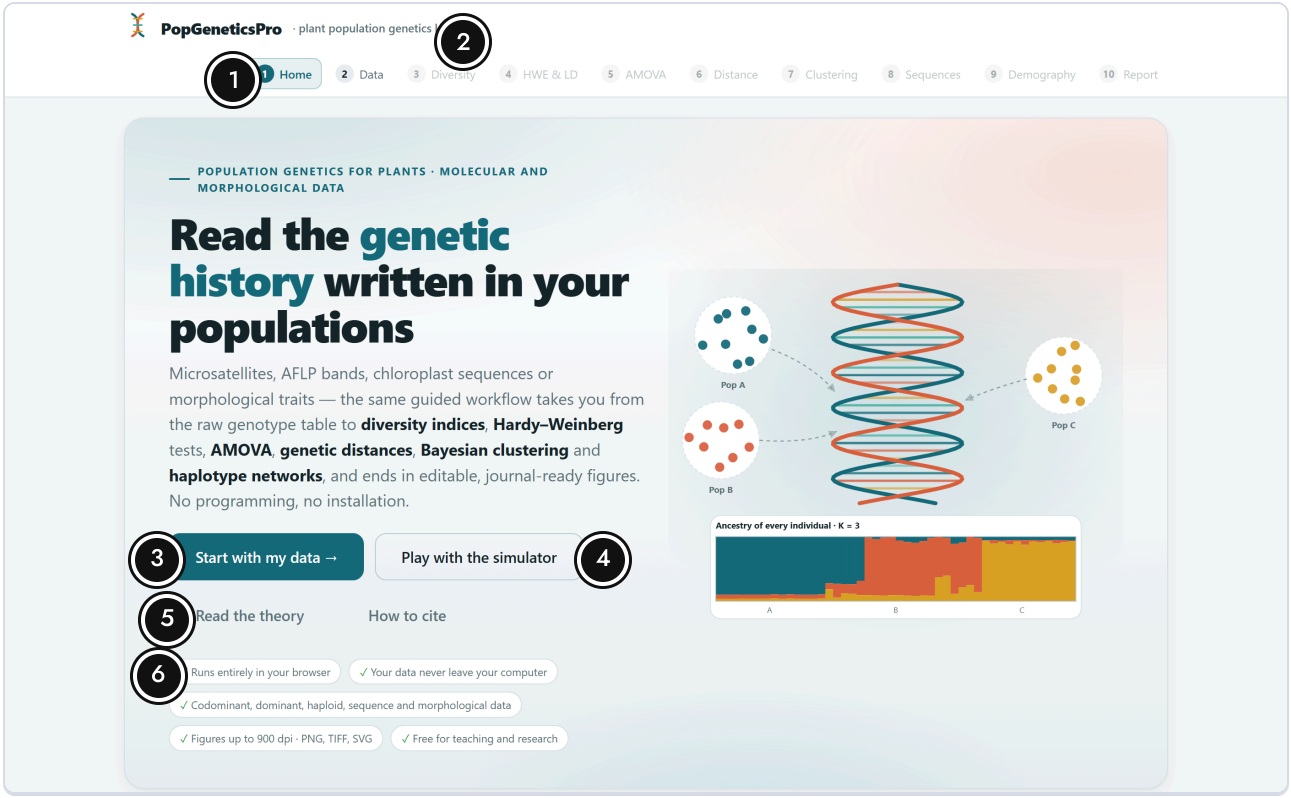



**Figure 1.1.** The app's home page (Block 1).

1. Bar with the ten blocks. The active block is highlighted.
2. Greyed-out blocks become active when you load data that support them.
3. `Start with my data` takes you to Block 2 to load your data.
4. `Play with the simulator` opens the drift and gene flow simulator.
5. Essential theory and how to cite the software.
6. Guarantees: everything runs on your computer, and five data types are supported.

**0** INSTALLATIONS

It opens with a double click on `index.html`. It needs neither administrator rights nor an internet connection.

**10** BLOCKS

From the data file to the final report, in the order in which a study is carried out.

## What it does, in one table

| Stage | What you get | Block |
|---|---|---|
| **Data input** | Reading of spreadsheets (.xlsx, .csv), genotype files (.str, .gen, .arp), FASTA and VCF; a data sheet for pasting; ready-to-fill templates. | 2 |
| **Quality control** | Missing data, monomorphic loci, clones, coding errors and sample size per population. | 2 |
| **Diversity** | $N_a$, $N_e$, $H_o$, $H_e$, $uH_e$, Shannon index, PIC, private alleles, allelic richness and F. | 3 |
| **Equilibrium** | Hardy–Weinberg tests, $F_{IS}$ with intervals, null alleles and linkage disequilibrium. | 4 |
| **Structure** | $G_{ST}$, θ, $G'_{ST}$, Jost's D, $\Phi_{PT}$, hierarchical AMOVA, pairwise matrices and gene flow. | 5 |
| **Relationships** | Genetic distances, PCoA, NJ and UPGMA trees with bootstrap, Mantel tests and isolation by distance. | 6 |
| **Clustering** | Bayesian admixture model, ΔK, DAPC, assignment of individuals and detection of migrants. | 7 |
| **Sequences** | Hd, π, $\theta_W$, Tajima's D, Fu's Fs, mismatch distribution, haplotype networks and dating with fossils or rates. | 8 |
| **History** | Bottlenecks, effective population size, spatial autocorrelation, Sp statistic, mating system and $P_{ST}$. | 9 |
| **Publication** | Editable figures up to 900 dpi (PNG, TIFF, SVG, JPG, WEBP), a full report and a ZIP package with tables and methods. | 10 |

## Who is it for?

For anyone who studies the genetic diversity of wild or cultivated plants and wants sound results without wrestling with file formats or command lines: undergraduate and graduate students starting out in population genetics, researchers in plant genetic resources and conservation, breeders characterising germplasm, and teachers who need to show in class how drift, gene flow or selection operate. Every result comes with a paragraph that interprets it in plain language, which also makes the app a teaching tool.

# 1.2 Questions it answers

A good analysis starts with a biological question. The following table matches the most common questions in a population genetics study with the analysis that answers each one and the block where it is found.

**Table 1.1.** From question to analysis.

| Question | Analysis that answers it | Block |
|---|---|---|
| Are my data clean and correctly coded? | Summary of missing data, monomorphic loci, clones and inconsistent codes | 2 |
| How much genetic variation does each population hold? | $H_e$, rarefied allelic richness, private alleles | 3 |
| Is there inbreeding or selfing? | $F_{IS}$ per locus and population; comparison of $H_o$ with $H_e$ | 4 |
| Does any locus have null alleles? | Estimated null allele frequency; high $F_{IS}$ at a single locus | 4 |
| Are the populations genetically distinct? | $F_{ST}$, $G'_{ST}$, Jost's D, $\Phi_{PT}$ with permutation tests | 5 |
| How much of the variation lies among regions, among populations and within populations? | Hierarchical AMOVA | 5 |
| Which materials are most similar to each other? | Genetic distances, PCoA, trees with bootstrap | 6 |
| Are more distant populations more different? | Mantel test and isolation by distance | 6 |
| How many genetic groups are there, without assuming them in advance? | Bayesian clustering with ΔK; DAPC | 7 |
| Are there individuals that come from another population? | Assignment tests and detection of migrants | 7 |
| How are the haplotypes related, and where are they found? | Median-joining haplotype network; $N_{ST}$ versus $G_{ST}$ | 8 |
| When did the lineages diverge? | Molecular dating with fossils or a substitution rate | 8 |
| Has the population gone through a bottleneck or expanded? | Heterozygosity excess, M ratio, mode shift; Tajima's D, mismatch distribution | 9<br>8 |
| How many breeders actually contribute to the next generation? | Effective population size ($N_e$) | 9 |
| Are close neighbours related? | Spatial autocorrelation, kinship coefficients and Sp statistic | 9 |
| Do morphological characters diverge more than expected from drift alone? | $P_{ST}$ compared with neutral $F_{ST}$ | 9 |

## 1.3 Five key ideas of population genetics

You do not need any prior coursework to use the app, but these five ideas come up in every chapter. Read them once and come back to them whenever you need to.

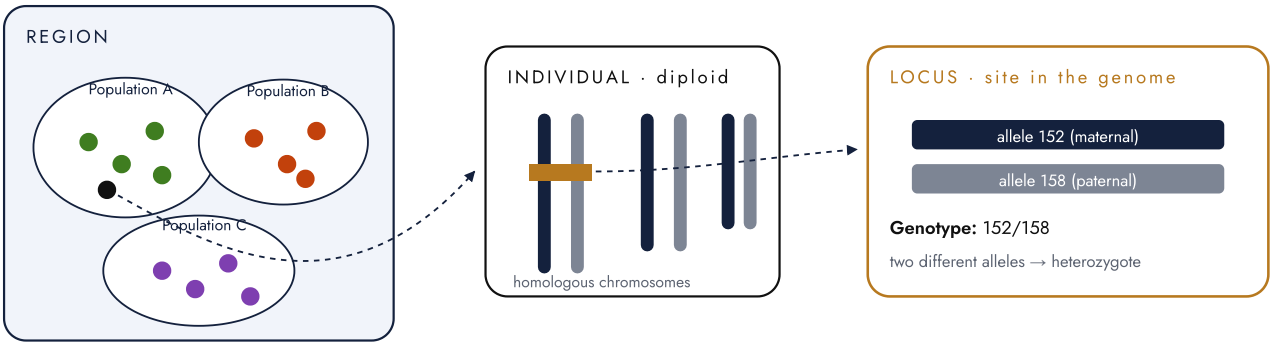



**Figure 1.2.** The levels of organisation in a study. A region contains populations, and each population contains individuals. At each *locus*, every diploid individual carries two copies of the gene, the **alleles**, which together make up its **genotype**.

### 1 ALLELE, LOCUS AND GENOTYPE

A **locus** (plural *loci*) is a specific place in the genome: a microsatellite, an AFLP band or a fragment of chloroplast DNA. The variants that exist at that place are the **alleles**. In a diploid plant each locus has two alleles; if they are identical the individual is **homozygous** (152/152), and if they differ it is **heterozygous** (152/158).

### 2 ALLELE FREQUENCIES

The unit of population genetics is not the individual but the **frequency** of each allele in the population. If allele 152 appears 35 times in 25 diploid plants (50 gene copies), its frequency is $p = 35/50 = 0.70$. Almost all the indices in the software are calculated from these frequencies.

### 3 HARDY–WEINBERG EQUILIBRIUM: THE BASELINE

If a population is large, mates at random and is subject to neither migration, mutation nor selection, genotype frequencies can be predicted from allele frequencies: with two alleles of frequencies $p$ and $q$, one expects $p^2$ homozygotes of one type, $2pq$ heterozygotes and $q^2$ homozygotes of the other. Departures from this baseline are informative: they point to inbreeding, hidden structure or technical problems.

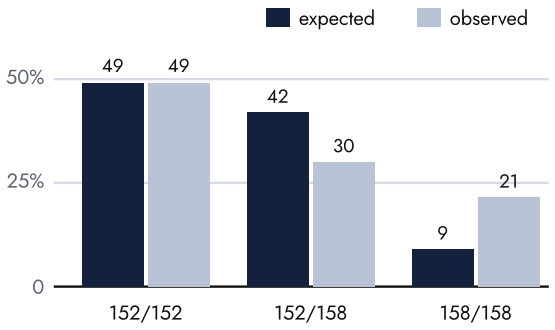
1

### WORKED EXAMPLE

With $p = 0.70$ and $q = 0.30$, the expected proportions are 49% homozygotes 152/152, **42% heterozygotes** and 9% homozygotes 158/158. If you find only **30% heterozygotes** in the field, heterozygotes are lacking. The fixation index measures this deficit:

$$F = 1 - H_o/H_e = 1 - 0.30/0.42 = \mathbf{0.29}$$

A positive value indicates a heterozygote deficit (inbreeding or selfing); a negative value, an excess.




## 4 THE FORCES THAT CHANGE FREQUENCIES

Five processes move populations away from equilibrium. Each leaves a distinct signature on the diversity within populations and on the differentiation among them, and recognising these signatures lies at the heart of interpretation.

**Table 1.2.** Signature of each evolutionary force. ↑ increases · ↓ decreases · ≈ small or variable effect.

| Force | Diversity within | Differentiation among | How to recognise it in your results |
|---|---|---|---|
| **Genetic drift** chance in small populations | ↓ | ↑ | Small populations with few alleles, loss of rare alleles, high $F_{ST}$ unrelated to distance. |
| **Gene flow** migration of pollen or seed | ↑ | ↓ | Low $F_{ST}$, admixed individuals in Bayesian clustering, migrants detected. |
| **Mutation** | ↑ | ≈ | Private alleles in old, stable populations; very slow on an ecological time scale. |
| **Divergent selection** | ↓ at loci under selection | ↑ | One locus or character with much greater differentiation than the rest; $P_{ST} > F_{ST}$. |
| **Selfing and inbreeding** | ↓ heterozygosity | ↑ | Positive $F_{IS}$, similar across all loci; $H_o$ well below $H_e$. |

## 5 GENETIC STRUCTURE: THE WHOLE AND ITS PARTS

When a species is divided into populations that exchange few migrants, the total variation is partitioned into two components: the variation that exists **within** each population and the variation that **separates** them. Wright's F-statistics and AMOVA measure this partition. An $F_{ST}$ of 0.15, for example, means that 15% of the genetic variation lies among populations and 85% within them.

## 1.4 Types of data it accepts

The first decision in any analysis is to establish what kind of marker you have, because that determines what can be computed. PopGeneticsPro recognises five types and adapts each block to them.

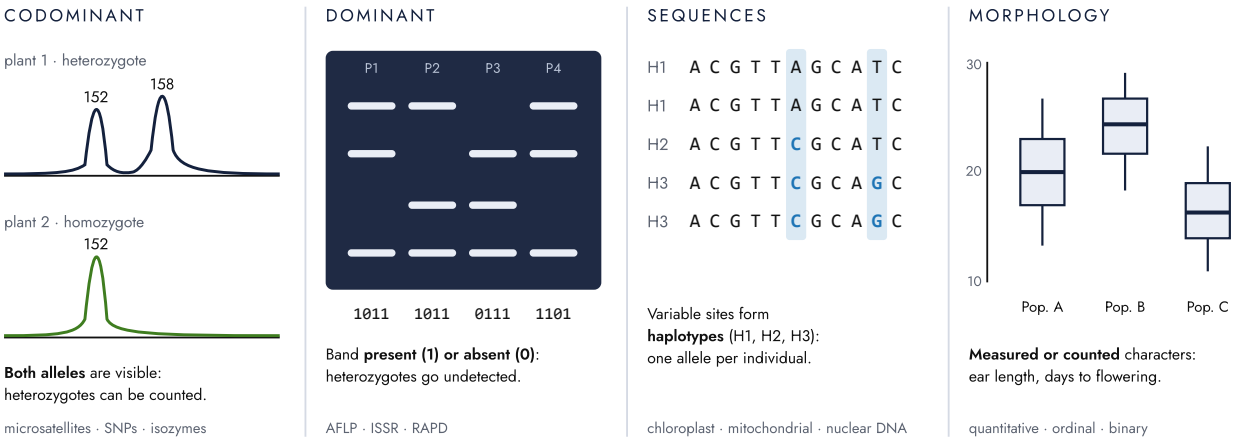



**Figure 1.3.** Four ways of observing variation. With **codominant** markers, both alleles of each individual are read; with **dominant** markers, only the presence or absence of a band; in **sequences**, the variable sites define haplotypes; in **morphology**, characters of the phenotype are measured.

**Table 1.3.** The five data types and what each one implies.

| Type | Examples | Advantage | Limitation |
|---|---|---|---|
| **Codominant** | Microsatellites (SSR), SNPs, isozymes | Heterozygotes are observed; allows all tests of equilibrium, inbreeding, null alleles, $N_e$ and bottlenecks. | Possible null alleles and allele-sizing errors. |
| **Dominant** | AFLP, ISSR, RAPD | Many loci at low cost and with no prior knowledge of the genome. | Heterozygotes cannot be distinguished: frequencies are estimated by assuming equilibrium, and there are no Hardy–Weinberg tests. |
| **Haploid** | Chloroplast or mitochondrial haplotype codes | Inherited from a single parent; ideal for tracing seed dispersal. | A single lineage per individual; less variation than nuclear markers. |
| **Sequences** | FASTA alignments of cpDNA, ITS, nuclear genes | Provide information on haplotype genealogy, demographic history and divergence times. | An alignment is a single locus; aligned sequences of equal length are required. |
| **Morphological** | Plant, flower, fruit and seed characters | Reflect the phenotype that matters to breeders and conservationists; allow $P_{ST}$ to be estimated. | Influenced by the environment; they do not measure genetic variation directly. |

1

**! IMPORTANT**

Declaring the wrong data type changes the results. A 0/1 ISSR matrix read as codominant would yield absurd heterozygosity values. Block 2 detects the type automatically, but **it always shows you what it has assumed** and lets you correct it before anything is computed.

## 1.5 What you need and how to open the software

### REQUIREMENTS

- A desktop or laptop computer with a **modern, up-to-date web browser**.
- A screen at least 1280 pixels wide; the app adapts to smaller screens, but tables and figures are easier to work with on a large one.
- 4 GB of memory is enough for most studies; Bayesian clustering and dating with thousands of iterations benefit from 8 GB.
- **No** internet connection, installation or administrator rights are needed.

### FOLDER CONTENTS

| | |
|---|---|
| `index.html` | The app. Open it with a double click. |
| `Open PopGeneticsPro.bat` | Optional launcher with a local server. |
| `js/`, `css/` | The program code. |
| `data/` | The example files. |
| `vendor/` | Third-party components and their licences. |

### Opening PopGeneticsPro in three steps

1. **Copy the whole folder** `PopGeneticsPro` to your computer, for example into Documents. Do not separate `index.html` from the other folders.
2. **Double-click** `index.html`. It will open in your default browser, showing the Block 1 home page.
3. **Click** Start with my data to load a file, or go straight to Block 2 and choose one of the examples.

**i NOTE · OPENING WITH A LOCAL SERVER**

Everything works with a double click, but some browsers restrict background processes when the page is opened as a local file. In that case Bayesian clustering and dating still run, although the window may stay busy while they compute. If you work with large studies, use `Open PopGeneticsPro.bat` : it starts a small server on your own computer (address `http://localhost:9000` ) and opens the app from there. Nothing is sent to the internet in this case either.

**NOTE · YOUR WORK IS REMEMBERED**

The app stores your figure style preferences and the contents of the built-in data sheet in your browser. If you switch browsers or clear your browsing data, those preferences are lost, but your original files are never modified.



## 1.6 A tour of the interface

All the analysis blocks share the same layout, so you only need to learn it once. From top to bottom, each block has three parts:

| Part | What it contains | How it appears on screen |
|---|---|---|
| **1 · Settings** | Analysis options, with default values that follow the most common conventions in the literature, and a button to run the calculation. | 1 · Settings · Compute … |
| **2 · What the numbers say** | Cards with the main values, and paragraphs that interpret the results in plain language. | 2 · What the numbers say |
| **3 · Tables and figures** | Downloadable tables and editable figures with style and export controls. | Edit figure · Download figure |

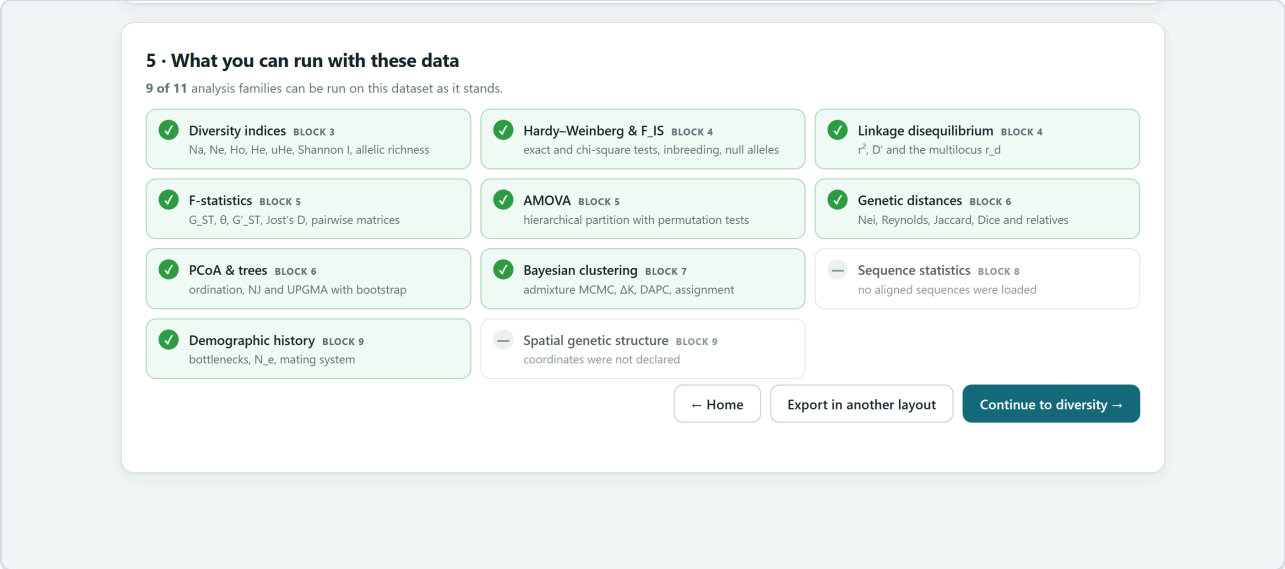



**Figure 1.4.** Once the data have been loaded, Block 2 shows What you can run with these data: which families of analyses your data support (green tick) and why the others are not available. Here, with the maize microsatellite example, 9 of the 11 analyses are available; sequences and spatial structure are missing because there are no alignments or coordinates.

**Figure 1.5.** A Block 3 figure (gene diversity by population, maize with microsatellites) with its editing panel open.

1. The figure is redrawn instantly with every change.
2. **Edit figure** opens and closes the editing panel.
3. Two tabs: content and colours; style and typography.
4. **Flip axes** rotates the chart: horizontal bars with legible names.
5. Colour palette; also titles, index, error bars and sort order.
6. Format, resolution, background and download, with the resulting size in pixels and centimetres.

## Interface terms

The app is in English, the language of scientific journals, and its labels are deliberately short. This table gathers the terms you will see in every block, with what each one refers to.

**Table 1.4.** Basic interface vocabulary.

| In the app | What it means | In the app | What it means |
|---|---|---|---|
| Settings | Analysis options | Download | Save a table or figure |
| Compute · Run | Start the calculation | Edit figure | Open the figure editor |
| What the numbers say | Plain-language interpretation | Content & colours | Editor tab for data and colours |
| Population · Region | Grouping levels of the samples | Style & typography | Editor tab for fonts and layout |
| Locus · Loci | Marker(s) analysed | Palette | Colour palette |
| Missing data | Absent genotypes or values | Flip axes | Swap the axes |

**Table 1.4.** Basic interface vocabulary. *(continued)*

| In the app | What it means | In the app | What it means |
|---|---|---|---|
| Replicates | Number of repetitions (bootstrap, permutations) | Resolution | Image resolution in dpi |
| Random seed | Starting value of the random number generator | Report | Automatic report |
| Continue to … | Go to the next block | How to cite | Citation of the software |

**NOTE · THE RANDOM SEED**

Analyses that use resampling, permutations or Markov chains depend on chance. The Random seed fixes that chance: with the same seed and the same data you will obtain exactly the same intervals and P values. Record the seed in the methods section of your paper so that others can reproduce your results; the Block 10 report does this for you.

## 1.7 The six example datasets

The software includes six ready-to-use datasets. They are embedded in the app, so they load even without a connection, and each one is designed for practising a different data type. The chapter on each block uses them in its step-by-step examples.

**Table 1.5.** Included examples. They are loaded in Block 2, under Or try an example.

| Button in the app | Type | Contents | Blocks for practice |
|---|---|---|---|
| Maize landraces · ISSR | Dominant | 31 ISSR bands in 15 Mexican maize landraces, one bulked sample per landrace. The loci are in rows: the app has to recognise this and transpose the table. | 2 6 10 |
| Maize populations · SSR | Codominant | 8 microsatellites in 3 maize populations (Cónico, Chalqueño and Palomero) of 25 plants each; two columns per locus. | 2 3 4 5 6 7 9 |
| Agave · AFLP | Dominant | 60 AFLP bands in 4 agave populations of 20 plants each, with geographical coordinates. | 2 3 4 5 6 7 9 |
| Pine · chloroplast DNA | Sequences | 48 aligned chloroplast DNA sequences (480 bp) of pine from 4 populations. | 2 5 6 8 |
| Maize · morphology | Morphological | 9 characters measured on the same plants as the microsatellite example. | 2 3 6 9 |
| Five genera · dating with fossils | Sequences | Synthetic: 16 species from five fictitious genera plus an outgroup; 1,600 sites simulated on a tree with known ages and fictitious fossils. | 2 6 8 |

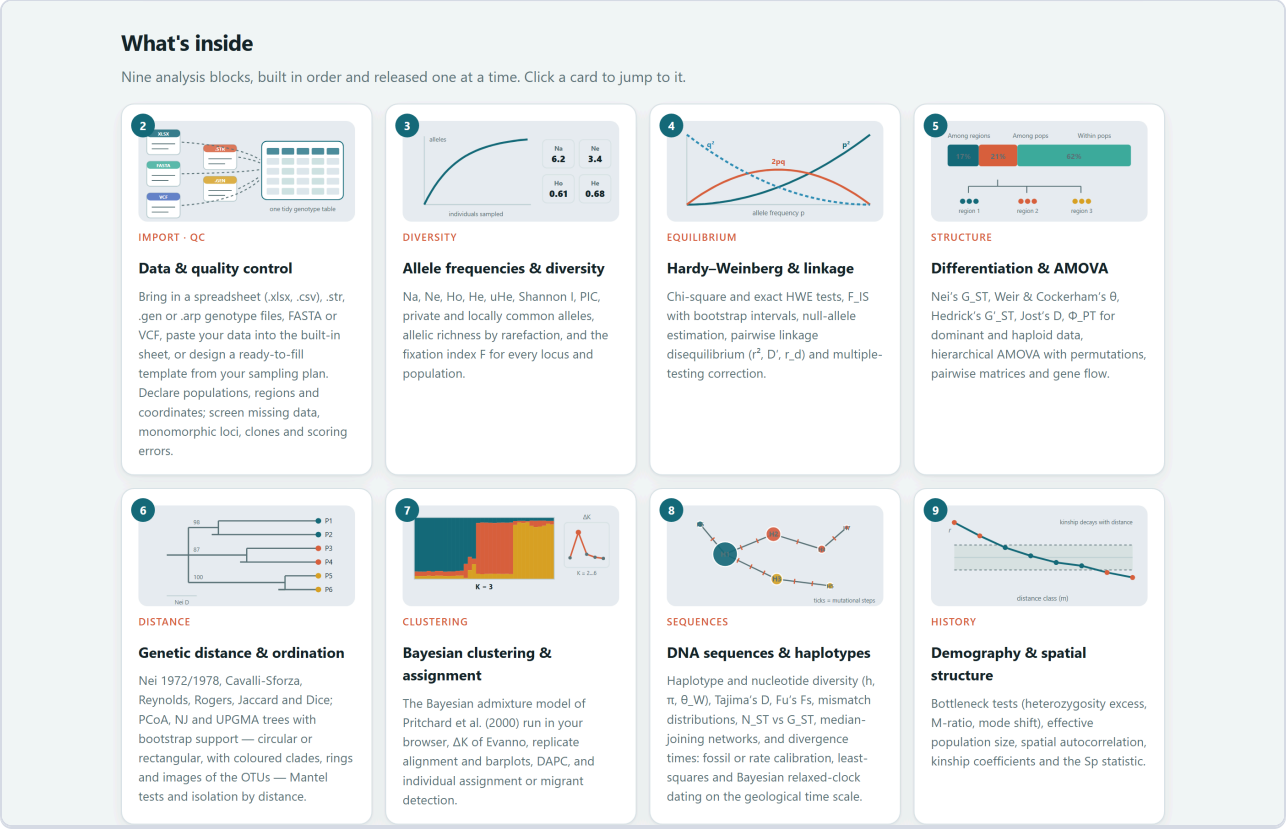



**Figure 1.6.** The What's inside section of the home page summarises the analysis blocks. Clicking any card takes you to its block; if no data have been loaded yet, the app takes you to Block 2.

### YOUR FIRST ANALYSIS IN TWO MINUTES

1. Open the app and click **Start with my data**.
2. Under **Or try an example**, click **Maize populations · SSR**. The app reads the file and shows how it has interpreted it: 75 individuals, 3 populations, 8 codominant loci.
3. Review the summary and click the button that builds the dataset. The quality control results and the available-analyses card shown in figure 1.4 appear.
4. Click **Continue to diversity** and then **Compute diversity**. In an instant you will see $H_e$ = 0.747 for the three populations combined, together with the interpretation and figure 1.5.

### NOTE · EXAMPLES AND REAL DATA

The ISSR matrix of maize landraces comes from a real study and is published with its author's permission. The microsatellite, AFLP, chloroplast and morphology datasets are realistic simulations created for teaching, and the five-genera dataset is entirely synthetic, with known true ages so that the dating can be checked.

# 1.8 Recommended workflow

The blocks are arranged in the order in which a study is carried out: first the quality of the data is ensured, then diversity is described, assumptions are checked, structure is measured and, finally, history is interpreted. Each data type follows its own route through this sequence, like metro lines that stop at only some of the stations.

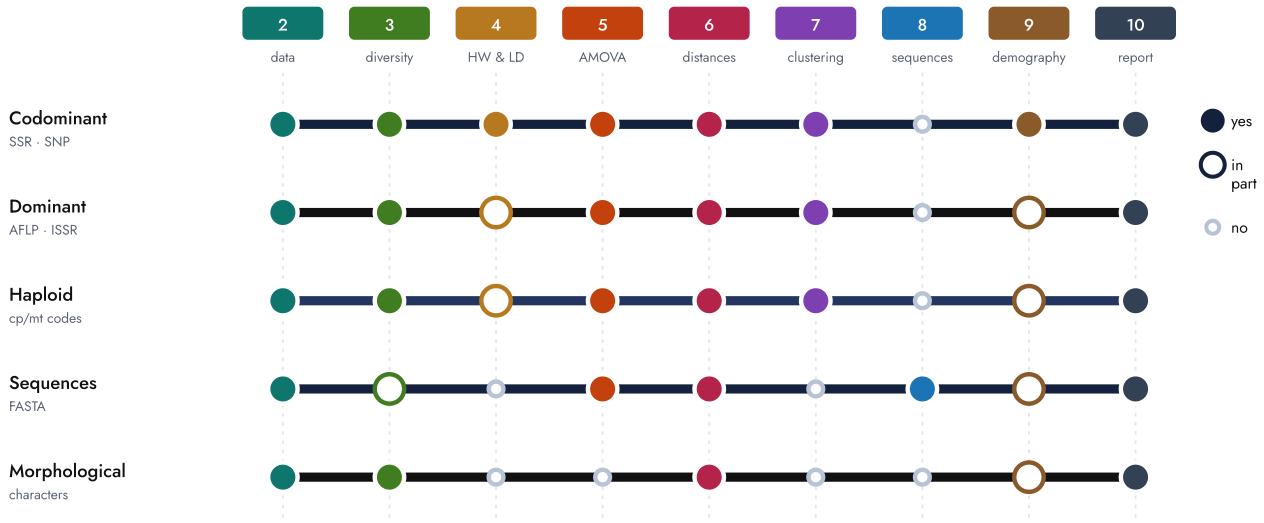



**Figure 1.7.** Analysis routes by data type. A filled station indicates that the block is available; a ring, that it is partly available or available under certain conditions (table 1.6); a small dot, that the data type passes straight through. All routes start at Block 2 and end at the report.

**Table 1.6.** Which blocks each data type activates, and under what conditions.

| Type | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Conditions |
|---|---|---|---|---|---|---|---|---|
| **Codominant** | ● | ● | ● | ● | ● | — | ● | Blocks 4 and 5 require more than one individual per population; Block 9, at least 10 individuals per population or coordinates. |
| **Dominant** | ● | ◐ | ● | ● | ● | — | ◐ | No Hardy–Weinberg tests (linkage only); in Block 9, spatial structure only, if coordinates are available. |
| **Haploid** | ● | ◐ | ● | ● | ● | — | ◐ | As for dominant markers; clustering requires at least three polymorphic loci. |
| **Sequences** | ◐ | — | ● | ● | — | ● | ◐ | An alignment is a single locus: no linkage analysis or Bayesian clustering. Dating requires at least three sequences. |
| **Morphological** | ● | — | — | ● | — | — | ◐ | Phenotypic diversity per character; in Block 9, $P_{ST}$ for each quantitative character. |

1

**DECISION RULE · HOW MANY PLANTS PER POPULATION?**

| Size per population | Interpretation | What to do |
|---|---|---|
| **25 to 30 or more** | Adequate | Sufficient to estimate allele frequencies reliably with microsatellites (Hale et al., 2012). All analyses are dependable. |
| **10 to 24** | Acceptable, with caution | Compare diversity using **rarefied allelic richness** rather than the raw number of alleles, and report confidence intervals. |
| **5 to 9** | Limited | Rare alleles are lost and $F_{IS}$ values are unstable. Use such samples for description, not for precise comparisons between populations. |
| **Fewer than 5** | Insufficient | The app flags this in quality control. Consider pooling neighbouring populations or restricting the analysis to distances and ordination. |

With dominant markers, more plants and more polymorphic bands are advisable, because each band carries less information than a microsatellite.

**IMPORTANT · ORDER DOES MATTER**

Although you can move freely between blocks, check **Block 4 before interpreting Blocks 5 and 7**. A locus with null alleles or a strongly inbred population distorts $F_{ST}$ and Bayesian clustering. Block 5 lets you exclude the suspect loci that Block 4 detects.

## 1.9 How to read P values and intervals

Almost every block reports P values and confidence intervals. Understanding them properly avoids the two most common mistakes: believing that a non-significant result proves there is no effect, and believing that a significant one is necessarily important.

**P THE P VALUE**

It is the probability of observing a result at least as extreme as yours **if there were in fact no effect**. A P of 0.003 in a Hardy–Weinberg test means that, if the population were in equilibrium, a heterozygote deficit like this would arise by chance 3 times in 1,000.

**CI THE CONFIDENCE INTERVAL**

It is the range of values compatible with your data. If the 95% interval for $F_{IS}$ runs from 0.08 to 0.21, inbreeding is real and moderate. If it runs from −0.05 to 0.30, the data cannot distinguish between no inbreeding and strong inbreeding.

**DECISION RULE · READING A P VALUE**

| P value | Evidence against the null hypothesis | How to word it |
|---|---|---|
| **P < 0.001** | Very strong | "The departure is highly significant". |
| **0.001 ≤ P < 0.01** | Strong | "The departure is significant". |
| **0.01 ≤ P < 0.05** | Moderate | "There is evidence of a departure"; confirm it with other loci or populations. |
| **P ≥ 0.05** | Insufficient | "No departure was detected". Do not claim that the population *is* in equilibrium: the test may have lacked statistical power. |

Always report the exact P value and the effect size (for example, $F_{IS}$ with its interval), not just whether the result is significant.

1 in 20

THE MULTIPLE-TESTING PROBLEM

At a significance level of 0.05, **one test in twenty comes out significant by pure chance**. If you test 8 loci in 3 populations (24 tests), expect roughly one false positive even if everything is in equilibrium. That is why Block 4 corrects P values with Holm's sequential method, the default, or with the Bonferroni, Benjamini–Hochberg or Benjamini–Yekutieli method.

**NOTE · BIOLOGICAL SIGNIFICANCE**

With large amounts of data, tiny differences become significant. An $F_{ST}$ of 0.01 can have $P < 0.001$ with hundreds of plants and yet indicate populations that exchange migrants frequently. Always ask yourself whether the magnitude matters for your species and your objective, not just whether the P value is small.

With these foundations you are ready to work through the blocks. The next chapter presents the app's home page, its built-in theory and the simulator, a quick way to see the forces described in section 1.3 in action.

# B1

CHAPTER TWO · BLOCK 1 · HOME

# Home, theory and simulator

The app's home page is more than a welcome screen. It is a map of the whole program, a laboratory where you can watch drift, gene flow, selection, mutation and selfing at work, and a summary of the theory with the references behind every calculation.


2.1 The app's home page

2.2 The Wright–Fisher simulator

2.3 Five exercises with the simulator

2.4 Essential theory in the app

2.5 How to cite and references


## 2.1 The app's home page

When you open `index.html`, Block 1 appears. It reads from top to bottom, like a brochure: first what the program does, then how to work through it, then a simulator to experiment with, the list of methods, the theory and, finally, how to cite it.

**Table 2.1.** Sections of the home page, in the order in which they appear.

| Section in the app | What it contains | What it is for |
|---|---|---|
| **Introduction** four buttons and five guarantees | Start with my data, Play with the simulator, Read the theory and How to cite; an illustration with three populations and an ancestry plot. | Going straight to loading data or to any of the sections below. |
| How it works | The seven steps of a study, from the data file to the paper (figure 2.1). | Placing each block within the logic of a research project. |
| Try it: watch drift, migration and selection at work | The Wright–Fisher simulator (sections 2.2 and 2.3). | Understanding, through play, what $H_S$, $H_T$ and $F_{ST}$ mean. |
| What's inside | One card per analysis block, with an illustration and a description. | Jumping to a block with a single click. |
| Every kind of data a plant study produces | The four main types of marker and what each one allows. | Confirming that your data fit the program (chapter 1, section 1.4). |
| Methods covered | A gallery of 18 methods grouped by family (table 2.3). | Seeing at a glance everything that can be calculated. |
| What it brings together | The capabilities the program combines in a single workflow. | Justifying its use in a project or a thesis. |
| The theory, in plain language | Eight expandable topics with formulae, rules of thumb and references (section 2.4). | Reviewing the theory before interpreting or defending results. |
| How to cite and References behind the calculations | The citation of the program, ready to copy, and the 25 references for the methods (section 2.5). | Citing correctly in your paper. |

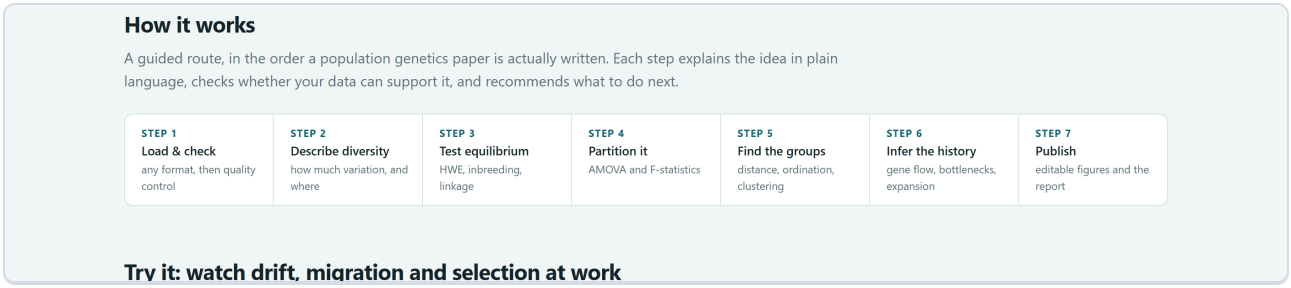



**Figure 2.1.** The How it works section: seven steps that follow the order in which a population genetics paper is written.

**Table 2.2.** The seven steps on the home page and the blocks that carry them out.

| Step | In the app | What is done | Blocks |
|---|---|---|---|
| 1 | Load & check | Load any format and check the quality of the data. | 2 |
| 2 | Describe diversity | How much variation there is and where it lies. | 3 |
| 3 | Test equilibrium | Hardy–Weinberg, inbreeding and linkage. | 4 |
| 4 | Partition it | Partition the variation with AMOVA and F-statistics. | 5 |

**Table 2.2.** The seven steps on the home page and the blocks that carry them out. *(continued)*

| Step | In the app | What is done | Blocks |
|---|---|---|---|
| 5 | Find the groups | Distances, ordination and clustering. | 6 7 |
| 6 | Infer the history | Gene flow, bottlenecks, expansion and divergence times. | 8 9 |
| 7 | Publish | Editable figures and report. | 10 |

**Table 2.3.** The Methods covered gallery, arranged by family.

| Family | Methods | Blocks |
|---|---|---|
| **Diversity** | Allele frequencies by locus and population · diversity indices ($N_a$, $N_e$, $H_o$, $H_e$, $uH_e$, I, PIC) · Hardy–Weinberg with $\chi^2$ and exact tests · linkage disequilibrium ($r^2$, D′, $\bar{r}_d$) | 3 4 |
| **Structure** | Pairwise differentiation ($F_{ST}$, $G'_{ST}$, Jost's D, $\Phi_{PT}$) · hierarchical AMOVA · isolation by distance · spatial autocorrelation and Sp | 5 6 9 |
| **Distance** | NJ and UPGMA trees with bootstrap · principal coordinates of individuals or populations | 6 |
| **Clustering** | Bayesian clustering with ΔK · assignment tests and first-generation migrants | 7 |
| **Sequences** | Haplotype networks (median-joining) · divergence times with fossils and a relaxed clock · mismatch and neutrality tests (Tajima's D, Fu's Fs, $R_2$) | 8 |
| **Demography** | Bottlenecks (heterozygosity excess, M ratio) · mating system (outcrossing rate and kinship) · effective population size | 9 |

**NOTE · NAVIGATING FROM THE HOME PAGE**

The buttons in the introduction take you to the corresponding section of the home page itself, except Start with my data, which opens Block 2. The What's inside cards open their block. If you have not loaded any data yet, the app takes you to Block 2 with the message *Load your data first*, because every block reads from there. To return to the home page from anywhere, click the program's name at the top left or the 1 Home button.

## 2.2 The Wright–Fisher simulator

The simulator reproduces, generation after generation, the evolution of a locus with two alleles in several populations. It is not a decorative animation: it applies the same equations found in textbooks, and the diversity and differentiation curves are calculated from the simulated frequencies.

### THEORY IN BRIEF · THE WRIGHT–FISHER MODEL

Imagine several populations of the same size that start with allele *A* at frequency $p = 0.5$. In each generation four things happen, always in the same order: **selection** favours one genotype, **migration** mixes each population with the average of all of them, **mutation** turns some alleles into others and, finally, **drift** randomly draws the $2N_e$ gene copies that will form the next generation. This last draw is what makes identical populations drift apart by pure chance.

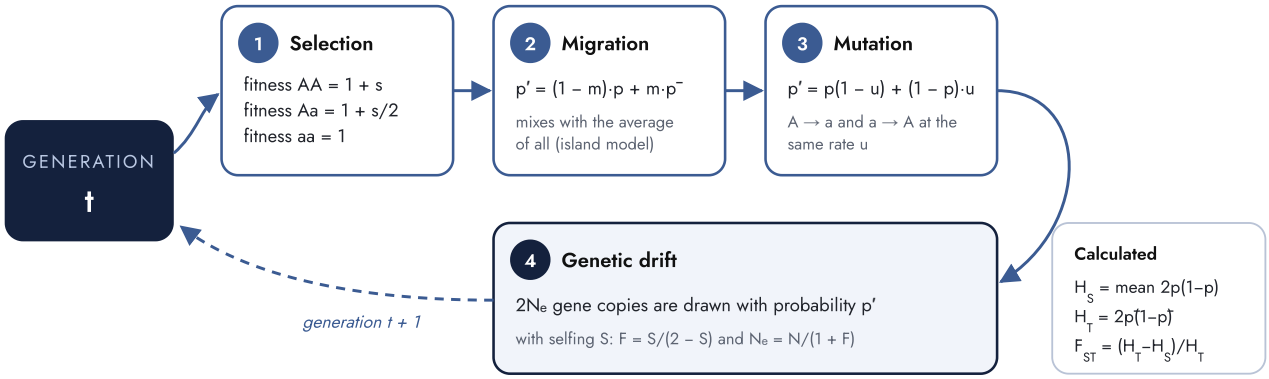



**Figure 2.2.** What the simulator does in each generation. The four steps transform the frequency *p* of each population; at the end, the diversity within populations ($H_S$), the diversity of the whole set ($H_T$) and the differentiation ($F_{ST}$) are calculated. Selection is additive: the heterozygote has half the advantage of the homozygote.

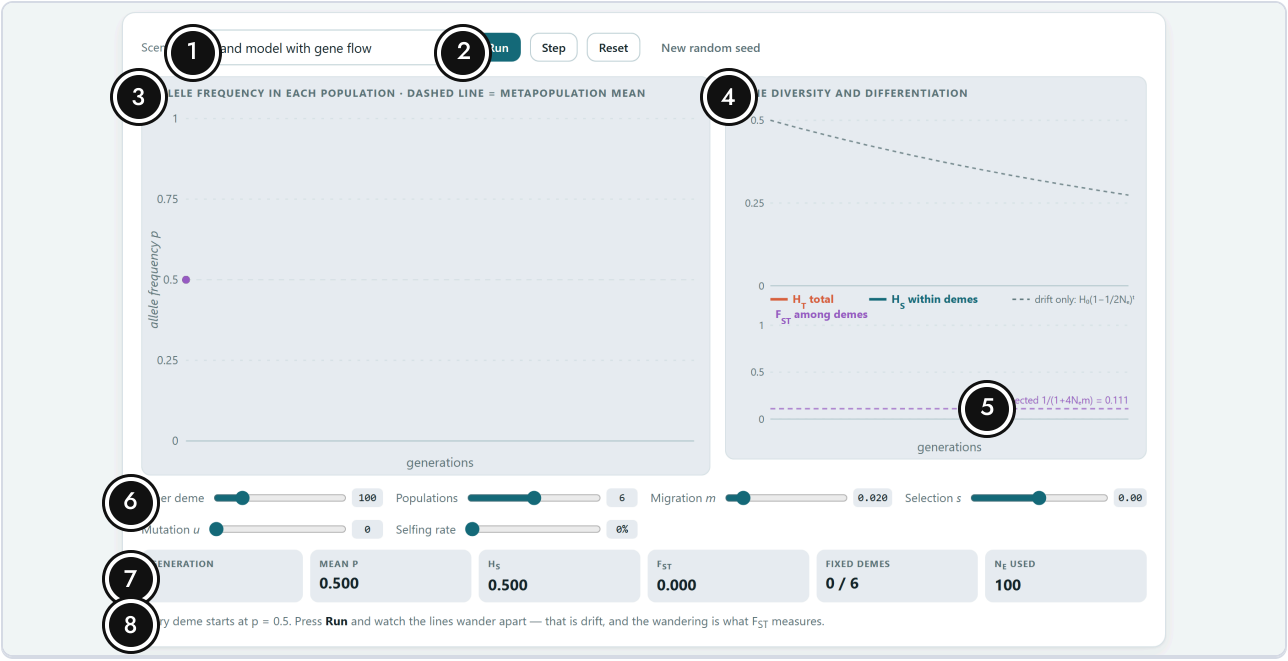



**Figure 2.3.** The simulator ready to run with the Island model with gene flow scenario.

1 Scenario: five preset scenarios that set all the sliders.

2. Run runs 120 generations (click again to pause), Step advances one, Reset restarts with the same seed and New random seed with a different one.
3. Allele frequency in each population; the dotted line is the average of all of them.
4. Gene diversity $H_T$ and $H_S$, with the curve expected if drift alone were acting.
5. $F_{ST}$ among populations and, dotted, the equilibrium value of the island model.
6. Sliders for the six parameters (table 2.5).
7. Cards with the values for the current generation (table 2.6).
8. A sentence describing what is happening.

## The controls

**Table 2.4.** The five preset scenarios in Scenario.

| Scenario | N | Pops. | m | s | u | Self. | What it shows |
|---|---|---|---|---|---|---|---|
| Small isolated populations (pure drift) | 25 | 6 | 0 | 0 | 0 | 0% | Drift alone |
| Island model with gene flow | 100 | 6 | 0.02 | 0 | 0 | 0% | Balance between drift and migration |
| A selfing plant (90% selfing) | 100 | 6 | 0.005 | 0 | 0 | 90% | Effect of selfing |
| Selection on a favourable allele | 100 | 6 | 0.01 | 0.10 | 0 | 0% | Directional selection |
| Mutation–drift balance | 40 | 6 | 0 | 0 | 0.001 | 0% | Mutation versus drift |

**Table 2.5.** The six parameters and what each one does.

| Slider | Range | Meaning | If you increase it… |
|---|---|---|---|
| N per deme | 10 to 500 | Number of diploid individuals in each population. | Drift weakens: the lines fluctuate less and $H_S$ declines more slowly. |
| Populations | 2 to 10 | Number of populations followed at the same time. | More lines; $F_{ST}$ is estimated with less noise. |
| Migration m | 0 to 0.2 | Fraction of each population that arrives, in each generation, from the pool of all the others. | The populations stay together: $F_{ST}$ falls. |
| Selection s | −0.2 to 0.2 | Advantage of the AA homozygote; with negative values, a disadvantage. | Allele A rises in all populations at once. |
| Mutation u | 0 to 0.01 | Probability that a gene copy changes allele per generation. | New variation appears; fixed populations become variable again. |
| Selfing rate | 0 to 98% | Proportion of seeds produced by self-fertilisation. | Inbreeding increases and the effective size $N_e$ decreases: drift speeds up. |

**Table 2.6.** The readout cards.

| Card | What it is | How to read it |
|---|---|---|
| Generation | Generations elapsed (from 0 to 120). | Time: compare it with $N_e$ to judge whether "a lot" of time has passed. |
| Mean p | Mean frequency of allele A across all populations. | Barely changes under drift alone; rises or falls steadily under selection. |
| $H_S$ | Mean gene diversity within populations. | Starts at 0.5, the maximum with two alleles; declines through drift. |
| $F_{ST}$ | Differentiation among populations. | 0 if all are identical; 1 if each has fixed a different allele (decision rule in section 2.2). |
| Fixed demes | Populations that have already lost one of the two alleles. | A fixed population cannot become variable again without migration or mutation. |
| $N_e$ used | Effective size used for the drift draw. | Equal to N without selfing; smaller with it. |

## The formulae you will see on screen

### 1 LOSS OF DIVERSITY THROUGH DRIFT

$$H_t = H_0\,(1 - 1/2N_e)^t$$

Each generation, a fraction $1/2N_e$ of the heterozygosity is lost. Half of it is lost in about **$1.39 \cdot N_e$ generations**. This is the dotted line in the diversity panel.

### 2 BALANCE BETWEEN DRIFT AND MIGRATION

$$F_{ST} \approx 1 / (1 + 4\,N_e\,m)$$

Wright showed that, in the island model, differentiation stabilises at this value. The product $N_em$ is the number of **migrants per generation**. This is the dotted line in the $F_{ST}$ panel.

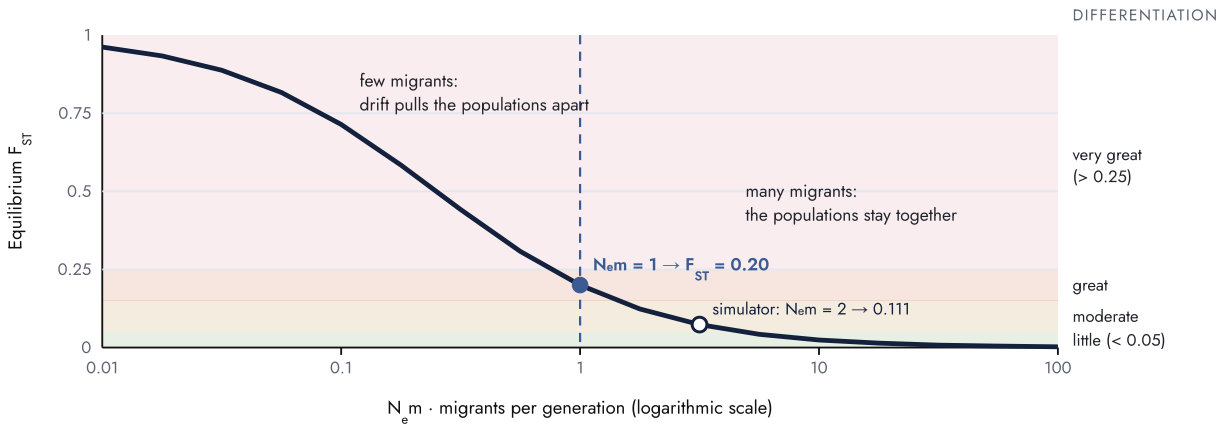



**Figure 2.4.** Equilibrium $F_{ST}$ in the island model as a function of the number of migrants per generation. With just one migrant per generation, differentiation levels off at 0.20. The coloured bands are Wright's (1978) categories of differentiation.

**DECISION RULE · HOW MUCH DIFFERENTIATION IS A LOT?**

| $F_{ST}$ | Differentiation | Usual biological interpretation |
|---|---|---|
| **0 to 0.05** | Little | The populations exchange genes frequently; they behave almost as a single population. |
| **0.05 to 0.15** | Moderate | There is structure, but gene flow is still important. This is common in outcrossing plants. |
| **0.15 to 0.25** | Great | Fairly isolated populations, or populations with limited pollen and seed dispersal. |
| **Over 0.25** | Very great | Strong isolation, selfing, or lineages that diverged long ago. |

Wright's (1978) categories. With highly polymorphic markers, such as microsatellites, $F_{ST}$ has a ceiling below 1; complement it with $G'_{ST}$ or Jost's D (Block 5 chapter).

$$N_em = 1$$

THE ONE-MIGRANT RULE

A single effective migrant per generation is enough to prevent drift from completely differentiating the populations, **regardless of their size**. That is why, in conservation, connecting fragments with just a few individuals has an enormous genetic effect.

**IMPORTANT · THE NUMBER OF MIGRANTS IS ONLY INDICATIVE**

Converting an observed $F_{ST}$ into $N_em$ assumes an island model at equilibrium, with no selection or mutation, and with populations of equal size that exchange migrants equally with all the others. Real populations almost never meet these conditions. Report $N_em$ as a rough guide, never as an actual count of migrants.

## 2.3 Five exercises with the simulator

Each exercise uses one of the preset scenarios. The values quoted as **expected** come from theory. The **simulated** values are the average of 500 replicates of the same model the app uses, calculated for this manual. Because drift is random, your individual run will differ; compare it with the averages, not with an exact number.

## Exercise 1 · Drift alone

**STEP BY STEP**

1. On the home page, click **Play with the simulator**.
2. In **Scenario**, choose **Small isolated populations (pure drift)**: six populations of 25 plants, with no migration, selection or mutation.
3. Click **Run** and watch the lines over the 120 generations.
4. When it finishes, note down $H_S$, $F_{ST}$ and **Fixed demes**. Click **New random seed** and **Run** again to see another possible history.

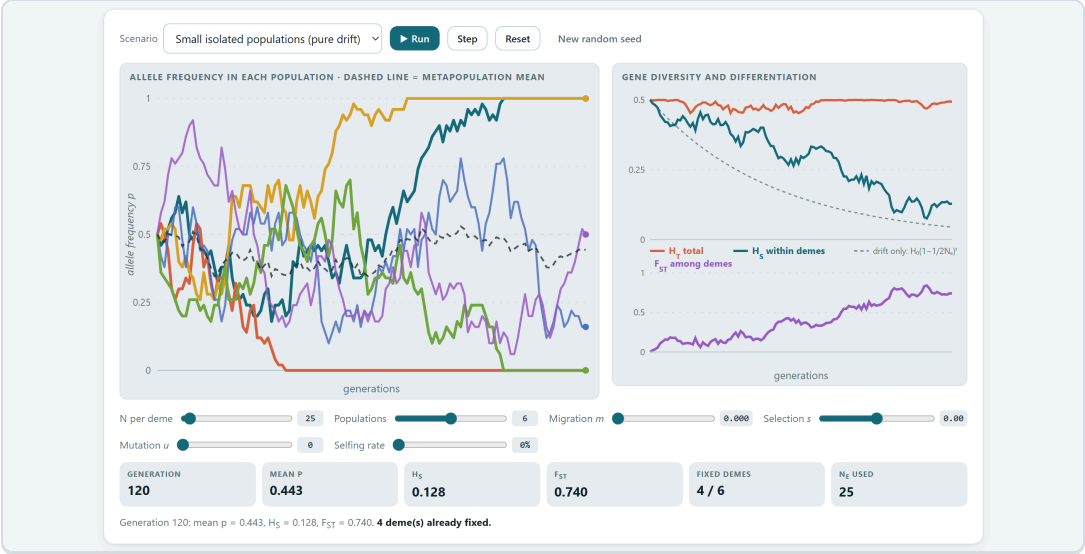



**Figure 2.5.** One run of the pure drift scenario. Four of the six populations fixed an allele, two for A and two for a; $H_S$ fell from 0.5 to 0.13 and $F_{ST}$ rose to 0.74. The overall average (dotted line) barely moved.

**Table 2.7.** Pure drift with $N_e$ = 25: theory versus simulation.

| Measure | Expected | Simulated (500 replicates) |
|---|---|---|
| $H_S$ at generation 50 | 0.182 | 0.181 |
| $H_S$ at generation 120 | 0.044 | 0.042 |
| Generations to lose half of the heterozygosity | 34 | — |
| Mean time to fixation or loss of the allele, starting from p = 0.5 | 69 | — |
| Populations fixed at generation 120 (out of 6) | nearly all | 5.3 |
| Runs with all six populations fixed before generation 120 | — | 44% |

B1

**WHAT YOU LEARNED**

Drift **does not change the average** of the frequencies, but it scatters the populations: each one wanders away from 0.5 in a random direction until it fixes an allele. That is why diversity **within** populations is lost ($H_S$ follows the dotted curve), whereas the diversity of the whole set is preserved, shared out among different populations ($H_T$ stays close to 0.5). The difference between the two is precisely $F_{ST}$. This explains why small, isolated populations of rare or fragmented species tend to have little internal diversity and to be strongly differentiated from one another.

**Try it yourself:** raise N per deme to 250 and repeat. With ten times as many plants, heterozygosity is lost ten times more slowly: after 120 generations $H_S$ has only fallen to 0.39, and it is rare for any population to become fixed.

## Exercise 2 · The island model

**STEP BY STEP**

1. Choose Island model with gene flow: 100 plants per population and $m$ = 0.02, that is, two migrants per generation.
2. Before running, look at the dotted line in the $F_{ST}$ panel: it reads *expected 1/(1+4N$_e$m) = 0.111*.
3. Click Run. The frequency lines move, but they do not drift apart: they chase one another.
4. Move Migration m to 0.005 and click Reset and Run. The expected line rises to 0.333 and the populations move further apart.

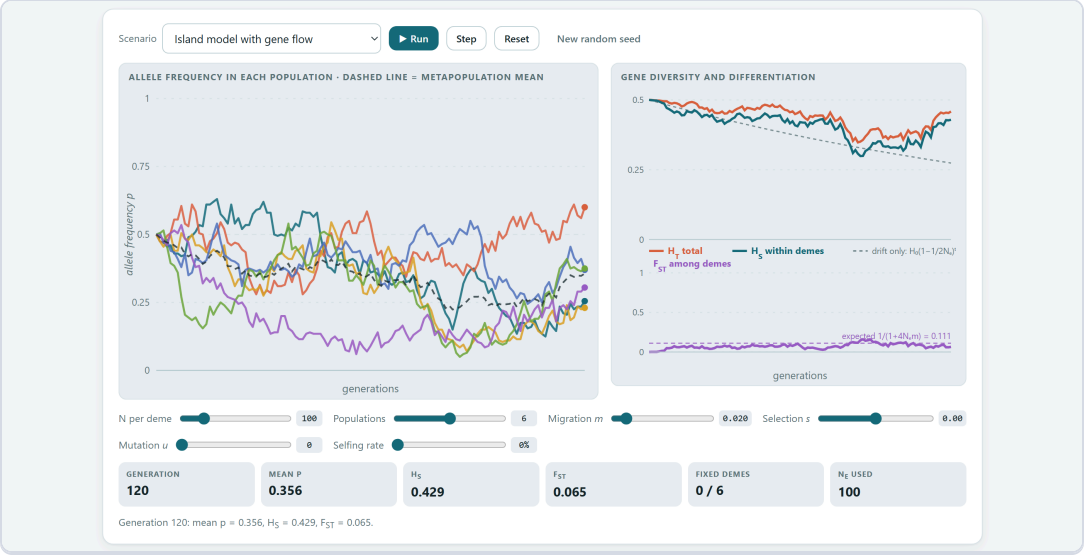



**Figure 2.6.** Island model with N = 100 and $m$ = 0.02. $H_S$ and $H_T$ decline together and $F_{ST}$ fluctuates below the expected line of 0.111, with no population becoming fixed.

**Table 2.8.** Expected $F_{ST}$ with 100 plants per population for different migration rates.

| Migration *m* | 0.001 | 0.005 | 0.01 | 0.02 | 0.05 | 0.10 |
|---|---|---|---|---|---|---|
| Migrants per generation ($N_e m$) | 0.1 | 0.5 | 1 | 2 | 5 | 10 |
| Equilibrium $F_{ST}$ | 0.714 | 0.333 | 0.200 | 0.111 | 0.048 | 0.024 |
| Differentiation (Wright) | very great | very great | great | moderate | little | little |



**NOTE · WHY THE SIMULATED VALUE FALLS SOMEWHAT SHORT**

In 500 replicates with $m = 0.02$, the mean $F_{ST}$ between generations 60 and 120 was **0.094**, slightly lower than the expected 0.111. Wright's formula assumes an infinite number of islands; with only six populations, each one receives part of its own genes back, and differentiation turns out lower. At low migration rates there is a further reason: equilibrium takes about 1/(2m) generations to be reached, so with $m = 0.005$ the simulator's 120 generations are not enough (the simulated mean was 0.21, against an expected 0.333).

## Exercise 3 · A selfing plant

**STEP BY STEP**

1. Choose **A selfing plant (90% selfing)**: 100 plants, $m = 0.005$ and 90% selfing.
2. Look at the **$N_e$ used** card: although there are 100 plants, drift acts as if there were **55**.
3. Click **Run** and watch $F_{ST}$. Then lower **Selfing rate** to 0%, click **Reset** and **Run**, and compare.

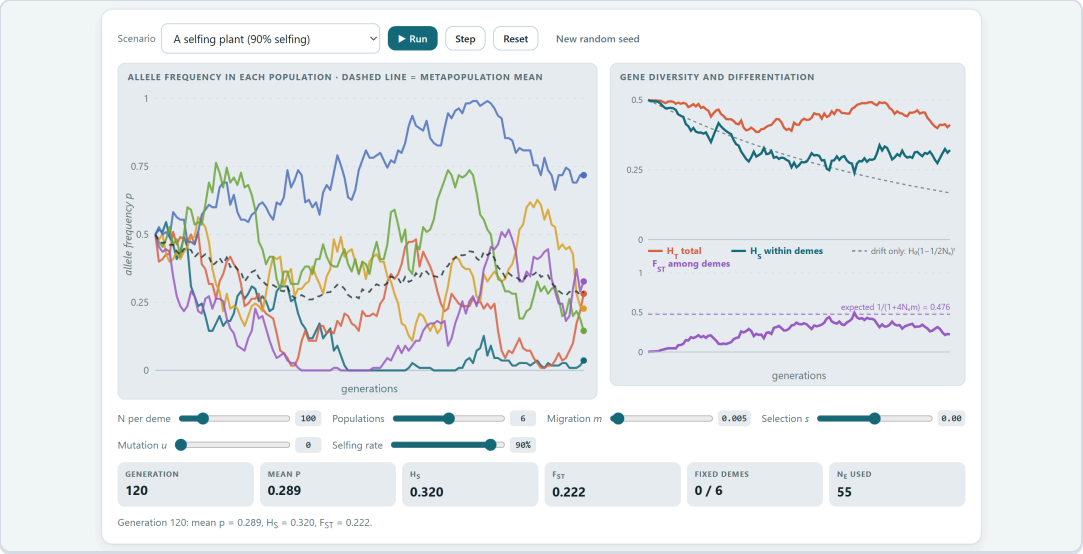



**Figure 2.7.** A plant with 90% selfing. The effective size falls to 55 and the expected $F_{ST}$ line rises to 0.476, compared with 0.333 for the same population without selfing.

**WHAT YOU LEARNED**

Selfing produces inbreeding: with a selfing rate S, the equilibrium inbreeding coefficient is **$F = S/(2 - S)$**, which equals 0.82 when S = 0.9. This inbreeding reduces the effective size to **$N_e = N/(1 + F)$**, because the gametes of a selfing plant are not independent samples of the population. With a smaller $N_e$, drift is stronger and differentiation greater. In 500 replicates, the mean $F_{ST}$ over generations 60 to 120 was **0.35 with selfing** versus **0.21 without it**. Neither has yet reached its equilibrium, but the difference is clear. That is why selfing species usually show more variation among populations than within them.

**NOTE · WHAT THE SIMULATOR DOES NOT SHOW**

The simulator tracks allele frequencies, so selfing acts only through $N_e$. The heterozygote deficit it causes, visible as a high $F_{IS}$ that is similar across all loci, is studied with real data in Block 4.

## Exercise 4 · Selection favouring one allele

**STEP BY STEP**

1. Choose Selection on a favourable allele: $s$ = +0.10, that is, the AA homozygote leaves 10% more offspring.
2. Click Run. All the lines rise together towards 1.
3. Change Selection s to −0.10 and repeat: now allele A disappears.
4. To see when chance wins, set N per deme to 10 and Selection s to +0.01, and repeat several times with New random seed.

**WHAT YOU LEARNED**

When selection pushes in the **same direction** in every population, it moves them together and does not differentiate them: in 500 replicates, the mean frequency of A reached **0.93 at generation 60 and 0.99 at generation 120**, while $F_{ST}$ averaged only 0.047. The selection that does differentiate populations is **divergent** selection, which favours different alleles in different environments. It leaves a typical signature in real data: a locus or a trait far more differentiated than the rest, which is sought by comparing $P_{ST}$ with $F_{ST}$ in Block 9.

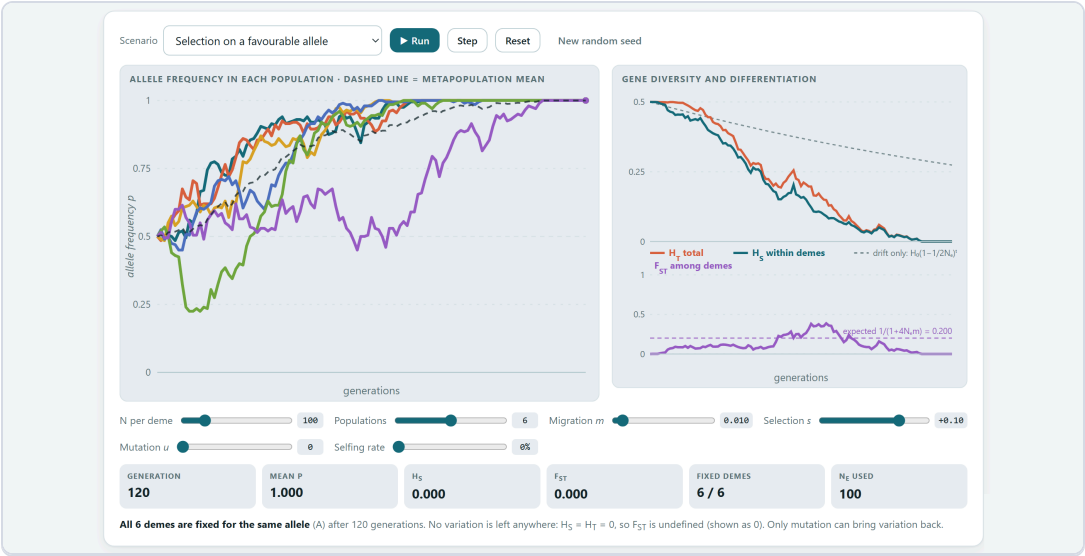



**Figure 2.8.** Directional selection with $s$ = +0.10. Within 120 generations all six populations fixed the favourable allele. $F_{ST}$ remained low, and at the end the app shows it as 0 because there is no variation left to partition.

**DECISION RULE · DOES SELECTION OR DRIFT WIN?**

| $4 N_e \lvert s \rvert$ | Which dominates | What to expect |
|---|---|---|
| **Less than 1** | Drift | The allele behaves almost as if it were neutral; it can be lost even though it is advantageous. Example: $N_e$ = 10 and $s$ = 0.01 give 0.4. |
| **1 to 10** | Both | Selection steers the change, but chance produces variable outcomes among populations. |
| **More than 10** | Selection | The change is almost deterministic. Example: $N_e$ = 100 and $s$ = 0.1 give 40. |

Practical consequence: in small populations, even adaptive variants can be lost by chance. Maintaining large effective sizes also maintains the capacity to adapt.

## Exercise 5 · Mutation–drift balance

**STEP BY STEP**

1. Choose **Mutation–drift balance**: 40 plants per population, no migration and $u$ = 0.001.
2. Click **Run** and watch the populations that reach 0 or 1: every so often they move away again, because mutation reintroduces the lost allele.
3. Raise **Mutation u** to 0.005 and repeat. Diversity within populations stays clearly higher.

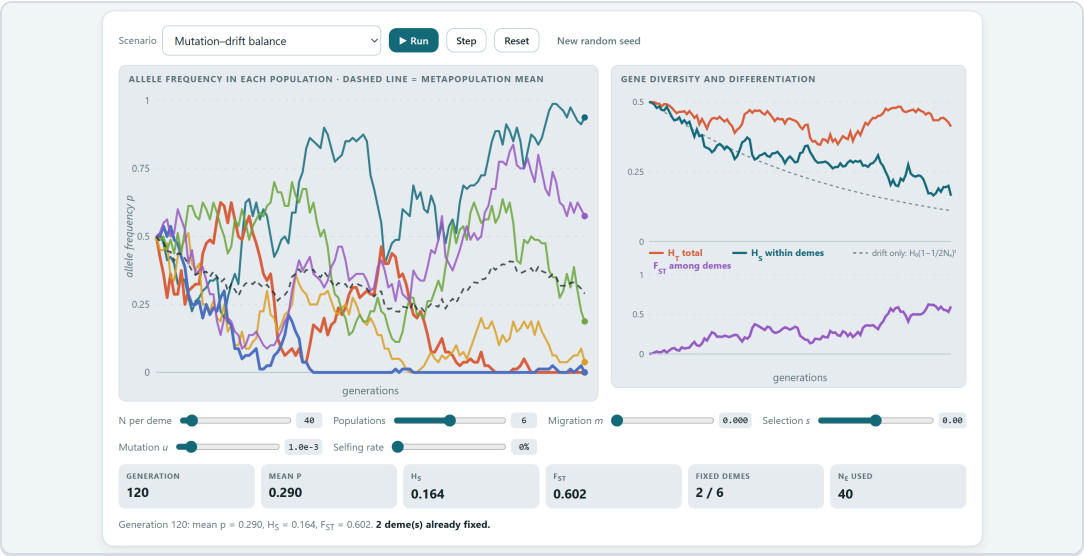



**Figure 2.9.** Mutation and drift with N = 40 and $u$ = 0.001. $H_S$ declines more slowly than the pure-drift curve, and fixed populations do not stay that way for ever.

### WHAT YOU LEARNED

Drift removes variation and mutation creates it; over time the two reach a balance. For a two-allele locus with symmetric mutation, the equilibrium diversity is $\mathbf{H = 4N_e u / (1 + 8N_e u)}$. With $N_e$ = 40 and $u$ = 0.001 it equals 0.12, and with $u$ = 0.005 it rises to 0.31. In the simulator, $H_S$ at generation 120 averaged 0.175, still on its way towards that equilibrium, whereas drift alone would have left it at 0.11. The lesson for real markers is that loci with high mutation rates, such as microsatellites ($10^{-5}$ to $10^{-3}$ per generation), retain far more diversity than sequences, whose rates are thousands of times lower.

### NOTE · CLASSROOM USE

The five exercises work well as a single-session workshop. A useful approach is to ask each team to predict, before clicking **Run**, whether $F_{ST}$ will end up above or below the expected line. The results of all the teams, which used different seeds, are then compared to discuss why drift is a random process and why several populations and several loci are needed to estimate it.

## 2.4 Essential theory in the app

The **The theory, in plain language** section brings together eight expandable topics. Each one presents the central idea, its formulae, a box with the most important rule of thumb and the references it is drawn from. It opens with the **Read the theory** button in the introduction, which automatically expands the first topic.

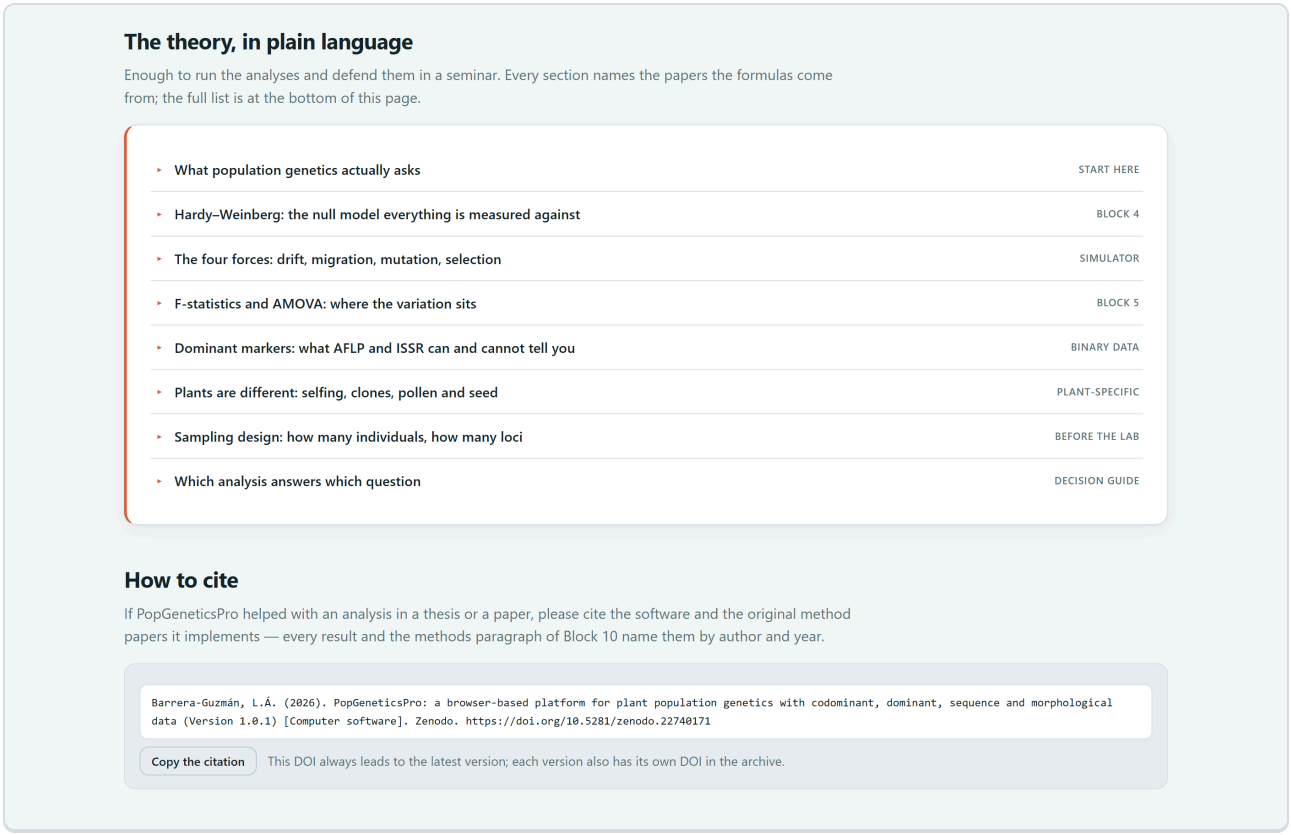



**Figure 2.10.** The eight theory topics. Each label on the right indicates the block or data type it applies to. The citation of the program appears below.

**Table 2.9.** The eight theory topics and where they are developed further in this manual.

| Topic in the app | Label | What you learn | Manual |
|---|---|---|---|
| What population genetics actually asks | start here | The four questions of the discipline, in order: how much variation there is, whether it is organised as random mating predicts, how it is partitioned and what history produced it. | 1 |
| Hardy–Weinberg: the null model everything is measured against | Block 4 | $p^2$, $2pq$, $q^2$; $H_e$, $H_o$ and F; the three causes of a positive F and how to tell them apart. | B4 |
| The four forces: drift, migration, mutation, selection | simulator | Loss of heterozygosity through drift, island equilibrium, mutation rates and why neutral markers are informative about demography. | B1 |
| F-statistics and AMOVA: where the variation sits | Block 5 | $F_{IS}$, $F_{ST}$ and $F_{IT}$; $G_{ST}$, θ, $G'_{ST}$ and Jost's D; when to use AMOVA. | B5 |
| Dominant markers: what AFLP and ISSR can and cannot tell you | binary data | How frequencies are estimated from bands, which analyses remain valid and which do not. | B3 B5 |
| Plants are different: selfing, clones, pollen and seed | plant-specific | Mixed mating systems, clonality, pollen and seed dispersal, fine-scale spatial structure. | B2 B9 |

**Table 2.9.** The eight theory topics and where they are developed further in this manual. *(continued)*

| Topic in the app | Label | What you learn | Manual |
|---|---|---|---|
| Sampling design: how many individuals, how many loci | before the lab | Minimum sample sizes and number of loci according to the objective (table 2.10). | 1 |
| Which analysis answers which question | decision guide | A quick guide from the question to the analysis and the block. | 1 |



80–95%

THE NUMBER WORTH REMEMBERING

In most **outcrossing** plants, between 80 and 95% of the genetic variation lies **within** populations and only 5 to 20% among them. In **selfing** plants the proportion is usually reversed. Knowing which kind of species you are studying changes how you read every result.

**Table 2.10.** Sampling design suggested in the app according to the objective of the study.

| Objective | Individuals per population | Loci |
|---|---|---|
| Diversity indices ($H_e$, $N_a$) | 20 to 30 | 8 to 10 microsatellites |
| Allelic richness and rare alleles | 30 to 50 | 10 to 15 microsatellites |
| $F_{ST}$ and AMOVA | 20 to 25, in at least 5 populations | 10 or more microsatellites, or more than 100 AFLP bands |
| Bayesian clustering | 25 to 30 | 15 or more microsatellites |
| Bottleneck tests | 30 or more | 15 to 20 microsatellites |
| Fine-scale spatial structure | 100 or more individuals with coordinates | 8 or more microsatellites |

✓ DECISION RULE · MORE LOCI OR MORE INDIVIDUALS?

| If your objective is... | ... invest in |
|---|---|
| **Estimating differentiation** ($F_{ST}$, AMOVA, distances) | **More loci.** Each locus is an independent replicate of the history of the populations, and the variance of $F_{ST}$ decreases faster when you add loci than when you add individuals. |
| **Finding rare or private alleles** | **More individuals.** An allele at a frequency of 2% has only a 64% probability of appearing in 25 diploid plants. |
| **Detecting kinship or spatial structure** | **Both**, with mapped individuals and highly polymorphic markers. |

## 2.5 How to cite and references

If PopGeneticsPro helped you with a thesis or a paper, cite two things: **the program**, so that readers know which tool and version were used for the calculations, and **the original papers describing the methods**, whose authors deserve the scientific credit for each index or test.

**Figure 2.11.** The How to cite section with the citation ready to copy and the beginning of References behind the calculations.

1. On the home page, click **How to cite**; the page scrolls down to the citation box.
2. Click **Copy the citation**. The button changes to ✓ *Copied* for a couple of seconds.
3. Paste the citation into your reference manager or your document and adapt it to the journal's style.
4. Also cite the methods you used (table 2.11). Each result in the app names its method, and the Block 10 report gathers them in the methods paragraph.

Barrera-Guzmán, L. Á. (2026). PopGeneticsPro: a browser-based platform for plant population genetics with codominant, dominant, sequence and morphological data (Version 1.0.1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.22740171

**NOTE · THE DOI**

The digital object identifier (DOI) in the citation, 10.5281/zenodo.22740171, is the **concept DOI**: it always resolves to the most recent version of the program. Each version also has its own DOI, shown in the record for that version in the archive; the one for version 1.0.0, for example, is 10.5281/zenodo.22740172. Cite the concept DOI together with the version number; if the journal requires that anyone be able to retrieve exactly the version you used, cite the DOI of that version.

**Table 2.11.** Which method to cite depending on the analysis you report. The complete list, with publication details, is in References behind the calculations and in appendix E.

| If you report… | Cite | Block |
|---|---|---|
| Gene diversity $H_e$ and Nei's distance | Nei (1973, 1978) | 3 6 |
| Sample size per population | Hale, Burg and Steeves (2012) | 2 |
| Frequencies with dominant markers | Lynch and Milligan (1994) | 3 5 |
| Weir and Cockerham's θ | Weir and Cockerham (1984) | 5 |
| $G'_{ST}$ and Jost's D | Hedrick (2005); Jost (2008) | 5 |
| AMOVA and $\Phi_{PT}$ | Excoffier, Smouse and Quattro (1992); Peakall, Smouse and Huff (1995) | 5 |
| Bayesian clustering and ΔK | Pritchard, Stephens and Donnelly (2000); Evanno, Regnaut and Goudet (2005) | 7 |
| Tajima's D and haplotype networks | Tajima (1989); Bandelt, Forster and Röhl (1999) | 8 |
| Sequence likelihood and among-site rate variation | Felsenstein (1981); Yang (1994) | 8 |
| Molecular dating | To et al. (2016); Drummond et al. (2006); Gernhard (2008); Parham et al. (2012) | 8 |
| Reference substitution rates | Wolfe, Li and Sharp (1987); Kay, Whittall and Hodges (2006) | 8 |
| Geological time scale | Cohen et al. (2013, updated) | 8 |
| Bottlenecks | Cornuet and Luikart (1996) | 9 |
| Spatial structure and the Sp statistic | Vekemans and Hardy (2004) | 9 |
| Mating system | Ritland (2002) | 9 |

**! IMPORTANT · REPORT THE SETTINGS AS WELL**

Citing the program is not enough for another researcher to repeat your analysis. In the methods, state the version of PopGeneticsPro, the settings you changed (number of permutations, bootstrap replicates, multiple-testing correction, substitution model) and the **random seed**. The automatic report in Block 10 drafts that paragraph with the values you actually used.

With the home page, the simulator and the theory at hand, the next step is to load your own data. The Block 2 chapter explains the formats the app accepts, the built-in data sheet, the template designer and quality control, with the six examples worked through step by step.

# B2



CHAPTER THREE · BLOCK 2 · DATA

# Data and quality control

Every analysis starts here. Block 2 reads your file, shows you how it was interpreted so that you can confirm it, checks the quality of the matrix before computing a single statistic and tells you which analyses your data support. If you are not sure how to arrange your information, it prepares a template for you; if a file will not upload, you can paste the data into its built-in sheet.


3.1 Input formats

3.2 The built-in data sheet

3.3 The template designer

3.4 Interpretation and the role of each column

3.5 Quality control

3.6 Which analyses your data allow

3.7 Step by step with the six examples

**THEORY IN BRIEF · WHY CHECK BEFORE COMPUTING**

Statistical programs compute whatever they are given, even if it is wrong. An allele typed as `180` (with the letter O) counts as a new allele and inflates diversity; a plant entered twice looks like a clone; a row without a population is left out of every group; a latitude of 190° moves a population to the other side of the planet. None of these errors stops the calculation, but all of them change the results. That is why Block 2 follows one rule: **nothing is assumed silently**. Every decision the app makes is shown, explained and can be corrected before you continue.

FASTA, VCF, .str, .gen and .arp files carry their own structure: they go straight from 1 to 4.

**Figure 3.1.** The five cards of Block 2, in the order in which they appear. Each one opens when you finish the previous one.

## 3.1 Input formats

There are four ways to get data into the program: drag in a file, paste the data into the built-in sheet, fill in a template made by the app or load one of the examples. All of them end in the same checks.

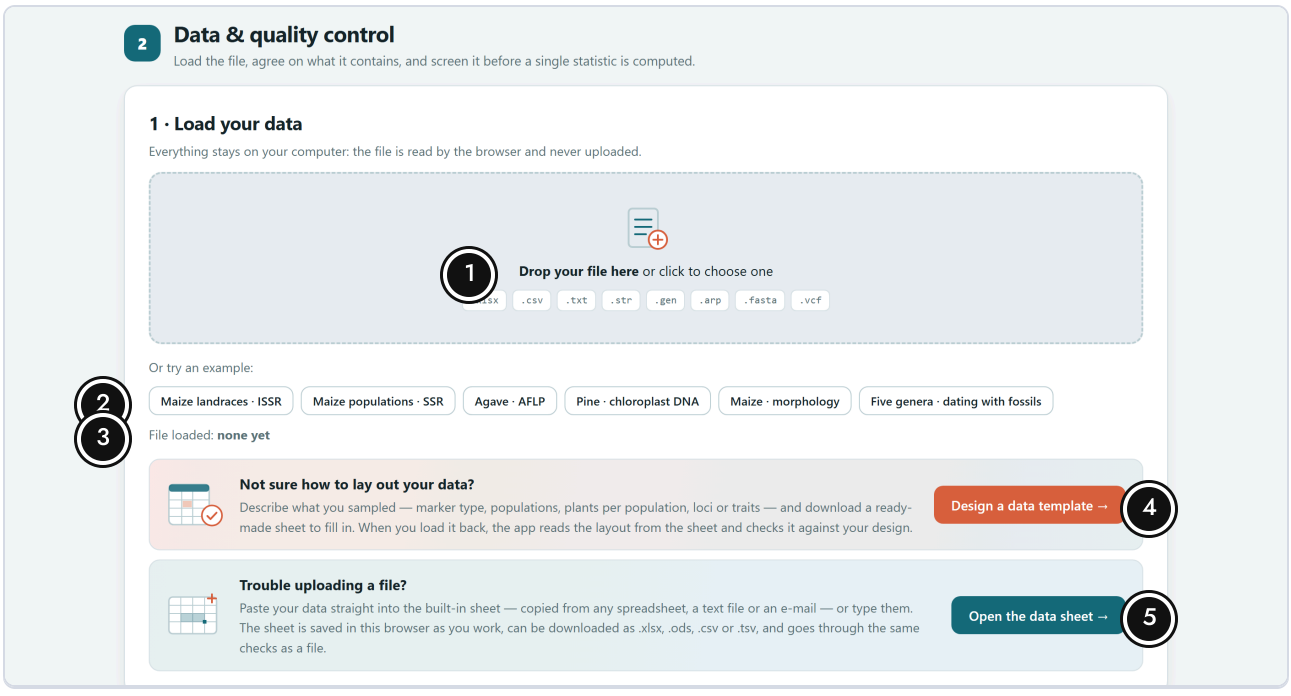



**Figure 3.2.** The `1 · Load your data` card.

1. Drag your file here or click to choose it. The accepted extensions are listed below.
2. Or try an example: the six built-in examples (section 3.7).
3. Name of the loaded file.
4. Design a data template: opens the template designer (section 3.3).
5. Open the data sheet: opens the built-in data sheet (section 3.2).

**Table 3.1.** Formats read by Block 2.

| Format | What it contains | What the app does | Pop. · coord. |
|---|---|---|---|
| **Spreadsheet** `.xlsx` `.xls` `.xlsm` | One table per sheet; any data type. | If the workbook has several sheets, the Sheet selector appears. The data then go through the interpretation and the role of each column. | ● ● |
| **Delimited text** `.csv` `.txt` `.tsv` | The same table, separated by commas, semicolons or tabs. | Detects the separator and follows the same path as a spreadsheet. | ● ● |
| **Genotype file for Bayesian clustering** `.str` `.stru` | One or two rows per individual, with the ID, the population and one allele per column; −9 = missing. | Detects whether there are one or two rows per individual and reads the locus names from the first row. | ● — |
| **Population file** `.gen` | Title, locus names and blocks separated by the word `POP`; genotypes such as `182186`. | Each `POP` block becomes a population. | ● — |
| **Project file** `.arp` | `[Profile]` and `SampleData` sections per population. | Reads each sample with its population name. | ● — |
| **Aligned sequences** `.fasta` `.fas` `.fa` | A `>name` line followed by the sequence. | Groups identical sequences into haplotypes. If the header is `>ID \| Population \| Region`, it also reads the population and the region. | ◐ — |
| **Variants** `.vcf` | SNPs with the genotypes of each sample. | Takes the first alternative allele at each site. | — — |

**NOTE · FORMATS THAT CARRY NO POPULATIONS OR COORDINATES**

FASTA, VCF, `.str`, `.gen` and `.arp` files go straight to quality control. Because they rarely include regions or coordinates (and VCF files do not even include populations), the Add sample information card appears. There you load a sheet with an **ID** column, holding the same names as the file, and any of **Population**, **Region**, **Latitude** and **Longitude**. The Download an empty sample sheet with these IDs button gives you that sheet with the IDs already filled in (figure 3.20).

## How to arrange a table

A table is read without trouble if it follows one simple rule: **one row per individual and one column per variable**, with the column names in the first row. Figure 3.3 shows the four layouts the app recognises.

**A · CODOMINANT, ONE ALLELE PER COLUMN**

| ID | Pop | SSR1 | SSR1 | SSR2 | SSR2 | … |
|---|---|---|---|---|---|---|
| P01 | Tlaxco | 182 | 186 | 140 | 142 | |
| P02 | Tlaxco | 184 | 184 | 0 | 0 | |
| P03 | Apizaco | 182 | 188 | 144 | 146 | |

two columns with the same name = one locus · 0 = missing · 184 | 184 = homozygote
the safest layout for entering data by hand

**B · CODOMINANT, GENOTYPE IN ONE CELL**

| ID | Pop | SSR1 | SSR2 | SSR3 |
|---|---|---|---|---|
| P01 | Tlaxco | 182/186 | 140/142 | 96/98 |
| P02 | Tlaxco | 184/184 | 0/0 | 96/96 |
| P03 | Apizaco | 182/188 | 144/146 | 100/102 |

accepted separators: 182/186 · 182|186 · 182-186 · 182 186 · 182186 · A/G
under `How genotypes are written` choose "Whole genotype in one cell"

**C · DOMINANT, INDIVIDUALS IN ROWS**

| Plant | Population | B01 | B02 | B03 | B04 | … |
|---|---|---|---|---|---|---|
| Oax_01 | Oaxaca | 1 | 0 | 1 | 1 | |
| Oax_02 | Oaxaca | 1 | 1 | 0 | 1 | |
| Pue_01 | Puebla | 0 | 1 | ? | 1 | |

one column per band · 1 present, 0 absent (also p/a, +/−, yes/no) · ? = missing

**D · DOMINANT, BANDS IN ROWS (TRANSPOSED)**

| ISSR | Ancho | Azul | Bofo | Conejo | … |
|---|---|---|---|---|---|
| 7945 | 0 | 1 | 1 | 1 | |
| 7945 | 0 | 0 | 0 | 0 | |
| P2 | 1 | 1 | 1 | 0 | |

the first column repeats names (primers) → the app infers that the rows are bands and turns the table round: each column (landrace, accession) becomes a row

**Figure 3.3.** The four table layouts the app recognises on its own. A and C are the recommended ones. D is the usual way of scoring a gel, and the app transposes it automatically; the maize landraces example with ISSR is written this way.

**Table 3.2.** How to write each data type.

| Type | How each cell is written | Missing values and common errors |
|---|---|---|
| **Codominant** | The fragment size in base pairs (182) or the number of repeats. A homozygote repeats the allele: 182 \| 182. SNPs are written as nucleotides: A \| G. | 0 or an empty cell. If only one allele amplified, leave **both** cells empty: partial genotypes are discarded. |
| **Dominant** | 1 if the band is present and 0 if it is absent. It also accepts p/a, present/absent, +/− and yes/no. | Empty, ?, −9 or −1. Any other value (2, "x1") is flagged as an error. |
| **Haploid** | One code per locus: the size of a chloroplast microsatellite or the name of the haplotype (H1, H2). | Empty, or the code you declare as missing. A cell containing "182/186" is flagged, because a haploid does not have two alleles. |
| **Sequences** | In FASTA, or one column per gene region with the complete sequence. IUPAC letters (A C G T N R Y S W K M B D H V); gaps as −. | ? or N. U is converted to T. Sequences of different lengths are flagged as unaligned. |
| **Morphological** | Numbers. The decimal comma is accepted (13,5). Class characters (colour) can be entered as text. | Empty or NA. Text in a numeric column becomes missing and is flagged. |

**VALUES THAT ARE ALWAYS MISSING**

An empty cell, `NA`, `n/a`, `NaN`, `?`, `.`, `..`, `x`, `null`, `missing`, `-9`, `-99` and `-999`. In addition, whatever code you enter under Code used for missing data (by default, 0 for codominant markers).

**COLUMN NAMES THE APP RECOGNISES**

**ID:** ID, Ind, Individual, Sample, Plant, Tree, Accession, Genotype, Name, Clave, Individuo, Muestra, Planta. **Population:** Pop, Population, Población, Site, Locality, Race, Raza, Landrace, Sitio, Localidad, Procedencia. **Region:** Region, Zone, Group, State, Estado, Area, Province, Country, País. **Coordinates:** Lat, Latitude, Latitud, Northing; Lon, Long, Longitude, Longitud, Easting. The whole name counts, with a plural or a unit in brackets: `Longitud_mazorca` is a trait.



**IMPORTANT · WHAT CONFUSES ANY PROGRAM**

- **Merged cells**, titles above the table or rows of totals at the bottom.
- **Colours that carry meaning** ("doubtful ones in red"): colour does not travel with the data. Use a column.
- **Formulas** instead of values, or several tables on the same sheet.
- **Repeated IDs** and population names written in two ways (*Tlaxco* and *tlaxco*), which the app would take as different populations.
- **The letter O instead of zero** (1O0) or the letter l instead of one.

## 3.2 The built-in data sheet

If a file will not upload, if you were sent the data in the body of an email or if you prefer to enter them directly, open the data sheet with Open the data sheet. It works like a simple spreadsheet: you paste or type, correct and, when it is ready, send it for analysis.

1. Click Open the data sheet. An empty grid appears, with columns A, B, C…
2. Copy the table from your spreadsheet, a text file or an email, including the row of names. Click cell **A1** and paste with Ctrl + V. Columns separated by tabs, commas or semicolons are split automatically.
3. Correct whatever is needed: type over a cell, clear it with Delete or insert rows and columns with the toolbar.
4. Type a name in File name and save a copy with one of the Save as buttons.
5. Click Use these data. The data follow the same path as a file: interpretation, column roles and quality control.

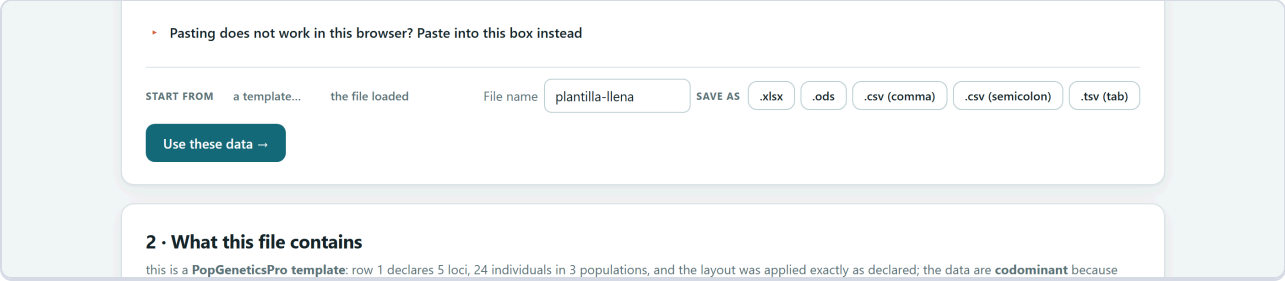



**Figure 3.4.** The foot of the data sheet: start from a template or from the loaded file, save in five formats and send the data for analysis with Use these data.

B2

**Table 3.3.** Keyboard in the sheet.

| Keys | Action |
|---|---|
| Ctrl + V / C / X | Paste, copy, cut |
| Ctrl + Z / Y | Undo, redo |
| Ctrl + A | Select the used range |
| Arrows · Tab · Enter | Move (with Shift, move back or extend the selection) |
| Home · End · PgUp · PgDn | Jump to the edges |
| F2 or double-click | Edit the cell |
| Delete | Clear the selection |
| Esc | Cancel editing |

**Table 3.4.** Formats for saving the sheet.

| Button | Use it when... |
|---|---|
| .xlsx | you want to keep working in a spreadsheet. |
| .ods | you use an open-format office suite. |
| .csv (comma) | you need plain text with a decimal point. |
| .csv (semicolon) | your computer uses a decimal comma, as is common in many countries. |
| .tsv (tab) | you are going to paste the data into another program. |

20,000

SHEET CAPACITY

Up to **20,000 rows and 5,000 columns**. The sheet is saved automatically in your browser while you work: if you close the app by mistake, the data are still there when you come back.

**Table 3.5.** The toolbar and the other controls of the sheet.

| Control | What it does |
|---|---|
| Undo · Redo | Undoes or redoes the last change, even a whole paste. |
| Rows: + above · + below · − delete · +100 | Inserts rows above or below the selection, deletes the selected rows or adds 100 empty rows at the end. |
| Columns: + left · + right · − delete · width | The same for columns; the slider sets their display width. |
| First row = column names | Untick it if your data start without a row of names. |
| Clear the sheet | Empties the sheet (this can be undone). |

**Table 3.5.** The toolbar and the other controls of the sheet. *(continued)*

| Control | What it does |
|---|---|
| Start from : a template... · the file loaded | Opens the template designer, or copies the file you loaded into the sheet so that you can correct it there. |
| Pasting does not work in this browser? | An alternative text box: paste there and click Put it in the sheet at the selected cell. Useful if the browser blocks the clipboard. |

> **NOTE · CORRECTING IN THE SHEET WHAT QUALITY CONTROL FOUND**
>
> When you load a file and quality control finds problematic values, the Correct them in the data sheet button opens your data in the sheet with those cells **marked in red**. Hovering the mouse over a marked cell shows the reason. You correct them, click Use these data and the checks are run again (section 3.5).



## 3.3 The template designer

The best way to avoid formatting errors is not to invent the format. With Design a data template you describe what you sampled and the app writes the sheet for you: IDs and populations already in place, one column per locus or trait and a first row that declares the layout, so that there is nothing to guess when you load it.

**Table 3.6.** The six data types of the designer (step What did you score?).

| Card | For | What step 3 asks for |
|---|---|---|
| Codominant markers | Microsatellites, isozymes | Number of loci or their names, prefix, ploidy, genotype layout (two columns or one cell) and missing-data code. |
| SNP genotypes | Biallelic SNPs, genotyping by sequencing | The same; alleles are written as nucleotides. |
| Dominant markers | AFLP, ISSR, RAPD, SRAP | Bands per primer (one name per primer and how many bands you scored) and whether the sheet has individuals or bands in rows. |
| Haplotype codes | Chloroplast microsatellites, PCR-RFLP | Loci with one code per cell. |
| DNA sequences | Aligned cpDNA, mtDNA, ITS | Gene regions and their aligned length; it proposes a FASTA skeleton with a sample sheet. |
| Morphological traits | Measurements, counts, scales | Name, unit and type of each trait (quantitative, discrete or ordinal). |

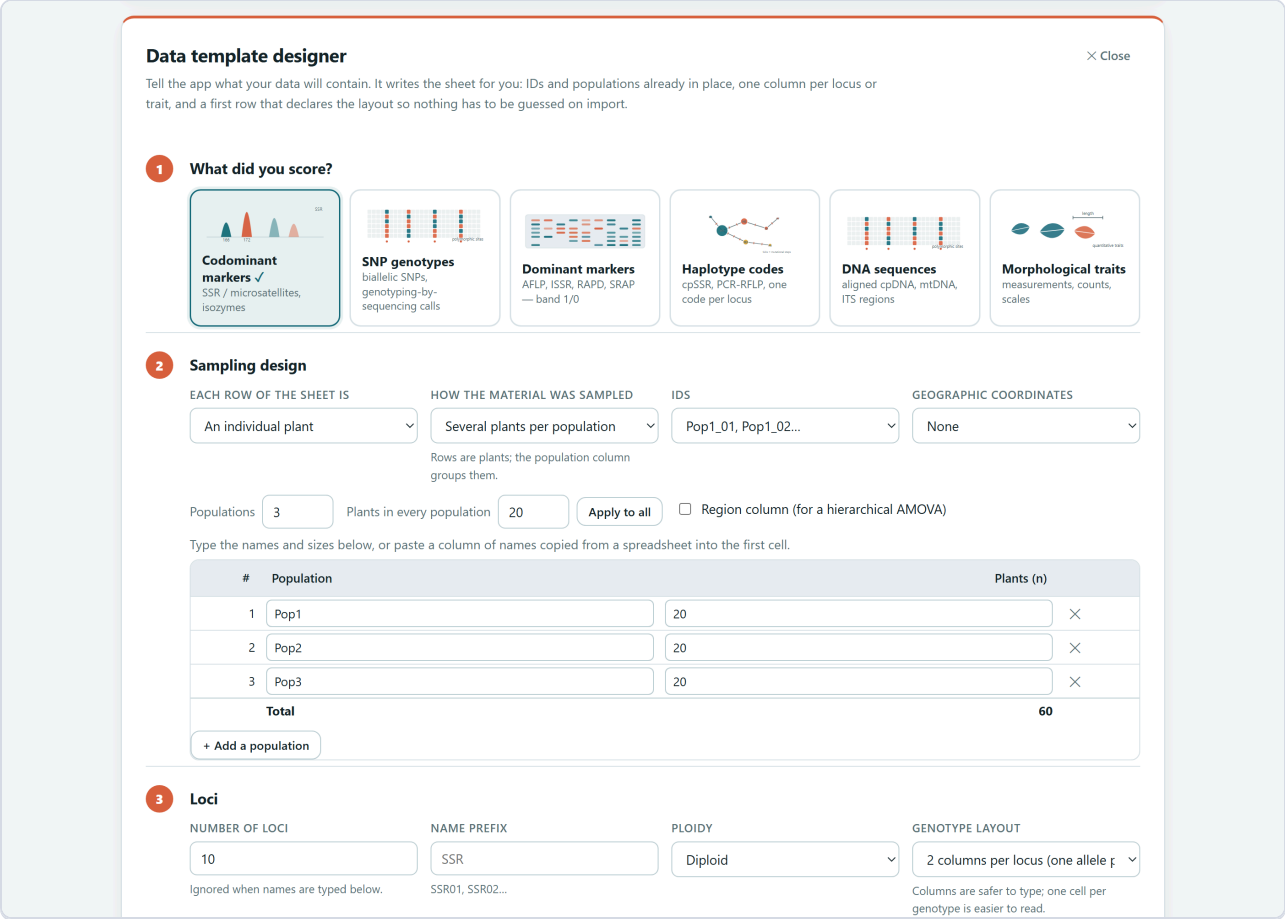



**Figure 3.5.** The template designer. **1** data type; **2** sampling design, with the table of populations and the number of plants in each; **3** loci, bands, regions or traits. The numbers in the circles belong to the app itself.

**Table 3.7.** Sampling design options (step Sampling design).

| Option | Meaning |
|---|---|
| Each row of the sheet is | What each row is: a plant, a landrace, an accession, a variety, a clone, a genotype, a sample, a bulked population or an OTU. The chosen name is used in every table and text. |
| How the material was sampled | Several plants per population, or one bulked sample per unit (*bulk*). With bulked samples there are no within-population analyses. |
| IDs | Pop1_01, Pop1_02… · 001, 002… · left empty for you to type in. |
| Geographic coordinates | None, one point per population or one point per plant. With one point per plant and at least 10 plants, fine-scale spatial structure becomes available (Block 9). |
| Region column · Group column | Add a column for a hierarchical AMOVA or for a group such as the racial complex. |
| Table of populations | Name and number of plants of each population. You can paste a column of names copied from a spreadsheet into the first cell; Apply to all gives them all the same size. |

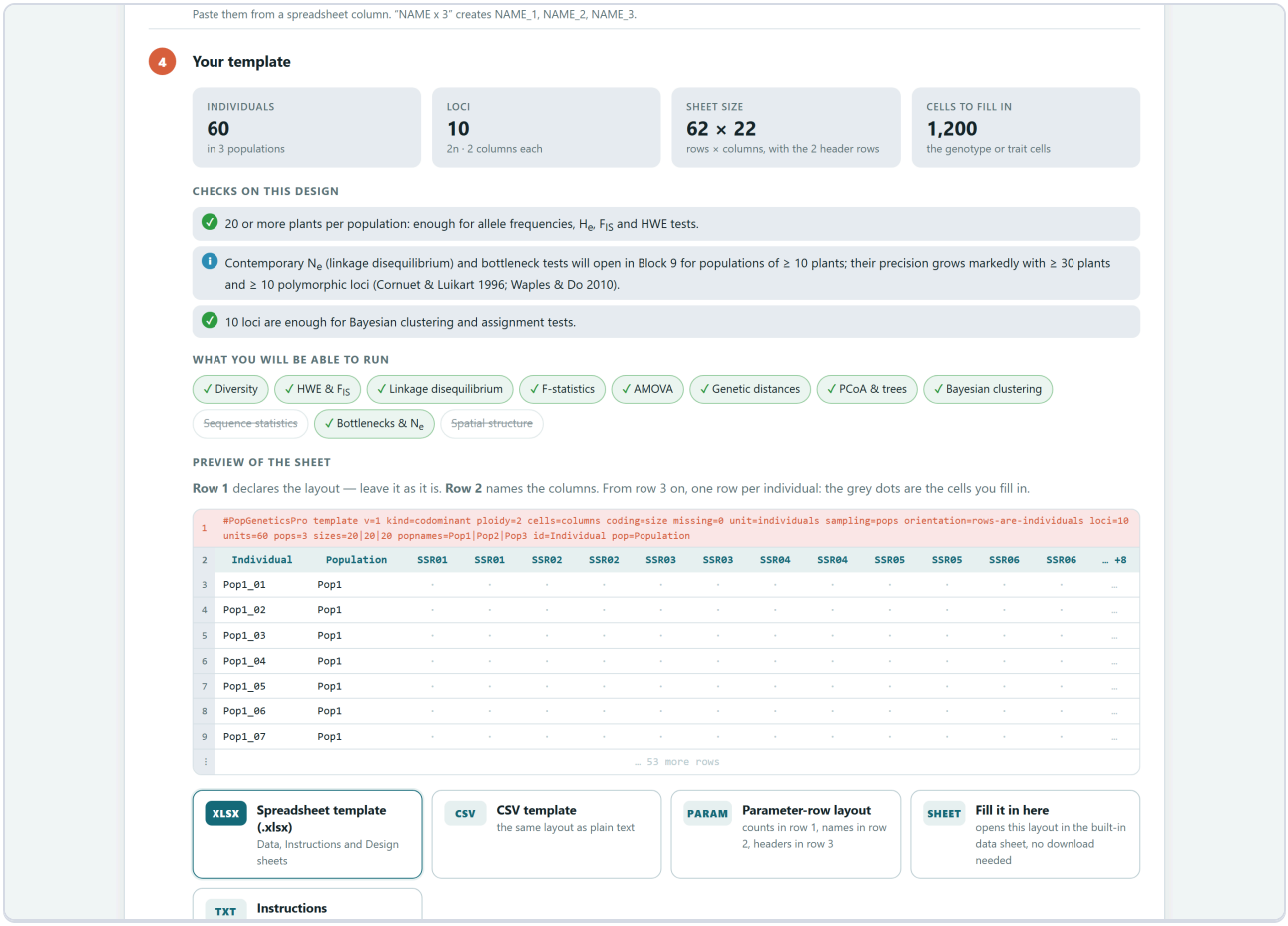



**Figure 3.6.** The Your template step: sheet size and cells to fill in, design checks, analyses you will be able to run (those you cannot are struck through), sheet preview and download buttons. In the preview, row 1 declares the layout and the grey dots are the cells to fill in.

**Table 3.8.** What you can download from the designer.

| Button | File | When to use it |
|---|---|---|
| XLSX | Spreadsheet with three sheets: data, instructions and design. | The recommended option for filling in on a computer. |
| CSV | The same template as text. | For programs that cannot read .xlsx. |
| PARAM | Layout with a parameter row: counts in row 1, names in row 2 and headers in row 3 (figure 3.7). | If you will also use programs that read that layout. Only enabled for designs that fit it. |
| SHEET | Opens the template in the built-in data sheet. | For entering data directly, without downloading anything. |
| FASTA | Skeleton with one header per sequence. | With sequences: paste each sequence below its name. |
| TXT | Filling-in instructions and design checks. | To share with whoever enters the data in the laboratory. |

## The row that declares the layout

The first row of a template starts with `#PopGeneticsPro template` and holds, cell by cell, `key=value` pairs. When the sheet is loaded, the app reads it and applies the layout exactly as declared, without guessing. **Do not delete or modify it.**

**Table 3.9.** Keys in row 1 of a template.

| Key | Meaning | Key | Meaning |
|---|---|---|---|
| `kind` | data type | `loci` | number of loci, bands or traits |
| `ploidy` | ploidy | `units` | number of data rows |
| `cells` | one allele per column or genotype in one cell | `pops` | number of populations |
| `coding` | sizes, nucleotides, 0/1 or values | `sizes` | plants per population (8\|8\|8) |
| `missing` | missing-data code | `popnames` | population names |
| `unit` · `sampling` · `orientation` | what each row is, how it was sampled and whether the rows are individuals or bands | `id` · `pop` · `region` | names of the label columns |

| | A1 · loci | B1 · individuals | C1 · populations | D1, E1 · plants per population | | F1 · regions | G1, H1 · plants per region | |
|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 5 | 2 | 3 | 2 | 2 | 3 | 2 |
| 2 | Title | | | SanJuan | LaCumbre | | Norte | Sur |
| 3 | Sample | Pop | SSR01 | | SSR02 | | SSR03 | |
| 4 | SanJuan_01 | SanJuan | 182 | 186 | 140 | 142 | 96 | 98 |
| 5 | SanJuan_02 | SanJuan | 184 | 184 | 138 | 142 | 96 | 96 |

…

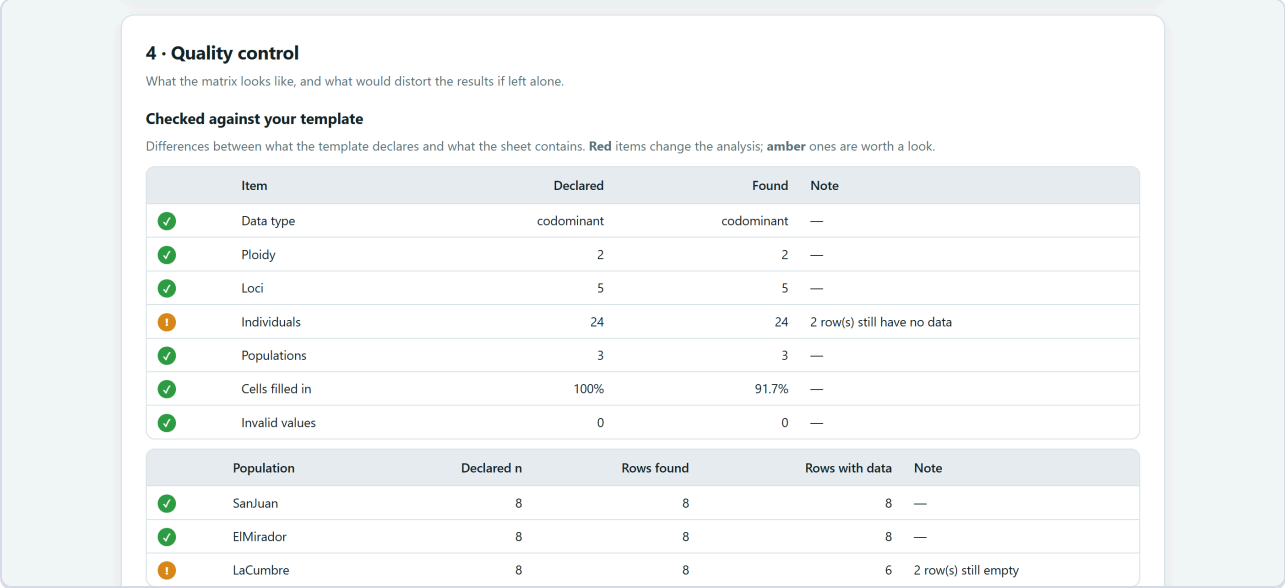
**Row 2:** title in A2; the names of the populations and regions go below their sizes.

**Row 3:** headers; each locus takes up two columns and only the first one is named. The data start in row 4.

**Figure 3.7.** The layout with a parameter row for 3 loci, 5 individuals in 2 populations (3 and 2 plants) and 2 regions. The app recognises it on its own when it is loaded and converts the regions into a column. In this layout, a missing dominant band is usually written as –1.

**EXAMPLE · FROM THE TEMPLATE TO THE VERIFICATION**

1. In the designer, choose **Codominant markers**, three populations (SanJuan, ElMirador and LaCumbre) with 8 plants each, and 5 loci.
2. In **Your template**, click **Fill it in here**. The template opens in the data sheet with 24 rows ready.
3. Enter the genotypes. For this example, leave the last two plants of LaCumbre blank.
4. Click **Use these data** and then **Build the dataset and run the checks**. The interpretation card says *this is a PopGeneticsPro template*: nothing was guessed.
5. Quality control shows **Checked against your template** (figure 3.8): what was declared against what was found.



**4 · Quality control**

What the matrix looks like, and what would distort the results if left alone.

**Checked against your template**

Differences between what the template declares and what the sheet contains. **Red** items change the analysis; **amber** ones are worth a look.

| | Item | Declared | Found | Note |
|---|---|---|---|---|
| ✓ | Data type | codominant | codominant | — |
| ✓ | Ploidy | 2 | 2 | — |
| ✓ | Loci | 5 | 5 | — |
| ! | Individuals | 24 | 24 | 2 row(s) still have no data |
| ✓ | Populations | 3 | 3 | — |
| ✓ | Cells filled in | 100% | 91.7% | — |
| ✓ | Invalid values | 0 | 0 | — |

| | Population | Declared n | Rows found | Rows with data | Note |
|---|---|---|---|---|---|
| ✓ | SanJuan | 8 | 8 | 8 | — |
| ✓ | ElMirador | 8 | 8 | 8 | — |
| ! | LaCumbre | 8 | 8 | 6 | 2 row(s) still empty |

**Figure 3.8.** Verification against the template. Everything matches except two LaCumbre rows that are still empty, marked in amber: the app detects them and does not mistake them for real plants. Items in **red** would change the analysis (for example, a number of loci different from the one declared); those in **amber** deserve a review.

**DECISION RULE · THE DESIGN CHECKS**

| If the design has... | the designer warns that... |
|---|---|
| **Fewer than 5 plants** in a population | this is not enough for any within-population test; 20 to 30 are needed to estimate frequencies well (Hale et al., 2012). |
| **20 or more plants** per population | this is enough for allele frequencies, $H_e$, $F_{IS}$ and Hardy–Weinberg tests. |
| **Fewer than 5 polymorphic loci** | multilocus analyses (Bayesian clustering, assignment, $\bar{r}_d$) are not reliable; clustering is usually run with 10 or more. |
| **Few populations per region** | the number of possible permutations limits the smallest P value of $\Phi_{CT}$ in the AMOVA (Fitzpatrick, 2009). With 2 + 2 populations there are only 3 distinct arrangements and P never falls below 0.33. |
| **10 or more plants** per population (codominant) | $N_e$ from linkage disequilibrium and the bottleneck tests of Block 9 become available; their precision improves greatly with 30 or more plants and 10 or more polymorphic loci. |

**NOTE · CONVERTING A DATASET YOU HAVE ALREADY LOADED**

At the end of Block 2, the Export in another layout button opens the designer with your data already loaded. This lets you save, for example, a `.str` or `.gen` file as a PopGeneticsPro template with its populations, in the layout with a parameter row, or an alignment as FASTA with its sample sheet.

## 3.4 Interpretation and the role of each column

A plain table does not say what it is. Before computing, the app works out three things: whether the rows are individuals or loci, what type of marker the table contains and what role each column plays. It shows them to you on two cards so that you can confirm them.

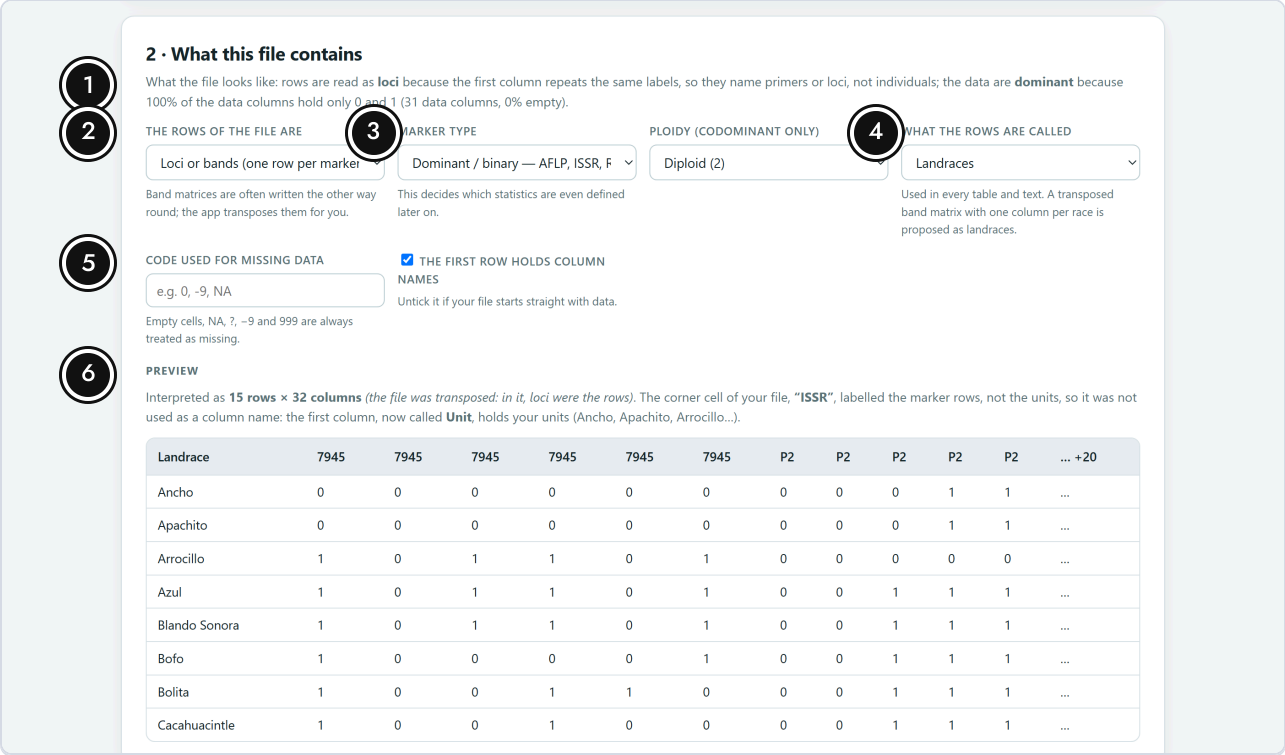



**Figure 3.9.** The 2 · What this file contains card with the maize landraces example with ISSR.

1. The explanation: why the rows were read as loci and why the data are dominant.
2. Orientation. The app transposed the table because the first column repeats primer names.
3. Marker type: it determines which statistics are available later on.
4. Name of the rows. With a transposed table it proposes *Landraces*.
5. Missing-data code, in addition to those that are always recognised.
6. Preview and a notice about the corner cell ("ISSR"), which is not used as a column name.

**Table 3.10.** Options of the What this file contains card.

| Option | What it decides | When to change it |
|---|---|---|
| The rows of the file are | Individuals (one row per plant) or loci and bands (one row per marker). | If the app transposed a table it should not have, or the other way round. |
| Marker type | Codominant, dominant, haploid, sequences or morphological. | Whenever what was read does not match your marker. |
| Ploidy | Diploid, haploid, triploid, tetraploid or hexaploid (codominant markers only). | In polyploid species, with one column per gene copy. |
| How genotypes are written | One allele per column or the whole genotype in one cell. | If your cells read 182/186. |
| What the rows are called | Individuals, landraces, accessions, varieties, populations, clones, genotypes, samples or OTUs. | So that tables, texts and the report use your term. |

**Table 3.10.** Options of the What this file contains card. *(continued)*

| Option | What it decides | When to change it |
|---|---|---|
| Code used for missing data | A missing-data code of your own. | If your laboratory uses, for example, 999. |
| The first row holds column names | Whether the first row contains names. | Untick it if the file starts directly with data. |

**DECISION RULE · HOW THE APP WORKS OUT THE MARKER TYPE**

| # | If it finds... | then the data are... |
|---|---|---|
| 1 | row 1 of a PopGeneticsPro template | the type declared in it |
| 2 | a parameter row with two columns per locus | codominant (with one column per locus: dominant or haploid) |
| 3 | more than 90% of the data columns containing only 0 and 1 | dominant |
| 4 | columns with complete DNA sequences | sequences |
| 5 | columns with genotypes in one cell (182/186, A/G) | codominant, genotype in one cell |
| 6 | column names that come in pairs | codominant, one allele per column |
| 7 | most columns with continuous measurements | morphological |
| 8 | most columns with a text code | haploid |
| 9 | none of the above | codominant (check it) |

Only the data columns are examined: a latitude or ID column does not turn a matrix of 0s and 1s into measurements. The orientation is worked out first: a matrix of 0s and 1s whose first column repeats names, or with many more rows than columns, is read with the bands in rows.

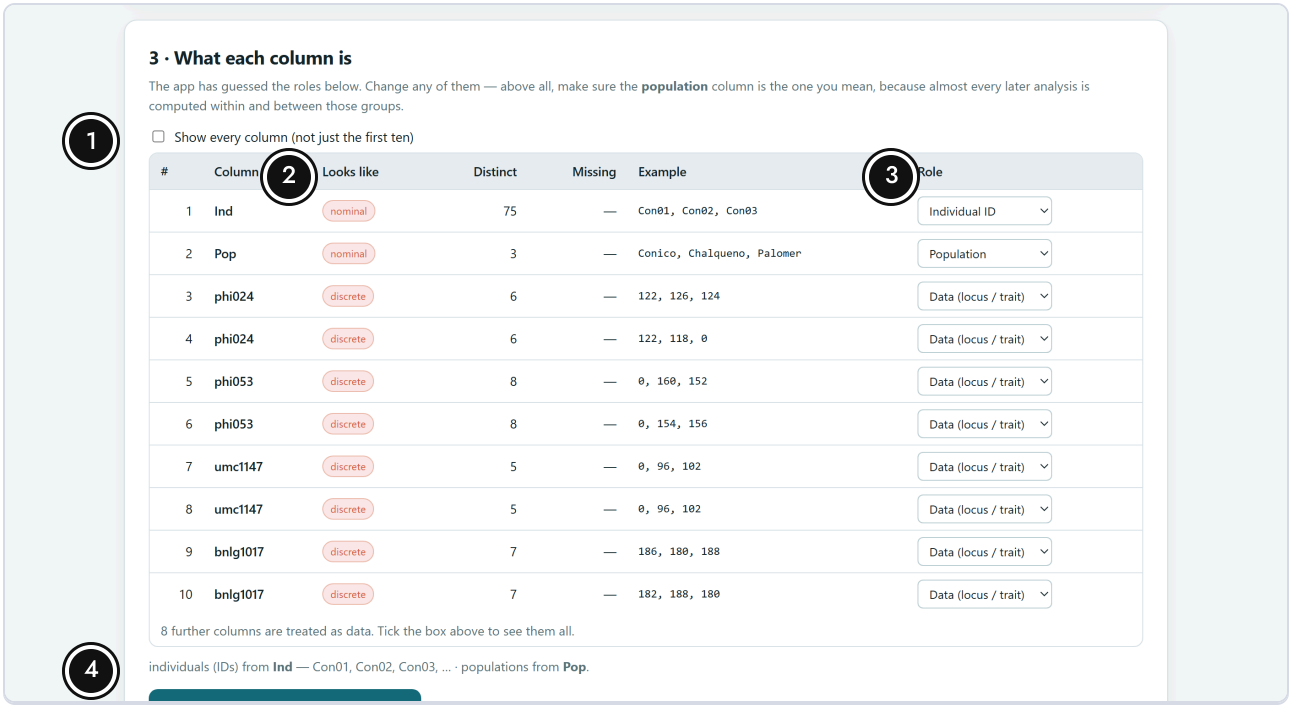



**Figure 3.10.** The `3 · What each column is` card with the maize microsatellite example.

1. Shows all the columns, not just the first ten.
2. What each column looks like: *nominal* (text), *discrete* (integers), *binary* (0/1) or *quantitative*, with distinct values, missing values and examples.
3. The assigned role; change it with the menu.
4. Summary: where the IDs, populations, regions and coordinates come from.

**Table 3.11.** Possible roles of a column.

| Role | Meaning |
|---|---|
| `Data (locus / trait)` | A data column: allele, band, sequence or trait. This is the default role. |
| `Individual ID` | The name of each row. If there is none, the app generates Ind1, Ind2… |
| `Population` | The group of each row. Without this column, each row is treated as a unit of its own. |
| `Region / group` | The upper level for a hierarchical AMOVA or for colouring figures. |
| `Latitude` · `Longitude` | Coordinates in decimal degrees (19.62, −98.12) or projected in metres, such as UTM (Northing, Easting). |
| `Ignore this column` | A column that should not be used: remarks, dates, gel number. |

**! IMPORTANT · THE POPULATION COLUMN**

Almost all later analyses are computed **within** and **among** the groups defined by the `Population` column. If the app took the wrong column as the population (for example, the plate number) or found none, correct it here before clicking `Build the dataset and run the checks`. A mistake at this step invalidates diversity by population, $F_{ST}$, the AMOVA and everything that follows.

# 3.5 Quality control

When you click **Build the dataset and run the checks**, the app assembles the dataset and checks it. The **4 · Quality control** card first shows six figures, then the warnings and, below them, tables and figures of the matrix.

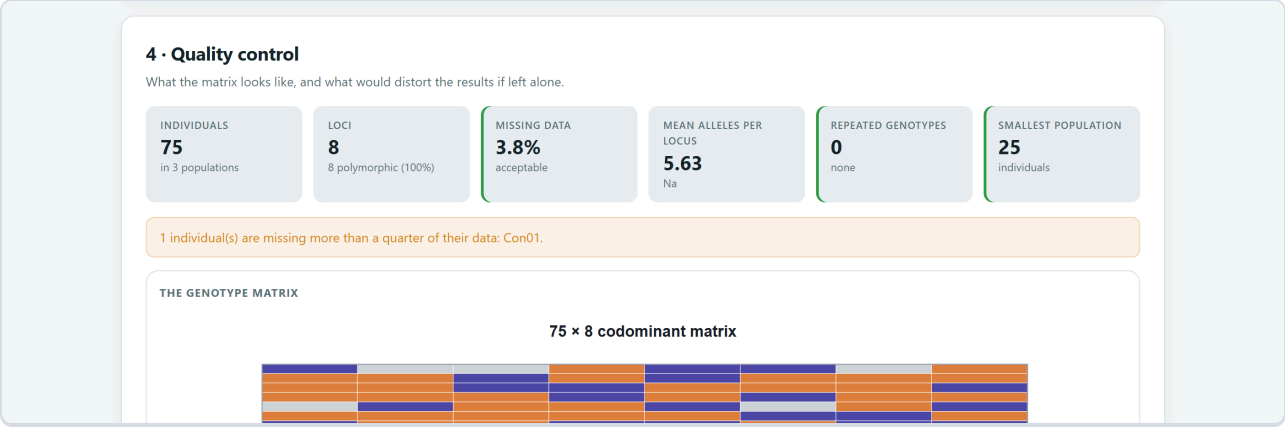



**Figure 3.11.** Quality control of the maize microsatellite example. Cards with a green border are within acceptable limits and those with an amber border need attention. The only warning points to Con01, which has no data at 3 of its 8 loci.

**Table 3.12.** The quality control figures and how to read them.

| Card | What it measures | Warning sign |
|---|---|---|
| Individuals (or the name of your units) | Rows read and number of populations. | Fewer rows than you entered: check the table of problems. |
| Loci · Bands · Haplotypes · Traits | Data columns and how many of them are polymorphic. | Many monomorphic loci: they provide no information. |
| Missing data | Percentage of data cells that are missing. | More than 10% (amber border). |
| Mean alleles per locus · Mean band frequency | Mean $N_a$ (codominant) or mean band frequency (dominant). | A very high $N_a$ at a single locus: possible scoring errors. |
| Repeated genotypes | Pairs of individuals with an identical multilocus genotype. | Any value greater than 0: possible clones or duplicated plants. |
| Smallest population | Size of the smallest population. It does not appear if there is no population column. | Fewer than 5 (amber border). |

**DECISION RULE · MISSING DATA**

| Missing data | Reading | What to do |
|---|---|---|
| **Less than 5%** | Very good | Carry on without changes. |
| **5 to 10%** | Acceptable | Carry on, but check whether the missing data are concentrated in one locus or one population. |
| **More than 10%** | Worrying | Remove the loci or individuals with the most missing data rather than letting the gaps drive the result. |
| **An individual with more than 25%** | Weak individual | The app names it. Genotype it again, or exclude it if its absence does not leave the population too small. |
| **A locus with far more missing data than the rest** | Problematic locus | It may have null alleles or amplify poorly. Check it in Block 4 as well. |



## Tables and figures of the matrix

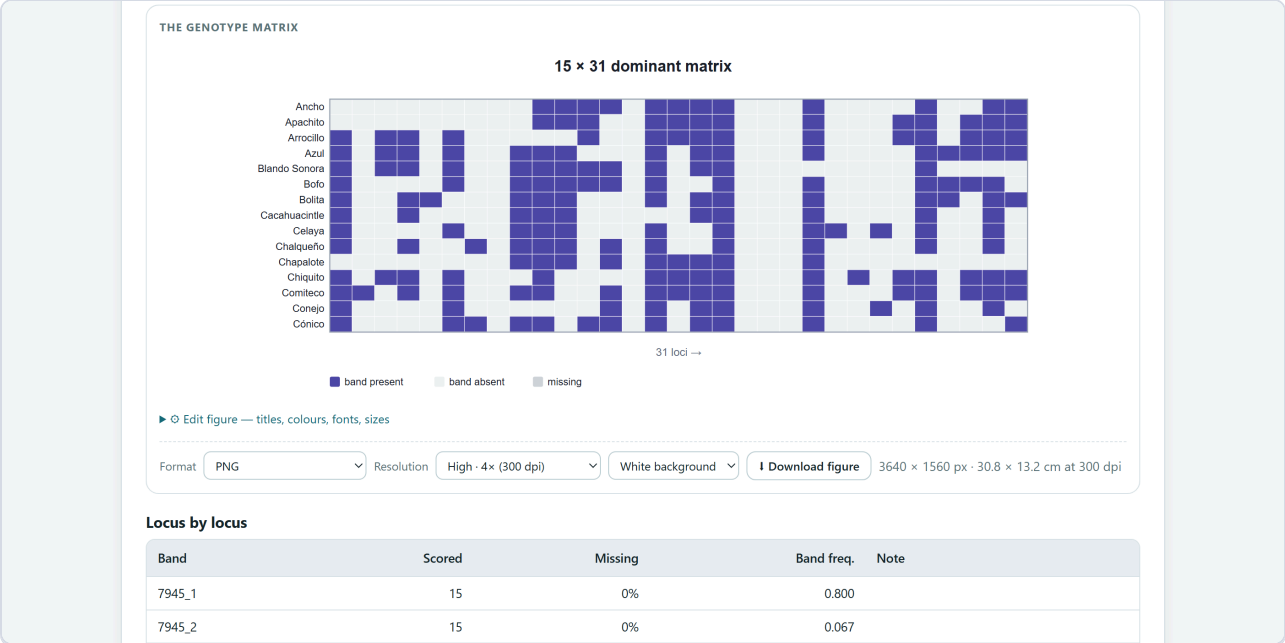



**Figure 3.12.** The genotype matrix with the 15 landraces and the 31 ISSR bands. Each cell is one individual at one locus: for dominant markers, band present or absent; for codominant markers, homozygote or heterozygote; in grey, missing. Columns that are full or empty from top to bottom are monomorphic bands. Under Edit figure you can show only the missing data, which reveals problematic loci or plants at a glance.

**Locus by locus**

| Locus | Scored | Missing | Alleles (Na) | $H_o$ | Note |
|---|---|---|---|---|---|
| phi024 | 70 | 6.7% | 5 | 0.643 | |
| phi053 | 71 | 5.3% | 7 | 0.648 | |
| umc1147 | 72 | 4% | 4 | 0.611 | |
| bnlg1017 | 74 | 1.3% | 6 | 0.662 | |
| phi072 | 74 | 1.3% | 8 | 0.689 | |
| umc1304 | 71 | 5.3% | 5 | 0.535 | |
| bnlg1621 | 70 | 6.7% | 6 | 0.657 | |
| phi096 | 75 | 0% | 4 | 0.640 | |

EVERY LOCUS, AND HOW VARIABLE IT IS

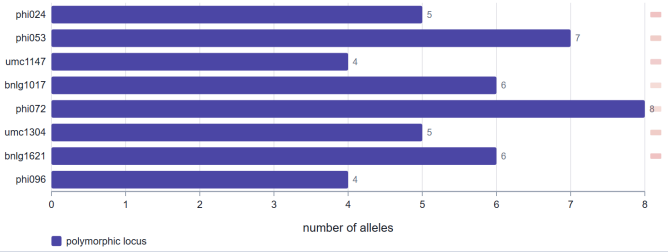
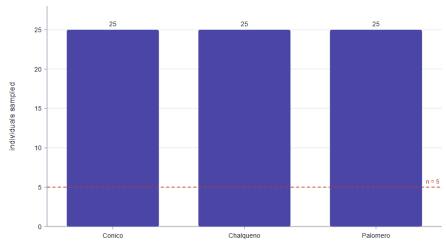



**Figure 3.13.** Locus by locus with the 8 microsatellites. The table gives the individuals with data, the missing values, the number of alleles $N_a$ and the observed heterozygosity $H_o$; the figure ranks the loci by variability. For dominant markers the table gives the frequency of each band; for morphological traits, the mean, standard deviation, coefficient of variation, minimum and maximum.

SAMPLE SIZE PER POPULATION



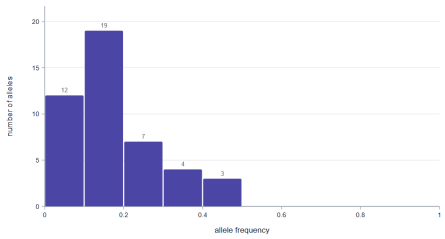

▸ ⚙ Edit figure — titles, colours, fonts, sizes

Format PNG Resolution High · 4× (300 dpi)

White background ⬇ Download figure

3080 × 1800 px · 26.1 × 15.2 cm at 300 dpi

DISTRIBUTION OF ALLELE FREQUENCIES




▸ ⚙ Edit figure — titles, colours, fonts, sizes

Format PNG Resolution High · 4× (300 dpi)

White background ⬇ Download figure

3080 × 1760 px · 26.1 × 14.9 cm at 300 dpi

**Figure 3.14.** Sampling and frequencies. Left: plants per population, with the red line marking the recommended minimum (5, adjustable). Right: how many of the 45 alleles fall into each frequency class. In a population at equilibrium the lowest class is expected to be the most numerous, giving an L shape; if the mode shifts to intermediate frequencies there may have been a recent bottleneck. Here the 0.1–0.2 class is the most frequent, but with the three populations pooled this is only a first look: the formal test is carried out by population in Block 9.

**DECISION RULE · MONOMORPHIC LOCI AND EXTREME BANDS**

| If a locus or band... | then... |
|---|---|
| **Is monomorphic** (a single allele, or a band present in all or in none) | It contributes nothing to differentiation or distances. The app keeps it and warns you; it counts towards the percentage of polymorphic loci (%P). |
| **Is a dominant band with a frequency below 5% or above 95%** | Its frequency is poorly estimated from a gel. It is common practice to remove such bands before analysis; the app counts them in a warning. |
| **Has an allele found in a single plant only** | It may be real (a rare allele) or a scoring error. Check the electropherogram before reporting it as a private allele. |



## Clones and repeated genotypes

**THEORY IN BRIEF · RAMETS AND GENETS**

Many plants also reproduce vegetatively: through rhizomes, stolons, bulbils or offsets, as agaves do. Each independent stem or plant is a **ramet**, but all those that arose from the same original individual form a single **genet** and share the same genotype. If 40 ramets of one genet are sampled as though they were 40 plants, the sample size is inflated, diversity is underestimated and $F_{IS}$ is distorted. The usual practice is to keep one representative per multilocus genotype. The app compares all pairs of individuals genotyped at at least half of the loci and reports those that are identical and those that match at 95% or more of their loci.

**DECISION RULE · REPEATED GENOTYPES**

| If you find... | the most likely explanation is... | what to do |
|---|---|---|
| **Identical pairs in a clonal species** | ramets of the same genet | keep one per genotype |
| **Identical pairs in a species without vegetative reproduction** | the same plant sampled or entered twice, or too few loci | check the labels and the field sheet |
| **Pairs that match at 95% or more without being identical** | close relatives, or the same plant with a scoring error | check the loci that differ |
| **Many identical pairs with few loci** | a lack of power to tell individuals apart | add loci before concluding that there is clonality |

With sequences, sharing a haplotype is normal, and with traits there is no multilocus genotype: in both cases this check does not apply.

## Values to fix: an example with deliberate errors

To see how quality control reacts, table 3.13 reproduces a small matrix of 20 plants and 4 microsatellites with six deliberately planted errors. You can enter it in the data sheet and follow the steps.

**Table 3.13.** Rows with errors in the practice matrix (the other rows are correct). In red, the wrong value; in row 8 the cell reads 18O, with a capital letter O instead of zero. The row numbers are those of the sheet, with the column names in row 1.

| Row | ID | Population | Latitude | Longitude | SSR1 | SSR1 | SSR2 | SSR2 | SSR3 | SSR3 | SSR4 | SSR4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | A02 | Tlaxco | 19.62 | -98.12 | 182 | 184 | 138 | 142 | 96 | 96 | 210 | 210 |
| 4 | A03 | Tlaxco | 19.62 | -98.12 | 182 | 184 | 138 | 142 | 96 | 96 | 210 | 210 |
| 6 | A05 | Tlaxco | 19.62 | -98.12 | 184 | 186 | 142 | 144 | 96 | 100 | 210 | 216 |
| 7 | A05 | Tlaxco | 19.62 | -98.12 | 182 | 182 | 138 | 140 | 100 | 102 | 212 | 214 |
| 8 | A07 | Tlaxco | 19.62 | -98.12 | 18O | 186 | 140 | 142 | 96 | 98 | 210 | 212 |
| 9 | A08 | Tlaxco | 19.62 | -98.12 | 184 | 184 | 142 | 142 | 98 | 98 | 0 | 214 |
| 11 | A10 | (empty) | 19.62 | -98.12 | 186 | 188 | 140 | 144 | 98 | 100 | 210 | 214 |
| 12 | B01 | Apizaco | 190.41 | -98.14 | 182 | 184 | 144 | 146 | 100 | 102 | 216 | 218 |
| 20 | B09 | Apizaco | 19.41 | -98.14 | · | · | · | · | · | · | · | · |

1. Open the data sheet, paste the complete matrix and click **Use these data**.
2. Confirm the interpretation (codominant, 4 loci in pairs) and click **Build the dataset and run the checks**.
3. Read the warnings and the **Values to fix in the sheet** table (figure 3.15).
4. Go back up to the data sheet: the problem cells are in red (figure 3.16). Correct each one and click **Use these data** again.

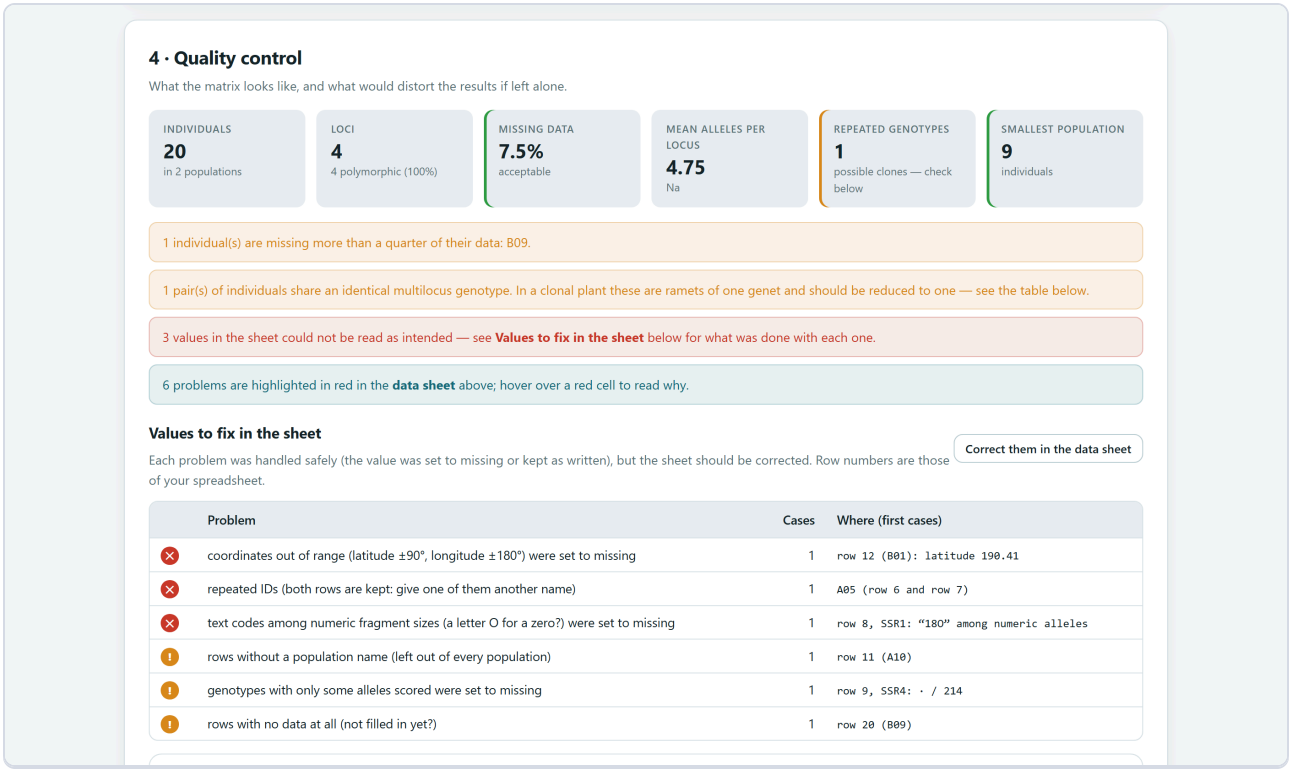



**Figure 3.15.** Quality control with the practice matrix. The warnings summarise what was found, and the Values to fix in the sheet table states, for each problem, what the app did and where the problem is. A red ✕ marks errors that change the data; an amber ! marks warnings. If you loaded a file instead of using the sheet, the Correct them in the data sheet button opens it in the sheet with the marks.

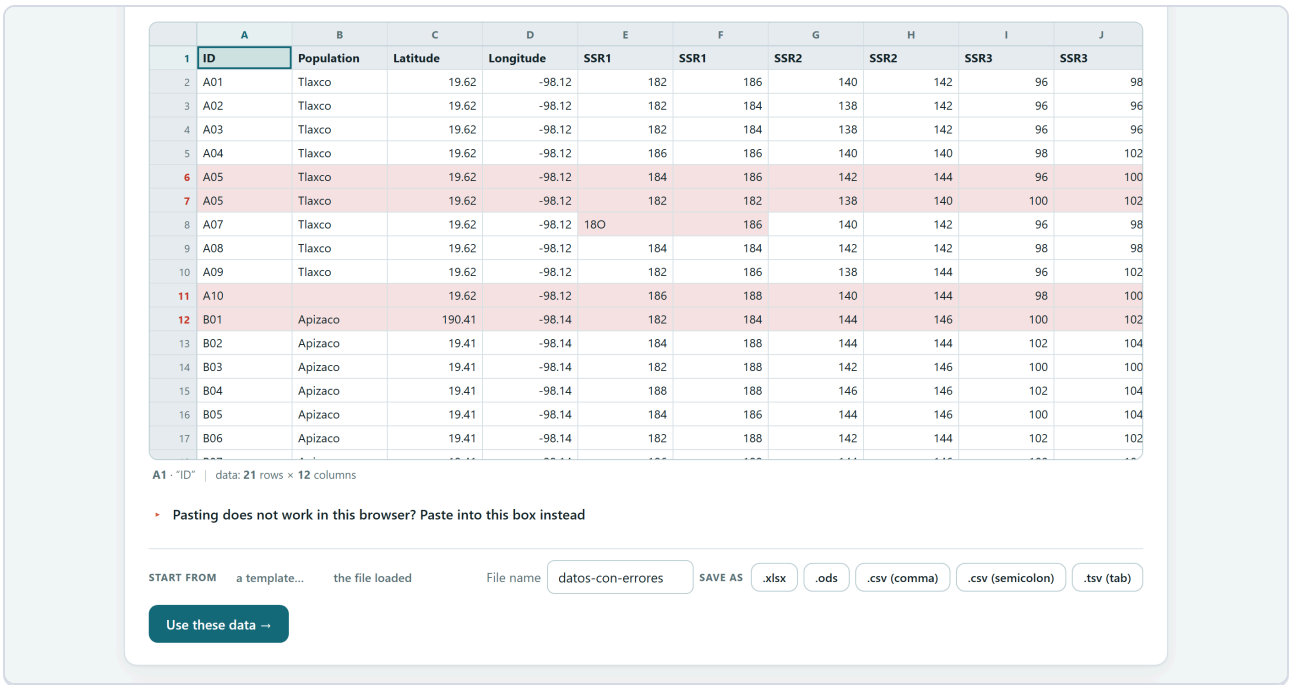



**Figure 3.16.** The data sheet with the problems marked: the 180 cell (with the letter O) and its partner in row 8, and the whole of rows 6 and 7 (repeated ID), 11 (no population) and 12 (impossible latitude). Hovering the mouse over a red cell shows the reason.

**Table 3.14.** What quality control detected in the practice matrix and how to correct it.

| Planted error | How the app reports it | What it did and how to correct it |
|---|---|---|
| `180` (with the letter O) in row 8 | × *text codes among numeric fragment sizes (a letter O for a zero?) were set to missing* | It removed the genotype so as not to inflate $N_a$ with a false allele. Type 180. |
| A05 twice (rows 6 and 7) | × *repeated IDs (both rows are kept: give one of them another name)* | It kept both rows. Rename the second one A06. |
| Latitude 190.41 (row 12) | × *coordinates out of range … were set to missing* | It deleted the latitude so as not to distort the distances. Type 19.41. |
| Empty population (row 11) | ! *rows without a population name (left out of every population)* | A10 is left out of every group. Type Tlaxco. |
| 0 \| 214 at SSR4 (row 9) | ! *genotypes with only some alleles scored were set to missing* | A partial genotype is not a homozygote. Score it again or leave both alleles empty. |
| B09 with no data (row 20) | ! *rows with no data at all (not filled in yet?)* and a warning of more than 25% missing data | Genotype the plant or delete the row. |
| A02 and A03 identical | **Repeated genotypes** = 1 and the table of repeated genotypes | A clone, or a plant entered twice? Check the field labels. |

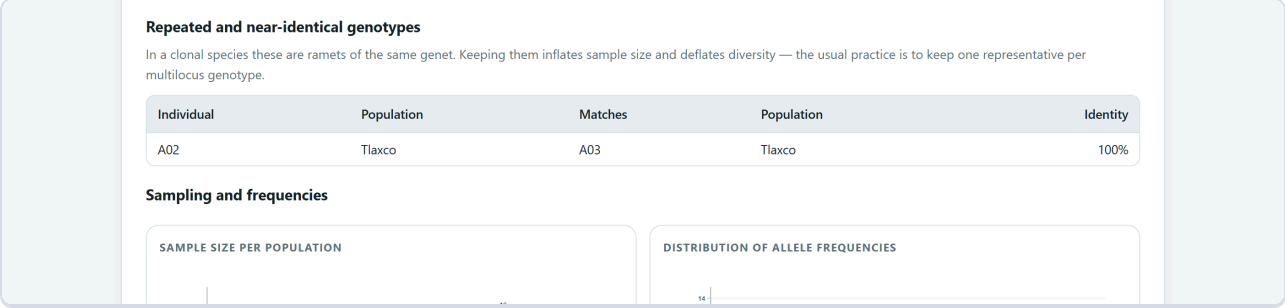



**Figure 3.17.** The **Repeated and near-identical genotypes** table: A02 and A03, from Tlaxco, match at 100% of their loci.

**Table 3.15.** All the problems that the **Values to fix in the sheet** table can report.

| | Problem | What the app does with the value |
|---|---|---|
| × | Band that is neither 0 nor 1 | Sets it to missing. |
| × | Genotype with a number of alleles different from the ploidy | Sets it to missing. |
| × | Whole genotype (182/186) in a one-allele-per-column layout | Sets it to missing; if the whole sheet is written that way, choose "whole genotype in one cell". |
| × | Number of genotype columns that does not match the ploidy | Ignores the extra columns at the end. |
| × | Non-numeric allele in a size template · text code among sizes | Sets it to missing. |
| × | Repeated ID | Keeps both rows. |

**Table 3.15.** All the problems that the Values to fix in the sheet table can report. *(continued)*

| | Problem | What the app does with the value |
|---|---|---|
| × | Coordinate out of range (±90°, ±180°) among coordinates in degrees | Sets it to missing. |
| × | Letters that are not IUPAC nucleotides · no data columns | Flags it for correction. |
| ! | Genotype with only some alleles scored | Sets it to missing. |
| ! | Haploid cell that appears to have two alleles | Keeps it and flags it. |
| ! | Row or locus with no data at all | Keeps it; this is usually an unfinished template. |
| ! | Row without a population name | Leaves it out of every population. |
| ! | Coordinate that is not a number | Sets it to missing. |
| ! | Sequence shorter than the longest in its region · sequence not entered | Flags it as unaligned · pads it with ? to keep the alignment. |
| ! | Text in a numeric trait column | Sets it to missing. |

**! IMPORTANT · THE APP PROTECTS THE ANALYSIS; IT DOES NOT CORRECT YOUR FILE**

Each problem is handled safely so that the calculations are not contaminated, but **your original file does not change**. Correct the source sheet or save the corrected version from the built-in data sheet. That way, the next time you load the data, or when someone else uses them, they will be clean.

## 3.6 Which analyses your data allow

The last card of the block, 5 · What you can run with these data, summarises which families of analyses the dataset supports as it now stands and, for those it does not, explains why.

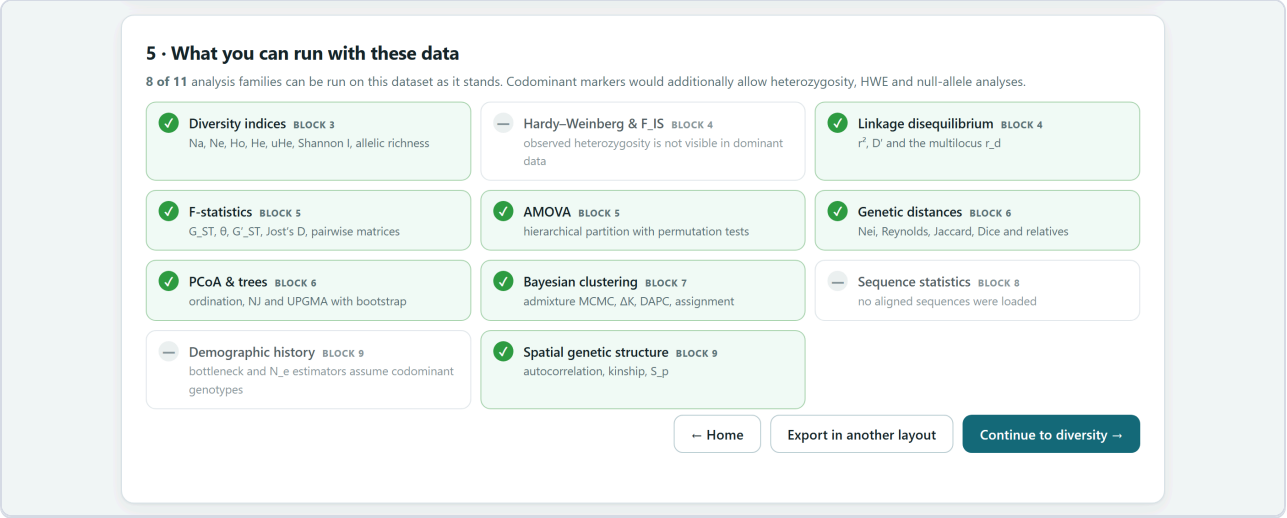



**Figure 3.18.** Analyses available with the agave AFLP example: 8 of 11. Hardy–Weinberg and demographic history are missing, because dominant markers do not reveal heterozygotes, as are the sequence statistics. Spatial structure is available, because the file includes coordinates.

**Table 3.16.** The 11 families of analyses and what each one requires.

| Family in the app | Block | Requirement |
|---|---|---|
| Diversity indices | 3 | Markers and at least one population with 2 or more individuals. |
| Hardy–Weinberg & F_IS | 4 | Codominant markers and at least one population with more than one individual. |
| Linkage disequilibrium | 4 | Two or more loci (not sequences or traits) and populations with more than one individual. |
| F-statistics · AMOVA | 5 | Markers and at least two populations with 2 or more individuals each. |
| Genetic distances | 6 | Three or more units, with at least one polymorphic locus or traits. |
| PCoA & trees | 6 | Three or more units. |
| Bayesian clustering | 7 | Multilocus markers (not sequences or traits), 6 or more individuals and 5 or more polymorphic loci. |
| Sequence statistics | 8 | Aligned sequences. |
| Demographic history | 9 | Codominant markers, two or more loci and populations with more than one individual. |
| Spatial genetic structure | 9 | Coordinates for 10 or more individuals. |

**NOTE · THE CARD RECOMMENDS; THE BLOCK BUTTONS OPEN EARLIER**

The card uses the recommended minimums. The buttons in the top bar are somewhat more permissive so that you can explore: Block 7 opens from 3 polymorphic loci, although the card asks for 5, and Block 3 opens even with one sample per unit to show the diversity of the whole dataset. When an analysis falls below the recommended minimum, the corresponding block warns you in its interpretation.

**Table 3.17.** Buttons at the end of the block.

| Button | What it does |
|---|---|
| ← Home | Returns to the home page; the data remain loaded. |
| Export in another layout | Opens the designer with the data so that you can save them as a template, a layout with a parameter row or FASTA with a sample sheet (section 3.3). |
| Continue to diversity → | Moves on to Block 3. |

## 3.7 Step by step with the six examples



Each example shows a different situation in Block 2. All of them are loaded from **Or try an example** and, except for the FASTA files, finished with **Build the dataset and run the checks**. Table 3.18 summarises what you should obtain.

**Table 3.18.** Result of Block 2 with each example.

| Example | Read as | Missing | Warnings | Analyses |
|---|---|---|---|---|
| Maize landraces · ISSR | Dominant, transposed; 15 landraces, 31 bands (25 polymorphic, 80.6%); no population column | 0% | 6 monomorphic bands | 3 of 11 |
| Maize populations · SSR | Codominant, one allele per column; 75 plants, 3 populations of 25, 8 loci; $N_a$ = 5.63 | 3.8% | Con01 has no data at 3 of 8 loci | 9 of 11 |
| Agave · AFLP | Dominant; 80 plants, 4 populations of 20, 60 polymorphic bands, with coordinates | 0% | None: matrix ready | 8 of 11 |
| Pine · chloroplast DNA | FASTA; 48 sequences of 480 bp, 4 populations of 12, 8 haplotypes | 0% | Offers to add sample information | 6 of 11 |
| Maize · morphology | Morphological; 75 plants, 3 populations, 9 traits | 0% | None: matrix ready | 2 of 11 |
| Five genera · dating with fossils | FASTA; 16 sequences of 1600 bp, 6 genera as populations, 16 haplotypes | 0% | Offers to add sample information | 6 of 11 |

## Example 1 · Maize landraces with ISSR: a table that has to be turned round

**STEP BY STEP**

1. Click **Maize landraces · ISSR**. The file is a spreadsheet workbook with the bands in rows and the landraces in columns, the way a gel is scored.
2. Read the interpretation card (figure 3.9): the app noticed that the first column repeats the primer names (7945, P2, P5, A8), transposed the table and proposes calling the rows *landraces*.
3. In the roles, the first column, now **Landrace**, is the **Individual ID** and there is no population. This is correct: each landrace is a bulked sample.
4. Click **Build the dataset and run the checks**.

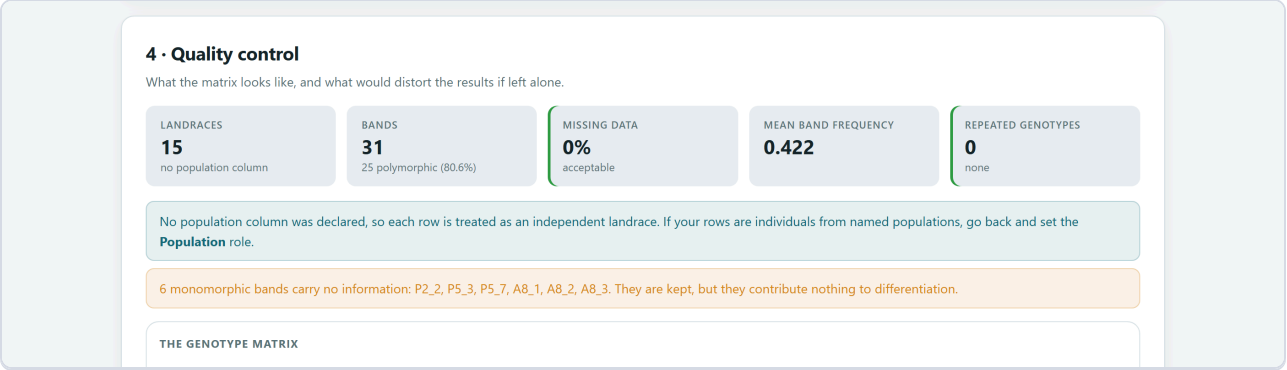



**Figure 3.19.** Quality control of the landraces with ISSR. Since there is no population column, each row is treated as an independent landrace and the smallest-population card does not appear. Six bands are monomorphic.

**WHAT IT MEANS FOR THE ANALYSIS**

With a single sample per landrace there is no variation **within** units, so diversity by population, Hardy–Weinberg equilibrium and the AMOVA do not exist. What can be studied are the **relationships among landraces**: genetic distances, principal coordinates, trees with bootstrap and Bayesian clustering (3 of 11 families). This is the path of Block 6, where this example is taken up again.

## Example 2 · Maize populations with microsatellites: the complete case

**STEP BY STEP**

1. Click **Maize populations · SSR**. It is a CSV with the columns `Ind`, `Pop` and two columns per locus.
2. The interpretation says *the column names come in 8 pairs, two per locus*: codominant, one allele per column, missing 0.
3. In the roles (figure 3.10), `Ind` is the ID and `Pop` the population, with 3 distinct values: Cónico, Chalqueño and Palomero.
4. Build the dataset and check it: 75 plants, 8 polymorphic loci, 3.8% missing data and a single warning (figure 3.11).
5. Scroll down to **Locus by locus** (figure 3.13): missing data range from 0% at phi096 to 6.7% at phi024 and bnlg1621, and $N_a$ from 4 to 8 alleles.

**HOW TO READ THIS RESULT**

Everything is within the recommended limits: 25 plants per population, less than 5% missing data and no locus with more than 7%. Plant Con01 lacks 3 of its 8 loci (37.5%); with 25 plants in its population, it is best to keep it and let each analysis use the loci it does have. With 9 of 11 families available, this dataset runs through almost the whole manual.

## Example 3 · Agave with AFLP: dominant markers with coordinates

**STEP BY STEP**

1. Click **Agave · AFLP**. The interpretation says *100% of the data columns hold only 0 and 1*: dominant.
2. In the roles, `Plant` is the ID, `Population` the population and two columns are latitude and longitude; the summary says *coordinates found*.
3. Build the dataset: 80 plants, 60 polymorphic bands, with no missing data or repeated genotypes.
4. Check the analyses card (figure 3.18).

**NOTE · WHY THERE ARE NO CLONES IN AN AGAVE**

Agaves produce offsets, so in a real study the table of repeated genotypes deserves a careful look. In this simulated example there are none. With 60 bands, two ramets of the same genet would show up as an identical pair.

# Example 4 · Pine with chloroplast DNA: a FASTA file

**STEP BY STEP**

1. Click **Pine · chloroplast DNA**. The message says *Read as a FASTA alignment: 48 individuals, 1 alignment, 4 populations*. There are no interpretation or role cards: the format already carries its own structure.
2. The population was taken from the header of each sequence: `>Sierra_Norte_01 | Sierra_Norte`.
3. **Add sample information** appears, so that you can add regions or coordinates with a sample sheet (figure 3.20).
4. Quality control reports 8 haplotypes in 48 sequences of 480 bp.

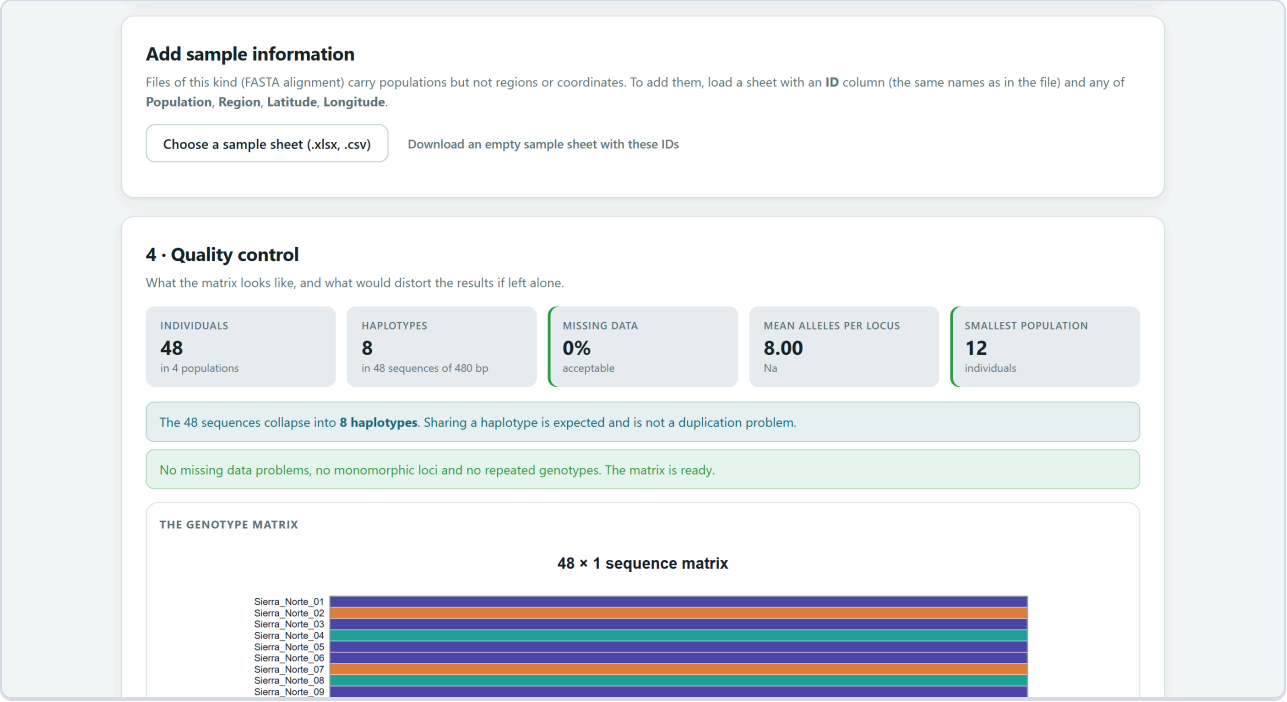



**Figure 3.20.** With a FASTA file, the app offers to complete the sample information, and quality control notes that sharing a haplotype is expected, not a duplication.

**NOTE · THE HEADER OF A FASTA FILE**

Write the headers as `>ID | Population | Region` so that you do not need a sample sheet. If you already have the file without populations, click **Download an empty sample sheet with these IDs**, fill in the columns and load it with **Choose a sample sheet**; the IDs must match those in the FASTA file (the app ignores case and leading or trailing spaces).

# Example 5 · Maize morphology: traits instead of markers

**STEP BY STEP**

1. Click **Maize · morphology**. The interpretation says *the data columns hold continuous measurements*: morphological.
2. Build the dataset. Instead of the locus table, **Trait by trait** appears, with the mean, standard deviation, coefficient of variation, minimum and maximum (figure 3.21).
3. The analyses card offers 2 of 11 families: distances and PCoA. In addition, the phenotypic diversity of Block 3 and $P_{ST}$ of Block 9 become available.



**Trait by trait**

| Trait | Type | n | Missing | Mean | SD | CV | Min | Max |
|---|---|---|---|---|---|---|---|---|
| PlantHeight_cm | quantitative | 75 | 0 | 227.41 | 39.24 | 17.3% | 155.10 | 319.60 |
| EarLength_cm | quantitative | 75 | 0 | 13.82 | 3.58 | 25.9% | 7.50 | 20.50 |
| EarDiameter_mm | quantitative | 75 | 0 | 40.53 | 10.47 | 25.8% | 21.10 | 61.80 |
| KernelRows | discrete | 75 | 0 | 13.57 | 2.09 | 15.4% | 9.00 | 18.00 |
| KernelsPerRow | quantitative | 75 | 0 | 29.25 | 5.01 | 17.1% | 20.00 | 42.00 |
| CobWeight_g | quantitative | 75 | 0 | 37.00 | 15.40 | 41.6% | 12.20 | 73.30 |
| Weight100Kernels_g | quantitative | 75 | 0 | 25.07 | 9.85 | 39.3% | 10.00 | 41.20 |
| DaysToFlowering | quantitative | 75 | 0 | 78.45 | 7.71 | 9.8% | 60.00 | 92.00 |
| LeafWidth_cm | quantitative | 75 | 0 | 8.67 | 1.45 | 16.7% | 5.80 | 11.90 |

**Figure 3.21.** The 9 traits of the morphological example. The number of kernel rows per ear is recognised as *discrete* (integer) and the rest as quantitative.

**DECISION RULE · READING THE COEFFICIENT OF VARIATION OF A TRAIT**

| CV | Variation | In the example |
|---|---|---|
| **Less than 10%** | Low | Days to flowering, 9.8%: a very stable trait across plants. |
| **10 to 30%** | Moderate | Plant height (17.3%), ear length (25.9%) and ear diameter (25.8%). |
| **More than 30%** | High | Cob weight (41.6%) and 100-kernel weight (39.3%): good candidates for differentiating populations or for selection. |

An indicative guide: the expected CV depends on the trait and the environment. A very high CV may also reveal data-entry errors, such as a weight in grams entered as kilograms; check the minimum and maximum.

## Example 6 · Five genera for dating with fossils

**STEP BY STEP**

1. Click **Five genera · dating with fossils**. It is read as FASTA with 16 sequences of 1600 bp, and the genus name in each header is taken as the population.
2. Quality control reports 16 haplotypes: each species is different, as expected in a comparison among species.
3. The smallest population has 2 sequences: this does not matter, because this example is used not for population statistics but for the dating in Block 8.



**NOTE · POPULATIONS THAT ARE REALLY GENERA**

In studies among species, the population column serves to group taxa (genera, sections, clades) and to colour the trees. The within-population analyses of Blocks 3 to 5 make no biological sense in that case, even though the app allows them; go straight to Blocks 6 and 8.

**5 min**

**BEFORE MOVING ON**

Spend five minutes on Block 2 every time you load new data: confirm the marker type and the population column, and check the missing data and the table of values to fix. It is the step that prevents more errors than any other in the whole study.

With the data clean and confirmed, the next chapter moves on to the first analysis proper: the allele frequencies and diversity indices of Block 3.

# B3

CHAPTER FOUR · BLOCK 3 · DIVERSITY

# Allele frequencies and diversity



How much genetic variation is there, and how is it distributed among populations? Block 3 calculates the frequency of every allele and, from those frequencies, the diversity indices reported in almost every paper: number of alleles, observed and expected heterozygosity, effective number of alleles, Shannon's index, rarefied allelic richness, polymorphic information content, private alleles and the fixation index.


4.1 Theory in brief

4.2 The indices and what they mean

4.3 Decision rules

4.4 Block settings, tables and figures

4.5 Step by step: maize with microsatellites

4.6 Dominant data, haplotypes and morphology


## 4.1 Theory in brief

Genetic diversity is the raw material of evolution and of plant breeding: a population with many alleles, evenly distributed, can respond to environmental change, to new pests or to selection by a breeder. Measuring it properly is the first result of any study.

### TWO WAYS OF MEASURING VARIATION: HOW MANY ALLELES AND HOW EVENLY DISTRIBUTED

Imagine two populations with the same four alleles. In the first, each allele has a frequency of 25%; in the second, one allele dominates at 85% and the other three are rare. Both have **four alleles** (the same **richness**), but the first is far more **diverse**: if you draw two copies of the gene at random, they are much more likely to be different. This is why population genetics uses indices from two families: those that count alleles ($N_a$, allelic richness, private alleles) and those that take their frequencies into account ($H_e$, $N_e$, Shannon's index, PIC).

**Figure 4.1.** Same richness, different diversity. Both populations have four alleles at one locus, but in B one of them dominates. $N_a$ does not see the difference; $H_e$, $N_e$, Shannon's index (I) and the PIC do. $N_e$ = 1.37 means that B behaves as if it had little more than one equally frequent allele.

### EXPECTED HETEROZYGOSITY IN ONE SENTENCE

$H_e$ is the **probability that two copies of the gene drawn at random from the population are different**. If the population mated at random, this would also be the proportion of heterozygous individuals; hence the name "expected" heterozygosity. Nei (1973) called it **gene diversity** and showed that it can be calculated with any type of marker, even in selfing or haploid species, where there are no heterozygotes to count. It is the most widely used diversity index and the one you should always report.

### 1 FREQUENCIES WITH DOMINANT MARKERS

With AFLP or ISSR you can only see whether the band is present or absent. Absence corresponds to the recessive homozygote, whose frequency is $q^2$ under Hardy–Weinberg equilibrium. Therefore $q$ is estimated as the square root of the proportion of plants without the band, with the Lynch and Milligan (1994) correction for small samples. Every index computed from dominant markers rests on that assumption.

### 2 HAPLOTYPES AND TRAITS

With sequences or chloroplast codes each individual carries a single haplotype: diversity is measured with **haplotype diversity** Hd, which is $H_e$ corrected for sample size (Nei, 1987). With morphological traits, each trait is divided into classes and the standardised Shannon index H', which ranges from 0 to 1, is calculated.

### 3 WHY RAREFACTION IS NEEDED

The number of alleles you find depends on how many plants you sample: with more plants, more rare alleles turn up. If one population has 40 plants and another 15, their $N_a$ values are not comparable. **Allelic richness** ($A_r$) solves the problem: it calculates how many alleles would be expected if the same number $g$ of gene copies were taken from each population, equal to that of the smallest sample (Hurlbert, 1971; El Mousadik and Petit, 1996). It is an exact calculation, not a random draw: it always gives the same value.

B3

## 4.2 The indices and what they mean

All indices are calculated locus by locus in each population and then averaged over loci. In the formulas, $p_i$ is the frequency of allele $i$, $n$ the number of individuals with data and $k$ the number of alleles.

**Table 4.1.** Indices of Block 3. "Marker" column: C codominant, D dominant, H haploid or sequences.

| Index | Formula | What it measures | Range | Marker |
|---|---|---|---|---|
| **$N_a$** number of alleles | $k$ | Distinct alleles observed in the sample. Sensitive to sample size. | 1 to ∞ | C D H |
| **$N_{a \geq 5\%}$** | alleles with $p_i \geq 0.05$ | Common alleles; leaves out rare ones, which depend heavily on sampling. | 0 to k | C D H |
| **$N_e$** effective alleles | $1 / \Sigma\, p_i^2$ | How many equally frequent alleles would give the same diversity. | 1 to k | C D H |
| **I** Shannon's index | $-\Sigma\, p_i \ln p_i$ | Diversity that gives more weight to rare alleles than $H_e$ does. | 0 to ln k | C D H |
| **$H_o$** observed heterozygosity | heterozygotes / n | Proportion of heterozygous individuals counted. | 0 to 1 | C |

**Table 4.1.** Indices of Block 3. "Marker" column: C codominant, D dominant, H haploid or sequences. *(continued)*

| Index | Formula | What it measures | Range | Marker |
|---|---|---|---|---|
| **$H_e$** expected heterozygosity | $1 - \Sigma\, p_i^2$ | Probability that two random copies are different (Nei's gene diversity). | 0 to 1 – 1/k | C D H |
| **$uH_e$** · ***h*** unbiased | $2n/(2n-1) \cdot H_c$ | $H_e$ corrected for sample size (Nei, 1978). With haplotypes: $n/(n-1) \cdot H_e$. | 0 to 1 | C D H |
| **F** fixation index | $(H_c - H_o) / H_c$ | Deficit (positive) or excess (negative) of heterozygotes. | –1 to 1 | C |
| **PIC** | $H_c - \Sigma_{i<j}\, 2p_i^2 p_j^2$ | Ability of a marker to distinguish individuals (Botstein et al., 1980). | 0 to 1 | C |
| **$A_r$** allelic richness | $\Sigma\,[1 - C(N\text{-}N_i, g) / C(N, g)]$ | Alleles expected in *g* gene copies; N copies in total, $N_i$ of allele *i*. | 1 to k | C H |
| **%P** | polymorphic loci / loci | Percentage of polymorphic loci; by default, the commonest allele does not exceed 95%. | 0 to 100% | C D H |
| **Private alleles** | — | Alleles present in only one population. With dominant markers, exclusive bands. | 0 to ∞ | C D H |

B3

### WORKED EXAMPLE · ONE LOCUS WITH THREE ALLELES

In a population of 25 plants, a microsatellite has three alleles with frequencies 0.5, 0.3 and 0.2.

| Index | Calculation | Result |
|---|---|---|
| $\Sigma\, p_i^2$ | $0.5^2 + 0.3^2 + 0.2^2 = 0.25 + 0.09 + 0.04$ | **0.38** |
| $H_e$ | 1 – 0.38 | **0.620** |
| $uH_e$ | (50/49) × 0.620 | **0.633** |
| $N_e$ | 1 / 0.38 | **2.63** |
| I | $-(0.5 \ln 0.5 + 0.3 \ln 0.3 + 0.2 \ln 0.2) = 0.347 + 0.361 + 0.322$ | **1.030** |
| PIC | $0.620 - 2(0.25{\times}0.09 + 0.25{\times}0.04 + 0.09{\times}0.04) = 0.620 - 0.072$ | **0.548** |

If 13 of the 25 plants were heterozygous, $H_o$ = 13/25 = 0.520 and F = (0.620 – 0.520)/0.620 = **0.161**: a heterozygote deficit of 16%.

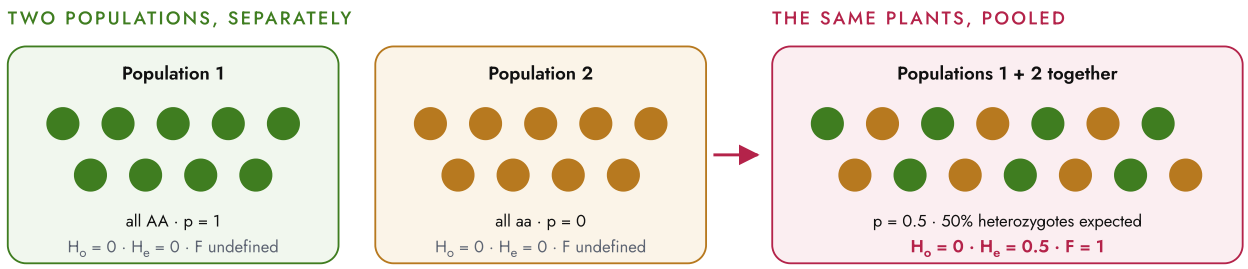



**Figure 4.2.** The Wahlund effect in its extreme form. This is why the fixation index of the pooled set (the Pooled row) is usually larger than the average over populations: it includes their differentiation. In the maize example of section 4.5, mean F = 0.104 and pooled F = 0.148.

B3

## 4.3 Decision rules

**DECISION RULE · IS DIVERSITY HIGH OR LOW?**

| Marker | Typical $H_e$ | Interpretation |
|---|---|---|
| **Microsatellites** in outcrossing plants | 0.45 to 0.85 | Below 0.45, low diversity (bottleneck, isolation, selfing); above 0.85, very high. |
| **Microsatellites** in selfing plants | below 0.45 | Low values are normal; compare with other species with a similar mating system. |
| **AFLP, ISSR, RAPD** | 0.15 to 0.40 | The theoretical maximum is 0.5, because each band has only two states. Never compare $H_e$ from dominant markers with $H_e$ from microsatellites. |
| Chloroplast **haplotypes** or sequences | highly variable | Depends on the marker and on history. Report it together with the number of haplotypes and the nucleotide diversity from Block 8. |

These are indicative ranges taken from the plant literature: they help you place a number, not judge a dataset. What is really informative is the comparison among your own populations with the same marker.

**DECISION RULE · DO TWO POPULATIONS DIFFER IN DIVERSITY?**

| If... | then... |
|---|---|
| **The 95% intervals of $H_e$ do not overlap** | The difference is supported by the loci sampled. |
| **The intervals overlap** | The difference is not resolved: treat the populations as equally diverse. The app says so explicitly. |
| **There are fewer than three loci** | Loci cannot be resampled and intervals cannot be calculated: the difference is not tested (the case of a single sequence alignment). |
| **Sample sizes differ by more than a factor of two** among populations | Do not compare $N_a$ or the number of private alleles; use $uH_e$ and $A_r$. The app warns you. |

The intervals are obtained by *bootstrap* over loci: loci are resampled with replacement 1000 times and the 2.5 and 97.5 percentiles of the mean are taken. They reflect how much the result would change with other loci, not with other plants.



**FIXATION INDEX F**

| F | Interpretation |
|---|---|
| **Greater than 0.1** | Clear heterozygote deficit. |
| **−0.05 to 0.1** | Close to equilibrium. |
| **Less than −0.05** | Heterozygote excess. |

F is a description. Whether there is inbreeding, null alleles or a Wahlund effect is decided in Block 4 with $F_{IS}$ and its tests.

**PIC OF A MARKER**

| PIC | Interpretation |
|---|---|
| **Greater than 0.5** | Highly informative. |
| **0.25 to 0.5** | Moderately informative. |
| **Less than 0.25** | Slightly informative. |

Categories of Botstein et al. (1980). Useful for choosing loci for variety identification or paternity testing.

**DECISION RULE · PRIVATE ALLELES AND LOCALLY COMMON ALLELES**

| If you find... | the interpretation is... |
|---|---|
| **A private allele with frequency ≥ 5%** (*frequent*) | Evidence of a distinct gene pool: restricted gene flow or an independent history. A priority for conservation. |
| **A rare private allele** (*rare*, < 5%) | It may be real or simply a sampling accident; with more plants it might turn up in other populations. |
| **A locally common allele**, ≥ 5% in 25% or fewer of the populations | Strong signal of restricted gene flow. |
| **A locally common allele**, ≥ 5% in half or fewer of the populations | Moderate signal. The app reports both thresholds when there are four or more populations. |

## 4.4 Block settings, tables and figures

When you enter Block 3 the app calculates everything with the default settings. If you change a setting, click Compute diversity and the whole block is recalculated.

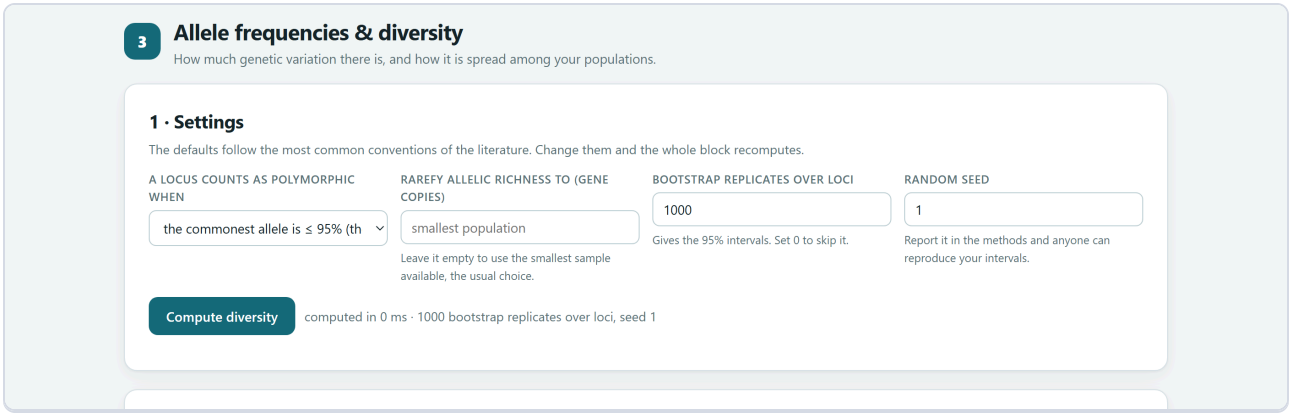



**Figure 4.3.** The `1 · Settings` card with codominant data. With dominant data, the method for estimating frequencies also appears and, with traits, the number of classes.

**Table 4.2.** Settings of Block 3.

| Setting | Options | Recommendation |
|---|---|---|
| A locus counts as polymorphic when | The commonest allele is ≤ 95% (default) · ≤ 99% · more than one allele is present. | 95%, the most widely used criterion. With samples of 20 plants or fewer the three usually agree. |
| Rarefy allelic richness to | Number of gene copies *g*. Empty: the smallest sample. | Leave it empty. Codominant and haploid data only. |
| Allele frequencies from dominant bands | Lynch and Milligan (default) · square root · Bayesian with a uniform prior. | Lynch and Milligan for outcrossing species; Bayesian with a high F for selfing species. |
| Inbreeding F assumed by the Bayesian estimator | 0 to 1. | 0 for outcrossing species; 0.8 to 0.95 for selfing species. |
| Classes per trait | 4 to 20 (default 10). | 10 classes half a standard deviation wide, the usual convention. |
| Bootstrap replicates over loci | 0 to 10,000 (default 1000). | 1000; 0 skips the intervals. |
| Random seed | Integer. | Report it in your methods so that the intervals are reproducible. |

**Table 4.3.** Result cards of Block 3.

| Card | Contents | Download |
|---|---|---|
| 2 · What the numbers say | Figures for the pooled set and interpretation paragraphs. | — |
| 3 · Diversity of every population | Table of indices by population with the pooled row (Pooled), standard errors over loci, bar chart and rarefaction curves. | Summary (CSV) |
| 4 · Locus by locus | The index chosen in Show for each locus in each population, and the $H_o$ against $H_e$ figure. | — |

**Table 4.3.** Result cards of Block 3. *(continued)*

| Card | Contents | Download |
|---|---|---|
| 5 · Allele frequencies | Frequency of each allele in each population, with private alleles marked; stacked bars by locus and a heat map. | All frequencies (CSV) |
| 6 · Private alleles | Private alleles with their frequency, and locally common alleles. Appears only if there are any. | — |

**Table 4.4.** The seven figures of the block. All of them are edited and exported as explained in the chapter on Block 10.

| Figure in the app | What it shows | When it appears |
|---|---|---|
| Diversity of every population | Bars of the chosen index by population, with intervals or standard errors and the line for the pooled set. | Two or more populations with several samples |
| Allelic richness against sampling effort | Rarefaction curves: expected alleles as a function of the gene copies sampled. | Codominant and haploid data |
| Observed against expected heterozygosity | Each locus in each population; points below the diagonal have a heterozygote deficit. | Codominant data |
| Allele frequencies, locus by locus | Stacked bars with the allelic composition of each population. | Molecular data |
| Every allele in every population | Heat map of all frequencies; blank cells are alleles absent from that population. | Molecular data |
| Private alleles | Private alleles by population, separating frequent and rare ones. | If there are private alleles |
| Phenotypic diversity by trait and population | Heat map of H′ by trait and population. | Morphological data |



## 4.5 Step by step: maize with microsatellites

The example of three maize populations with 8 microsatellites goes through every card of the block. The values in this chapter are those the app produces with the default settings and seed 1.

**STEP BY STEP**

1. In Block 2, load **Maize populations · SSR** and click **Build the dataset and run the checks**.
2. Click **Continue to diversity**. The block is calculated automatically in a fraction of a second, with 1000 bootstrap replicates.
3. Read the figures and the interpretation in **2 · What the numbers say** (figure 4.4).
4. Review the table by population and the bar chart (figures 4.5 and 4.6).
5. In **4 · Locus by locus**, change **Show** to **F · fixation index** to look for problematic loci.
6. Explore the allele frequencies of each locus: open **Edit figure** below the stacked bars and choose the locus in the **Locus** menu.



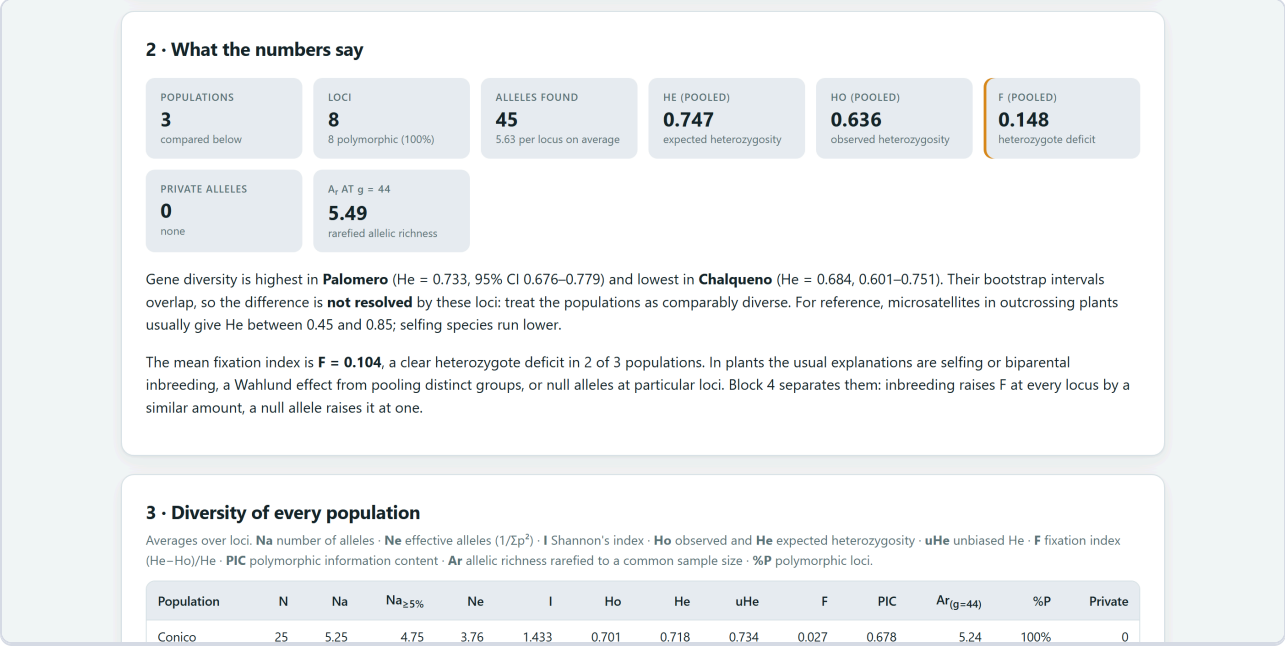
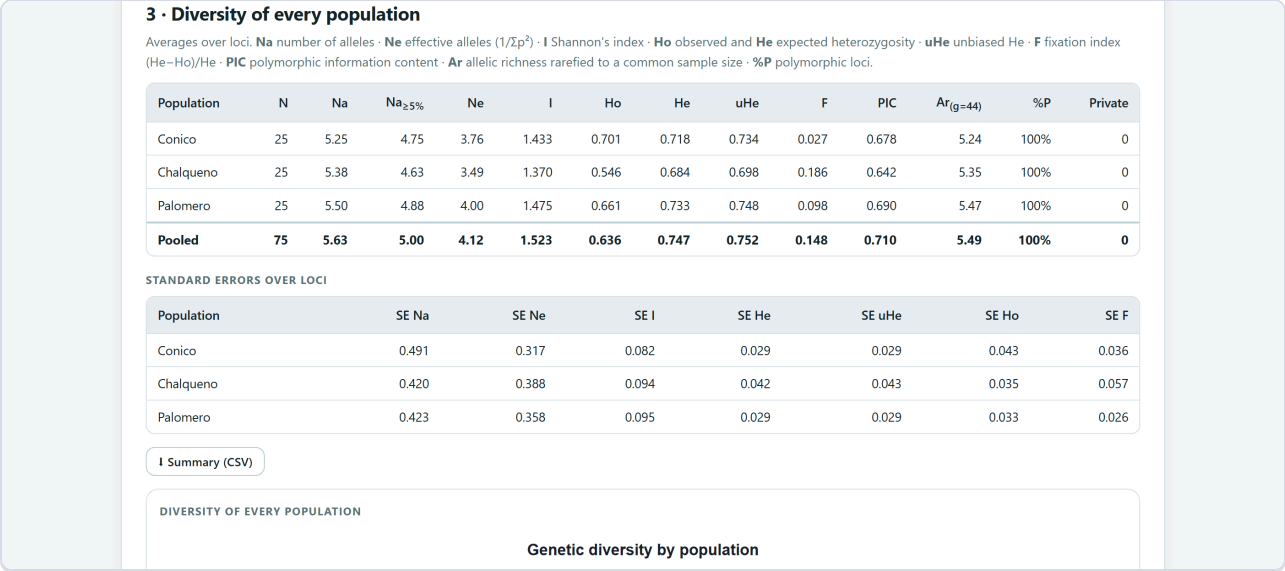



**Figure 4.4.** **What the numbers say** for maize: 45 alleles at 8 loci, pooled $H_e$ 0.747, $H_o$ 0.636 and pooled F 0.148 (amber card: heterozygote deficit). The interpretation compares the most and the least diverse populations and explains what F means.

**Table 4.5.** Diversity of the three maize populations (8 microsatellites, 25 plants each).

| Population | $N_a$ | $N_e$ | I | $H_o$ | $H_e$ (95% CI) | $uH_e$ | F | PIC | $A_r$ |
|---|---|---|---|---|---|---|---|---|---|
| Cónico | 5.25 | 3.76 | 1.433 | 0.701 | 0.718 (0.661–0.763) | 0.734 | 0.027 | 0.678 | 5.24 |
| Chalqueño | 5.38 | 3.49 | 1.370 | 0.546 | 0.684 (0.601–0.751) | 0.698 | **0.186** | 0.642 | 5.35 |
| Palomero | 5.50 | 4.00 | 1.475 | 0.661 | 0.733 (0.676–0.779) | 0.748 | 0.098 | 0.690 | 5.47 |
| **Pooled** | **5.63** | **4.12** | **1.523** | **0.636** | **0.747** | **0.752** | **0.148** | **0.710** | **5.49** |



**3 · Diversity of every population**

Averages over loci. **Na** number of alleles · **Ne** effective alleles ($1/\Sigma p^2$) · **I** Shannon's index · **Ho** observed and **He** expected heterozygosity · **uHe** unbiased He · **F** fixation index (He−Ho)/He · **PIC** polymorphic information content · **Ar** allelic richness rarefied to a common sample size · **%P** polymorphic loci.

| Population | N | Na | $Na_{\geq 5\%}$ | Ne | I | Ho | He | uHe | F | PIC | $Ar_{(g=44)}$ | %P | Private |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Conico | 25 | 5.25 | 4.75 | 3.76 | 1.433 | 0.701 | 0.718 | 0.734 | 0.027 | 0.678 | 5.24 | 100% | 0 |
| Chalqueno | 25 | 5.38 | 4.63 | 3.49 | 1.370 | 0.546 | 0.684 | 0.698 | 0.186 | 0.642 | 5.35 | 100% | 0 |
| Palomero | 25 | 5.50 | 4.88 | 4.00 | 1.475 | 0.661 | 0.733 | 0.748 | 0.098 | 0.690 | 5.47 | 100% | 0 |
| **Pooled** | **75** | **5.63** | **5.00** | **4.12** | **1.523** | **0.636** | **0.747** | **0.752** | **0.148** | **0.710** | **5.49** | **100%** | **0** |

STANDARD ERRORS OVER LOCI

| Population | SE Na | SE Ne | SE I | SE He | SE uHe | SE Ho | SE F |
|---|---|---|---|---|---|---|---|
| Conico | 0.491 | 0.317 | 0.082 | 0.029 | 0.029 | 0.043 | 0.036 |
| Chalqueno | 0.420 | 0.388 | 0.094 | 0.042 | 0.043 | 0.035 | 0.057 |
| Palomero | 0.423 | 0.358 | 0.095 | 0.029 | 0.029 | 0.033 | 0.026 |

I Summary (CSV)

DIVERSITY OF EVERY POPULATION

Genetic diversity by population

**Figure 4.5.** Diversity of every population: indices averaged over the 8 loci and, below them, their standard errors.

### HOW TO READ THESE RESULTS

- **The three landraces are equally diverse.** $H_e$ ranges from 0.684 in Chalqueño to 0.733 in Palomero, but the intervals overlap: with these 8 loci the difference is not resolved. The values fall within the usual range for microsatellites in outcrossing plants.
- **Allelic richness confirms the same.** Rarefied to $g$ = 44 gene copies, the size of the locus with the fewest data, $A_r$ ranges from 5.24 to 5.47. Because all three populations have 25 plants, $N_a$ and $A_r$ almost coincide.
- **Chalqueño has a heterozygote deficit.** F = 0.186, compared with 0.027 in Cónico. It may be inbreeding in that population or a problem with some locus; Block 4 tells them apart.
- **The pooled F is larger than the mean.** 0.148 versus 0.104: the difference is the Wahlund effect of figure 4.2, and it anticipates that the landraces are somewhat differentiated (Block 5).
- **The markers are highly informative.** PIC between 0.642 and 0.690, above 0.5 in all three populations.

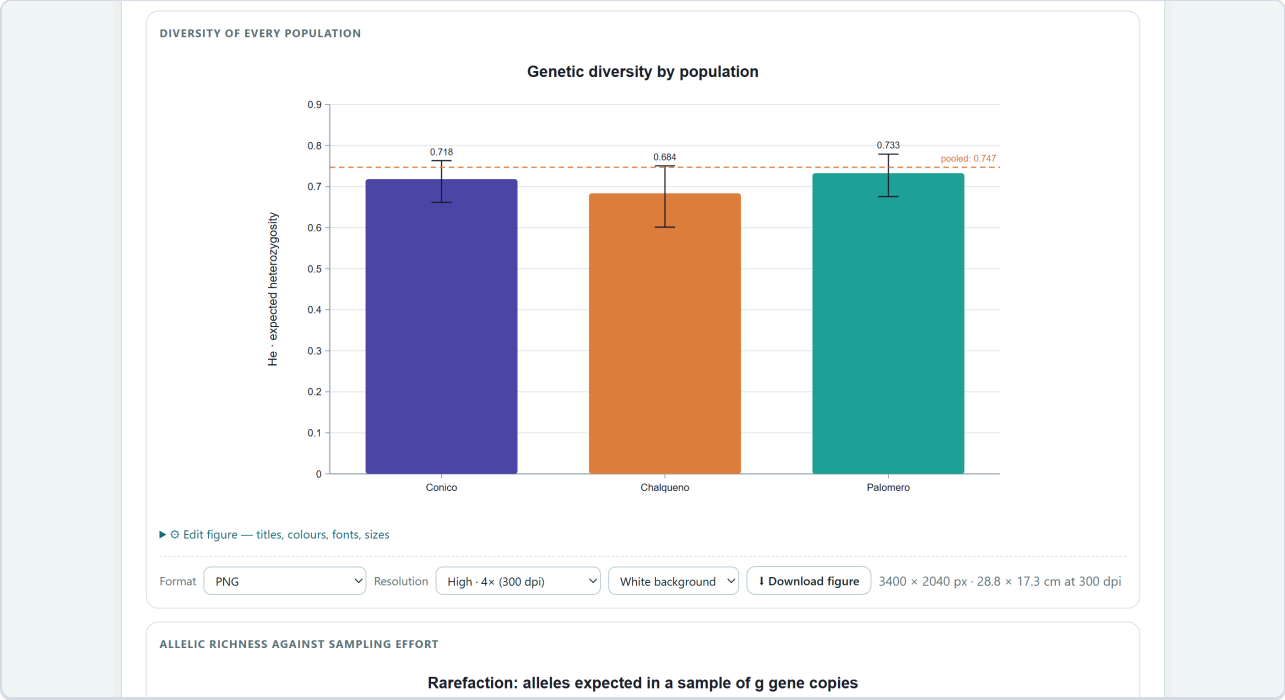



**Figure 4.6.** $H_e$ by population with 95% *bootstrap* intervals and the line for the pooled set. The three intervals overlap widely. The pooled line (0.747) lies above all three bars: when the landraces are pooled, their frequencies are averaged and the alleles become more evenly distributed, which happens only if the landraces differ in their frequencies. This difference between total diversity and the diversity within each population is the basis of $F_{ST}$ (Block 5).

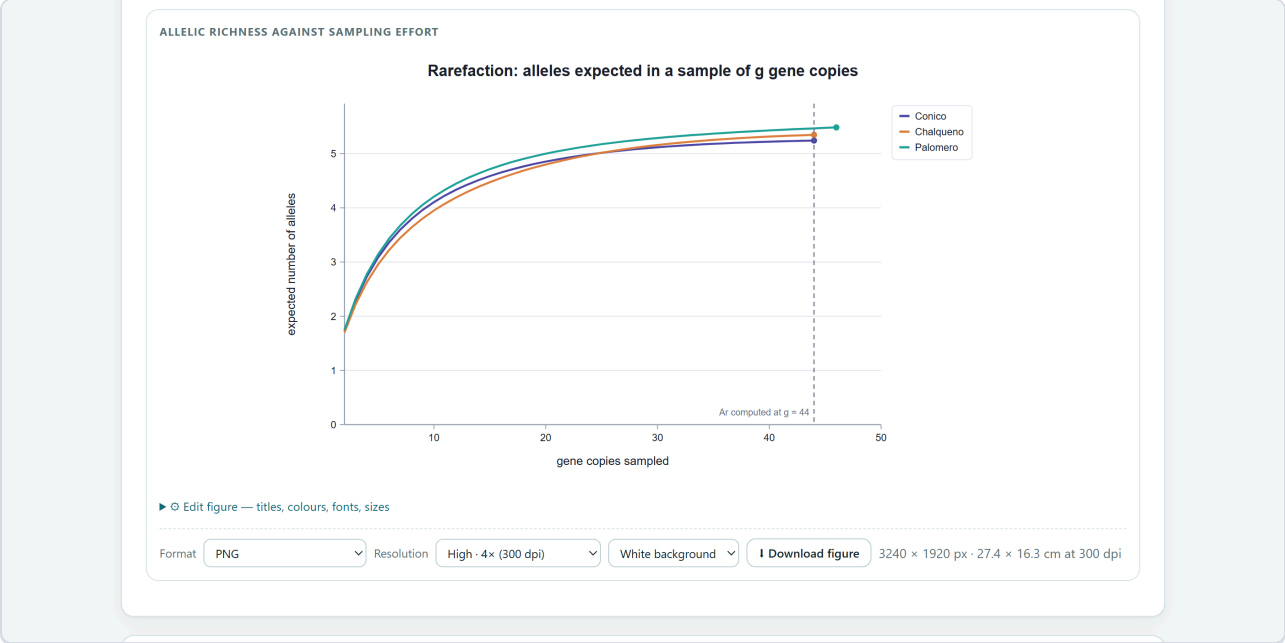
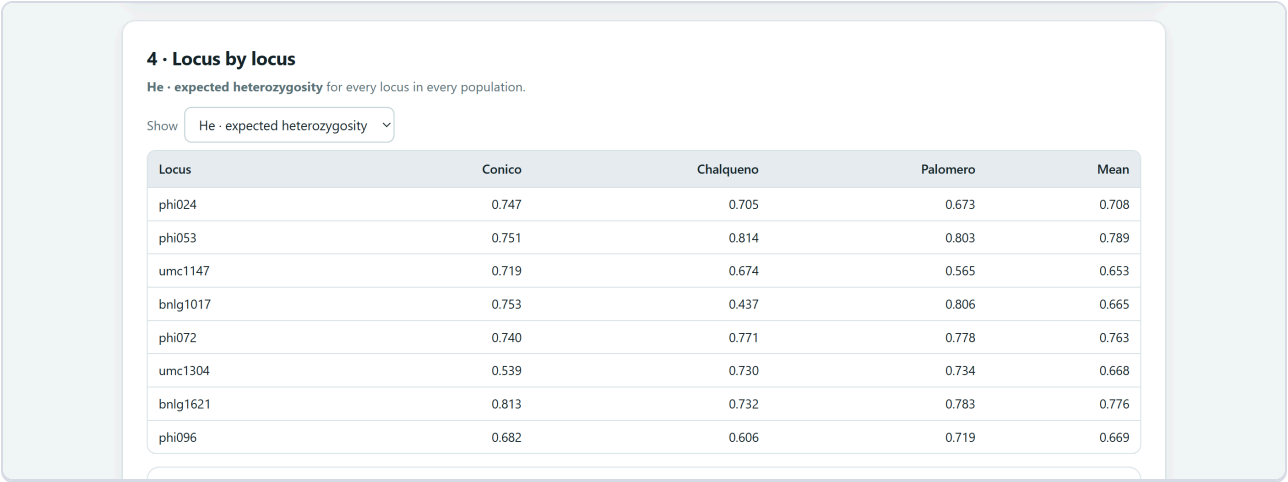



**Figure 4.7.** Rarefaction curves. Each curve gives the expected number of alleles, averaged over loci, as an increasing number of gene copies is sampled; the vertical line marks $g$ = 44. The curves almost level off before 44: sampling more plants would add few new alleles. A curve that is still rising at the end indicates that the population has undiscovered rare alleles.



**4 · Locus by locus**

**He · expected heterozygosity** for every locus in every population.

Show: He · expected heterozygosity

| Locus | Conico | Chalqueno | Palomero | Mean |
|---|---|---|---|---|
| phi024 | 0.747 | 0.705 | 0.673 | 0.708 |
| phi053 | 0.751 | 0.814 | 0.803 | 0.789 |
| umc1147 | 0.719 | 0.674 | 0.565 | 0.653 |
| bnlg1017 | 0.753 | 0.437 | 0.806 | 0.665 |
| phi072 | 0.740 | 0.771 | 0.778 | 0.763 |
| umc1304 | 0.539 | 0.730 | 0.734 | 0.668 |
| bnlg1621 | 0.813 | 0.732 | 0.783 | 0.776 |
| phi096 | 0.682 | 0.606 | 0.719 | 0.669 |

**Figure 4.8.** Locus by locus with $H_e$. bnlg1017 stands out in Chalqueño (0.437), well below its value in the other landraces.

B3

**! IMPORTANT · INBREEDING OR A FAULTY LOCUS?**

With **Show** set to **F · fixation index** you can see that in Chalqueño F ranges from −0.007 at bnlg1017 to **0.440** at umc1304, with 0.317 at bnlg1621, 0.305 at phi053 and 0.229 at umc1147. This variation alone decides nothing: with 25 plants, single-locus F fluctuates a great deal by chance. Inbreeding raises F at most loci; a **null allele** raises it at the same locus in almost every population. Block 4 makes that comparison with formal tests and, in this example, concludes that the deficit is general (chapter 5). Do not draw conclusions about inbreeding before checking it.

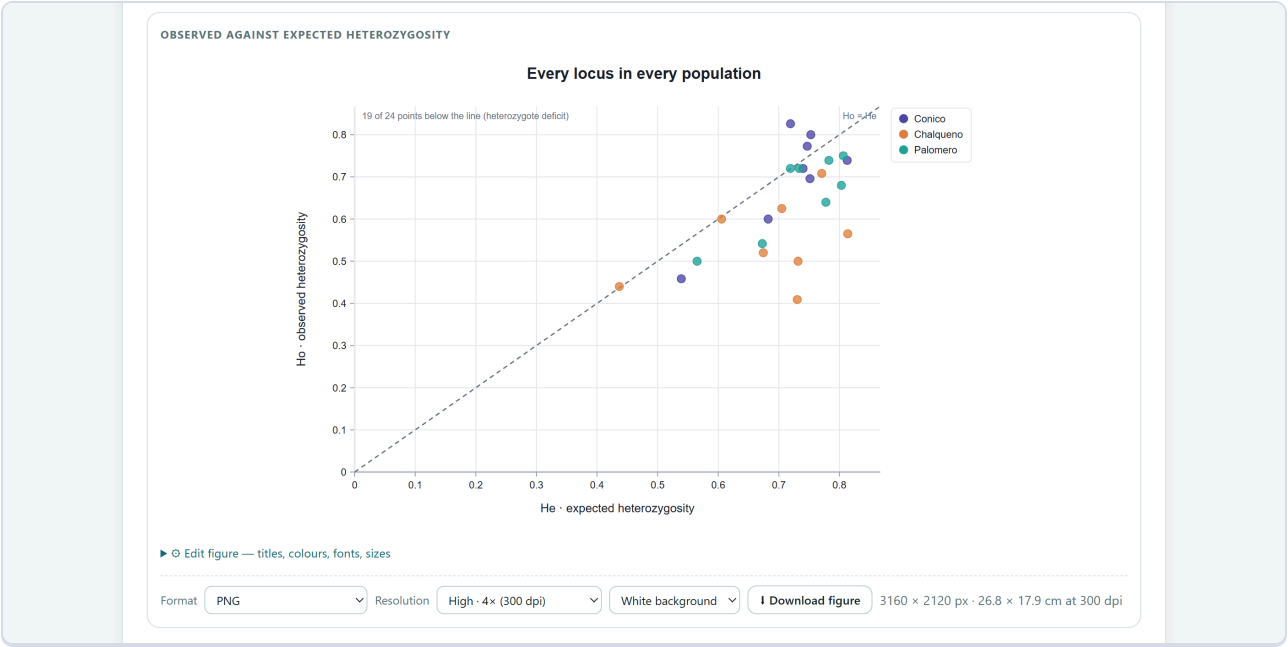



**Figure 4.9.** $H_o$ against $H_e$ for the 8 loci in the 3 populations: 19 of the 24 points lie below the diagonal. The lowest orange point (Chalqueño, umc1304) has $H_o$ = 0.41 with $H_e$ = 0.73, that is, F = 0.44.

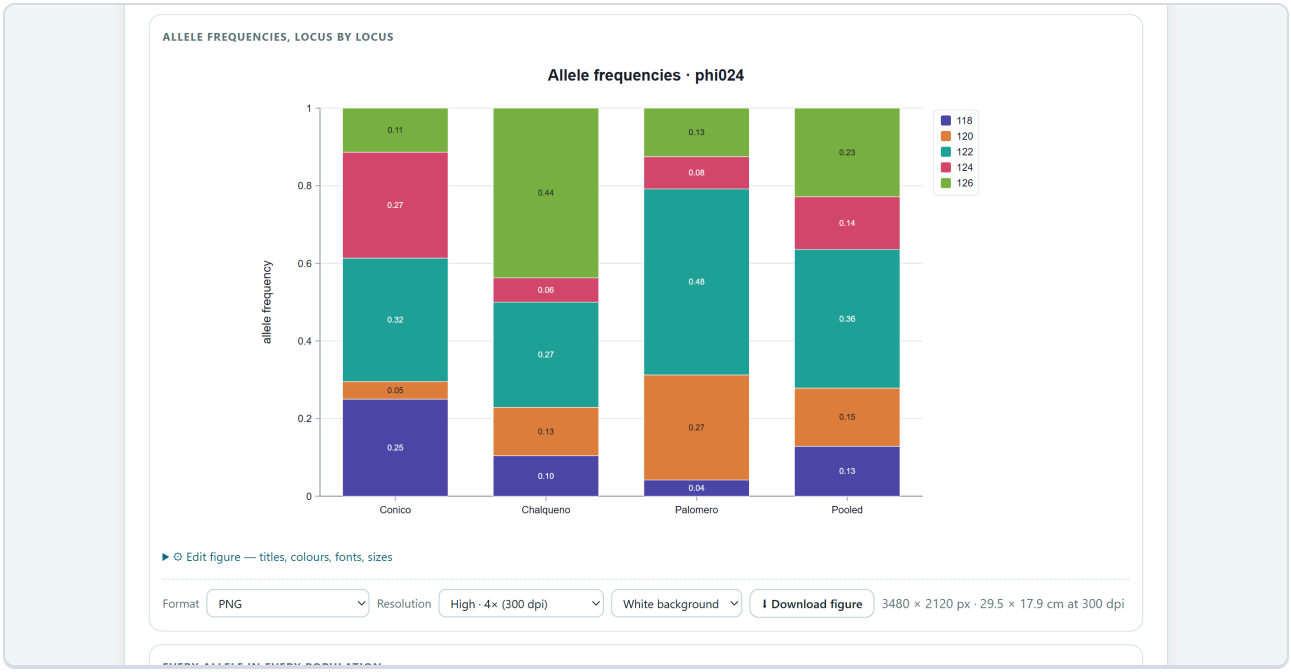



**Figure 4.10.** Allele frequencies at phi024. All five alleles occur in the three landraces, but in very different proportions: allele 126 dominates in Chalqueño (0.44) and allele 122 in Palomero (0.48). Differences like these, repeated across several loci, are what $F_{ST}$ and genetic distances measure.

**Figure 4.11.** The 45 allele frequencies of the example in a single heat map. Blank cells are alleles absent from a population, such as phi053·158 in Cónico or phi072·222 in Palomero. **Hide alleles below** hides rare alleles so that you can see the main pattern.

**NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, the indices of the example were recalculated from the genotypes with an independently written program: $H_e$, $H_o$, $uH_e$, $N_e$, I, PIC, F and $A_r$ agree with the app to the 15th decimal place. Allelic richness was also checked with 40,000 random subsamples, which differ from the exact formula in the fourth decimal place (4.9879 and 4.9878), and the Lynch and Milligan estimator was checked against its published formula.

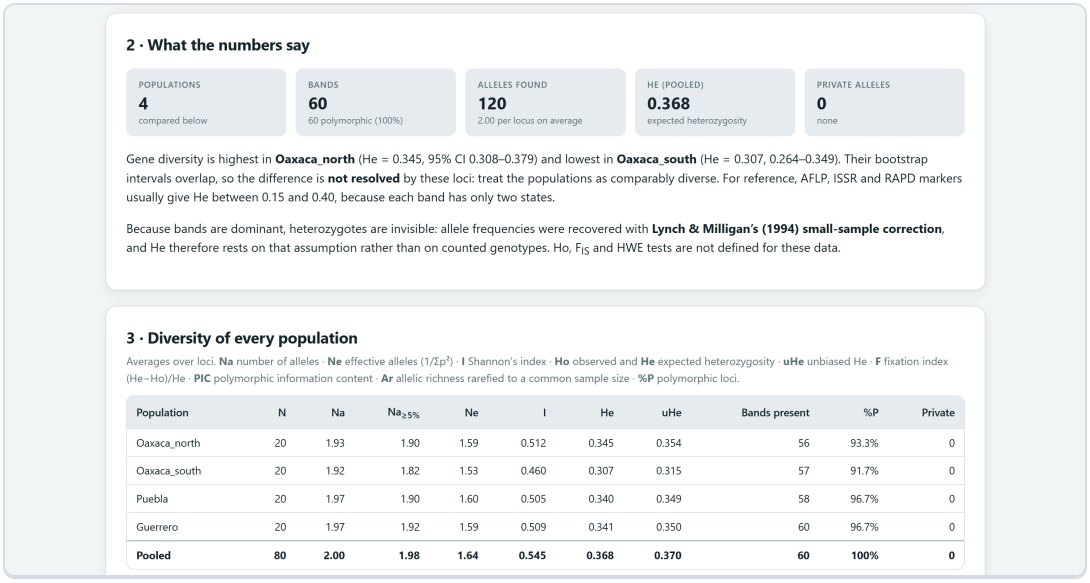
# 4.6 Dominant data, haplotypes and morphology

## Agave with AFLP: frequencies estimated from bands

B3

**STEP BY STEP**

1. Load **Agave · AFLP** in Block 2, build the dataset and go to Block 3.
2. The settings now include **Allele frequencies from dominant bands**, with Lynch and Milligan as the default option.
3. Read the summary (figure 4.12). There is no $H_o$, F, PIC or $A_r$: heterozygotes cannot be seen with bands.
4. Scroll down to **6 · Private alleles**: there are no private bands, but there are six locally common bands (figure 4.13).



**2 · What the numbers say**

| POPULATIONS | BANDS | ALLELES FOUND | HE (POOLED) | PRIVATE ALLELES |
|---|---|---|---|---|
| 4 | 60 | 120 | 0.368 | 0 |
| compared below | 60 polymorphic (100%) | 2.00 per locus on average | expected heterozygosity | none |

Gene diversity is highest in **Oaxaca_north** (He = 0.345, 95% CI 0.308–0.379) and lowest in **Oaxaca_south** (He = 0.307, 0.264–0.349). Their bootstrap intervals overlap, so the difference is **not resolved** by these loci: treat the populations as comparably diverse. For reference, AFLP, ISSR and RAPD markers usually give He between 0.15 and 0.40, because each band has only two states.

Because bands are dominant, heterozygotes are invisible: allele frequencies were recovered with **Lynch & Milligan's (1994) small-sample correction**, and He therefore rests on that assumption rather than on counted genotypes. Ho, $F_{IS}$ and HWE tests are not defined for these data.

**3 · Diversity of every population**

Averages over loci. **Na** number of alleles · **Ne** effective alleles ($1/\Sigma p^2$) · **I** Shannon's index · **Ho** observed and **He** expected heterozygosity · **uHe** unbiased He · **F** fixation index (He−Ho)/He · **PIC** polymorphic information content · **Ar** allelic richness rarefied to a common sample size · **%P** polymorphic loci.

| Population | N | Na | $Na_{\geq 5\%}$ | Ne | I | He | uHe | Bands present | %P | Private |
|---|---|---|---|---|---|---|---|---|---|---|
| Oaxaca_north | 20 | 1.93 | 1.90 | 1.59 | 0.512 | 0.345 | 0.354 | 56 | 93.3% | 0 |
| Oaxaca_south | 20 | 1.92 | 1.82 | 1.53 | 0.460 | 0.307 | 0.315 | 57 | 91.7% | 0 |
| Puebla | 20 | 1.97 | 1.90 | 1.60 | 0.505 | 0.340 | 0.349 | 58 | 96.7% | 0 |
| Guerrero | 20 | 1.97 | 1.92 | 1.59 | 0.509 | 0.341 | 0.350 | 60 | 96.7% | 0 |
| **Pooled** | **80** | **2.00** | **1.98** | **1.64** | **0.545** | **0.368** | **0.370** | **60** | **100%** | **0** |

**Figure 4.12.** Diversity with 60 AFLP bands in 4 agave populations. $H_e$ ranges from 0.307 in Oaxaca south to 0.345 in Oaxaca north, with overlapping intervals, and the pooled set reaches 0.368, within the usual range of 0.15 to 0.40 for dominant markers.

**Table 4.6.** $H_e$ of agave according to the method used to estimate frequencies from bands.

| Method | Oaxaca north | Oaxaca south | Puebla | Guerrero | Pooled |
|---|---|---|---|---|---|
| Square root | 0.345 | 0.305 | 0.340 | 0.342 | 0.368 |
| Lynch and Milligan (default) | 0.345 | 0.307 | 0.340 | 0.341 | 0.368 |
| Bayesian, F = 0 | 0.361 | 0.331 | 0.356 | 0.364 | 0.372 |
| Bayesian, F = 0.9 | 0.379 | 0.329 | 0.365 | 0.389 | 0.394 |

**✓ DECISION RULE · WHICH ESTIMATOR TO USE WITH BANDS?**

| If your species... | use... |
|---|---|
| **Is outcrossing** and you have 20 or more plants per population | Lynch and Milligan. With large samples it coincides with the square root; with few plants it corrects the square root's bias. |
| **Is outcrossing** but some bands are absent from almost every plant | Bayesian with F = 0: it does not leave frequencies at 0 or 1 when a band is missing from the whole sample. |
| **Is selfing** or has known inbreeding | Bayesian with F between 0.8 and 0.95: without this correction the frequency of the recessive allele is underestimated. |

Whichever method you choose, report it: $H_e$ from dominant markers depends on it. Comparisons among populations hardly change, as table 4.6 shows.

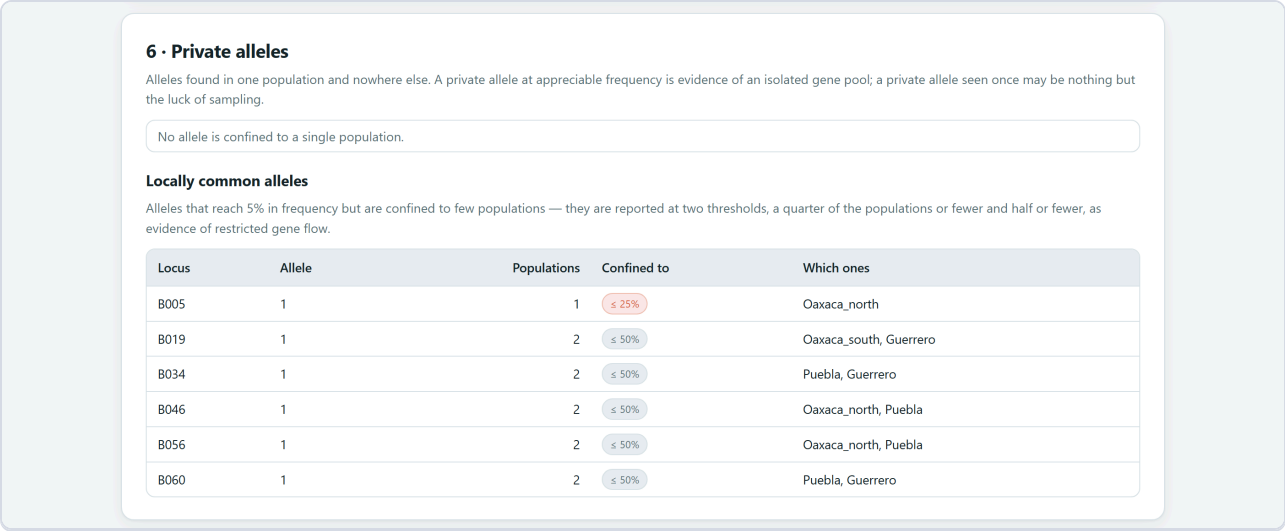
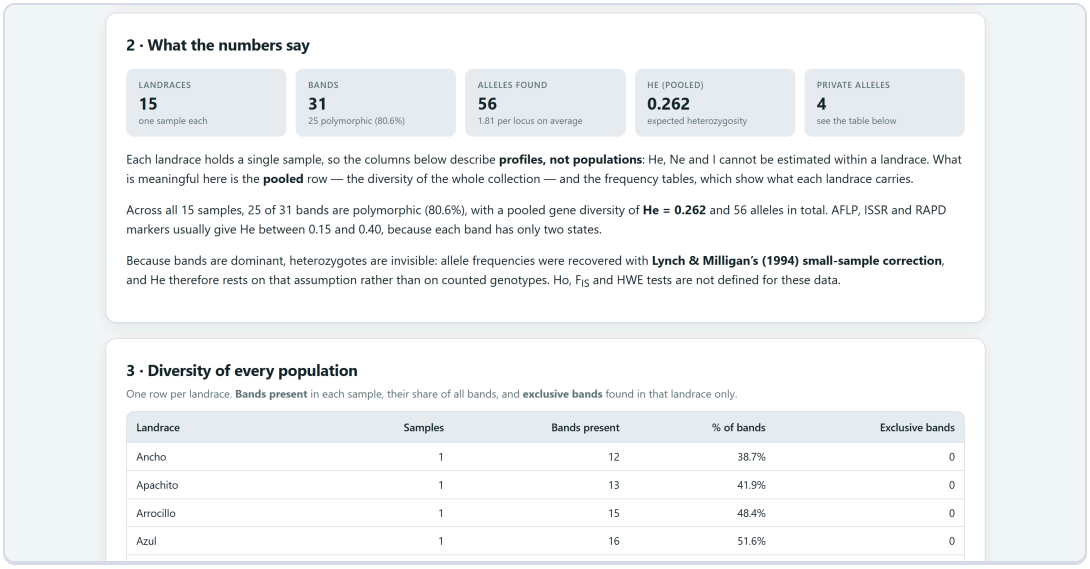



**Figure 4.13.** Locally common bands in agave. B005 reaches 5% only in Oaxaca north (one of four populations, 25% or fewer); the other five do so in two populations. They point to restricted gene flow even though there are no private bands.

## Maize landraces with ISSR: one sample per unit

In the maize landraces example, each row is a bulked sample of one landrace. There is no diversity **within** landraces to measure, so the block changes its presentation: the table by population shows how many bands each landrace has and which ones are exclusive to it, and the interpretation focuses on the pooled set.



**2 · What the numbers say**

| LANDRACES | BANDS | ALLELES FOUND | HE (POOLED) | PRIVATE ALLELES |
|---|---|---|---|---|
| 15 | 31 | 56 | 0.262 | 4 |
| one sample each | 25 polymorphic (80.6%) | 1.81 per locus on average | expected heterozygosity | see the table below |

Each landrace holds a single sample, so the columns below describe **profiles, not populations**: He, Ne and I cannot be estimated within a landrace. What is meaningful here is the **pooled** row — the diversity of the whole collection — and the frequency tables, which show what each landrace carries.

Across all 15 samples, 25 of 31 bands are polymorphic (80.6%), with a pooled gene diversity of **He = 0.262** and 56 alleles in total. AFLP, ISSR and RAPD markers usually give He between 0.15 and 0.40, because each band has only two states.

Because bands are dominant, heterozygotes are invisible: allele frequencies were recovered with **Lynch & Milligan's (1994) small-sample correction**, and He therefore rests on that assumption rather than on counted genotypes. Ho, $F_{IS}$ and HWE tests are not defined for these data.

**3 · Diversity of every population**

One row per landrace. **Bands present** in each sample, their share of all bands, and **exclusive bands** found in that landrace only.

| Landrace | Samples | Bands present | % of bands | Exclusive bands |
|---|---|---|---|---|
| Ancho | 1 | 12 | 38.7% | 0 |
| Apachito | 1 | 13 | 41.9% | 0 |
| Arrocillo | 1 | 15 | 48.4% | 0 |
| Azul | 1 | 16 | 51.6% | 0 |

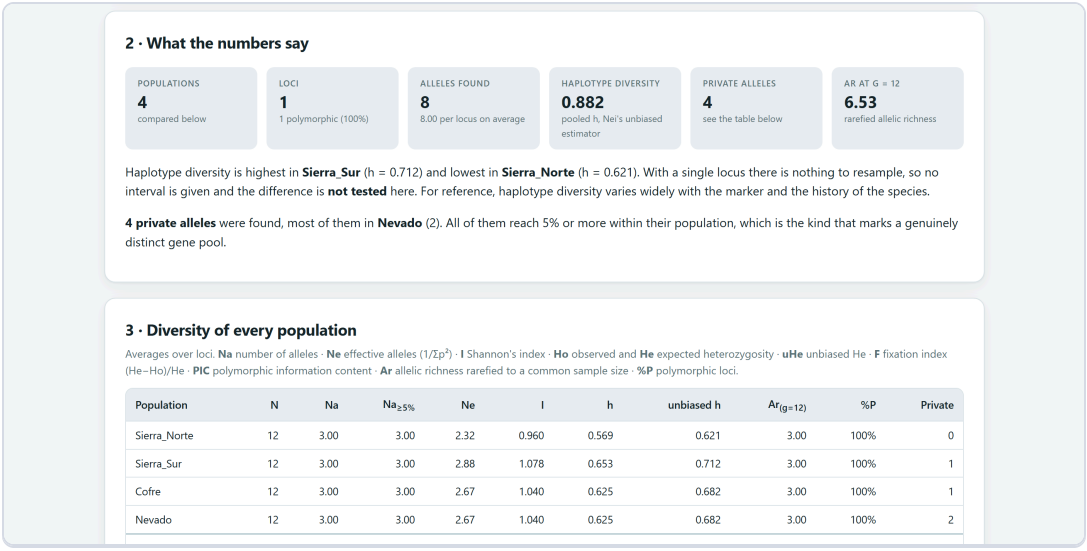


**Figure 4.14.** The 15 landraces with ISSR. For the pooled set: 25 of 31 bands polymorphic (80.6%), $H_e$ = 0.262 and 4 bands exclusive to a single landrace (Comiteco, Bolita, Celaya and Chiquito). For each landrace, the table gives the bands present (from 9 in Chapalote to 17 in Comiteco), their percentage and the exclusive bands.

**NOTE · EXCLUSIVE BANDS IN BULKED SAMPLES**

A band exclusive to one landrace is a good candidate identity marker for that landrace, but with a single bulked sample you cannot tell how often its plants carry it. Confirming it requires several plants per landrace. The relationships among the landraces are studied in Block 6.

# Pine with chloroplast DNA: haplotype diversity



**2 · What the numbers say**

| POPULATIONS | LOCI | ALLELES FOUND | HAPLOTYPE DIVERSITY | PRIVATE ALLELES | AR AT G = 12 |
|---|---|---|---|---|---|
| 4 compared below | 1 1 polymorphic (100%) | 8 8.00 per locus on average | 0.882 pooled h, Nei's unbiased estimator | 4 see the table below | 6.53 rarefied allelic richness |

Haplotype diversity is highest in **Sierra_Sur** (h = 0.712) and lowest in **Sierra_Norte** (h = 0.621). With a single locus there is nothing to resample, so no interval is given and the difference is **not tested** here. For reference, haplotype diversity varies widely with the marker and the history of the species.

**4 private alleles** were found, most of them in **Nevado** (2). All of them reach 5% or more within their population, which is the kind that marks a genuinely distinct gene pool.

**3 · Diversity of every population**

Averages over loci. **Na** number of alleles · **Ne** effective alleles ($1/\Sigma p^2$) · **I** Shannon's index · **Ho** observed and **He** expected heterozygosity · **uHe** unbiased He · **F** fixation index (He−Ho)/He · **PIC** polymorphic information content · **Ar** allelic richness rarefied to a common sample size · **%P** polymorphic loci.

| Population | N | Na | $Na_{\geq 5\%}$ | Ne | I | h | unbiased h | $Ar_{(g=12)}$ | %P | Private |
|---|---|---|---|---|---|---|---|---|---|---|
| Sierra_Norte | 12 | 3.00 | 3.00 | 2.32 | 0.960 | 0.569 | 0.621 | 3.00 | 100% | 0 |
| Sierra_Sur | 12 | 3.00 | 3.00 | 2.88 | 1.078 | 0.653 | 0.712 | 3.00 | 100% | 1 |
| Cofre | 12 | 3.00 | 3.00 | 2.67 | 1.040 | 0.625 | 0.682 | 3.00 | 100% | 1 |
| Nevado | 12 | 3.00 | 3.00 | 2.67 | 1.040 | 0.625 | 0.682 | 3.00 | 100% | 2 |

**Figure 4.15.** With haplotypes, the block reports the unbiased haplotype diversity *h*. Sierra Sur is the most diverse (*h* = 0.712) and Sierra Norte the least (0.621), but with a single alignment there are no intervals and the difference is not tested. The 48 sequences together give Hd = 0.882 with 8 haplotypes.

**6 · Private alleles**

Alleles found in one population and nowhere else. A private allele at appreciable frequency is evidence of an isolated gene pool; a private allele seen once may be nothing but the luck of sampling.

| Locus | Allele | Only in | Copies | Frequency there | |
|---|---|---|---|---|---|
| Haplotype | H6 | Cofre | 6 | 0.500 | frequent |
| Haplotype | H7 | Nevado | 6 | 0.500 | frequent |
| Haplotype | H4 | Sierra_Sur | 5 | 0.417 | frequent |
| Haplotype | H8 | Nevado | 3 | 0.250 | frequent |

PRIVATE ALLELES



**Figure 4.16.** Four private haplotypes, all of them frequent: H6 in Cofre and H7 in Nevado (half of the sequences of each population), H4 in Sierra Sur and H8 in Nevado. Nevado holds two haplotypes that exist in no other population.

### WHAT PRIVATE HAPLOTYPES MEAN IN THE CHLOROPLAST

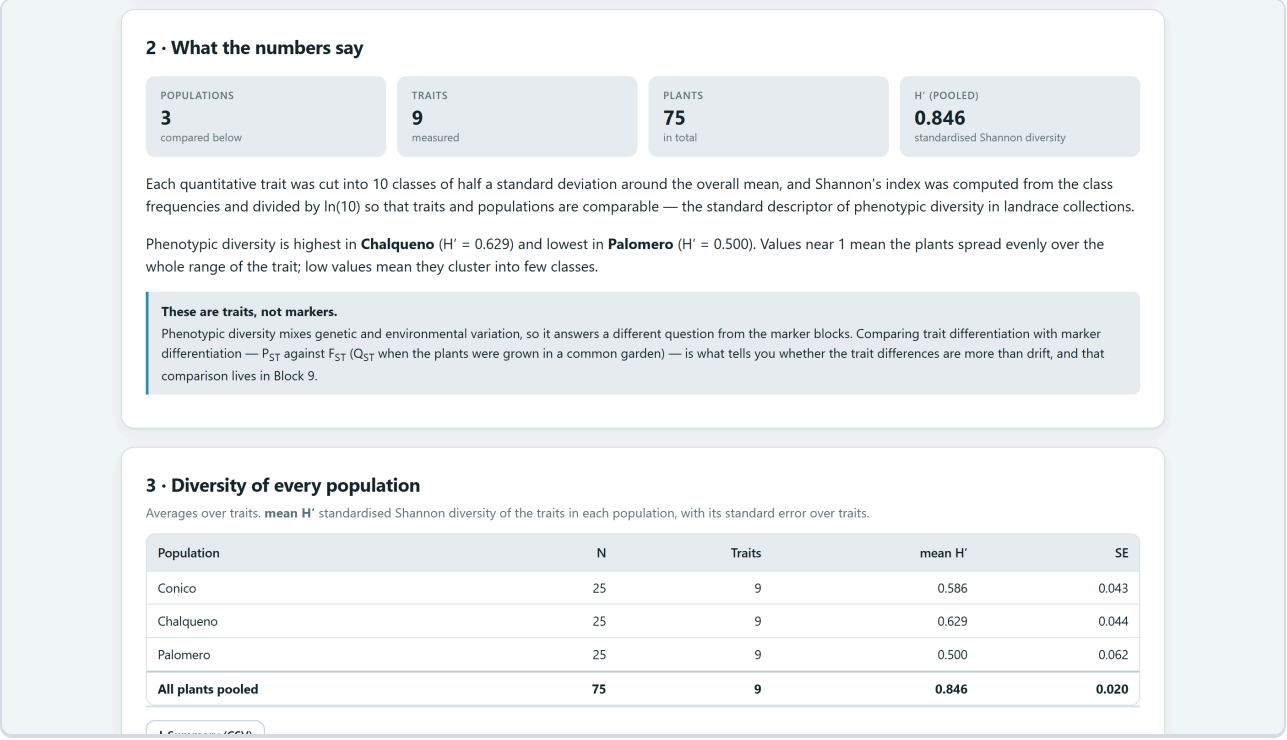
Chloroplast DNA is inherited from a single parent. In most angiosperms it comes from the mother and travels only in seeds; in pines and other conifers it comes from the father and travels in both pollen and seeds. So, in a pine, the fact that Nevado and Cofre retain frequent haplotypes of their own indicates that gene flow between these populations is scarce through both pollen and seeds; in an angiosperm it would speak only of seed movement. The geographical distribution and genealogy of the haplotypes are studied with the network in Block 8.

## Maize morphology: phenotypic diversity



### STEP BY STEP

1. Load **Maize · morphology**, build the dataset and go to Block 3.
2. The settings now include **Classes per trait**, set to 10 classes.
3. Read the interpretation and the table of H' by population (figure 4.17).
4. Review the heat map by trait and population (figure 4.18).



**2 · What the numbers say**

| POPULATIONS | TRAITS | PLANTS | H′ (POOLED) |
|---|---|---|---|
| 3 | 9 | 75 | 0.846 |
| compared below | measured | in total | standardised Shannon diversity |

Each quantitative trait was cut into 10 classes of half a standard deviation around the overall mean, and Shannon's index was computed from the class frequencies and divided by ln(10) so that traits and populations are comparable — the standard descriptor of phenotypic diversity in landrace collections.

Phenotypic diversity is highest in **Chalqueno** (H′ = 0.629) and lowest in **Palomero** (H′ = 0.500). Values near 1 mean the plants spread evenly over the whole range of the trait; low values mean they cluster into few classes.

**These are traits, not markers.**
Phenotypic diversity mixes genetic and environmental variation, so it answers a different question from the marker blocks. Comparing trait differentiation with marker differentiation — $P_{ST}$ against $F_{ST}$ ($Q_{ST}$ when the plants were grown in a common garden) — is what tells you whether the trait differences are more than drift, and that comparison lives in Block 9.

**3 · Diversity of every population**

Averages over traits. **mean H′** standardised Shannon diversity of the traits in each population, with its standard error over traits.

| Population | N | Traits | mean H′ | SE |
|---|---|---|---|---|
| Conico | 25 | 9 | 0.586 | 0.043 |
| Chalqueno | 25 | 9 | 0.629 | 0.044 |
| Palomero | 25 | 9 | 0.500 | 0.062 |
| **All plants pooled** | **75** | **9** | **0.846** | **0.020** |

**Figure 4.17.** Phenotypic diversity of 9 traits. Chalqueño is the most diverse (H′ = 0.629 ± 0.044) and Palomero the least (0.500 ± 0.062). The pooled set reaches 0.846 because, when the landraces are pooled, the plants span a much wider range of each trait.

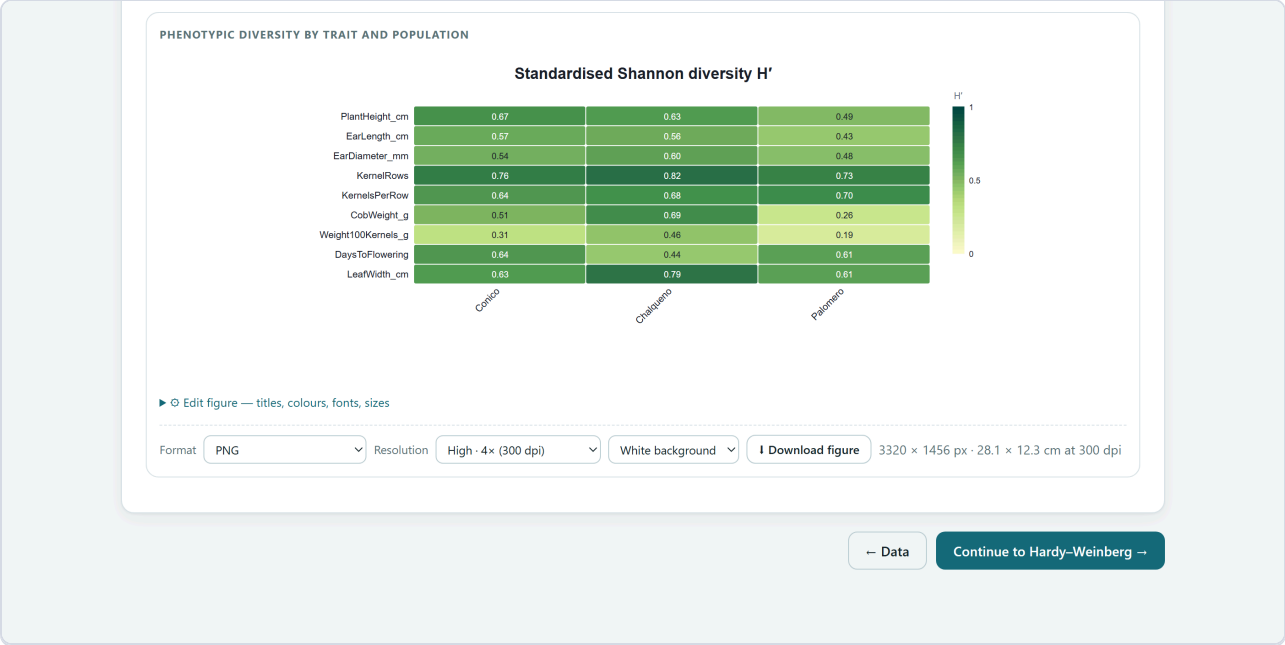



**Figure 4.18.** H′ by trait and population. Number of kernel rows (0.73 to 0.82) is the most variable trait within landraces, and 100-kernel weight (0.19 to 0.46) the most uniform. Cob weight clearly separates Chalqueño (0.69) from Palomero (0.26).

### HOW H′ IS CALCULATED

Each quantitative trait is divided into 10 classes half a standard deviation wide, centred on the mean of all plants; values beyond 2.5 standard deviations fall into the extreme classes. In each population, the proportion $p_i$ of plants in each class is counted, $H = -\Sigma\, p_i \ln p_i$ is calculated and divided by ln 10, so that H' ranges from 0 (all plants in one class) to 1 (plants evenly spread over the 10 classes). Text traits use their own categories.

### DECISION RULE · PHENOTYPIC H′

| H′ | Diversity | Interpretation |
|---|---|---|
| **0.6 or more** | High | The plants cover almost the whole range of the trait. Useful material as a source of variation. |
| **0.4 to 0.6** | Medium | Moderate variation, common within a landrace or local variety. |
| **Less than 0.4** | Low | Trait uniform in that population: fixed by selection or with a narrow genetic base. |

An indicative guide used in studies of local varieties. H' mixes genetic and environmental variation: to find out whether differences among populations exceed what drift would produce, $P_{ST}$ is compared with $F_{ST}$ in Block 9.

**NOTE · WHAT TO REPORT IN A PAPER**

A typical diversity table includes, for each population, N, $N_a$, $A_r$ with its *g*, $H_o$, $uH_e$, F and the number of private alleles, with their standard errors or intervals. Download it with **Summary (CSV)**: it contains the indices and their standard errors to four decimal places. With dominant markers, report $H_e$, %P and the frequency estimation method; with haplotypes, *h* and the number of haplotypes; with traits, H' by trait. In your methods, state the polymorphism criterion, the value of *g*, the number of bootstrap replicates and the seed.

$uH_e + A_r$

THE TWO INDICES WORTH COMPARING

If you could report only two indices to compare populations, choose **unbiased expected heterozygosity** and **rarefied allelic richness**: one measures how evenly the alleles are distributed and the other how many there are, and both are corrected so that sample size does not favour any population.



The next chapter takes the fixation index, which was only described here, and puts it to the test: Block 4 checks Hardy–Weinberg equilibrium, separates inbreeding from null alleles and measures linkage disequilibrium between loci.

# B4

CHAPTER FIVE · BLOCK 4 · HWE & LD

# Hardy–Weinberg and linkage disequilibrium



Are the genotypes the ones predicted by random mating? And if they are not, is it inbreeding, a null allele or a mixture of populations? Block 4 tests Hardy–Weinberg equilibrium locus by locus, estimates the inbreeding coefficient $F_{IS}$ and the frequency of null alleles, measures whether loci are inherited independently, and corrects the P values when many tests are run at once.


5.1 Hardy–Weinberg equilibrium

5.2 $F_{IS}$, inbreeding and null alleles

5.3 Linkage disequilibrium

5.4 Correction for multiple tests

5.5 Block settings, tables and figures

5.6 Step by step: maize and agave

**F = 0.16**

AN EXAMPLE WITH A KNOWN ANSWER

The maize microsatellite dataset was simulated with a heterozygote deficit of **F = 0.16 in all three landraces, no null alleles** and 4% of data missing at random. It is used throughout the chapter to check whether the block reaches the right conclusion.

## 5.1 Hardy–Weinberg equilibrium

If plants mate at random, genotype frequencies are fixed by allele frequencies: with two alleles at frequencies $p$ and $q$, you expect $p^2$ AA homozygotes, $2pq$ heterozygotes and $q^2$ aa homozygotes. This is Hardy–Weinberg equilibrium (HWE). A single generation of random mating is enough to reach it.



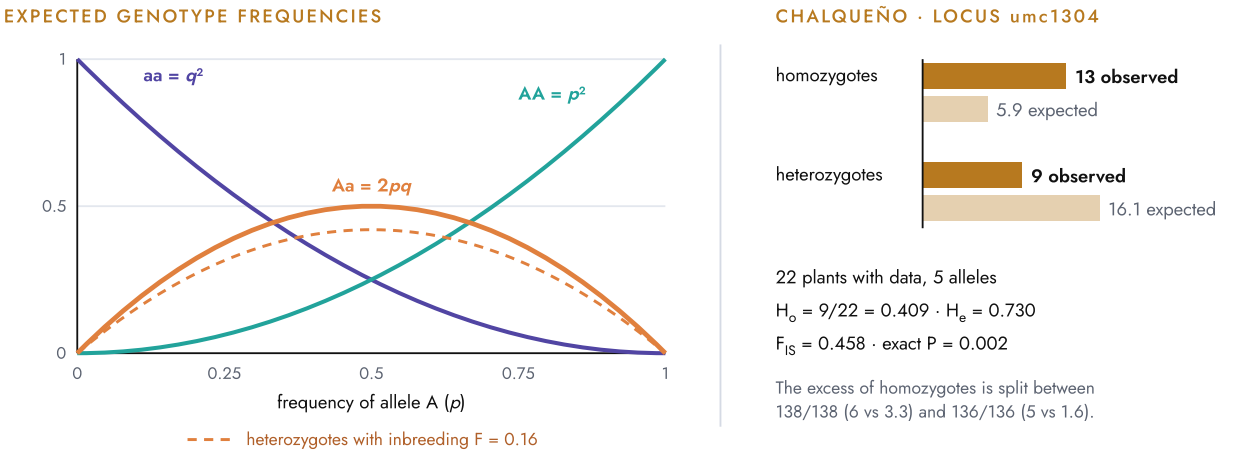



**Figure 5.1.** Left: Hardy–Weinberg proportions for a locus with two alleles. Heterozygotes reach their maximum, 0.5, when both alleles are equally frequent; with inbreeding (dashed line, F = 0.16) the heterozygote curve drops by 16% at every frequency and the homozygotes rise by the same amount. Right: the locus with the largest deficit in the maize example, with the genotypes counted by the app against those expected.

### WHAT EQUILIBRIUM ASSUMES AND WHAT A DEPARTURE FROM IT MEANS

Equilibrium holds if mating is random, the population is large and there is no selection, migration or mutation changing the frequencies. No real population meets all of these conditions, yet genotypes usually stay very close to expectation. That is why a departure **is information**: an excess of homozygotes points to selfing, matings between relatives, a mixture of different populations in a single sample, or marker problems; an excess of heterozygotes points to clonal reproduction, a recent cross between different sources, or selection favouring heterozygotes.

### Two tests: $\chi^2$ and the exact test

The **$\chi^2$** test compares observed and expected genotypes with $\Sigma(O - E)^2/E$ and $k(k - 1)/2$ degrees of freedom, where $k$ is the number of alleles. It works when every expected genotype has at least 5 individuals, which almost never happens with microsatellites: with 5 alleles there are 15 possible genotypes among which to distribute 25 plants. The **exact test** (Guo and Thompson, 1992) does not depend on that assumption. It keeps the allele counts fixed, shuffles them at random among the plants thousands of times and asks how unusual the observed table of genotypes is.

**Table 5.1.** The two tests at the three loci in the example where they differ most. In all 24 cases (3 landraces × 8 loci) at least one expected genotype falls below 5.

| Landrace · locus | Alleles | $\chi^2$ P | Exact P | Reading |
|---|---|---|---|---|
| Chalqueño · phi053 | 7 | 0.0001 | 0.043 | $\chi^2$ overstates the evidence hundreds of times over. |
| Palomero · phi072 | 7 | 0.0016 | 0.024 | With many rare genotypes, $\chi^2$ shoots up. |
| Chalqueño · umc1304 | 5 | 0.0023 | 0.002 | When the departure is large, both agree. |

**DECISION RULE · WHICH HARDY–WEINBERG TEST TO READ**

| If… | then… |
|---|---|
| **You use microsatellites or SNPs with rare alleles** | Read the exact test. The app flags unreliable $\chi^2$ values with †. |
| **You want to know the direction** of the departure | Use the one-tailed tests: one-sided: heterozygote deficit or excess (Rousset and Raymond, 1995). They are more powerful than the global test when you already suspect inbreeding. |
| **The exact P is close to α** | Raise the permutations to 10,000. With 1000, a P of 0.05 has an error of ±0.007; with 10,000, of ±0.002. |
| **You are comparing with an older paper** | Show $\chi^2$ for reference only and explain that the conclusion rests on the exact test. |

A Monte Carlo test never gives P = 0: with B permutations and b tables as unusual as the observed one, P = (b + 1)/(B + 1). The smallest possible value with 1000 permutations is 0.001.

B4

## 5.2 $F_{IS}$, inbreeding and null alleles

The **inbreeding coefficient** $F_{IS}$ measures what proportion of the expected heterozygosity is missing: $F_{IS} = 1 - H_o/H_e$. It equals 0 under random mating, is positive when heterozygotes are lacking and negative when they are in excess. For a single population the app computes it in two equivalent ways, Nei and Chesser's (1983) locus by locus and Weir and Cockerham's (1984) *f* for all loci together, with a *bootstrap* interval over loci.

**Table 5.2.** Block 4 statistics. *n* is the number of individuals with data at the locus, *k* the number of alleles and *L* the number of loci.

| Statistic | Formula | What it measures |
|---|---|---|
| Unbiased $\mathbf{H_s}$ | $n/(n-1)\cdot(1-\Sigma p_i^2 - H_o/2n)$ | Expected heterozygosity corrected for sample size (Nei and Chesser, 1983). |
| $\mathbf{F_{IS}}$ per locus | $1 - H_o / H_s$ | Deficit (positive) or excess (negative) of heterozygotes at one locus. |
| Multilocus ***f*** | $1 - \Sigma_L H_o / \Sigma_L H_s$ | $F_{IS}$ of the whole population: a ratio of sums, not a mean of ratios (Weir and Cockerham, 1984). |
| Equilibrium ***s*** | $2F / (1 + F)$ | Selfing rate that would produce that F in a mixed-mating plant. |
| HWE $\chi^2$ | $\Sigma\,(O-E)^2 / E$, $df = k(k-1)/2$ | Classical test; unreliable with expected genotypes below 5. |
| **Exact P** | $\Sigma\ \Pr(\text{tables with } \Pr \le \Pr_{obs})$ | Probability of a table of genotypes as unusual as the observed one, with alleles held fixed (Levene, 1949; Guo and Thompson, 1992). |
| **Null allele** · Chakraborty | $(H_e - H_o) / (H_e + H_o)$ | Frequency of the allele that fails to amplify (Chakraborty et al., 1992). |
| **Null allele** · Brookfield 1 | $(H_e - H_o) / (1 + H_e)$ | Brookfield's (1996) variant for when every plant amplifies something. |
| **Null allele** · EM | iterative maximum likelihood | Separates apparent A/null homozygotes from true A/A ones (Dempster et al., 1977). |

B4

### THREE CAUSES OF THE SAME SYMPTOM

A positive $F_{IS}$ can arise from three very different sources, and it is worth telling them apart before writing "inbreeding" in a paper:

- **Inbreeding** (selfing or matings between relatives). It affects the whole genome: it raises $F_{IS}$ at almost every locus by a similar amount.
- **Wahlund effect**. The sample pools plants from two or more distinct gene pools. It also raises $F_{IS}$ at many loci and, in addition, creates linkage disequilibrium (section 5.3).
- **Null alleles**. A mutation at the primer site prevents an allele from amplifying, so its heterozygotes are read as homozygotes. It affects **one locus**, and does so in almost every population.

**Table 5.3.** Selfing rate corresponding to each value of F at equilibrium, $s = 2F/(1 + F)$.

| F | 0.05 | 0.10 | 0.16 | 0.22 | 0.33 | 0.50 | 0.80 | 1 |
|---|---|---|---|---|---|---|---|---|
| *s* | 10% | 18% | 28% | 36% | 50% | 67% | 89% | 100% |

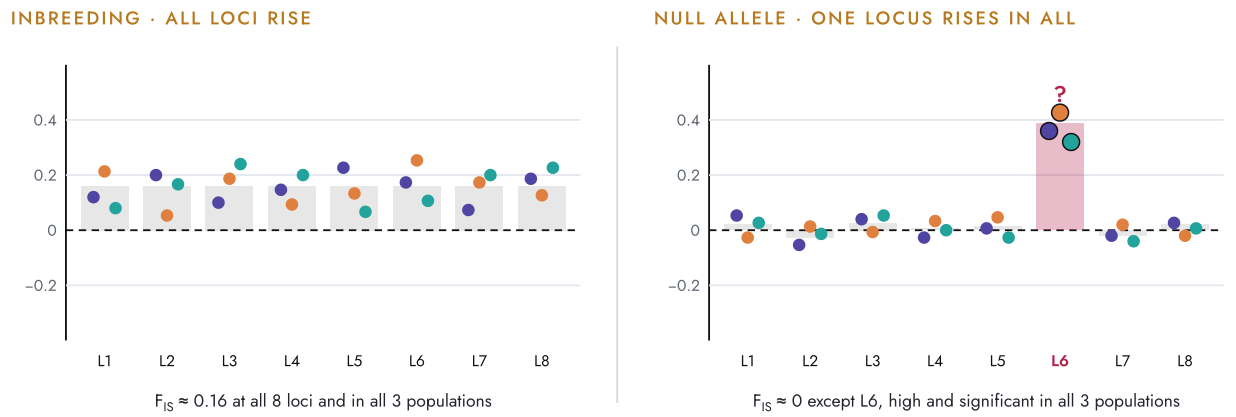



**Figure 5.2.** The two patterns the block looks for in the figure Inbreeding or null alleles?, with illustrative data. Each point is the $F_{IS}$ of one locus in one population; the grey bar is the locus mean. Outlined points are significant departures from equilibrium.

### ✓ DECISION RULE · INBREEDING, WAHLUND OR NULL ALLELES?

| Pattern | Likely cause | What to do |
|---|---|---|
| **Positive $F_{IS}$ at most loci** and an interval for *f* that excludes 0 | Inbreeding | Report *f* with its interval and the selfing rate *s* for reference. Keep the loci. |
| **One locus with a significant deficit in half or more of the populations** and a null frequency above 5% | Null allele | The app flags the locus as suspected null allele. Recompute $F_{ST}$ without it in Block 5 and compare. |
| **General deficit and linkage disequilibrium within the population** | Wahlund or selfing | Look for subgroups with Block 7. If they appear, the "population" mixes gene pools. |
| **Both patterns at once** | Inbreeding plus a null | Estimate *f* from the unflagged loci and report both results. |
| **Scattered departures**, with no repeated locus or population | Chance or scoring errors | This does not justify removing loci. Check the gels for the extreme cases. |

The automatic diagnosis in card 2 applies this rule. A significant excess of heterozygotes (*f* with an interval below 0) suggests clonality, a recent cross between different sources or, in small samples, mere chance.

### ! IMPORTANT · NULL ALLELE ESTIMATORS CANNOT TELL CAUSES APART

All three estimators attribute **the entire** heterozygote deficit to null alleles. In the maize example, simulated without nulls, EM reaches 0.183 at Chalqueño · umc1304 because it absorbs the inbreeding. That is why the app does not flag a locus on its estimated frequency alone: it also requires the deficit to recur in at least half of the populations. And the option Treat missing genotypes as null homozygotes should be switched on only if the missing data are amplification failures: in the example, whose missing data are random, it raises the mean EM estimate from 0.026, 0.081 and 0.041 to 0.104, 0.139 and 0.092.

# 5.3 Linkage disequilibrium

Two loci are in **linkage disequilibrium** (LD) when their alleles do not combine at random: knowing which allele a plant carries at one locus helps predict the allele at the other. Recombination erases that association generation after generation; what creates or maintains it is **physical proximity** between the loci, **selfing** and **clonality**, which reduce effective recombination, **admixture** of different populations, drift in small populations, and selection.

### THE INDEX OF ASSOCIATION IN A NUTSHELL

For each pair of plants, count at how many allele copies they differ at each locus. If the loci are independent, the differences add up like independent variables and the variance of the total, $V_D$, equals the sum of the per-locus variances. If there is association, plants that differ at one locus tend to differ at the others, and $V_D$ grows. The **index of association** is $I_A = V_D/\Sigma V_j - 1$ (Brown et al., 1980). Because it depends on the number of loci, the app also reports its standardised version, $\bar{r}_d = (V_D - \Sigma V_j) / (2\,\Sigma_{j<k}\,\sqrt{(V_j V_k)})$ (Agapow and Burt, 2001), which lies between 0 and 1 apart from small negative fluctuations due to chance and can be compared across studies. It works with codominant, dominant and haploid markers.

B4

Significance is obtained by permutation: the genotypes at each locus are shuffled among plants separately, which destroys any association, and the app counts how often the shuffled $\bar{r}_d$ equals or exceeds the observed one. For each **pair of loci**, $\bar{r}_d$ is the correlation between their distances; when every locus has two alleles, the app adds $r^2$ and D' (Excoffier and Slatkin, 1995, to resolve the phases of double heterozygotes). With dominant markers, $r^2$ and D' are computed on band presence and absence, which are not haplotypes: treat them as a correlation between bands.

### DECISION RULE · READING $\bar{r}_d$

| If… | then… |
|---|---|
| $\bar{r}_d$ **≈ 0 and not significant** in every population | The loci behave as independent. The Block 7 methods, which assume this, can be used without reservation. |
| $\bar{r}_d$ **significant and below 0.05** | Detectable but weak association; it does not bias analyses that assume independence. |
| $\bar{r}_d$ **significant and above 0.05** within populations | Selfing, clonality or recent admixture. Read Bayesian clustering and assignments with caution. |
| **Significant only in the pooled sample** | A sign of differentiation between populations, not of physical linkage. |
| **One particular pair stands out** in several populations | Possible physical linkage between those two loci. Consider using only one of them in analyses that require independence. |

The 0.05 threshold is a guide: in outcrossing plants with unlinked markers, $\bar{r}_d$ usually stays below 0.01; in clonal or highly selfing species, values of 0.1 to 0.5 are common.

# 5.4 Correction for multiple tests

With α = 0.05, one test in twenty comes out significant even when nothing is going on. The maize example runs 24 Hardy–Weinberg tests, so 1.2 "rejections" would be expected by pure chance; for linkage, 28 pairwise tests are run per landrace. Corrections adjust the P values so that the set of conclusions as a whole remains reliable.

**Table 5.4.** Corrections available in Correction for multiple tests, and how many of the example's 24 exact tests remain significant with each.

| Method | What it controls and how | When to use it | Significant |
|---|---|---|---|
| **None** | Nothing: each test at level α. | For exploration only. | 7 |
| **Bonferroni** | The probability of at least one false positive; multiplies each P by the number of tests *m*. | Few tests and critical conclusions. | 1 |
| **Holm** (default) | The same as Bonferroni, but sorts the P values and multiplies the smallest by *m*, the next by *m* − 1, and so on (Holm, 1979). | The general-purpose choice: it never rejects fewer than Bonferroni. | 1 |
| **Benjamini–Hochberg** | The expected proportion of false positives among the rejections (FDR). | Many tests, such as pairs of loci; screening for candidates. | 1 |
| **Benjamini–Yekutieli** | The FDR even when tests are correlated (Benjamini and Yekutieli, 2001). | Tests that depend on one another. | 0 |

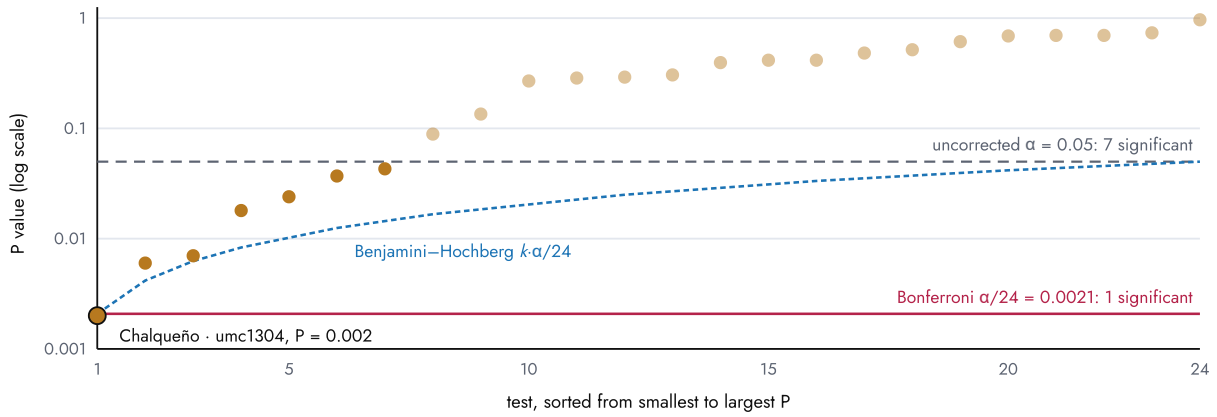



**Figure 5.3.** The example's 24 exact P values, sorted. A test is significant if its point lies below the method's line. Only Chalqueño · umc1304 survives the correction; its Holm-adjusted P is 0.002 × 24 = 0.048.

### COMBINING INSTEAD OF CORRECTING: FISHER'S METHOD

Correcting answers the question "which particular tests are significant?". Sometimes the question is different: "does this population, as a whole, depart from equilibrium?". Fisher's method pools the evidence from several independent tests, $X = -2 \Sigma \ln P$, which is compared with a $\chi^2$ with $2k$ degrees of freedom. In the example, no Chalqueño locus except umc1304 survives the correction, yet combining its 8 loci gives **P < 0.0001**: the landrace as a whole has fewer heterozygotes than expected. The app shows this combination per population (Fisher (all loci)) and per locus (Fisher (all pops)).

### DECISION RULE · WHICH CORRECTION TO USE

| If... | use... |
|---|---|
| **You have dozens of tests** and every rejection matters (HWE per locus) | Holm, the default option. |
| **You have hundreds or thousands of tests** (pairs of loci with dominant markers) | Benjamini–Hochberg: Holm would be so strict that it would detect nothing. |
| **You are interested in the population as a whole**, not each locus | Fisher's method, in addition to the per-locus correction. |
| **No individual test survives** but there are many small P values | Examine the pattern (do they recur at one locus or in one population?) before concluding that nothing is going on. |



## 5.5 Block settings, tables and figures

Block 4 opens when at least one population has five or more individuals and there are two or more loci. It does not apply to morphological characters or to a single sequence alignment. With diploid codominant markers everything is computed; with dominant or haploid markers, only linkage. On entry, the tests run with the default settings; if you change any of them, click Run the tests.

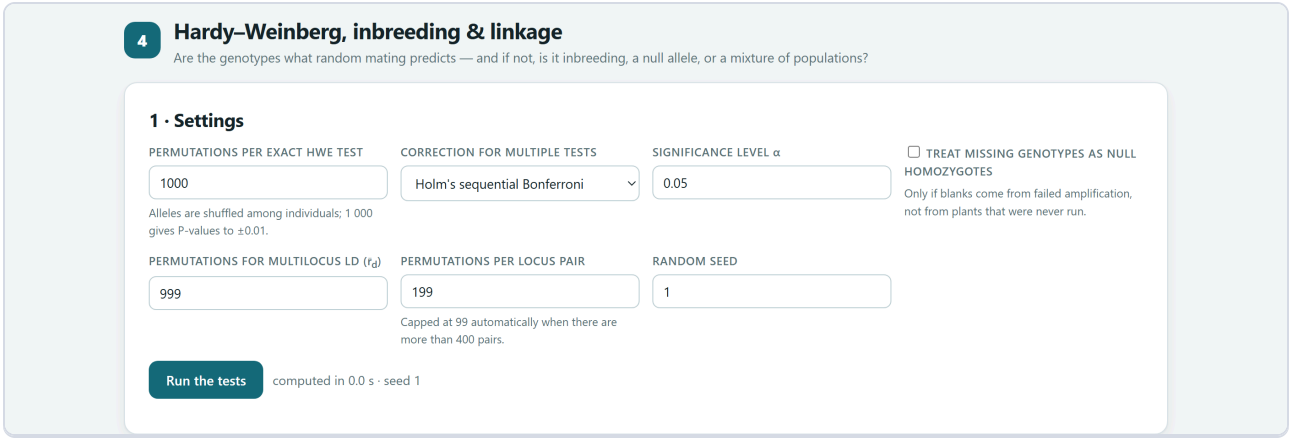



Figure 5.4. The 1 · Settings card with codominant data.

**Table 5.5.** Block 4 settings.

| Setting | Options | Recommendation |
|---|---|---|
| Permutations per exact HWE test | 0 to 100,000 (default 1000). Codominant only. | 1000 for exploration; 10,000 for the paper. |
| Correction for multiple tests | Holm (default) · Bonferroni · Benjamini–Hochberg · Benjamini–Yekutieli · none. | Holm; Benjamini–Hochberg with many pairs of loci. |
| Significance level α | 0.001 to 0.2 (default 0.05). | 0.05. |
| Treat missing genotypes as null homozygotes | Off by default. Codominant only. | Switch it on only if the missing data are amplification failures. |
| Permutations for multilocus LD | 0 to 9999 (default 999). | 999. |
| Permutations per locus pair | 0 to 9999 (default 199); drops to 99 with more than 400 pairs. | 199; with few pairs, 999. |
| Random seed | Integer. | Report it in the methods. |

**Table 5.6.** Block 4 result cards. Cards 3 to 5 appear only with codominant data.

| Card | Contents | Download |
|---|---|---|
| 2 · What the tests say | Summary values for the whole dataset, automatic diagnosis and interpretation of linkage. | — |
| 3 · Hardy–Weinberg tests, locus by population | The test chosen in Show for each locus and population, with – (deficit) or + (excess), Fisher's combination per locus and the map of departures. | HWE, $F_{IS}$ and null alleles (CSV) |
| 4 · Inbreeding coefficient $F_{IS}$ | $F_{IS}$ per population with interval and selfing rate; the locus-by-locus diagnosis with its table and figure. | — |
| 5 · Null alleles | Null allele frequency with the estimator chosen in Estimator, and its heat map. | — |
| 6 · Linkage disequilibrium | $I_A$ and $\bar{r}_d$ per population and for the whole dataset; the pairs of loci for the sample chosen in Sample (the 40 with the highest $\bar{r}_d$), the triangle of pairs and the permutation distribution. | All pairs (CSV) |

**Table 5.7.** The six figures of the block. They are edited and exported as explained in the Block 10 chapter.

| Figure in the app | What it shows |
|---|---|
| Hardy–Weinberg tests at a glance | Locus × population grid coloured by significance level, with – or +. |
| Inbreeding coefficient of every population | Multilocus *f* with its 95% interval and the per-locus $F_{IS}$ values as faint points. |
| Inbreeding or null alleles? $F_{IS}$ locus by locus | The diagnosis of figure 5.2 with your data. |
| Estimated null allele frequencies | Locus × population heat map of the chosen estimator. |

**Table 5.7.** The six figures of the block. They are edited and exported as explained in the Block 10 chapter. *(continued)*

| Figure in the app | What it shows |
|---|---|
| Pairwise linkage disequilibrium | Triangle of $\bar{r}_d$, $r^2$ or D′ between pairs, with asterisks on those significant after correction; in grey, pairs left undefined by a monomorphic locus. |
| Is the multilocus association more than chance? | Histogram of $\bar{r}_d$ across permutations, with the observed value. |

# 5.6 Step by step: maize and agave

## Maize with microsatellites: the full analysis

STEP BY STEP

1. In Block 2, load **Maize populations · SSR**, click **Build the dataset and run the checks** and open Block 4 from the step bar or with **Continue to Hardy–Weinberg** at the end of Block 3.
2. The tests run on their own in under a second. Read the diagnosis in **2 · What the tests say** (figure 5.5).
3. Review the table of exact tests and its map (figures 5.6 and 5.7). Change **Show** to see uncorrected P values, the one-tailed tests or $\chi^2$.
4. In **4 · Inbreeding coefficient $F_{IS}$**, read *f* with its interval and the locus-by-locus diagnosis (figures 5.8 to 5.10).
5. Check the null alleles with the **Estimator** menu (figure 5.11) and finish with linkage; **Sample** changes the landrace whose pairs are shown (figures 5.12 and 5.13).

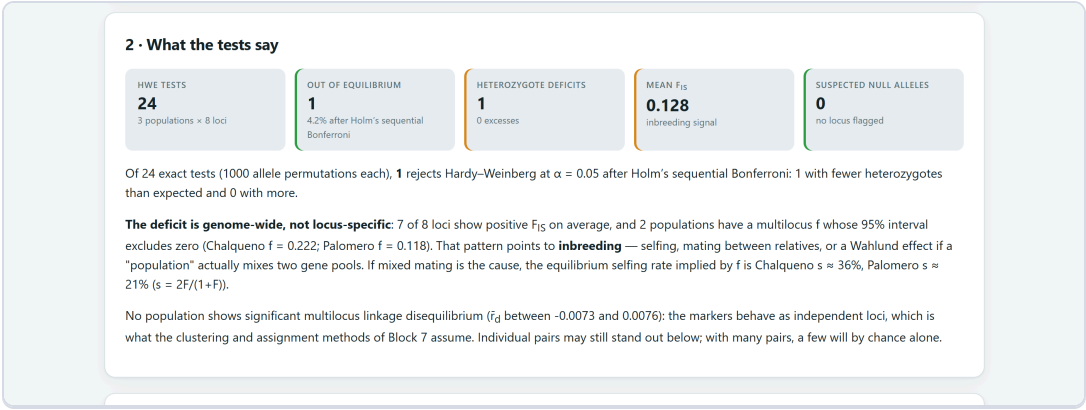



**Figure 5.5.** **What the tests say**. Of 24 exact tests, one rejects equilibrium after Holm's correction. The diagnosis concludes that the deficit is general: 7 of 8 loci have a positive mean $F_{IS}$, and two landraces have an *f* whose interval excludes zero. This is the correct conclusion for these data, simulated with inbreeding and without null alleles.

**NOTE · WHY CÓNICO DOES NOT REVEAL THE SIMULATED INBREEDING**

All three landraces were simulated with F = 0.16, yet only two show a clear deficit. With 25 plants and 8 loci there are about 190 genotypes per landrace, and the number of heterozygotes varies from one sample to the next: Cónico came out with 133 where about 117 were expected, almost 2.5 standard deviations above. That is unlikely, but with three landraces it is not unusual for one to stray that far. This is why you should always report the interval, and never conclude "there is no inbreeding" from an interval that includes zero: with more loci or more plants, the interval narrows (table 5.8).

**3 · Hardy–Weinberg tests, locus by population**

The exact test conditions on the allele counts and asks how unusual the observed genotype table is (Guo & Thompson 1992, by Monte Carlo). The chi-square test is shown for comparison with older papers; it is unreliable whenever an expected count falls below five, which with microsatellites is almost always.

Show: exact test P, corrected

| Locus | Conico | Chalqueno | Palomero | Fisher (all pops) |
|---|---|---|---|---|
| phi024 | 1.0000 + | 1.0000 – | 1.0000 – | 0.6779 |
| phi053 | 1.0000 – | 0.7732 – | 1.0000 – | 0.1032 |
| umc1147 | 1.0000 + | 0.1379 – | 1.0000 – | 0.0657 |
| bnlg1017 | 1.0000 + | 1.0000 – | 1.0000 – | 0.1431 |
| phi072 | 1.0000 – | 0.7023 – | 0.4795 – | 0.0175 |
| umc1304 | 0.3776 – | 0.0480 – | 1.0000 – | 0.0011 |
| bnlg1621 | 1.0000 – | 0.1538 – | 1.0000 – | 0.0253 |
| phi096 | 1.0000 – | 1.0000 – | 1.0000 – | 0.7013 |

exact test P-value corrected by Holm's sequential Bonferroni. – fewer heterozygotes than expected, + more. The last column combines the populations with Fisher's method.

HWE, $F_{IS}$ and null alleles (CSV)

**Figure 5.6.** Holm-corrected exact P values. Only Chalqueño · umc1304 falls below 0.05 (in red). The Fisher column combines the three landraces: umc1304 (0.0011), phi072 (0.0175) and bnlg1621 (0.0253) have recurring departures, even though no individual test shows it.

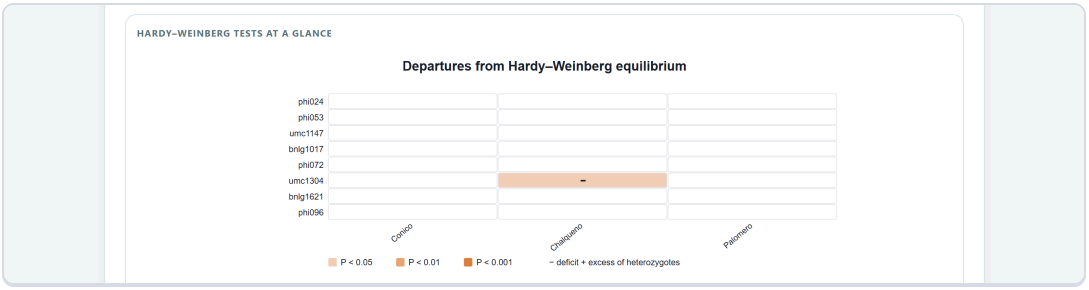
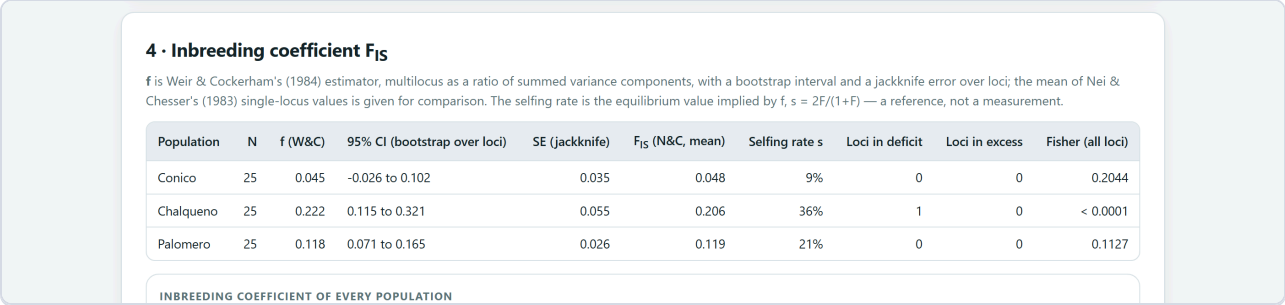



**Figure 5.7.** Hardy–Weinberg tests at a glance. Each cell is one test; the colour shows the significance level and the – sign a deficit. Test shown switches to uncorrected P values, where the 7 nominal departures appear.

B4



**4 · Inbreeding coefficient $F_{IS}$**

**f** is Weir & Cockerham's (1984) estimator, multilocus as a ratio of summed variance components, with a bootstrap interval and a jackknife error over loci; the mean of Nei & Chesser's (1983) single-locus values is given for comparison. The selfing rate is the equilibrium value implied by f, s = 2F/(1+F) — a reference, not a measurement.

| Population | N | f (W&C) | 95% CI (bootstrap over loci) | SE (jackknife) | $F_{IS}$ (N&C, mean) | Selfing rate s | Loci in deficit | Loci in excess | Fisher (all loci) |
|---|---|---|---|---|---|---|---|---|---|
| Conico | 25 | 0.045 | -0.026 to 0.102 | 0.035 | 0.048 | 9% | 0 | 0 | 0.2044 |
| Chalqueno | 25 | 0.222 | 0.115 to 0.321 | 0.055 | 0.206 | 36% | 1 | 0 | < 0.0001 |
| Palomero | 25 | 0.118 | 0.071 to 0.165 | 0.026 | 0.119 | 21% | 0 | 0 | 0.1127 |

INBREEDING COEFFICIENT OF EVERY POPULATION

**Figure 5.8.** $F_{IS}$ by landrace. Chalqueño (0.222, interval 0.115 to 0.321) and Palomero (0.118, 0.071 to 0.165) have a clear deficit; their equilibrium selfing rates would be 36% and 21%. Cónico (0.045) cannot be distinguished from zero.

**Table 5.8.** Multilocus $F_{IS}$ in the example against the value used to simulate the data.

| Landrace | *f* (Weir and Cockerham) | 95% CI | Jackknife SE | Mean $F_{IS}$ (N&C) | Simulated | Does it include the true value? |
|---|---|---|---|---|---|---|
| Cónico | 0.045 | –0.026 to 0.102 | 0.035 | 0.048 | 0.16 | No: this sample happened to have few homozygotes by chance. |
| Chalqueño | **0.222** | 0.115 to 0.321 | 0.055 | 0.206 | 0.16 | Yes. |
| Palomero | **0.118** | 0.071 to 0.165 | 0.026 | 0.119 | 0.16 | Yes. |

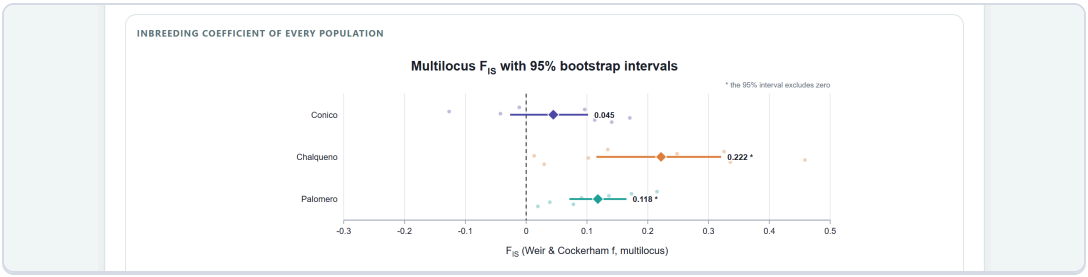



**Figure 5.9.** Inbreeding coefficient of every population. The diamond is *f*, the line its 95% interval and the faint points the $F_{IS}$ of each locus; the asterisk marks intervals that exclude zero.

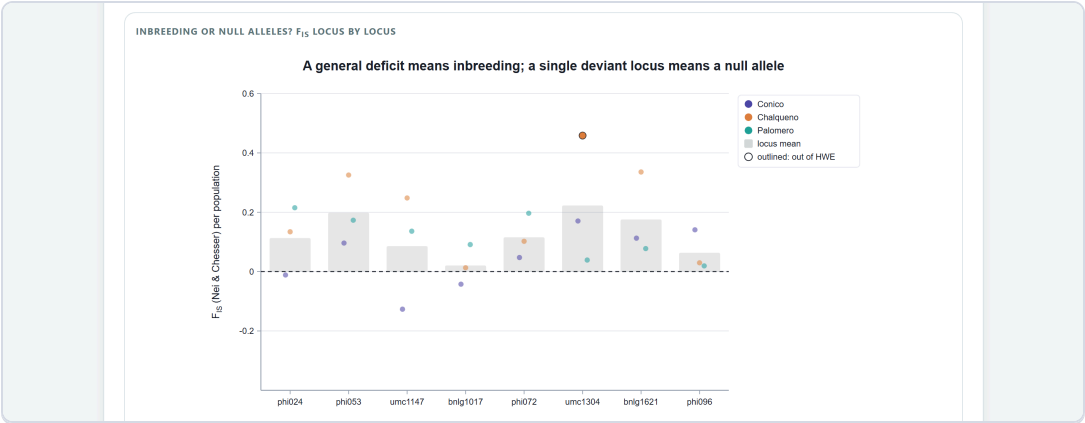



**Figure 5.10.** The diagnosis with the example data. Almost every locus has bars above zero, the inbreeding pattern of figure 5.2. The only outlined point, Chalqueño · umc1304 (0.458), is significant but does not recur in Cónico (0.170) or Palomero (0.039), so it is not flagged as a null allele.

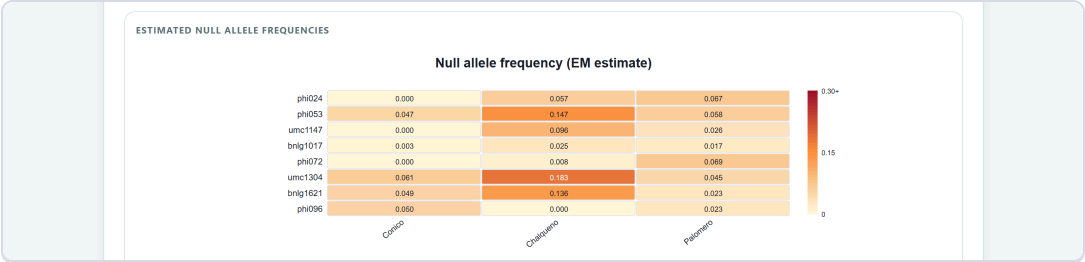



**Figure 5.11.** EM null allele frequencies. The highest values are in Chalqueño (0.183 at umc1304, 0.147 at phi053, 0.136 at bnlg1621), precisely the landrace with the most inbreeding: the estimators read the general deficit as if it were due to nulls (Important box in section 5.2).

### 6 · Linkage disequilibrium

The index of association measures whether individuals that differ at one locus tend to differ at the others. $\bar{r}_d$ (Agapow & Burt 2001) is its version corrected for the number of loci, so it can be compared between studies. It works for codominant, dominant and haploid data alike, and is tested by shuffling each locus independently among individuals. In an outcrossing plant with unlinked markers it is close to zero; selfing, clonality and admixture push it up.

| Sample | N | $I_A$ | $\bar{r}_d$ | P (permutation) | Permutations |
|---|---|---|---|---|---|
| Conico | 25 | 0.053 | 0.0076 | 0.2940 | 999 |
| Chalqueno | 25 | 0.046 | 0.0067 | 0.3140 | 999 |
| Palomero | 25 | -0.051 | -0.0073 | 0.7030 | 999 |
| **All samples pooled** | **75** | **0.058** | **0.0084** | **0.0740** | **999** |

**Locus pairs**

Sample Conico (n = 25)

**28 pairs** in **Conico** (n = 25): 0 significant after Holm's sequential Bonferroni (199 permutations per pair).

| Locus A | Locus B | $\bar{r}_d$ | P | P adjusted |
|---|---|---|---|---|
| umc1147 | umc1304 | 0.1739 | 0.0350 | 0.9800 |

**Figure 5.12.** Multilocus linkage. $\bar{r}_d$ ranges from −0.0073 to 0.0076 within landraces and is 0.0084 in the pooled sample; no value is significant. Among the pairs, 7 of 112 tests (28 pairs in each landrace and in the pooled sample) have uncorrected P < 0.05, close to the 5.6 expected by chance, and none survives Holm's correction.

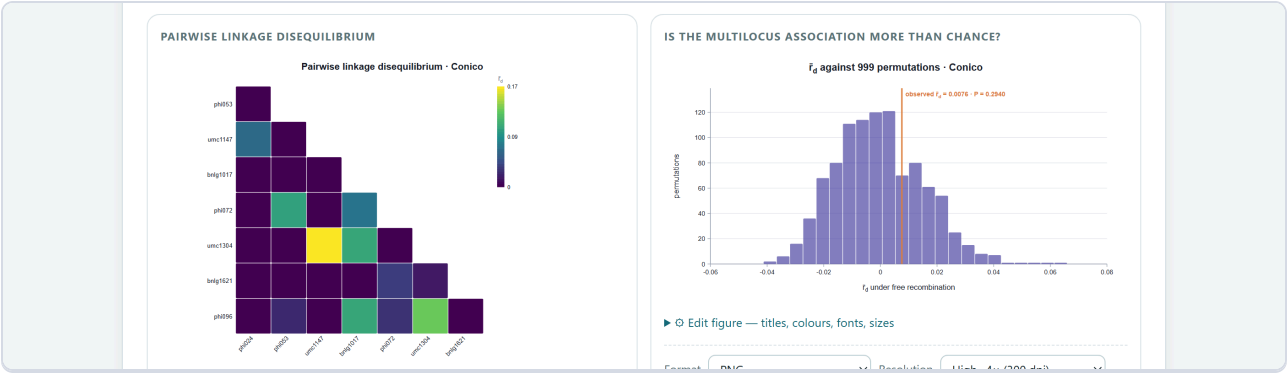



**Figure 5.13.** Left: $\bar{r}_d$ between the 28 pairs of loci in Cónico; the highest, umc1147–umc1304 (0.174, P = 0.035), does not survive the correction and carries no asterisk. Right: Cónico's observed $\bar{r}_d$ (0.0076) falls in the middle of the distribution of 999 permutations (P = 0.294).

### WHAT THE EXAMPLE CONFIRMS

- **The simulated inbreeding is detected as a general deficit**, not as a problem with particular loci: no locus is flagged as carrying null alleles.
- **Inbreeding does not create linkage** in these data, because the genotypes were simulated locus by locus. In a real selfing species, selfing would increase it, and that is one more criterion for distinguishing it from the effect of a null allele.
- **Individual tests have little power** with 25 plants per landrace: with a true F of 0.16, only one of 24 tests survives the correction. That is why the multilocus $F_{IS}$ with its interval and Fisher's combination matter.

## Agave with AFLP: linkage only

### STEP BY STEP

1. Load **Agave · AFLP** in Block 2, build the dataset and open Block 4.
2. The **Dominant or haploid data** notice explains that there are no Hardy–Weinberg tests, $F_{IS}$ or null alleles (figure 5.14).
3. The calculation takes a few seconds: 60 bands give 1770 pairs per sample.
4. Read the interpretation and the linkage table (figure 5.15) and explore the pairs in each population (figure 5.16).

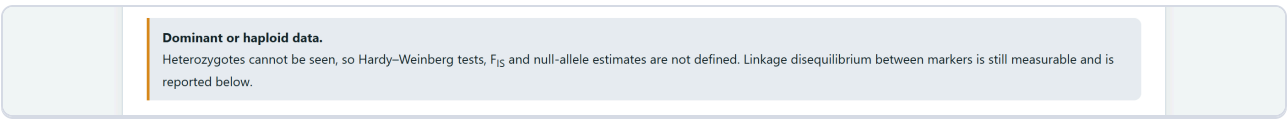
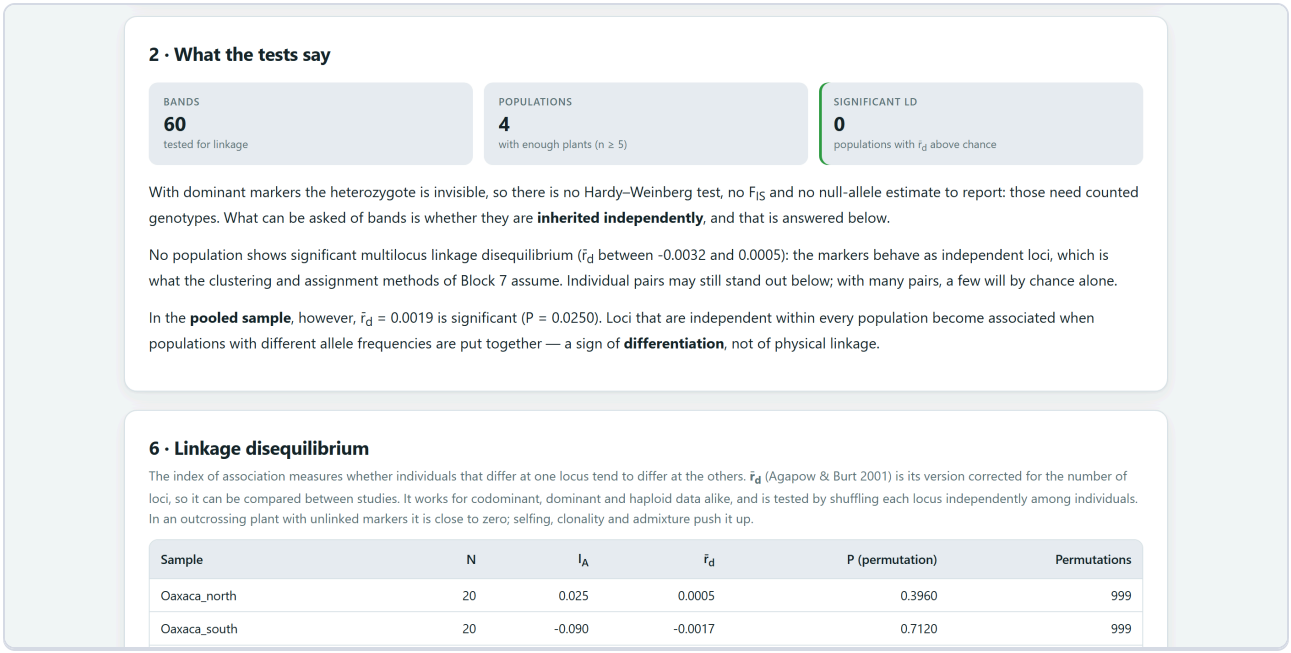



**Figure 5.14.** The notice that appears in **1 · Settings** with bands. The settings for the exact test and for treating missing data as nulls, which do not apply, also disappear.



**2 · What the tests say**

| BANDS | POPULATIONS | SIGNIFICANT LD |
|---|---|---|
| 60 | 4 | 0 |
| tested for linkage | with enough plants (n ≥ 5) | populations with $\bar{r}_d$ above chance |

With dominant markers the heterozygote is invisible, so there is no Hardy–Weinberg test, no $F_{IS}$ and no null-allele estimate to report: those need counted genotypes. What can be asked of bands is whether they are **inherited independently**, and that is answered below.

No population shows significant multilocus linkage disequilibrium ($\bar{r}_d$ between -0.0032 and 0.0005): the markers behave as independent loci, which is what the clustering and assignment methods of Block 7 assume. Individual pairs may still stand out below; with many pairs, a few will by chance alone.

In the **pooled sample**, however, $\bar{r}_d$ = 0.0019 is significant (P = 0.0250). Loci that are independent within every population become associated when populations with different allele frequencies are put together — a sign of **differentiation**, not of physical linkage.

**6 · Linkage disequilibrium**

The index of association measures whether individuals that differ at one locus tend to differ at the others. $\bar{r}_d$ (Agapow & Burt 2001) is its version corrected for the number of loci, so it can be compared between studies. It works for codominant, dominant and haploid data alike, and is tested by shuffling each locus independently among individuals. In an outcrossing plant with unlinked markers it is close to zero; selfing, clonality and admixture push it up.

| Sample | N | $I_A$ | $\bar{r}_d$ | P (permutation) | Permutations |
|---|---|---|---|---|---|
| Oaxaca_north | 20 | 0.025 | 0.0005 | 0.3960 | 999 |
| Oaxaca_south | 20 | -0.090 | -0.0017 | 0.7120 | 999 |

**Figure 5.15.** Within the four agave populations, $\bar{r}_d$ ranges from −0.0032 to 0.0005 and none is significant: the bands are inherited as independent loci. In the pooled sample, by contrast, $\bar{r}_d$ = 0.0019 is significant (P = 0.025), because pooling populations with different frequencies creates association between bands. It is a sign of differentiation, which is measured in Block 5.

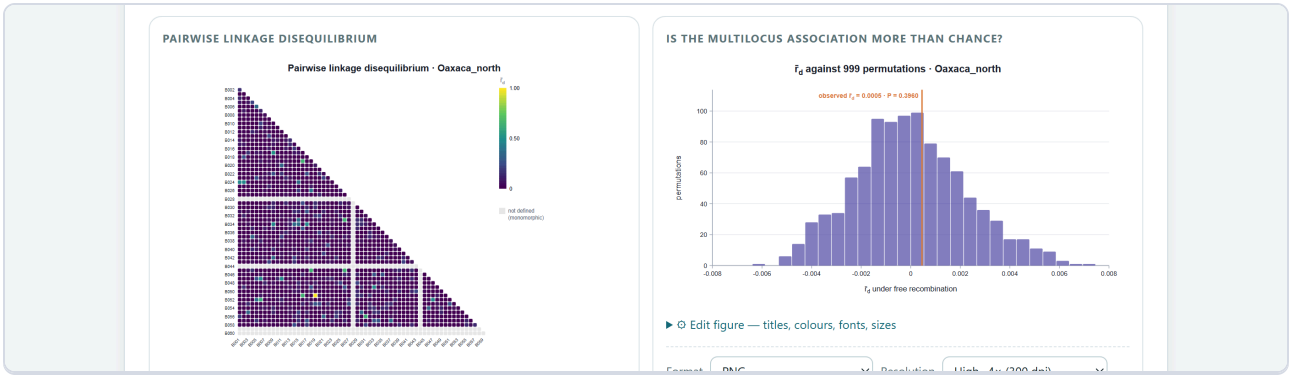



**Figure 5.16.** The 1770 pairs of bands in Oaxaca north. The grey rows and columns correspond to the 4 bands that are monomorphic in that population, whose $\bar{r}_d$ is undefined (230 pairs). No pair is significant after correction; with so many pairs, it is advisable to switch to Benjamini–Hochberg if you are looking for linkage candidates.

**NOTE · WHAT TO REPORT IN A PAPER**

For each population: the number of loci in disequilibrium after correction (state which correction you used), the multilocus $F_{IS}$ with its interval and, if you interpret it as a mating system, the selfing rate for reference. Mention the loci with possible null alleles and whether you excluded them from subsequent analyses. For linkage, $\bar{r}_d$ per population with its P. In the methods, the number of permutations and the seed. The HWE, $F_{IS}$ and null alleles (CSV) download contains every value per locus and population to five decimal places.

B4

**NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, every statistic in the maize example was recomputed with independent code: Nei and Chesser's $F_{IS}$, Weir and Cockerham's *f* with its jackknife error, $\chi^2$, the Chakraborty and Brookfield estimators, Holm's correction, $I_A$ and $\bar{r}_d$ agree to the 12th decimal place. The exact test was compared against full enumeration of every possible table of genotypes in the 14 cases with five alleles or fewer, and against 20,000 permutations in the other 10. The largest difference, at Cónico · bnlg1017 (0.089 against an exact 0.065), is due to the randomness of the 1000 permutations: with 100,000, the app gives between 0.064 and 0.067. The EM estimator agrees with an independent implementation of our own to the ninth decimal place.

With the genotypes reviewed and the problematic loci identified, the next chapter moves from the individual population to the set as a whole: Block 5 measures how much populations differ with $F_{ST}$, $G_{ST}$, Jost's D and the analysis of molecular variance (AMOVA).

# B5

CHAPTER SIX · BLOCK 5 · AMOVA

# Differentiation and AMOVA



How much of the genetic variation separates the populations? Which pairs differ? Do the regions you defined really group the populations? Block 5 computes the classical and modern estimators of differentiation ($F_{ST}$, $G_{ST}$, $G'_{ST}$ and Jost's D), the analysis of molecular variance (AMOVA) with or without regions, the pairwise matrices with their tests, and an indicative conversion to number of migrants.


6.1 Measuring differentiation

6.2 Hierarchical AMOVA

6.3 Pairwise matrices and gene flow

6.4 Decision rules

6.5 Block settings, tables and figures

6.6 Step by step: maize, agave and pine


## 6.1 Measuring differentiation

When populations exchange few migrants, drift changes their allele frequencies in different directions and they gradually drift apart. Differentiation measures how far apart they have moved: it compares the diversity of each population with that of the whole set.

### THE IDEA OF $F_{ST}$ IN ONE SENTENCE

If the populations were a single one, the diversity within them ($H_S$) would equal that of the whole set ($H_T$). The more their frequencies differ, the more diversity "appears" when they are pooled: $\mathbf{F_{ST} = (H_T - H_S)/H_T}$ is the fraction of the total diversity due to differences among populations. It is 0 if all of them have the same frequencies and 1 if each is fixed for a different allele. This is the same reasoning as the Wahlund effect in chapter 4.

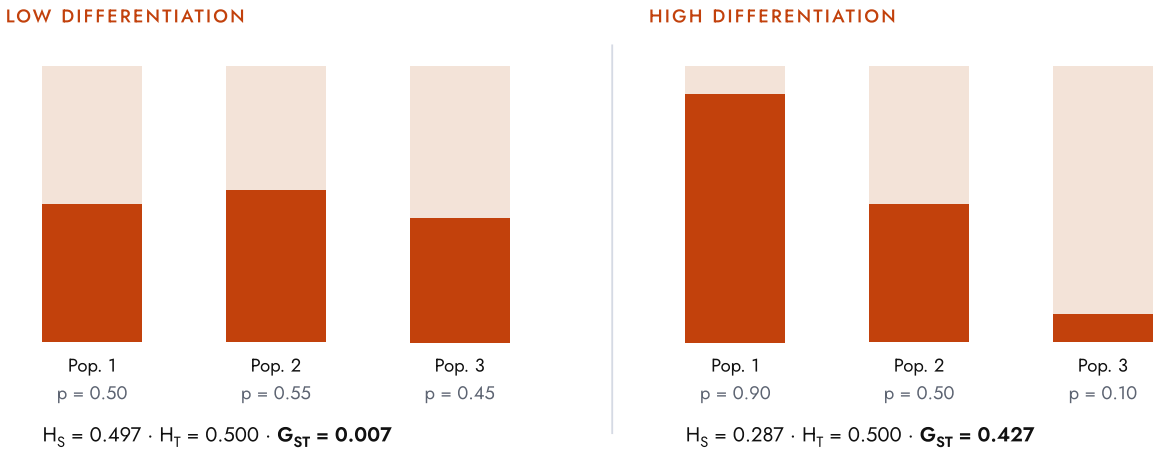



**Figure 6.1.** Three populations with one two-allele locus; the dark bar is the frequency *p* of allele A. In both cases the mean frequency is 0.5, so $H_T$ = 0.5. On the left, each population retains almost all of that diversity; on the right, two populations are close to fixing opposite alleles and $H_S$ drops to 0.287.

B5

## Four estimators and why they disagree

$G_{ST}$ has a ceiling: it can never exceed 1 – $H_S$, so with highly diverse markers such as microsatellites it stays close to zero even when the populations do not share a single allele (figure 6.2). Hedrick (2005) proposed $G'_{ST}$, which divides $G_{ST}$ by its maximum possible value, and Jost (2008) proposed D, which directly measures how different the sets of alleles are. None of them replaces the others: they measure related but different things (Meirmans and Hedrick, 2011).

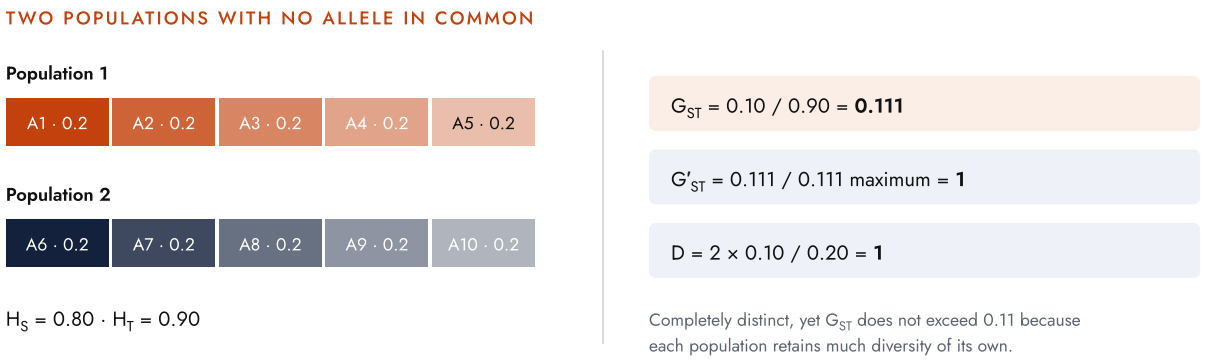



**Figure 6.2.** The ceiling of $G_{ST}$. With five equally frequent alleles in each population and none shared, $G_{ST}$ (and $F_{ST}$ as well) equals 0.111, whereas $G'_{ST}$ and D equal 1, their maximum. This is why microsatellites, with high $H_S$, give small $F_{ST}$ values even when the populations are very different.

**Table 6.1.** Statistics of Block 5. *r* is the number of populations; $H_S$ and $H_T$ include the sample-size corrections of Nei and Chesser (1983).

| Statistic | Formula | What it measures and when to use it | Range |
|---|---|---|---|
| **θ · $F_{ST}$** Weir and Cockerham (1984) | $\sigma^2_{among} / \sigma^2_{total}$, with variance components summed over alleles and loci | Differentiation as the correlation between alleles from the same population. Corrects for sample size and number of samples. **The estimator to report**, and the one related to gene flow and drift. | ≈ 0 to 1 |
| **$\theta_B$** dominant bands | θ for haploid units applied to band presence or absence | The same idea with AFLP, ISSR or RAPD. Equal to $\Phi_{PT}$ from AMOVA. | ≈ 0 to 1 |
| **$G_{ST}$** Nei (1973) | $(H_T - H_S) / H_T$ | Fraction of the total diversity found among populations. Useful for comparison with older studies; limited by the ceiling $1 - H_S$. | 0 to $1 - H_S$ |
| **$G'_{ST}$** Hedrick (2005) | $G_{ST}\,(r - 1 + H_S) / [(r - 1)(1 - H_S)]$ | $G_{ST}$ divided by its maximum possible value: comparable across markers with different diversity. | 0 to 1 |
| **D** Jost (2008) | $r/(r - 1) \cdot (H_T - H_S) / (1 - H_S)$ | How different the sets of alleles are: 0 if identical, 1 if they share none. Says nothing about migration. | 0 to 1 |
| **$\Phi_{ST}$ · $\Phi_{PT}$** AMOVA | $\sigma^2_{among\ populations} / \sigma^2_{total}$ | Differentiation from AMOVA; $\Phi_{PT}$ with bands. Without regions, equal to θ or $\theta_B$ (section 6.2). | ≈ 0 to 1 |
| **Nm** Wright | $(1 - \theta) / (4\theta)$; $(1 - \theta) / (2\theta)$ for haplotypes | Migrants per generation that would produce that θ under the island model. An order of magnitude only (section 6.3). | 0 to ∞ |

B5

**NOTE · HOW LOCI ARE COMBINED**

The multilocus θ is a ratio of sums: the variance components of all loci are summed first and then divided, so the more informative loci carry more weight. $G_{ST}$, $G'_{ST}$ and D are computed from $H_S$ and $H_T$ averaged over loci, so the three are comparable with one another. Some studies instead average the per-locus D values, using the arithmetic or the harmonic mean; the app shows both in the note below the per-locus table so that you can compare like with like. The 95% intervals are obtained by resampling loci (*bootstrap*) and the standard errors by *jackknife* over loci.

## 6.2 Hierarchical AMOVA

The analysis of molecular variance (AMOVA; Excoffier, Smouse and Quattro, 1992) partitions the squared genetic distances among the levels of the sampling design, just as an analysis of variance partitions the sum of squares among treatments and replicates. Each level yields a **variance component** and a **Φ statistic**, which is the fraction of the variation explained by that level.

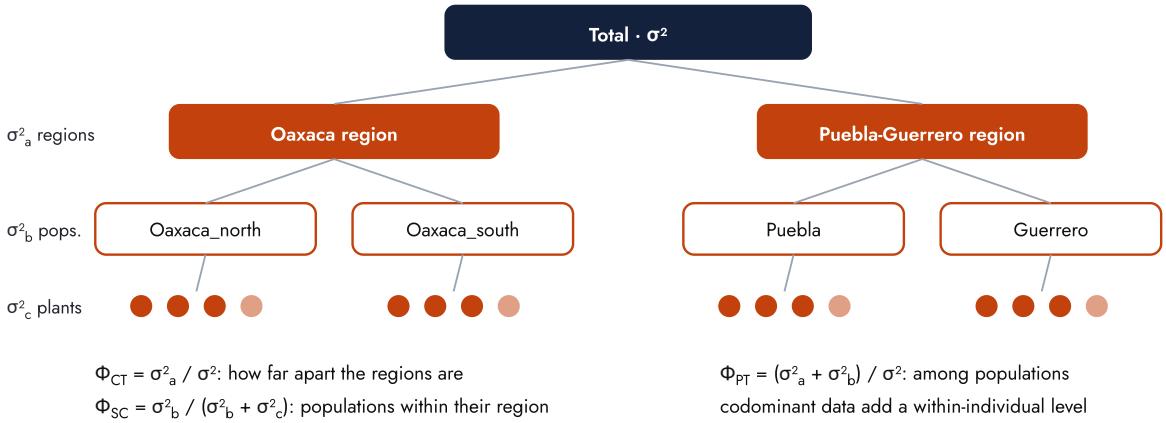



**Figure 6.3.** The levels of the hierarchical AMOVA with the design of the agave exercise: two regions, two populations in each and 20 plants per population. With codominant data, the plant level is split into "among individuals" and "within individuals", which gives $\Phi_{IS}$ and $\Phi_{IT}$.

**Table 6.2.** AMOVA levels in the app. N is the number of individuals, P the number of populations and G the number of regions.

| Source of variation | d.f. | Φ statistic | Test: what is permuted |
|---|---|---|---|
| Among regions | G – 1 | $\Phi_{CT} = \sigma^2_a / \sigma^2$ | Whole populations among regions. |
| Among populations within regions | P – G | $\Phi_{SC} = \sigma^2_b / (\sigma^2_b + \sigma^2_c + \sigma^2_d)$ | Individuals among the populations of the same region. |
| Among populations (no regions) | P – 1 | $\Phi_{ST}$ ($\Phi_{PT}$ with bands) | Individuals among all populations. |
| Among individuals within populations | N – P | $\Phi_{IS} = \sigma^2_c / (\sigma^2_c + \sigma^2_d)$ | Gene copies among individuals of the same population. |
| Within individuals | N | $\Phi_{IT} = (\sigma^2_a + \sigma^2_b + \sigma^2_c) / \sigma^2$ | — |

### THE DISTANCE AMOVA USES FOR EACH MARKER TYPE

- **Codominant:** between two gene copies, 1 if they carry different alleles and 0 if they carry the same one, summed over loci. Without regions, $\Phi_{ST}$, $\Phi_{IS}$ and $\Phi_{IT}$ are exactly equal to Weir and Cockerham's $\theta$, $f$ and F: they are two routes to the same number.
- **Dominant:** between two plants, the number of bands in which they differ (Peakall, Smouse and Huff, 1995). It gives $\Phi_{PT}$, equal to $\theta_B$.
- **Haplotypes:** 1 if two sequences carry different haplotypes. The AMOVA that weights how many mutations separate the haplotypes is in Block 8.

**NOTE · NEGATIVE COMPONENTS**

A variance component can come out negative when a level contributes no variation and sampling chance pushes the estimate below zero. This is not a computational error: it is interpreted as zero. The app keeps it in the table, so that the sum adds up, and draws it as zero in the figure.

**WORKED EXAMPLE · FROM MEAN SQUARES TO COMPONENTS**

In the agave exercise (section 6.6) the design is balanced: 2 regions, 2 populations per region, 20 plants per population and no missing data. The mean squares are 45.60 among regions, 38.65 among populations within regions and 10.653 within populations. Each mean square contains the components of the levels below it, multiplied by the number of plants in each unit:

| Component | Calculation | Result |
|---|---|---|
| $\sigma^2_c$ within populations | MS within | **10.653** |
| $\sigma^2_b$ populations within regions | (38.65 − 10.653) / 20 | **1.400** |
| $\sigma^2_a$ regions | (45.60 − 10.653 − 20 × 1.400) / 40 | **0.174** |
| $\Phi_{CT} \cdot \Phi_{SC} \cdot \Phi_{PT}$ | $\sigma^2_a/\sigma^2 \cdot \sigma^2_b/(\sigma^2_b + \sigma^2_c) \cdot (\sigma^2_a + \sigma^2_b)/\sigma^2$, with $\sigma^2$ = 12.226 | **0.014 · 0.116 · 0.129** |

With unbalanced designs the multipliers (20 and 40) are replaced by the coefficients $n$, $n'$ and $n''$ of Excoffier et al. (1992), and with missing data the app computes them locus by locus.

**IMPORTANT · WITH FEW POPULATIONS, $\Phi_{CT}$ CANNOT BE SIGNIFICANT**

The test of $\Phi_{CT}$ permutes **whole populations** among regions. With few populations there are few distinct ways to assign them, and the P value can never drop below 1 divided by that number of ways, however many permutations you request. The app computes that number and warns you about it (table 6.3). With fewer than about 8 populations, judge the regional level by the size of its component, not by its P.

**Table 6.3.** Distinct ways to assign the populations to regions of equal size, and minimum possible P of $\Phi_{CT}$.

| Design | 4 pops. in 2 + 2 | 4 in 3 + 1 | 6 in 3 + 3 | 8 in 4 + 4 | 10 in 5 + 5 | 12 in 4 + 4 + 4 |
|---|---|---|---|---|---|---|
| Distinct ways | 3 | 4 | 10 | 35 | 126 | 5775 |
| Minimum P | 0.33 | 0.25 | 0.10 | 0.029 | 0.008 | 0.0002 |

## 6.3 Pairwise matrices and gene flow

Global differentiation tells you how much the populations differ as a whole; pairwise matrices tell you **which ones** differ. For each pair the app computes θ, the AMOVA Φ, $G_{ST}$, $G'_{ST}$, D and Nm, and tests θ and Φ by permuting the individuals of the two populations. Since there are as many tests as pairs, the P values are corrected with Holm's method (chapter 5).

### FROM $F_{ST}$ TO MIGRANTS: WRIGHT'S ISLAND MODEL

If many populations of equal size exchange migrants at random, at the same rate, for long enough to reach equilibrium between drift and migration, then $F_{ST} \approx 1/(1 + 4N_em)$ for nuclear genes of a diploid organism, with codominant or dominant markers. Rearranging gives $N_em \approx (1 - F_{ST})/(4F_{ST})$. For uniparentally inherited haploid genomes, such as the chloroplast, $F_{ST} \approx 1/(1 + 2N_em)$. The classical rule says that with more than one migrant per generation, drift cannot differentiate the populations much. Chapter 2 shows this relationship with the simulator.

**Table 6.4.** Number of migrants per generation corresponding to each $F_{ST}$ under the island model.

| $F_{ST}$ | 0.01 | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 | 0.33 | 0.50 |
|---|---|---|---|---|---|---|---|---|
| Nm, diploid nuclear genes | 24.8 | 4.75 | 2.25 | 1.42 | 1.00 | 0.75 | 0.51 | 0.25 |
| Nm, haplotypes | 49.5 | 9.50 | 4.50 | 2.83 | 2.00 | 1.50 | 1.02 | 0.50 |

### IMPORTANT · NM IS A CONVERSION, NOT A MEASUREMENT

No real population meets the island model: sizes differ, migration is higher between neighbours, differentiation may stem from ancient isolation rather than from low present-day migration, and equilibrium takes many generations to be reached. Nm therefore serves to say "several migrants per generation" or "fewer than one", not to give an exact number of seeds or pollen grains (Whitlock and McCauley, 1999). The app labels it indicative. For the geographical pattern of differentiation, use the isolation-by-distance analysis of Block 6.

B5

## 6.4 Decision rules

### DECISION RULE · HOW MUCH DIFFERENTIATION IS A LOT?

| $F_{ST}$ or θ | Differentiation | Interpretation |
|---|---|---|
| **Below 0.05** | Low | The populations share almost all their frequencies; abundant gene flow or recent separation. |
| **0.05 to 0.15** | Moderate | Clear but shallow structure. Common in outcrossing plants with nuclear markers. |
| **0.15 to 0.25** | High | Well-differentiated populations; little exchange or a separate history. |
| **Above 0.25** | Very high | Common in selfing species, isolated populations and chloroplast markers. |

Wright's (1978) scale. It was designed for $F_{ST}$ with markers that have few alleles: with microsatellites, whose $H_S$ is high, a θ of 0.05 can conceal quite different populations (figure 6.2), so read it together with $G'_{ST}$ or D.

**DECISION RULE · WHICH ESTIMATOR SHOULD YOU REPORT?**

| If your question is... | report... |
|---|---|
| **How much structure there is** and how it relates to drift and migration | θ ($F_{ST}$) with its interval, and $\Phi_{ST}$ or $\Phi_{PT}$ from AMOVA with its P. |
| **How different the allele pools are**, for example for conservation | Jost's D or $G'_{ST}$, in addition to θ. |
| **Comparison with another study or another marker** | $G'_{ST}$ or D, which do not depend on the $H_S$ ceiling, and the same kind of locus average used in the other study. |
| **Comparison with an older study** that only gave $G_{ST}$ | $G_{ST}$ as a reference, noting its ceiling. |

The general recommendation is not to report a single estimator: θ plus D or $G'_{ST}$ covers both interpretations (Meirmans and Hedrick, 2011).

**DECISION RULE · READING AN AMOVA WITH REGIONS**

| If... | then... |
|---|---|
| **$\Phi_{CT}$ is significant** and the region component is larger than the population component | The grouping into regions is real and captures most of the structure. |
| **$\Phi_{CT}$ and $\Phi_{SC}$ are significant**, with populations weighing more | The regions differ, but within each one the populations differ even more. |
| **$\Phi_{SC}$ is significant and $\Phi_{CT}$ is not** | The populations differ, but they do not group according to the regions you defined. Try a grouping suggested by the data (Block 7). |
| **Few populations** (table 6.3) | The test of $\Phi_{CT}$ cannot reach significance: read the size of the component. |
| **None is significant** | At this scale, the species behaves almost as a single large population. |

**DECISION RULE · POPULATION PAIRS AND MIGRANTS**

| If... | then... |
|---|---|
| **The corrected P of a pair is below 0.05** | The two populations differ in frequencies; treat them as distinct units. |
| **Nm is above 4** | Gene flow is sufficient to keep them similar. |
| **Nm is between 1 and 4** | Moderate exchange; drift leaves a mark but does not separate them completely. |
| **Nm is below 1** | Drift dominates and the populations diverge; conserving each one is a priority. |
| **All corrected P values are equal** and small | This is the minimum possible with those permutations (with 199 and 3 pairs, 0.005 × 3 = 0.015). Increase the permutations to tell them apart. |

The Nm thresholds are indicative and inherit all the limitations of the island model.

# 6.5 Block settings, tables and figures

Block 5 opens when there are at least two populations with two or more individuals each and molecular markers; it does not apply to morphological traits or to landraces with a single sample. On entry, everything is computed with the default settings; if you change any of them, press **Compute differentiation**.

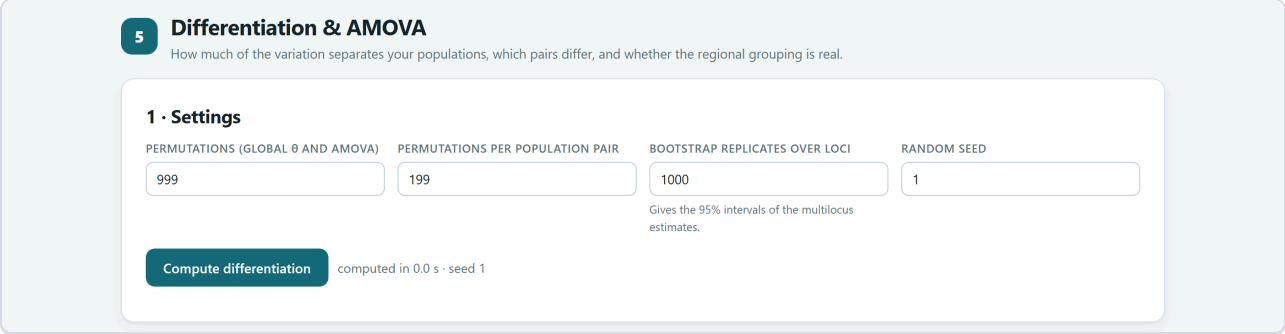



**Figure 6.4.** The **1 · Settings** card with the maize data, which have no regions. The check boxes for the hierarchical AMOVA and for leaving out loci appear only when they apply (figure 6.13).

**Table 6.5.** Block 5 settings.

| Setting | Options | Recommendation |
|---|---|---|
| **Permutations (global θ and AMOVA)** | 0 to 99,999 (default 999). | 999; 9999 for the paper. |
| **Permutations per population pair** | 0 to 9999 (default 199). | 199 for exploration; 999 if there are few pairs. |
| **Bootstrap replicates over loci** | 0 to 10,000 (default 1000). | 1000. Requires three or more loci. |
| **Hierarchical AMOVA (regions › populations)** | On by default. Appears if Block 2 has two or more regions. | Turn it off to see the single-level AMOVA. |
| **Leave out the loci flagged for null alleles in Block 4** | Appears if Block 4 flagged any locus. | Run with and without the flagged loci and compare. |
| **Random seed** | An integer. | Report it in the methods. |

**NOTE · HOW TO DECLARE REGIONS**

The app takes the regions from a column of your data. The column is recognised automatically if its name starts with `Region`, `Zone`, `Group`, `State`, `Estado`, `Area`, `Cluster`, `Subregion`, `Watershed`, `Province`, `País` or `Country`; otherwise, assign it the **Region / group** role in Block 2. Each population must belong to a single region. In FASTA files, the region can go in the sequence name, as in `ID|population|region`, or in a sample sheet.

**Table 6.6.** Block 5 result cards.

| Card | Content | Download |
|---|---|---|
| 2 · What the numbers say | Key figures and interpretation: level of differentiation, $G_{ST}$ ceiling, regions, inbreeding and pairs. | — |
| 3 · Analysis of molecular variance | AMOVA table with degrees of freedom, sums of squares, components, percentages, Φ and P, and the figure of the partition. | — |
| 4 · Global F-statistics, locus by locus | $H_O$, $H_S$, $H_T$, $G_{ST}$, $G'_{ST}$, D, θ, *f* and F per locus, the multilocus value with its interval, and the per-locus figure. | Per-locus statistics (CSV) |
| 5 · Pairwise differentiation | Matrix of the statistic chosen in Statistic below the diagonal and corrected P values above it; heat map and comparison of estimators by pair. | All pairs, all statistics (CSV) |

**Table 6.7.** The four figures of the block. They are edited and exported as explained in the chapter on Block 10.

| Figure in the app | What it shows |
|---|---|
| AMOVA: where the molecular variance sits | Bar with the percentage of each level and its Φ values with P. |
| Differentiation locus by locus | θ, $G_{ST}$, $G'_{ST}$ or D for each locus, with the multilocus value and its shaded interval. A locus well above the interval is highlighted in another colour. |
| Pairwise differentiation | Heat map of the pairwise statistic, with the corrected P values in the upper half. |
| Four estimators, pair by pair | $G_{ST}$, $G'_{ST}$, D and θ side by side for each pair, sorted by D. |



## 6.6 Step by step: maize, agave and pine

### Maize with microsatellites: differentiation without regions

**STEP BY STEP**

1. In Block 2, load Maize populations · SSR, build the dataset and open Block 5 from the step bar or with Continue to F-statistics & AMOVA at the end of Block 4.
2. The computation takes about a second. Read the figures and the interpretation (figure 6.5).
3. Review the AMOVA table and figure (figures 6.6 and 6.7).
4. In card 4, compare the estimators by locus and look for extreme loci (figures 6.8 and 6.9).
5. Finish with the pairs: change Statistic to see each estimator (figures 6.10 to 6.12).

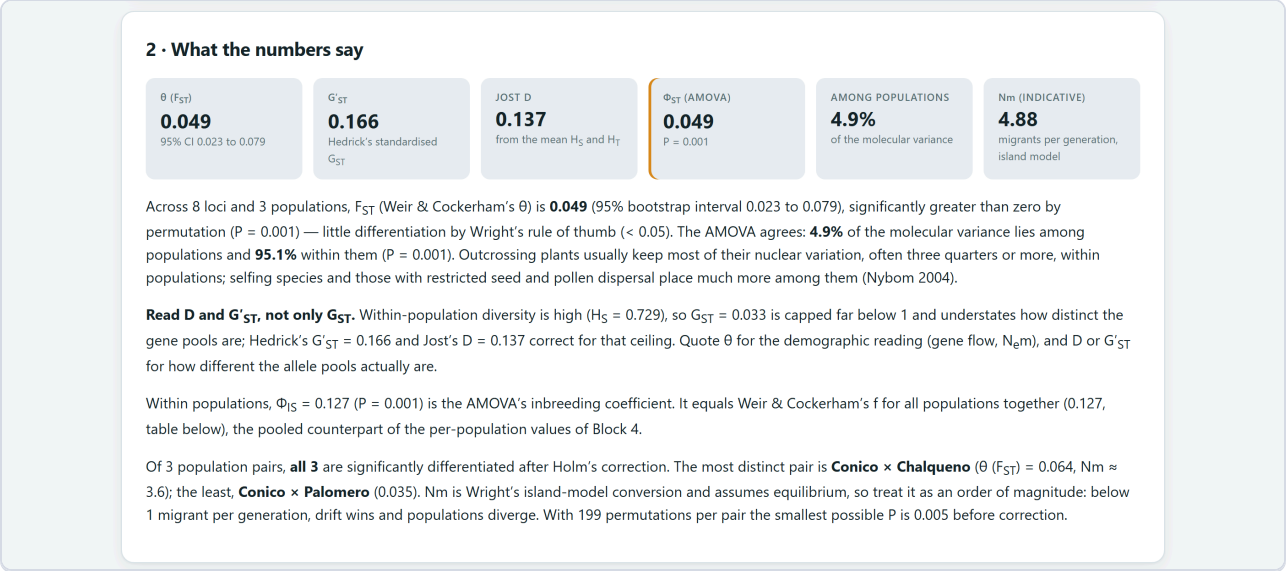



**Figure 6.5.** What the numbers say with the three maize landraces. θ = 0.049 is significant but falls in the low-differentiation category; the interpretation warns that, with $H_S$ = 0.729, the $G_{ST}$ of 0.033 is limited by its ceiling, and that $G'_{ST}$ and D are several times larger.

**Table 6.8.** Multilocus differentiation of the three maize landraces (8 microsatellites, 75 plants).

| Estimator | Value | 95% CI (*bootstrap*) | SE (*jackknife*) | Interpretation |
|---|---|---|---|---|
| θ ($F_{ST}$) | **0.049** | 0.023 to 0.079 | 0.016 | Low to moderate; P = 0.001. |
| $G_{ST}$ | 0.033 | 0.016 to 0.054 | 0.011 | Limited by $H_S$ = 0.729. |
| $G'_{ST}$ | 0.166 | 0.082 to 0.262 | 0.051 | Five times $G_{ST}$ once the ceiling is corrected. |
| Jost's D | 0.137 | 0.067 to 0.218 | 0.043 | The allele pools differ considerably more than θ suggests. |
| *f* ($F_{IS}$) · F ($F_{IT}$) | 0.127 · 0.169 | 0.083 to 0.169 · 0.136 to 0.204 | 0.024 · 0.019 | Overall inbreeding, equal to $\Phi_{IS}$ from AMOVA. |
| Nm | 4.9 | — | — | Several migrants per generation. |

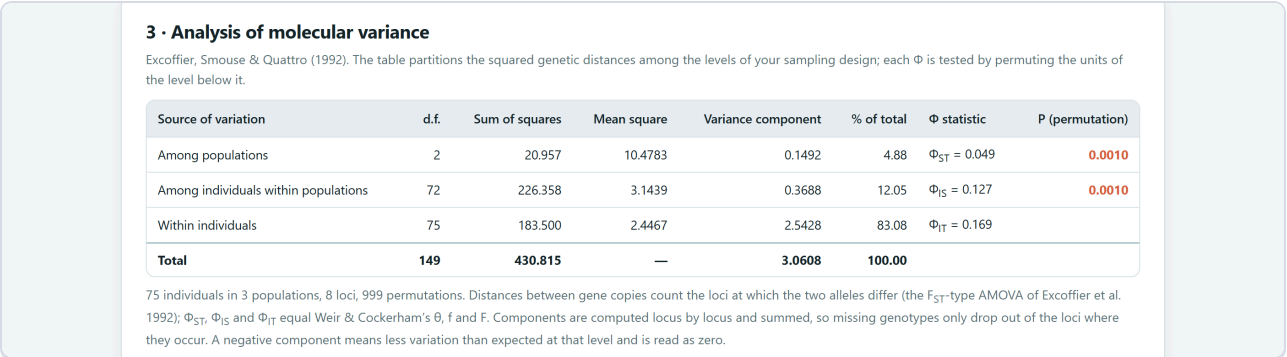



**Figure 6.6.** Three-level AMOVA. 4.88% of the variation lies among landraces, 12.05% among plants of the same landrace and 83.08% within each plant. $\Phi_{ST}$ = 0.049, $\Phi_{IS}$ = 0.127 and $\Phi_{IT}$ = 0.169 match θ, *f* and F in table 6.8 exactly.

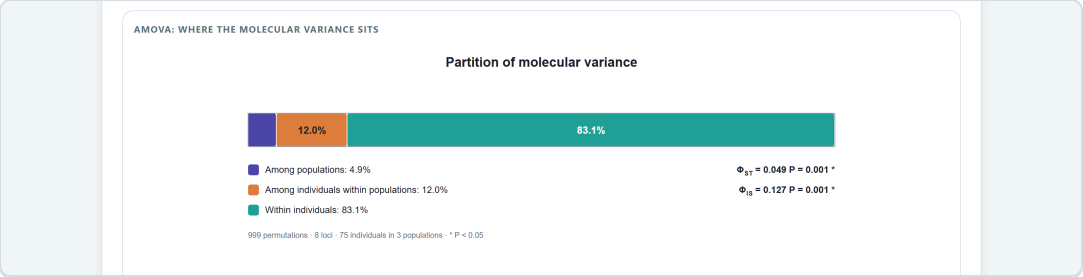



**Figure 6.7.** AMOVA: where the molecular variance sits. The small band on the left is the variation among landraces; the asterisk marks the significant Φ values.

### HOW TO READ THESE RESULTS

- **Most of the variation lies within plants.** This is expected in an outcrossing plant with microsatellites: each plant carries two different alleles at most of its loci.
- **The landraces differ, albeit little in terms of $F_{ST}$.** The 4.9% among landraces is significant; $G'_{ST}$ = 0.166 and D = 0.137 show that their allele pools are more different than the small θ value suggests.
- **The among-plant level confirms the inbreeding found in chapter 5.** $\Phi_{IS}$ = 0.127 is the overall equivalent of the per-landrace *f* values of 0.045, 0.222 and 0.118.
- **Nm ≈ 5** would indicate abundant gene flow if the landraces were at equilibrium; in maize landraces, whose differentiation depends on selection and on seed exchange among farmers, take it only as an order of magnitude.

B5

#### 4 · Global F-statistics, locus by locus

**θ**, **f** and **F** are Weir & Cockerham's (1984) estimators of $F_{ST}$, $F_{IS}$ and $F_{IT}$. **$G_{ST}$** is Nei's (1973) with the Nei & Chesser (1983) sample-size corrections; **$G'_{ST}$** is Hedrick's (2005) standardisation; **D** is Jost's (2008). A locus far above the shaded multilocus interval is a candidate for selection or a scoring artefact.

| Locus | Alleles | $H_O$ | $H_S$ | $H_T$ | $G_{ST}$ | $G'_{ST}$ | Jost D | θ ($F_{ST}$) | f ($F_{IS}$) | F ($F_{IT}$) |
|---|---|---|---|---|---|---|---|---|---|---|
| phi024 | 5 | 0.646 | 0.725 | 0.770 | 0.057 | 0.285 | 0.242 | 0.084 | 0.112 | 0.187 |
| phi053 | 7 | 0.647 | 0.810 | 0.831 | 0.025 | 0.185 | 0.164 | 0.036 | 0.201 | 0.230 |
| umc1147 | 4 | 0.615 | 0.668 | 0.682 | 0.020 | 0.081 | 0.062 | 0.029 | 0.084 | 0.111 |
| bnlg1017 | 6 | 0.663 | 0.679 | 0.753 | 0.098 | 0.408 | 0.344 | 0.141 | 0.023 | 0.160 |
| phi072 | 8 | 0.689 | 0.780 | 0.794 | 0.017 | 0.105 | 0.090 | 0.025 | 0.117 | 0.139 |
| umc1304 | 5 | 0.529 | 0.686 | 0.705 | 0.028 | 0.119 | 0.094 | 0.042 | 0.218 | 0.251 |
| bnlg1621 | 6 | 0.659 | 0.796 | 0.798 | 0.003 | 0.020 | 0.017 | 0.004 | 0.174 | 0.177 |
| phi096 | 4 | 0.640 | 0.684 | 0.695 | 0.016 | 0.070 | 0.054 | 0.025 | 0.064 | 0.087 |
| **Multilocus** | — | **0.636** | **0.729** | **0.753** | **0.033** | **0.166** | **0.137** | **0.049** | **0.127** | **0.169** |
| 95% CI (bootstrap over loci) | — | — | — | — | 0.016 to 0.054 | 0.082 to 0.262 | 0.067 to 0.218 | 0.023 to 0.079 | 0.083 to 0.169 | 0.136 to 0.204 |

Multilocus θ, f and F are ratios of variance components summed over loci; $G_{ST}$, $G'_{ST}$ and D come from the mean $H_S$ and $H_T$, so the three are directly comparable. Averaging the per-locus D instead gives 0.133 (arithmetic mean) or 0.062 (harmonic mean, which is pulled towards the least differentiated locus); if you compare with a study that averaged per locus, quote the same kind of mean. Jackknife SE over loci: θ ($F_{ST}$) ± 0.016 · f ± 0.024 · F ± 0.019 · $G_{ST}$ ± 0.011 · $G'_{ST}$ ± 0.051 · Jost's D ± 0.043.

**Figure 6.8.** Global F-statistics, locus by locus. Below the table, the note shows that averaging the per-locus D values would give 0.133 (arithmetic mean) or 0.062 (harmonic mean); the harmonic mean is pulled down by bnlg1621, with D = 0.017.

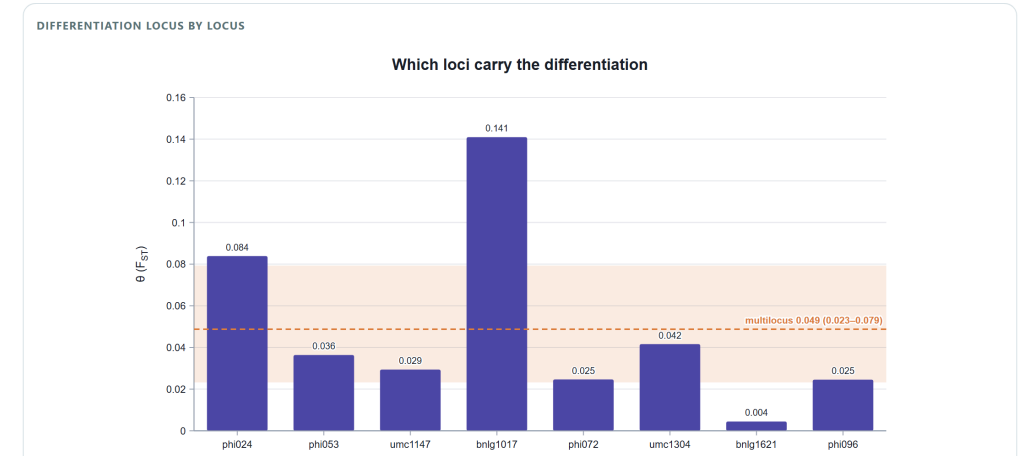



**Figure 6.9.** θ by locus with the multilocus value (dashed line) and its shaded interval. bnlg1017 (0.141) is almost three times the multilocus value and lies outside the interval; bnlg1621 (0.004) barely differentiates the landraces.

**! IMPORTANT · AN EXTREME LOCUS DOES NOT PROVE SELECTION**

A locus with $F_{ST}$ far above the rest may be linked to a gene under selection, but it may also reflect null alleles, scoring errors or plain chance: with 8 loci, one always ends up at an extreme. In this simulated example, bnlg1017 is not under selection. Claiming selection requires tens or hundreds of loci and dedicated outlier tests; here it is enough to check its gels and confirm that the conclusions do not change when it is excluded.

B5

**5 · Pairwise differentiation**

Statistic θ (FST)

| | Conico | Chalqueno | Palomero |
|---|---|---|---|
| Conico | — | 0.015 | 0.015 |
| Chalqueno | 0.064 | — | 0.015 |
| Palomero | 0.035 | 0.047 | — |

Below the diagonal: **θ ($F_{ST}$)**. Above: P-value of the permutation test (199 permutations of individuals between the two populations, Holm-corrected over 3 pairs) for θ ($F_{ST}$).

**Figure 6.10.** Pairwise θ below the diagonal and Holm-corrected P values above it. All three pairs have P = 0.015, the minimum possible with 199 permutations and three pairs.

**Table 6.9.** Differentiation between pairs of maize landraces.

| Pair | θ | $G_{ST}$ | $G'_{ST}$ | D | Nm | P (Holm) |
|---|---|---|---|---|---|---|
| Cónico × Chalqueño | **0.064** | 0.033 | 0.202 | 0.175 | 3.6 | 0.015 |
| Chalqueño × Palomero | 0.047 | 0.024 | 0.152 | 0.131 | 5.0 | 0.015 |
| Cónico × Palomero | 0.035 | 0.018 | 0.119 | 0.103 | 7.0 | 0.015 |

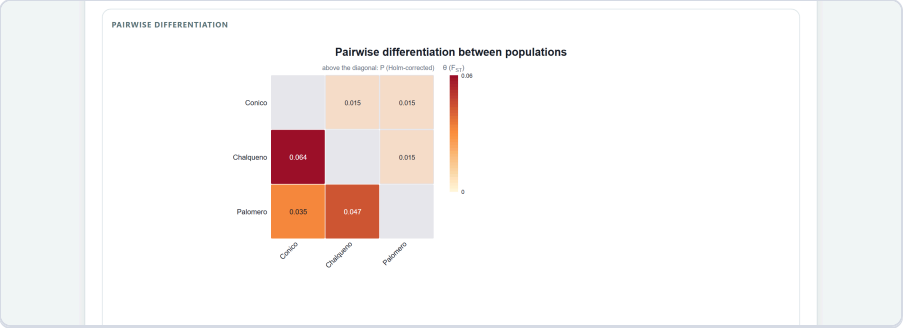



**Figure 6.11.** Pairwise differentiation. Chalqueño is the landrace most different from the other two; Cónico and Palomero are the most similar.

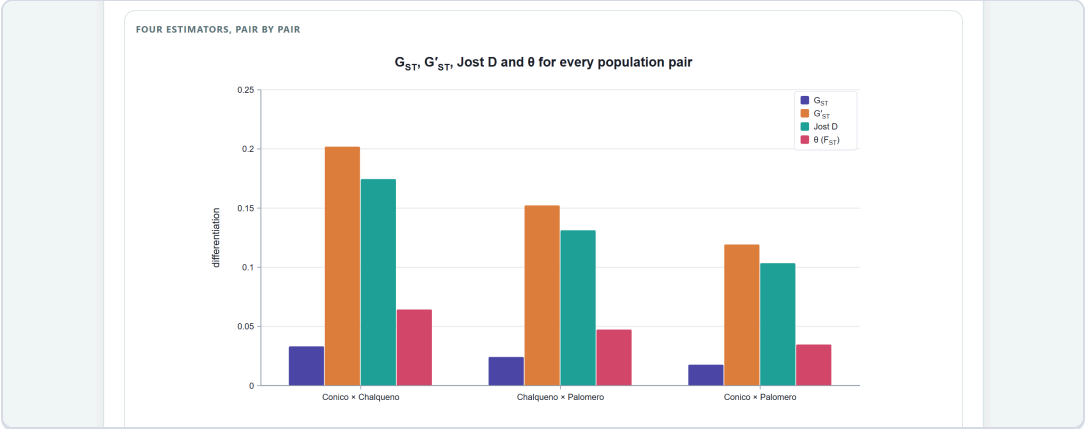



**Figure 6.12.** Four estimators, pair by pair. The four estimators rank the pairs in the same order, but their values are very different: $G_{ST}$ and θ stay low because of the diversity ceiling, while $G'_{ST}$ and D are several times higher.

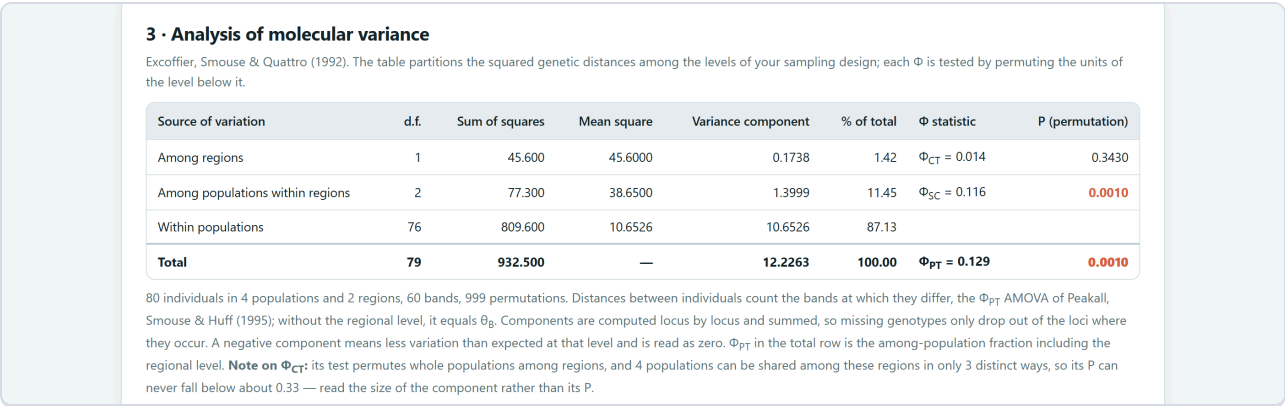
## Agave with AFLP and regions: the hierarchical AMOVA

None of the bundled examples includes regions, but they are easy to add to the agave example with the built-in data sheet. The plants are sorted by population: rows 2 to 41 are the two Oaxaca populations and rows 42 to 81 are Puebla and Guerrero.

### STEP BY STEP · ADDING A REGION COLUMN

1. In Block 2, load **Agave · AFLP**, press **Open the data sheet** and, in the sheet, **the file loaded**.
2. Click any cell in column B (`Population`) and press **+ right**: an empty column C is inserted.
3. Type `Region` in C1 and `Oaxaca` in C2. Copy C2 with Ctrl + C, select C2 to C41 (click C2 and Shift +click C41) and paste with Ctrl + V: a value pasted over a range fills the whole range.
4. Repeat with `Puebla-Guerrero` from C42 to C81.
5. Press **Use these data →**, check that the column has the **Region / group** role, build the dataset and open Block 5.

**Figure 6.13.** With two regions declared, the **Hierarchical AMOVA (regions › populations)** check box appears, ticked, with the names of the regions.



**3 · Analysis of molecular variance**

Excoffier, Smouse & Quattro (1992). The table partitions the squared genetic distances among the levels of your sampling design; each Φ is tested by permuting the units of the level below it.

| Source of variation | d.f. | Sum of squares | Mean square | Variance component | % of total | Φ statistic | P (permutation) |
|---|---|---|---|---|---|---|---|
| Among regions | 1 | 45.600 | 45.6000 | 0.1738 | 1.42 | $\Phi_{CT}$ = 0.014 | 0.3430 |
| Among populations within regions | 2 | 77.300 | 38.6500 | 1.3999 | 11.45 | $\Phi_{SC}$ = 0.116 | 0.0010 |
| Within populations | 76 | 809.600 | 10.6526 | 10.6526 | 87.13 | | |
| **Total** | **79** | **932.500** | — | **12.2263** | **100.00** | **$\Phi_{PT}$ = 0.129** | **0.0010** |

80 individuals in 4 populations and 2 regions, 60 bands, 999 permutations. Distances between individuals count the bands at which they differ, the $\Phi_{PT}$ AMOVA of Peakall, Smouse & Huff (1995); without the regional level, it equals $\theta_B$. Components are computed locus by locus and summed, so missing genotypes only drop out of the loci where they occur. A negative component means less variation than expected at that level and is read as zero. $\Phi_{PT}$ in the total row is the among-population fraction including the regional level. **Note on $\Phi_{CT}$:** its test permutes whole populations among regions, and 4 populations can be shared among these regions in only 3 distinct ways, so its P can never fall below about 0.33 — read the size of the component rather than its P.

**Figure 6.14.** Hierarchical AMOVA of the agave. The regions explain 1.42% of the variation ($\Phi_{CT}$ = 0.014, P = 0.343), populations within regions 11.45% ($\Phi_{SC}$ = 0.116, P = 0.001), and the rest lies within populations. The note warns that with 4 populations there are only 3 ways to assign them to two regions.

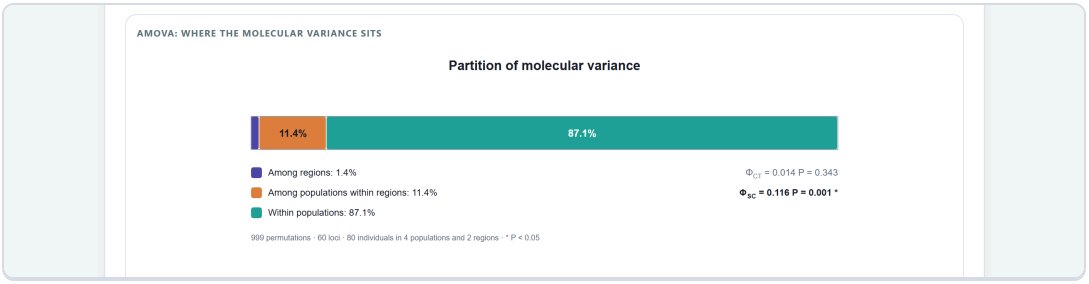



**Figure 6.15.** The three-level partition. With dominant bands there is no within-individual level.

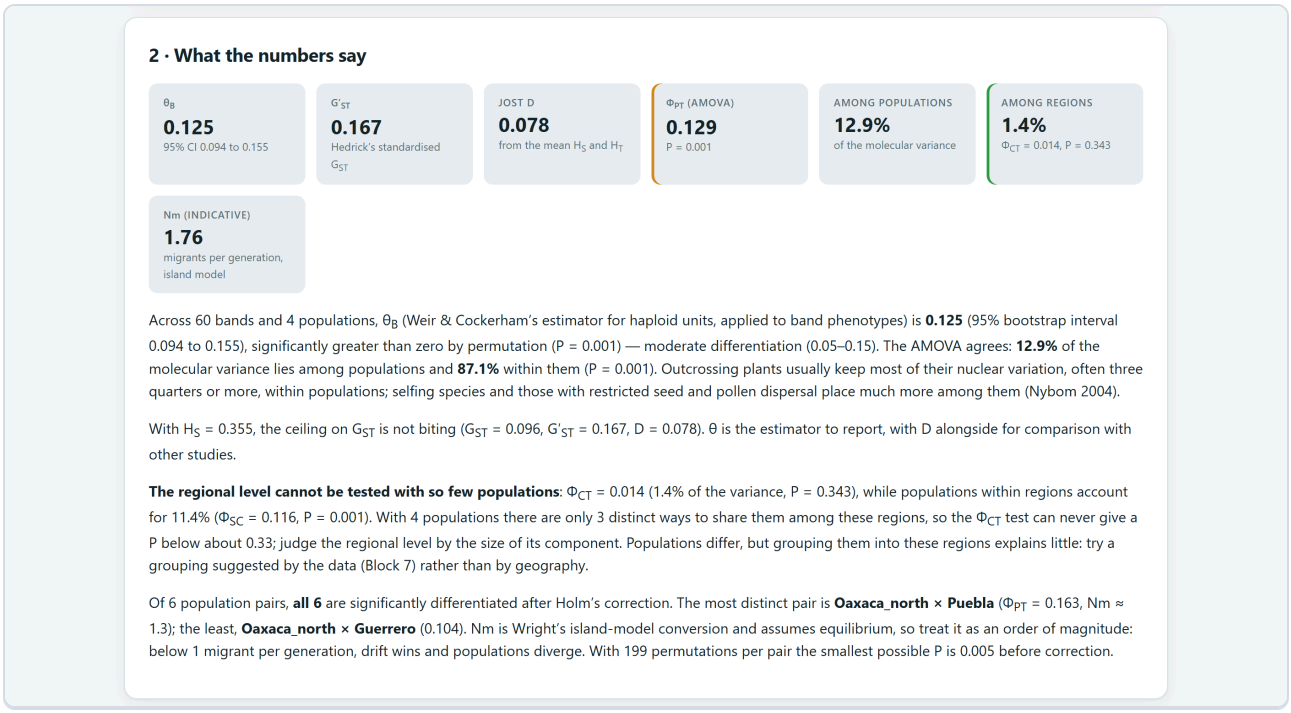



**Figure 6.16.** The interpretation with regions. $\theta_B$ = 0.125 indicates moderate differentiation among the four populations; all six pairs differ. Regarding the regions, the app concludes that the regional level cannot be tested with so few populations and that, judging by the size of its component, the geographical grouping explains little.

### HOW TO READ THESE RESULTS

- **The Oaxaca versus Puebla-Guerrero grouping does not capture the structure.** Populations from the same region differ from each other almost as much as those from different regions: the most different pair, Oaxaca north × Puebla ($\theta_B$ = 0.163), spans regions, but so does the least different, Oaxaca north × Guerrero (0.104).
- **Even with more permutations, the P of $\Phi_{CT}$ would not drop below 0.33.** Testing regions requires more populations per region.
- **With bands, D (0.078) falls below $G_{ST}$ (0.096).** $H_S$ = 0.355 is low and the $G_{ST}$ ceiling does not come into play; D and $G'_{ST}$ have nothing to correct here.
- **$\Phi_{PT}$ changes slightly with regions** (0.129 versus 0.125 without them), because the design with regions estimates the components with different coefficients; without regions, $\Phi_{PT}$ equals $\theta_B$.
- **Nm ≈ 1.8** overall and between 1.3 and 2.2 by pair: moderate exchange.

## Pine with chloroplast DNA: haplotypes



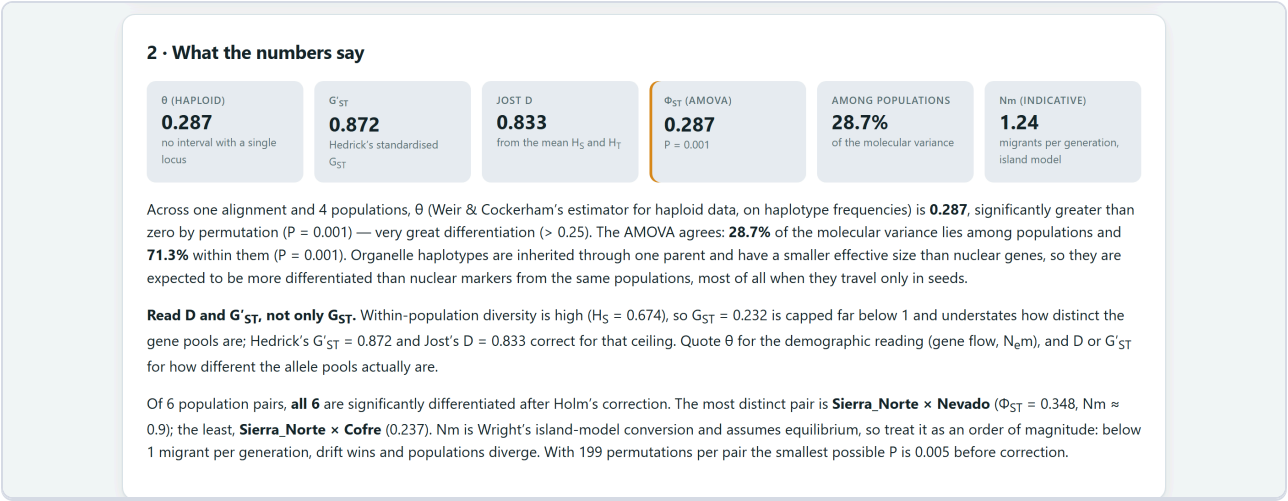



**Figure 6.17.** With haplotypes, θ = 0.287 is very high differentiation, and AMOVA places 28.7% of the variation among populations. Since there is a single alignment, there is no *bootstrap* interval. $G'_{ST}$ = 0.872 and D = 0.833 show that the populations share almost no haplotypes.

**Figure 6.18.** Pairwise θ in the pine; all six pairs differ (P = 0.030 after Holm, the minimum with 199 permutations and six pairs).

**Table 6.10.** Pairs of pine populations: θ, D and shared haplotypes.

| Pair | θ | D | Nm (haploid) | Haplotypes in common |
|---|---|---|---|---|
| Sierra Norte × Nevado | **0.348** | 1 | 0.9 | None |
| Sierra Sur × Cofre | 0.303 | 1 | 1.2 | None |
| Sierra Sur × Nevado | 0.303 | 1 | 1.2 | None |
| Cofre × Nevado | 0.273 | 0.804 | 1.3 | H5 |
| Sierra Norte × Sierra Sur | 0.244 | 0.646 | 1.5 | H2 and H3 |
| Sierra Norte × Cofre | 0.237 | 0.582 | 1.6 | H1 |

**Θ WHY D REACHES 1 WHILE Θ STAYS BELOW 0.35**

Three pairs share no haplotype at all, so their pools are completely different and D = 1. θ does not reach 1 because each population retains three haplotypes: two sequences from the same population are often different too, and θ measures the difference between populations **relative to** the difference within them. This is the ceiling of figure 6.2 with real data. For the chloroplast, which is inherited through a single parent and has a smaller effective size than nuclear genes, higher differentiation is expected than with microsatellites from the same populations.

**I NOTE · WHAT TO REPORT IN A PAPER**

The complete AMOVA table (d.f., sums of squares, components, percentages, Φ and P, with the number of permutations); θ or $\theta_B$ with its interval, together with D or $G'_{ST}$; the pairwise matrix with corrected P values, stating the correction method; and, if you use it, Nm as an indicative value. With regions, state how many populations there are per region and how many distinct ways of assigning them exist. The Per-locus statistics (CSV) and All pairs, all statistics (CSV) downloads contain all values to five decimal places.

**I NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, θ, *f*, F, $H_S$, $H_T$, $G_{ST}$, $G'_{ST}$ and D were recomputed with independent code for the maize data, by locus, overall and by pair, and for the agave data with the haploid version and with an analysis of variance on the bands: the differences do not exceed $10^{-12}$. The AMOVA sums of squares were compared with the direct sum of all distances between gene copies or between plants, and the hierarchical AMOVA of the agave with the formulas for the balanced design. The coefficients for unbalanced designs were checked by simulating 300 datasets with known components: the mean estimates were 0.0143, 0.0064 and 0.229 against true values of 0.0144, 0.0064 and 0.229. Under the null hypothesis, with populations shuffled at random, the θ test gave P ≤ 0.05 in 3 of 60 datasets, as expected by chance.

The next chapter moves from how much the populations differ to how they are related: Block 6 computes genetic distances, places populations and individuals in a PCoA ordination, builds trees with *bootstrap* support and tests isolation by distance.

# B6

CHAPTER SEVEN · BLOCK 6 · DISTANCE

# Distances, ordination and trees

How far apart, genetically, are populations and individuals? Block 6 measures this with distance coefficients and draws it in three ways: as a map (principal coordinates analysis, PCoA), as a tree with *bootstrap* support and, if coordinates are available, against geography. Its trees can carry images of the OTUs and be made ready for publication.




7.1 Genetic distances

7.2 Principal coordinates (PCoA)

7.3 NJ and UPGMA trees with bootstrap

7.4 Trees with images of the OTUs

7.5 Mantel and isolation by distance

7.6 Block settings, tables and figures

7.7 Step by step: maize landraces with ISSR


## 7.1 Genetic distances

A genetic distance sums up in a single number how different two individuals or two populations are. All the pairwise distances together form a matrix, and from that matrix come the map, the tree and the comparison with geography.

**Table 7.1.** Distances between populations. $x_i$ and $y_i$ are the frequencies of allele $i$ in the two populations; $J_X = \Sigma x_i^2$, $J_Y = \Sigma y_i^2$ and $J_{XY} = \Sigma x_i y_i$, averaged over loci.

| Coefficient | Formula | When to use it |
|---|---|---|
| **Nei (1972)** standard distance | $D = -\ln [J_{XY} / \sqrt{(J_X J_Y)}]$ | The classic one. It grows with the time since separation if there is mutation–drift equilibrium. The ratio inside the logarithm is the genetic identity I. |
| **Nei (1978)** unbiased (default) | same, with $J_X$ corrected: $(2nJ_X - 1)/(2n - 1)$ | The general-purpose option: it corrects the bias of small samples. |
| **Cavalli-Sforza and Edwards (1967)** | $(2/\pi)\sqrt{[2(1 - \Sigma\sqrt{(x_i y_i)})]}$, averaged over loci | For reconstructing the branching order when populations diverged by drift; it frequently recovers the correct tree (Takezaki and Nei, 1996). |
| **Reynolds, Weir and Cockerham (1983)** | $\sqrt{\{\Sigma\Sigma(x_i - y_i)^2 / [2\Sigma(1 - \Sigma x_i y_i)]\}}$, sums over loci and alleles | Based on the coancestry coefficient; suitable for recent divergence by drift, without mutation. |
| **Rogers (1972)** | $\sqrt{[\Sigma(x_i - y_i)^2/2]}$, averaged over loci | Euclidean and bounded between 0 and 1; useful with dominant markers. |
| **Prevosti et al. (1975)** | $\Sigma\lvert x_i - y_i\rvert/2$, averaged over loci | The proportion of alleles not shared; very intuitive. |
| **Net $D_A$** sequences only (default) | $d_{XY} - (d_X + d_Y)/2$ | Nucleotide divergence between populations minus the divergence within each of them (Nei and Li, 1979). With a substitution rate, it estimates separation times. |
| **$d_{XY}$** sequences only | mean of the $p$ distances between sequences of X and of Y | Total divergence, without subtracting the internal variation. |

B6

### WORKED EXAMPLE · ONE LOCUS WITH TWO ALLELES

Population X has frequencies 0.8 and 0.2; population Y, 0.3 and 0.7. Then $J_X = 0.8^2 + 0.2^2 = 0.68$, $J_Y = 0.58$ and $J_{XY} = 0.8 \times 0.3 + 0.2 \times 0.7 = 0.38$.

| Coefficient | Calculation | Result |
|---|---|---|
| Identity I | $0.38 / \sqrt{(0.68 \times 0.58)} = 0.38 / 0.628$ | **0.605** |
| Nei (1972) | $-\ln 0.605$ | **0.502** |
| Cavalli-Sforza | $\sqrt{(0.8{\times}0.3)} + \sqrt{(0.2{\times}0.7)} = 0.864$; $(2/\pi)\sqrt{[2(1 - 0.864)]}$ | **0.332** |
| Reynolds | $(0.5^2 + 0.5^2) = 0.5$; $\sqrt{[0.5 / (2 \times (1 - 0.38))]}$ | **0.635** |
| Rogers · Prevosti | $\sqrt{(0.5/2)} \cdot (0.5 + 0.5)/2$ | **0.500 · 0.500** |

The values differ from one another because each coefficient uses a different scale: never compare numbers from different coefficients. With two-allele loci or bands, Rogers and Prevosti always give the same result.

### TWO LEVELS, TWO KINDS OF DISTANCE

**Between populations**, allele frequencies are compared: two populations with the same alleles in the same proportions are at distance zero, even if every plant is different. **Between individuals**, genotypes or band profiles are compared plant by plant. The app computes both when there are three or more populations of several individuals; with one sample per unit (the ISSR landraces), everything is computed between the units, and with morphological characters, between the plants.

## Distances between individuals

**Table 7.2.** Distances between individuals by data type.

| Data | Coefficient | What it counts |
|---|---|---|
| Codominant | **Smouse and Peakall (1999)** (default) | Squared genotypic distance, locus by locus (table 7.3), summed and rescaled if any locus is missing. It is the AMOVA distance. |
| | Shared alleles (Bowcock et al., 1994) | 1 – (shared alleles / 2), averaged over loci. |
| Dominant | **Jaccard** (default) | Differing bands / bands present in at least one plant. It ignores shared absences. |
| | Dice (Nei and Li, 1979) | Like Jaccard, but gives double weight to shared bands. |
| | Simple matching | Proportion of differing bands, also counting shared absences. |
| | Squared Euclidean | Number of differing bands; the AMOVA distance with bands. |
| Sequences | ***p* distance** | Proportion of differing sites; sites with a gap or an ambiguous base are omitted pair by pair. |
| Characters | **Gower (1971)** | Difference in each character divided by its range; 0 or 1 for text characters; averaged over characters. |

**Table 7.3.** Smouse and Peakall distance between two diploid genotypes at one locus. Different letters are different alleles.

| Genotypes | AA · AA | AB · AB | AA · AB | AB · AC | AB · CD | AA · BC | AA · BB |
|---|---|---|---|---|---|---|---|
| Distance | 0 | 0 | 1 | 1 | 2 | 3 | 4 |

### SEQUENCES: HAPLOTYPE IDENTITY OR NUCLEOTIDE DIVERGENCE

Frequency-based distances only ask whether two haplotypes are identical: if two populations share none, Nei's distance is infinite, even if the haplotypes differ by a single base. With sequences, the app uses the **net nucleotide divergence $D_A$** by default, which does measure how many sites separate the populations, and resamples the **alignment sites** for the *bootstrap*. In the pine, 3 of the 6 pairs of populations share no haplotypes: Nei's distance is infinite, whereas $D_A$ ranges from 0.0033 to 0.0068 (from 1.6 to 3.3 net sites out of 480).

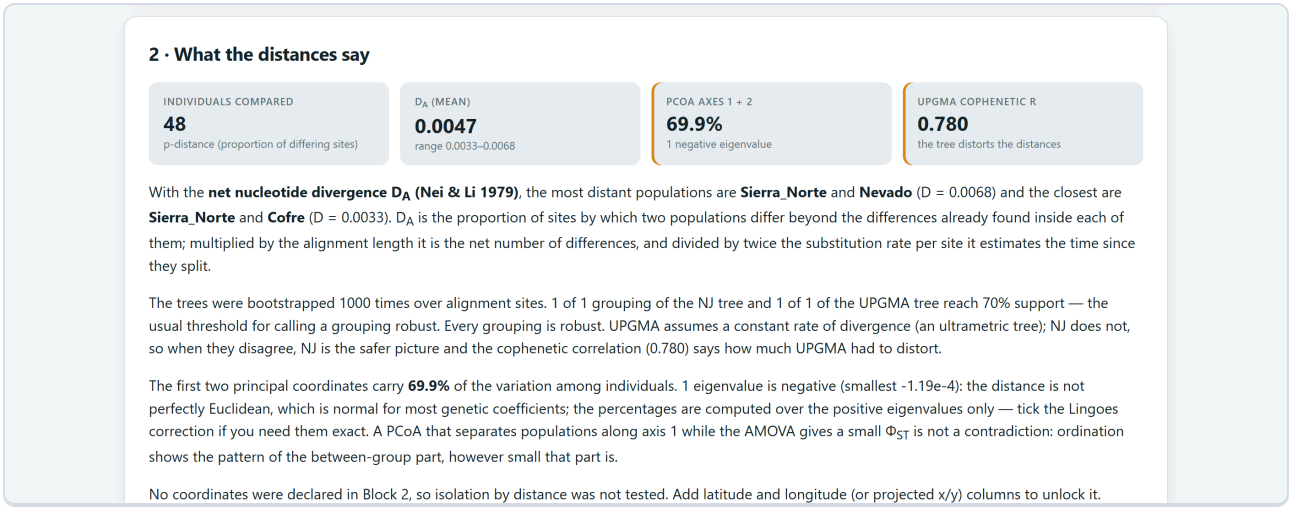



**Figure 7.1.** With the pine, the app chooses $D_A$. Sierra Norte and Nevado are the most divergent populations (0.0068) and Sierra Norte and Cofre the closest (0.0033); that grouping has 87% support when the sites are resampled.

## 7.2 Principal coordinates (PCoA)

Principal coordinates analysis (Gower, 1966) turns a distance matrix into a map: it looks for the axes on which the distances between points reproduce the genetic distances as closely as possible. The first axis captures most of the variation, the second most of what remains, and so on. Unlike principal component analysis, it works with any coefficient, which is why it is the usual ordination method in population genetics.

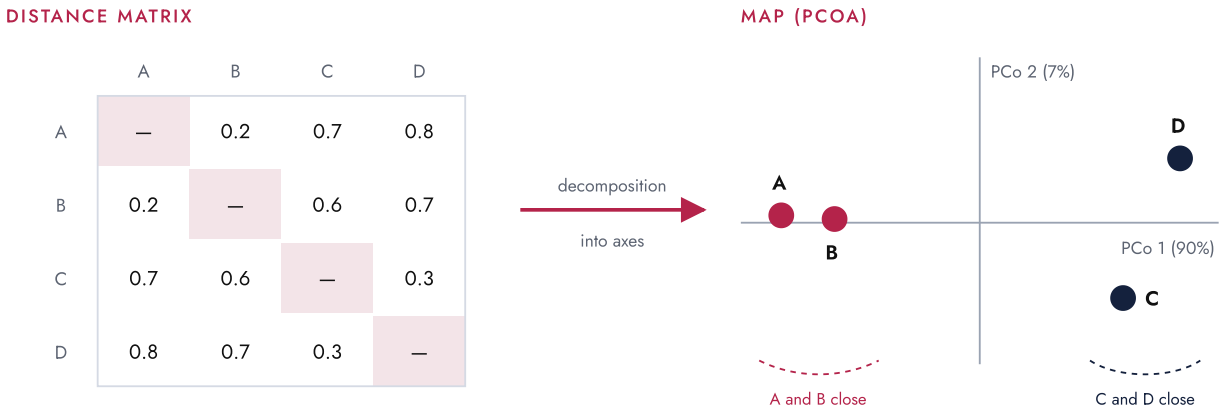



**Figure 7.2.** PCoA of four hypothetical units, computed with the app. The first axis (90% of the variation) separates the two pairs, the largest difference in the matrix; the second (7%) mainly separates C from D, which are farther apart (0.3) than A and B (0.2).

### NEGATIVE EIGENVALUES AND THE LINGOES CORRECTION

If a distance is Euclidean, it can be drawn without distortion and all the PCoA eigenvalues are positive. Many genetic distances are not (Jaccard, Nei, Gower), and some negative eigenvalues appear: they are variation that no map can represent. The app computes the percentages over the positive eigenvalues and warns you about it. The **Lingoes correction** adds a constant to the squared distances to make them Euclidean; the percentages are then exact, although somewhat smaller. The Smouse and Peakall distances and the squared Euclidean distance for bands are already squared, and the app uses them as they are.

### DECISION RULE · READING A PCOA

| If... | then... |
|---|---|
| **Axes 1 and 2 add up to more than 50%** | The map is a good summary: closeness on the plane reflects the matrix. |
| **They add up to less than 25%** | This is usual with individuals and highly variable markers. The map shows trends; two nearby points may differ on other axes. Check axis 3 as well. |
| **The population ellipses overlap a great deal** | Low differentiation, consistent with a small $F_{ST}$ (chapter 6). A significant AMOVA is not a contradiction. |
| **An individual falls within the cloud of another population** | Possible migrant, labelling error or admixture; check it with the assignment in Block 7. |
| **There are many large negative eigenvalues** | Turn on the Lingoes correction before reporting percentages. |

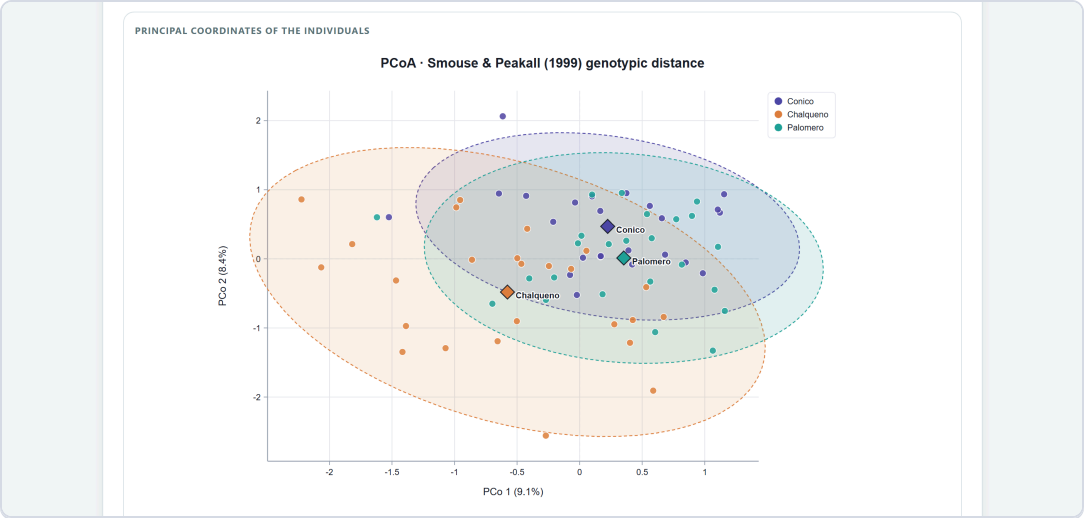
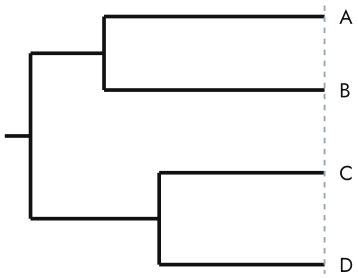



**Figure 7.3.** PCoA of the 75 maize plants (Smouse and Peakall), with 95% ellipses and centroids. Axes 1 and 2 add up to 17.5%, which is normal with individuals. Chalqueño stands apart on the left; Cónico and Palomero, the closest landraces (Nei's D 0.114), overlap.

# 7.3 NJ and UPGMA trees with bootstrap

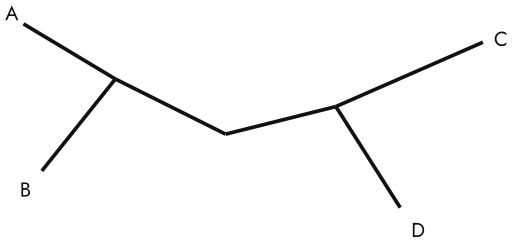
A tree draws the matrix as a hierarchy of groupings. The app builds two, for populations and for individuals: **UPGMA** (Sokal and Michener, 1958) joins the closest pairs first and averages the distances; it assumes that all branches evolve at the same rate, so it gives a rooted tree with all the tips at the same height. **Neighbour-joining** (NJ; Saitou and Nei, 1987) makes no such assumption: it allows branches of different lengths and produces an unrooted tree, which the app roots at the midpoint or on an outgroup.

UPGMA · ROOT AND CONSTANT RATE




all tips at the same distance from the root

NEIGHBOUR-JOINING · FREE BRANCHES




C evolved faster: its branch is longer

**Figure 7.4.** The two methods with the same four OTUs. UPGMA forces all of them to reach the same height; NJ lets each lineage have its own rate. When they disagree, NJ gives the safer picture.

### WHAT BOOTSTRAP SUPPORT MEANS

The *bootstrap* (Felsenstein, 1985) repeats the analysis 1000 times with loci resampled with replacement (bands for dominant markers and alignment sites for sequences). In each replicate a new tree is built and the groupings that appear are recorded. The number on a branch is the percentage of replicates in which the same grouping reappears: it measures how much that grouping depends on a few loci, **not** the probability that it is true. The app computes it for the population trees and also for the individual trees when there are 60 individuals or fewer, such as the landraces with one sample each or the species in an alignment.

### DECISION RULE · SUPPORT AND FIT OF THE TREE

| If… | then… |
|---|---|
| **Support of 70% or more** | The grouping is robust to the loci sampled; it can be described. |
| **Support of 50 to 70%** | Weak grouping; mention it as a trend. |
| **Support below 50%** | Do not interpret the branching order. The app hides it by default. |
| **Only three populations** | An unrooted three-tip tree has no internal branches to test: it only shows lengths. The app says so. |
| **UPGMA cophenetic correlation greater than 0.85** | The tree reproduces the distances well. If it is lower, the tree distorts them; prefer NJ and the PCoA. |

With few loci, low support is normal and does not mean the tree is wrong: it means the data are not enough to resolve it. Zero-length branches (identical units) receive no support, because the order in which they are joined is arbitrary.

## A tree with a known answer

The five-genus example was simulated on a known tree: the outgroup *Exterocarpus* and, within the ingroup, (*Fictaria*, *Modelanthus*) and (*Demoxylon*, (*Probocarpa*, *Testiflora*)). The NJ tree of genera with $D_A$ recovers exactly that topology, with support values of 94, 100 and 99%.

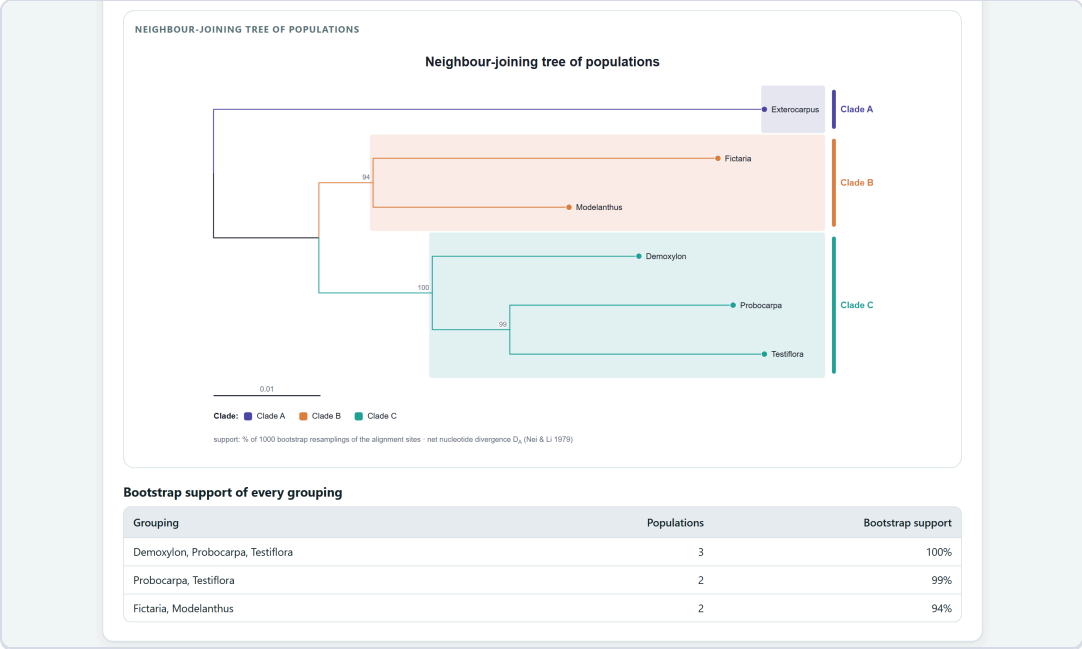



**Figure 7.5.** NJ tree of the genera with the net nucleotide divergence $D_A$, midpoint-rooted, and its support table. The three groupings match the tree on which the data were simulated; the UPGMA cophenetic correlation is 0.943.

B6

**NOTE · THE ROOT OF THE TREE**

NJ produces an unrooted tree. The app displays it rooted at the **midpoint** of the longest path between two tips, which is a good choice when rates are similar. If you know an outgroup, type it in Outgroup tips with Root set to outgroup; the root of a tree of populations of the same species hardly ever has evolutionary meaning, and in that case the unrooted layout is advisable.

# 7.4 Trees with images of the OTUs

The trees in this block are edited in a full studio: rectangular, circular or unrooted layout; clades coloured by population, by region or cut from the tree; rings with the population, the region or the ancestry from Block 7; and **images** of each tip or group next to its branch. The images are stored in the browser, linked to the data file, and never leave the computer.

### STEP BY STEP · ADDING IMAGES TO A TREE

1. In the **6 · Trees** card, choose the tree in **Tree** and click **Images of the OTUs...**.
2. The list shows the groups (populations or regions) and the tips. Choose a name and drag its image onto the box, or click **Add several images...** to load several at once: they are assigned by file name.
3. Adjust the framing (zoom, offset, rotation), the tone and, if the photo has a plain background, **Remove a plain background**.
4. Enter the source and the licence, and tick the confirmation that the image shows exactly that taxon. Without that tick, the image is not drawn.
5. Close with **Done**, open **Edit figure** and, under **Images**, choose **one per tip** or **one per group or clade**.

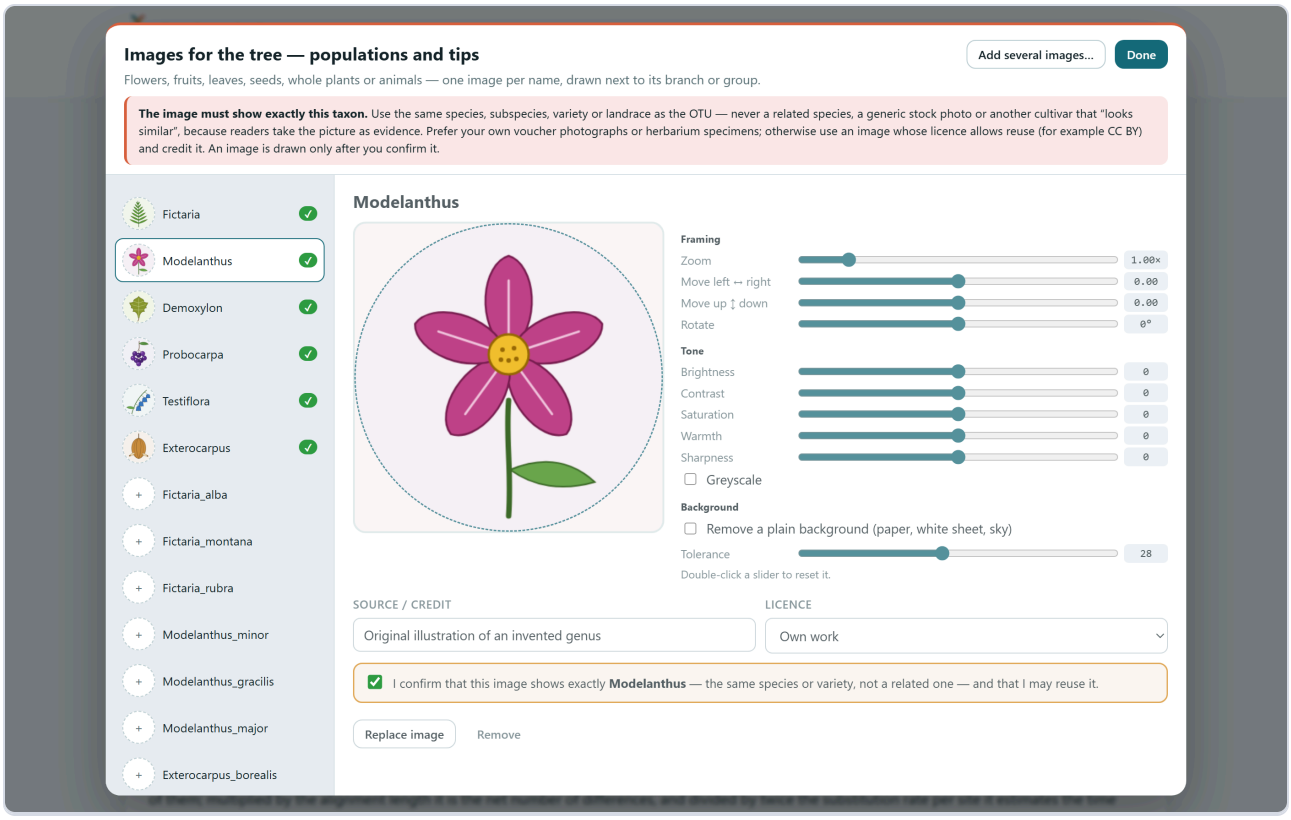



**Figure 7.6.** The image manager. On the left, the names with their status (✓ confirmed); on the right, the framing, the tone and background adjustments, the source, the licence and the confirmation. The illustrations in this example are original drawings made for the manual, because the genera in the example are fictitious.

> **IMPORTANT · THE IMAGE IS EVIDENCE**
>
> Anyone reading a tree with photographs understands that each photo shows that taxon. Use only images of the same species, variety or landrace: your own photos, herbarium specimens or images whose licence allows reuse (for example CC BY), and give credit. Never a "similar" species or a generic photo. The app asks you to confirm this for each image and stores the source and licence so that you can cite them.

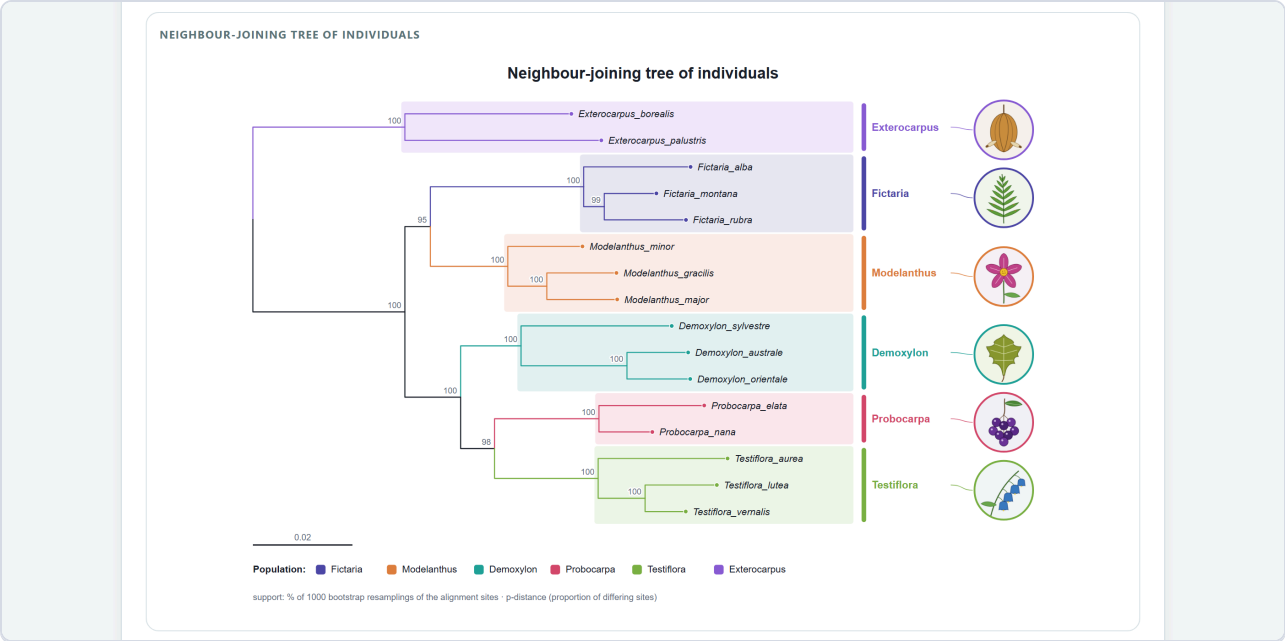



**Figure 7.7.** NJ tree of the 16 species, rooted on the outgroup *Exterocarpus*, coloured by genus and with one image per genus. All the groupings in the tree have 95% support or more when the 1600 alignment sites are resampled.

**Table 7.4.** Main options of the tree studio (Edit figure).

| Option | What it is for |
|---|---|
| Layout | Rectangular (default up to 25 tips), circular or unrooted (equal angle). |
| Root · Outgroup tips | Root as computed, at the midpoint or on an outgroup (NJ trees). |
| Colour · Number of clades | Colour by population or region, or cut the tree into *k* clades with names of your own. |
| Ring: Population · Ring: Region · Ring: Ancestry (Block 7) | Rings with the population or region of each individual, or with its ancestry proportions from Block 7. |
| Images · Image size · Image frame | One image per tip or per group, its size, frame and leader lines. |
| Support · Hide support below (%) | Support as dots or numbers, and the threshold below which it is hidden. |
| Italic names · Branch lengths | Scientific names in italics; proportional branches or aligned tips (cladogram). |

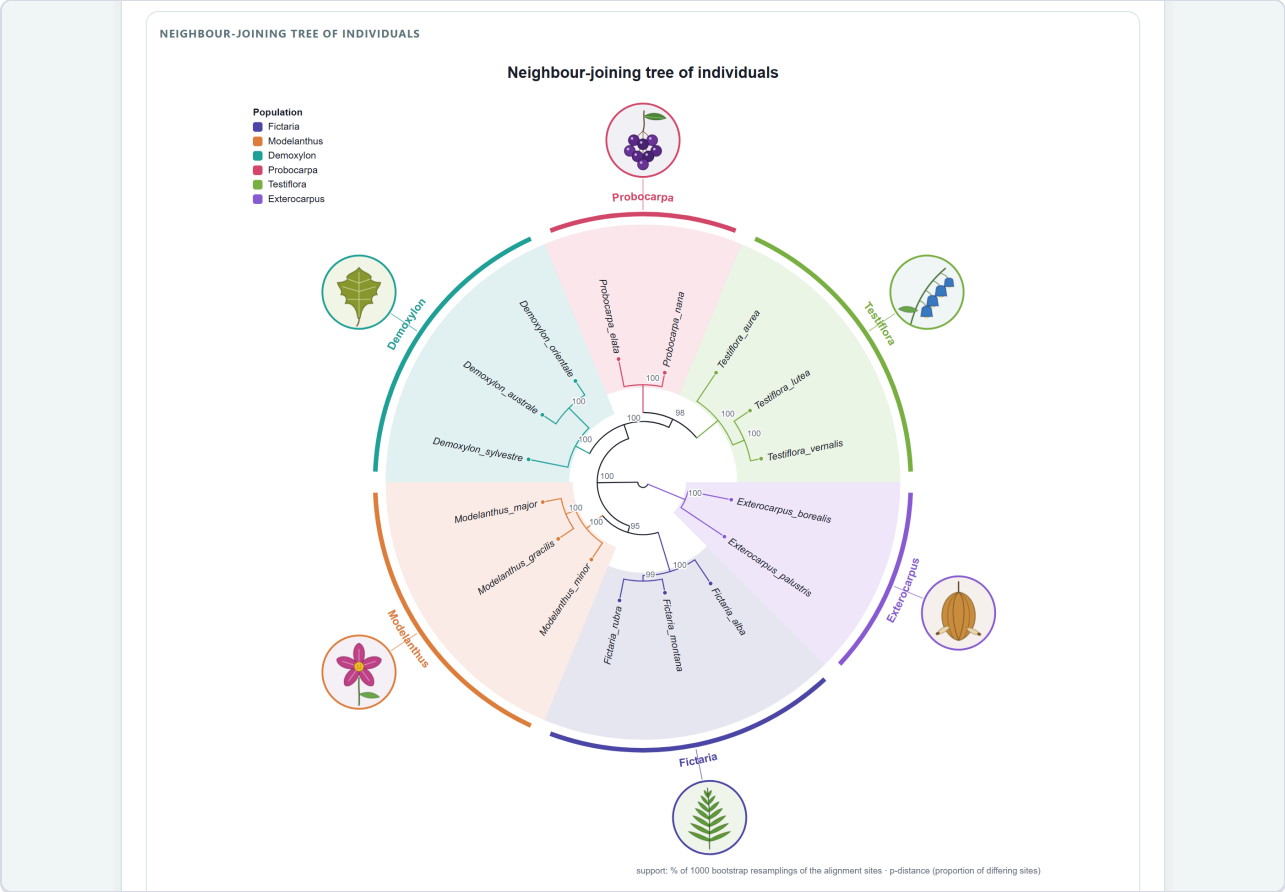



**Figure 7.8.** The same tree in circular layout and midpoint-rooted, with shaded clades, arcs bearing the name of each genus and names in italics. It is exported like any other figure (PNG, TIFF, SVG, JPG or WEBP) with the images included.

B6

## 7.5 Mantel and isolation by distance

If seeds and pollen travel only short distances, nearby populations exchange more migrants than distant ones and are more alike. This pattern is called **isolation by distance**. The **Mantel** test (1967) measures the correlation between the genetic distance matrix and the geographic distance matrix, and tests it by permuting the rows and columns of one of them. The app computes great-circle geographic distances from latitude and longitude.

### ROUSSET'S REGRESSION

Rousset (1997) showed that, in two dimensions, $F_{ST}/(1 - F_{ST})$ increases linearly with the logarithm of distance, and that the slope is $1/(4D\pi\sigma^2)$, where D is the plant density and $\sigma^2$ the mean squared parent–offspring dispersal distance. The app therefore plots $\theta/(1 - \theta)$ for each pair of populations against ln(distance) and tests that relationship with Mantel. A positive, significant slope indicates isolation by distance; its value allows the genetic neighbourhood to be estimated if the density is known.

**DECISION RULE · ISOLATION BY DISTANCE**

| If… | then… |
|---|---|
| **Positive slope and significant Mantel test** between populations | Isolation by distance: gene flow decreases with distance. |
| **Not significant** | Distance does not predict differentiation at this scale: extensive gene flow, structure due to another cause (habitat, altitude, history) or too few populations. |
| **Few populations** (6 or fewer) | With *n* populations there are only *n*! ways to reorder them; with 4, the smallest possible P is 1/24 ≈ 0.04. Read the slope and the figure, not the P value. |
| **Mantel significant only between individuals** | Pairs from the same population are close together and alike, so the correlation mainly reflects the differences between populations. Fine-scale spatial structure is studied in Block 9. |

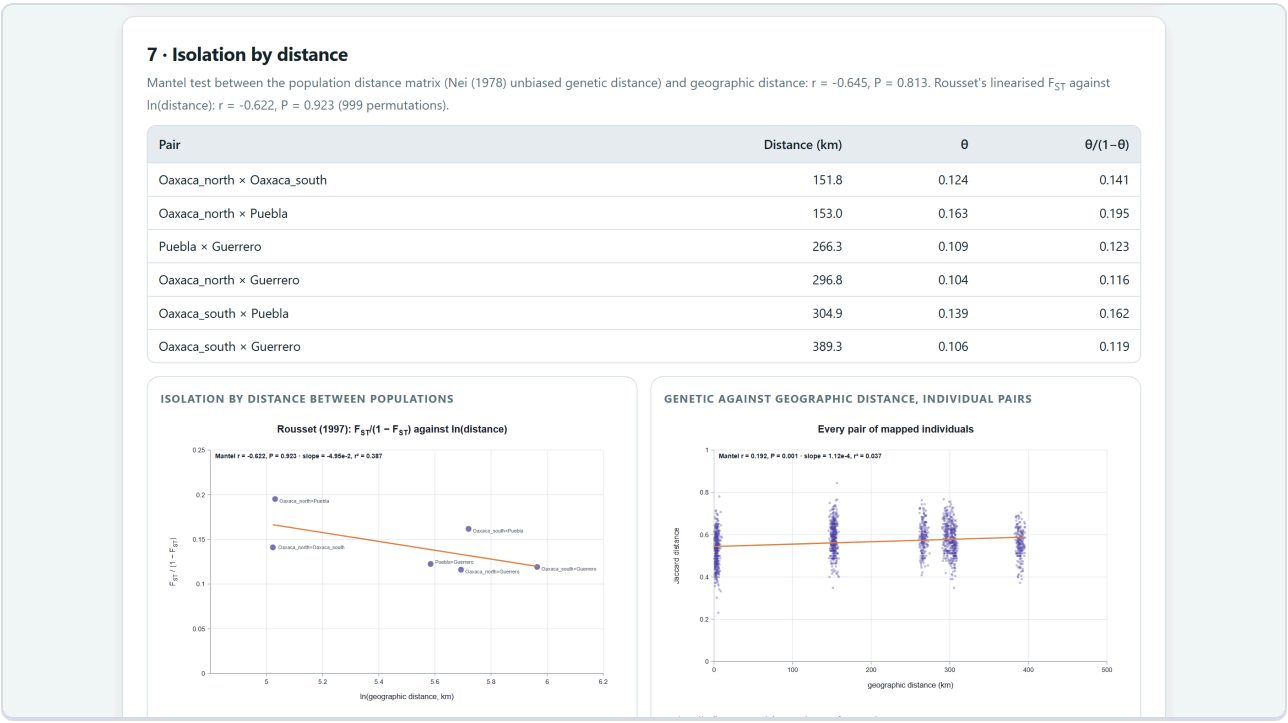



**Figure 7.9.** Isolation by distance in the four agave populations, which are between 152 and 389 km apart. Rousset's slope is negative (–0.050; Mantel r = –0.62, P = 0.92): there is no isolation by distance. Between individual plants the correlation is positive (r = 0.19, P = 0.001), but this is because plants from the same population share coordinates and are alike; on the right you can see the columns of pairs at each distance.

**NOTE · COORDINATES**

The 7 · Isolation by distance card appears when the file includes latitude and longitude (Block 2) for at least 4 individuals. With decimal degrees, the great-circle distance in kilometres is used; if the values do not fit as degrees (projected coordinates, for example UTM), they are used as Euclidean distances in map units. The test between populations uses the centroid of each population and requires all of them to have coordinates.

## 7.6 Block settings, tables and figures

Block 6 opens with three or more units, with any data type. On entering, everything is computed with the default settings; if you change any of them, click **Compute distances, ordination and trees**.

**Table 7.5.** Block 6 settings.

| Setting | Options | Recommendation |
|---|---|---|
| Distance between populations | The coefficients in table 7.1. It appears with three or more replicated populations. | Nei (1978); $D_A$ with sequences. |
| Distance between individuals | Depending on the data type (table 7.2). | The default for each type. |
| Bootstrap replicates over loci | 0 to 10,000 (default 1000); it reads *bands* or *alignment sites* depending on the data. | 1000. |
| Lingoes correction for negative eigenvalues | Off by default. | Turn it on to report exact percentages. |
| Mantel permutations | 0 to 9999 (default 999). | 999; 9999 for the paper. |
| Random seed | Integer. | Report it in the methods. |

**Table 7.6.** Block 6 result cards.

| Card | Contents | Download |
|---|---|---|
| 2 · What the distances say | Figures and interpretation: most and least distant pairs, tree support, PCoA and isolation by distance. | — |
| 3 · Distances between populations | Matrix of the chosen coefficient (with Nei's identity I), all the coefficients pair by pair and the heat map. | Population matrix (CSV) |
| 4 · Distances between individuals | Summary and, for up to 30 units, the full matrix. | Individual matrix (CSV) |
| 5 · Principal coordinates | Eigenvalues and percentages, PCoA of individuals and of populations, and the eigenvalue plot. | Scores of every unit (CSV) |
| 6 · Trees | NJ and UPGMA trees of populations and of individuals, with the tree studio and the support table. | Newick |
| 7 · Isolation by distance | Geographic distances and θ per pair, and the plots between populations and between individuals. | — |

## 7.7 Step by step: maize landraces with ISSR

The 15 maize landraces have one bulked sample each and 31 ISSR bands. There are no populations with replicates, so Block 6 works between the landraces: it is the central analysis for this kind of data.

**STEP BY STEP**

1. In Block 2, load **Maize landraces · ISSR**, build the dataset and open Block 6.
2. A notice explains that the analysis is carried out between landraces; the distance is Jaccard and the *bootstrap* is done over bands (figure 7.10).
3. Read the figures and the interpretation (figure 7.11) and check the distance matrix (figure 7.12).
4. Explore the PCoA and the eigenvalue plot (figures 7.13 and 7.14).
5. Check the NJ tree and its support table (figures 7.15 and 7.16); in **Tree**, compare it with UPGMA.

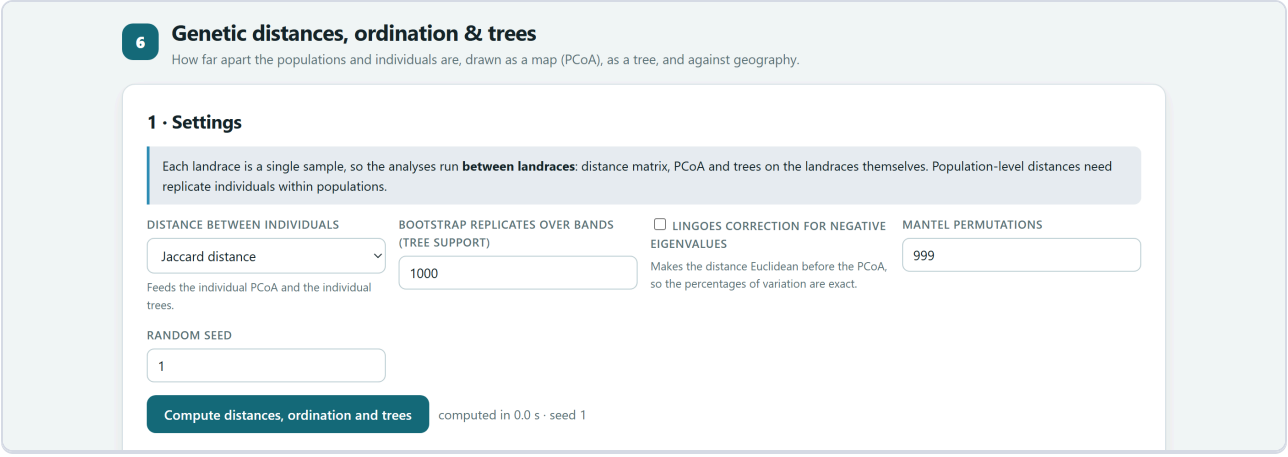



**Figure 7.10.** Settings with the maize landraces. The distance between populations does not appear, and the *bootstrap* is done over the bands.

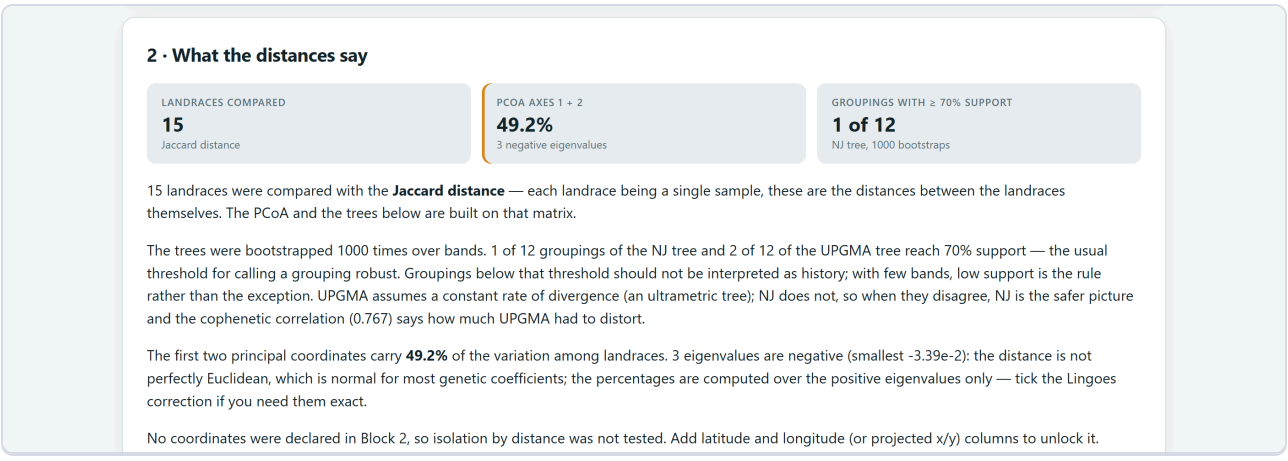



**Figure 7.11.** **What the distances say**. 15 landraces compared with Jaccard; the first two PCoA axes add up to 49.2%, and only 1 of the 12 groupings in the NJ tree reaches 70% support.

**4 · Distances between individuals**

**15 × 15** matrix of the **Jaccard distance** between landraces: mean 0.488, range 0.176–0.737.

| | Ancho | Apachito | Arrocillo | Azul | Blando Sonora | Bofo | Bolita | Cacahuacintle | Celaya | Chalqueño | Chapalote | Chiquito | Con |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ancho | — | 0.214 | 0.500 | 0.526 | 0.529 | 0.471 | 0.471 | 0.533 | 0.588 | 0.500 | 0.385 | 0.526 | |
| Apachito | 0.214 | — | 0.353 | 0.474 | 0.632 | 0.500 | 0.500 | 0.563 | 0.611 | 0.611 | 0.533 | 0.389 | |
| Arrocillo | 0.500 | 0.353 | — | 0.368 | 0.526 | 0.550 | 0.550 | 0.611 | 0.650 | 0.650 | 0.737 | 0.176 | |
| Azul | 0.526 | 0.474 | 0.368 | — | 0.389 | 0.333 | 0.235 | 0.375 | 0.444 | 0.444 | 0.611 | 0.316 | |
| Blando Sonora | 0.529 | 0.632 | 0.526 | 0.389 | — | 0.412 | 0.500 | 0.467 | 0.529 | 0.438 | 0.533 | 0.550 | |
| Bofo | 0.471 | 0.500 | 0.550 | 0.333 | 0.412 | — | 0.444 | 0.500 | 0.375 | 0.375 | 0.563 | 0.571 | |
| Bolita | 0.471 | 0.500 | 0.550 | 0.235 | 0.500 | 0.444 | — | 0.286 | 0.471 | 0.375 | 0.563 | 0.500 | |
| Cacahuacintle | 0.533 | 0.563 | 0.611 | 0.375 | 0.467 | 0.500 | 0.286 | — | 0.429 | 0.308 | 0.538 | 0.556 | |

**Figure 7.12.** Part of the Jaccard matrix (mean 0.488). The most similar pair is Arrocillo–Chiquito (0.176) and the most different Arrocillo–Chapalote (0.737).

**Table 7.7.** Nearest neighbour of each landrace according to Jaccard.

| Landrace | Most similar | Distance | Landrace | Most similar | Distance |
|---|---|---|---|---|---|
| Arrocillo | Chiquito | 0.176 | Cacahuacintle | Bolita | 0.286 |
| Ancho | Apachito | 0.214 | Chalqueño | Cacahuacintle | 0.308 |
| Azul | Bolita | 0.235 | Bofo | Azul | 0.333 |
| Comiteco | Chiquito | 0.263 | Blando Sonora | Cónico | 0.375 |
| Celaya | Bofo | 0.375 | Chapalote | Ancho | 0.385 |

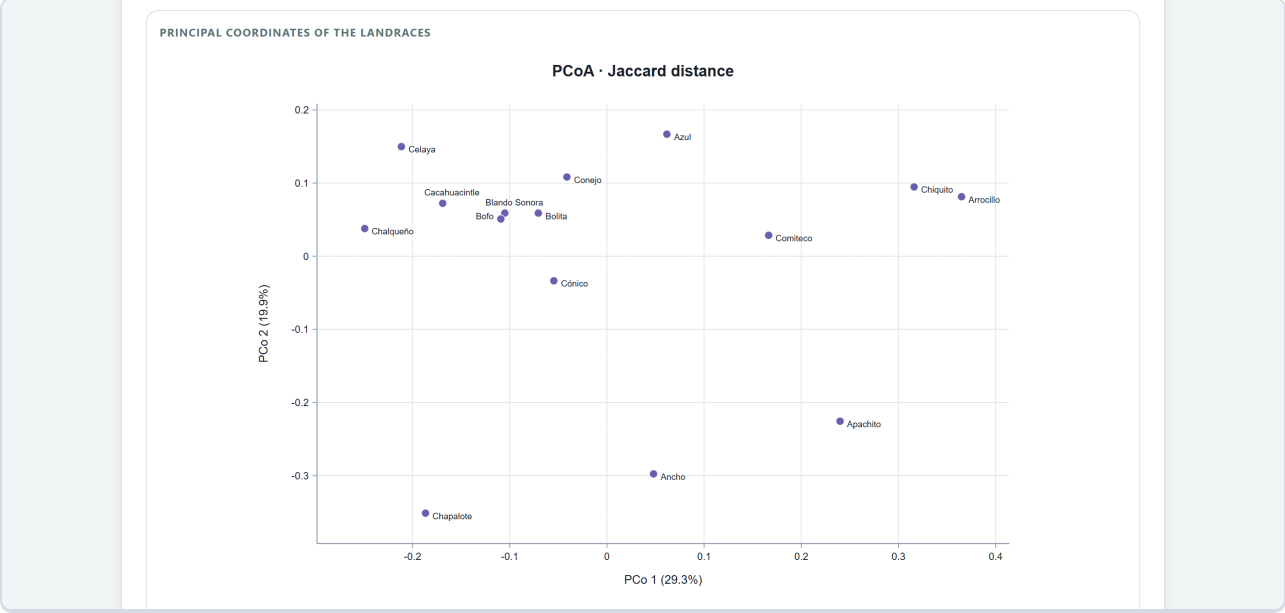



**Figure 7.13.** PCoA of the landraces. The first axis (29.3%) runs from Chalqueño, Celaya and Cacahuacintle on the left to Arrocillo, Chiquito and Apachito on the right; the second (19.9%) separates Chapalote, Ancho and Apachito towards the bottom. The labels are spread out automatically so that they do not overlap.

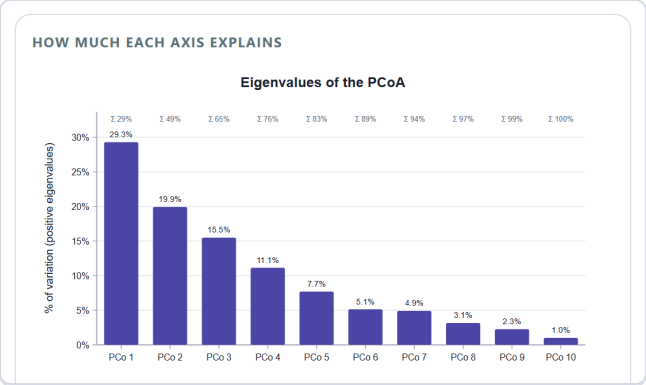



**Figure 7.14.** Variation of each axis and cumulative variation (Σ). There are 3 small negative eigenvalues (the smallest, −0.034); with the Lingoes correction the first three axes become 25.3, 17.7 and 14.1%.

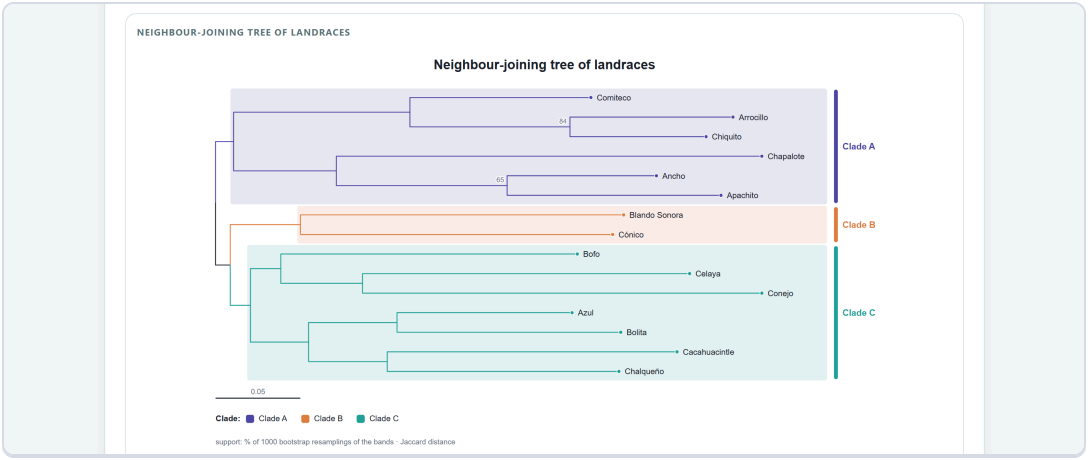



**Figure 7.15.** NJ tree rooted at the midpoint and cut into three clades. Clade A brings together Comiteco, Arrocillo, Chiquito, Chapalote, Ancho and Apachito, the same landraces as at the right-hand and lower end of the PCoA. Only Arrocillo–Chiquito (84%) and Ancho–Apachito (65%) exceed 50% support.

**Bootstrap support of every grouping**

| Grouping | Landraces | Bootstrap support |
|---|---|---|
| Arrocillo, Chiquito | 2 | 84% |
| Ancho, Apachito | 2 | 65% |
| Comiteco, Arrocillo, Chiquito | 3 | 46% |
| Celaya, Conejo | 2 | 42% |
| Chapalote, Ancho, Apachito | 3 | 41% |
| Azul, Bolita | 2 | 36% |
| Blando Sonora, Cónico | 2 | 31% |
| Cacahuacintle, Chalqueño | 2 | 30% |
| Comiteco, Arrocillo, Chiquito, Chapalote, Ancho, Apachito | 6 | 17% |
| Bofo, Celaya, Conejo | 3 | 12% |
| Azul, Bolita, Cacahuacintle, Chalqueño | 4 | 7% |
| Azul, Bofo, Bolita, Cacahuacintle, Celaya, Chalqueño, Conejo | 7 | 4% |

**Figure 7.16.** Bootstrap support of every grouping, sorted from highest to lowest. Each row names the smaller side of the grouping. The large clades have almost no support: A, 17%, and C, 4%.

**HOW TO READ THESE RESULTS**

- **The PCoA and the tree tell the same story.** Arrocillo, Chiquito and Comiteco form one group, and Ancho and Apachito another, in both representations.
- **Only two pairs are robust.** Arrocillo–Chiquito has 84% in NJ and 70% in UPGMA, and Ancho–Apachito 65 and 72%. The rest of the branching is not resolved with 31 bands.
- **The three clades are a way of arranging the figure, not demonstrated groups.** Their support is low; proposing groups of landraces requires the clustering of Block 7 or more markers.
- **UPGMA somewhat distorts the distances** (cophenetic correlation 0.767): NJ and the PCoA are the reference.

**NOTE · WHAT TO REPORT IN A PAPER**

The distance coefficient and why it was chosen; the PCoA with the percentage of the axes shown, stating whether the Lingoes correction was applied; the tree method, how it was rooted and the number of *bootstrap* replicates, showing only support values of 50% or more; and, if coordinates are available, the Mantel correlation, its P value and the number of permutations. If the tree carries images, cite the source and licence of each one. The Newick download saves the tree with the support values for other tree-editing programs.

**NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, the six between-population coefficients were recomputed with independent code for the maize (differences smaller than $10^{-15}$), as were the Smouse and Peakall, shared-allele, Jaccard, Dice, simple matching, Euclidean, $D_A$ and $d_{XY}$ distances. The PCoA reproduces the Euclidean band distance exactly when all its axes are used (error of $10^{-12}$). NJ recovered the topology and branch lengths of 20 randomly generated additive trees, UPGMA gave ultrametric trees, the great-circle distance from Mexico City to Oaxaca was 365 km, and under the null hypothesis the Mantel test was significant in 7 out of 100 cases. The five-genus tree recovered the simulated topology.

The next chapter looks for structure that was not declared in advance: Block 7 clusters individuals with a Bayesian admixture model, chooses the number of groups, performs discriminant analysis of principal components and detects possible migrants.

# B7

CHAPTER EIGHT · BLOCK 7 · CLUSTERING

# Bayesian clustering and assignment

Up to this point, the populations were the ones you declared. Block 7 asks the data how many genetic clusters they support and which one each individual belongs to, using two complementary methods: a Bayesian admixture model and discriminant analysis of principal components (DAPC). It then assigns each genotype to the population where it is most likely and flags possible migrants.




8.1 The admixture model

8.2 How many clusters? ΔK, Ln P(D) and Puechmaille

8.3 DAPC

8.4 Assignment and migrants

8.5 Block settings, tables and figures

8.6 Step by step: agave with AFLP


## 8.1 The admixture model

If the plants you sampled come from several gene pools, each gene pool has its own allele frequencies. The model of Pritchard, Stephens and Donnelly (2000) searches for those gene pools without knowing in advance where each plant came from.

### THE IDEA IN ONE SENTENCE

The model assumes **K clusters**, each with its own allele frequencies **P**, and, for each individual, the fraction of its genome that comes from each cluster, **Q**. The algorithm distributes the allele copies among the clusters so that, within each cluster, the genotypes fit Hardy–Weinberg equilibrium as closely as possible and the loci are independent. An individual with Q = (1, 0, 0) comes entirely from cluster 1; one with Q = (0.6, 0.4, 0) is admixed between clusters 1 and 2.

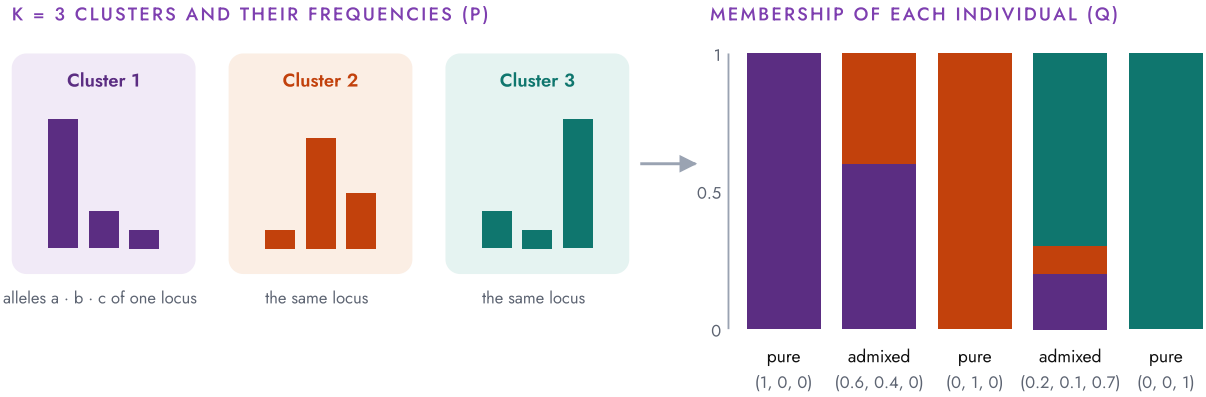



**Figure 8.1.** On the left, three clusters with different frequencies for the alleles of one locus. On the right, each bar is an individual and its colours add up to 1: the fraction of its genome that comes from each cluster. This is how to read the app's membership bar plot (*barplot*).

**Table 8.1.** The elements of the model and what the app reports for each of them.

| Element | What it is | Where it appears |
|---|---|---|
| **K** | Number of clusters tested. The app runs several values, from `K from` to `K to`. | K table and plots in card 2. |
| **Q** (membership) | Fraction of each individual's genome that comes from each cluster; average over all the sampled iterations. | Bar plot, table by population and `Q matrix (CSV)`. |
| **P** (frequencies) | Allele frequencies of each cluster, with a Dirichlet prior with λ = 1 (frequencies independent among clusters). | Used in the calculation; not shown. |
| **α** (admixture) | How admixed the individuals are: small (< 0.2), almost all pure; close to 1 or larger, widespread admixture or an excessive K. It is estimated at each iteration. | Interpretation in card 2. |
| **Ln P(D)** | Estimate of the likelihood of the model with K clusters: mean of ln L minus half its variance during sampling. | K table and the `Ln P(D) against K` plot. |
| **Burn-in** and **iterations** | Iterations discarded while the chain stabilises, and those averaged afterwards. | `Burn-in iterations` · `Sampled iterations`. |
| **Replicates** | Independent runs for each K, with different seeds; they make it possible to measure the variation of Ln P(D) and the agreement between solutions. | `Replicates per K`; H′ column. |

**NOTE · DETAILS OF THIS IMPLEMENTATION**

The app offers the admixture model (default) and the no-admixture model, in which each individual belongs to a single cluster; in both, allele frequencies are independent among clusters. Dominant bands are treated as haploid loci, one copy per individual. Missing data simply contribute nothing. The runs are carried out in a background thread, so you can keep using the page; when the app is opened by double-clicking, some browsers do not allow that thread and the runs are carried out in the page itself, with the same results.

**IMPORTANT · WHAT THE MODEL ASSUMES**

The model expects Hardy–Weinberg equilibrium and independent loci within each cluster. In plants this often fails: **selfing** and **clonal propagation** create heterozygote deficits and linkage; **families** sampled together form clusters that are not populations; **isolation by distance** turns a continuous gradient into artificial clusters; and **uneven sampling** makes ΔK underestimate K. Before interpreting, check $F_{IS}$ and the linkage results of Block 4 and the isolation by distance of Block 6.

## 8.2 How many clusters? ΔK, Ln P(D) and Puechmaille

No single number decides K on its own. The app runs the model with several values of K and several replicates, and brings together four lines of evidence: the likelihood Ln P(D), Evanno's ΔK, Puechmaille's estimators and the agreement between replicates. The final decision is made by looking at the bar plots as well.

**Table 8.2.** The criteria for choosing K. L(K) is the mean Ln P(D) of the replicates with K clusters and s its standard deviation.

| Criterion | Calculation | How to read it |
|---|---|---|
| **Ln P(D)**<br>Pritchard et al. (2000) | mean(ln L) − variance(ln L)/2 | It usually rises up to the best-supported K and then levels off or falls, while the replicates spread out (s increases). |
| **ΔK**<br>Evanno et al. (2005) | $L'(K) = L(K) - L(K-1)$<br>$\lvert L''(K)\rvert = \lvert L'(K+1) - L'(K)\rvert$<br>$\Delta K = \lvert L''(K)\rvert / s$ | It measures how abrupt the change in Ln P(D) is. It finds the uppermost level of structure; it cannot choose K = 1 or the last K, and it requires two or more replicates. |
| **MedMeaK · MaxMeaK · MedMedK · MaxMedK**<br>Puechmaille (2016) | clusters to which at least one population belongs with a mean (Mea) or median (Med) $Q \geq 0.5$; median (Med) or maximum (Max) across replicates | Designed for uneven sampling. Their highest value estimates K. They require declared populations. |
| **H′**<br>Jakobsson and Rosenberg (2007) | $1 - \lVert Q_1 - Q_2\rVert / \sqrt{(2N)}$, averaged over pairs of aligned replicates | 1 = identical replicates. Below 0.90, the runs reach different solutions. |
| **Individuals with Q ≥ 0.8** | individuals with at least 80% of their genome in one cluster (consensus Q) | If none exceeds 0.8, the clusters do not separate individuals: the signal is weak. |

**WORKED EXAMPLE · ΔK OF THE AGAVE AT K = 3**

The mean Ln P(D) values of the three replicates are −2768.9 with K = 2, −2699.7 with K = 3 and −2671.5 with K = 4, and the standard deviation with K = 3 is 4.04.

1. L'(3) = −2699.7 − (−2768.9) = 69.2 and L'(4) = −2671.5 − (−2699.7) = 28.2.
2. |L"(3)| = |28.2 − 69.2| = 41.0.
3. ΔK(3) = 41.0 / 4.04 = 10.1. The app shows 10.18 because it works with the unrounded values.

With K = 4, Ln P(D) is higher, but its standard deviation (10.94) almost triples and ΔK(4) drops to 5.43: ΔK prefers K = 3 even though the best fit is at K = 4.

**DECISION RULE · CHOOSING K**

| If... | then... |
|---|---|
| **ΔK, Ln P(D) and Puechmaille agree** | That K is well supported. Confirm it in the bar plot. |
| **ΔK gives a smaller K than Ln P(D)** | ΔK marks the uppermost level of structure; the K from Ln P(D) may be the subdivision within it. Choose the largest K at which every cluster has individuals with high Q and the replicates agree. |
| **Ln P(D) is highest at K = 1** | The ΔK peak is not evidence of clusters on its own, because ΔK never chooses K = 1. Accept K = 2 only if there are individuals with Q ≥ 0.8 in both clusters. |
| **H′ falls below 0.90** | Replicates are in different solutions: lengthen the chains, increase the replicates and do not interpret the consensus plot for that K. |
| **A Puechmaille estimator counts a cluster whose mean Q does not reach 0.6** in any population | The count is fragile and the app warns about it: confirm it with the bar plot and with the individuals with Q ≥ 0.8. |



## The agave: four clusters

The 80 agave plants with 60 AFLP bands, with the default settings (K from 1 to 6, 3 replicates, 5000 + 10,000 iterations), give a typical case in which ΔK and Ln P(D) disagree.

### 2 · How many clusters?

18 MCMC runs (admixture model, independent allele frequencies, 5,000 burn-in + 10,000 sampled iterations each) for K = 1 to 6, 3 replicates per K.

**Evanno's ΔK peaks at K = 3** (ΔK = 10.2), while the mean Ln P(D) is highest at K = 4. ΔK measures how sharply Ln P(D) changes and finds the uppermost level of structure; by construction it cannot return K = 1 and tends to favour K = 2. Ln P(D) usually rises up to the best-supported K and then levels off or falls while the runs spread apart. When the two disagree, look at the barplots: the right K is the largest one at which every cluster still has individuals assigned mostly to it.

Puechmaille's (2016) estimators, designed for uneven sampling, count at each K the clusters to which at least one population is assigned with mean (MeaK) or median (MedK) membership ≥ 0.5, as the median (Med) or the maximum (Max) over runs. Their largest values estimate K: **MedMeaK = 4**, MaxMeaK = 4, MedMedK = 4, MaxMedK = 4 (MedMeaK first reaches 4 at K = 4). At K = 4, though, the weakest counted cluster reaches a mean membership of only 0.52 in its best population, barely above the 0.5 threshold, so the count is fragile. At K = 6 only 2 of the 6 clusters are used: the surplus clusters are empty or spread thinly, the classic sign of a K set too high.

Replicate similarity H′ (Jakobsson & Rosenberg 2007) drops below 0.90 at K = 6: at those K the runs converge to different solutions (multimodality), and the consensus barplot averages solutions that do not agree — do not over-read it.

The admixture parameter α at K = 3 is 0.52 (mean of the runs): moderate, so many individuals are admixed.

| K | Runs | mean Ln P(D) | SD | L′(K) | \|L″(K)\| | ΔK | H′ | Q ≥ 0.8 | MedMeaK | MaxMeaK | MedMedK | MaxMedK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | -2792.0 | 0.29 | — | — | — | 1.000 | 80 | 1 | 1 | 1 | 1 |
| 2 | 3 | -2768.9 | 5.14 | 23.1 | 46.2 | 8.99 | 0.982 | 14 | 2 | 2 | 2 | 2 |
| 3 | 3 | -2699.7 | 4.04 | 69.2 | 41.1 | **10.18** | 0.981 | 15 | 3 | 3 | 3 | 3 |
| 4 | 3 | -2671.5 | 10.94 | 28.1 | 59.4 | 5.43 | 0.974 | 14 | 4 | 4 | 4 | 4 |
| 5 | 3 | -2702.8 | 17.47 | -31.3 | 8.0 | 0.46 | 0.911 | 9 | 2 | 3 | 2 | 3 |
| 6 | 3 | -2726.1 | 55.68 | -23.3 | — | — | 0.884 | 7 | 2 | 2 | 2 | 2 |

**Figure 8.2.** How many clusters? with the agave. Ln P(D) rises up to K = 4 (–2671.5) and then falls while the standard deviation grows (17.47 and 55.68). ΔK peaks at K = 3 (10.18). The Puechmaille estimators reach 4 at K = 4 and drop afterwards; the app warns that the least-used cluster, that of Guerrero, barely reaches a mean membership of 0.52. H′ stays above 0.97 up to K = 4, and 14 to 15 plants have Q ≥ 0.8 with K from 2 to 4.

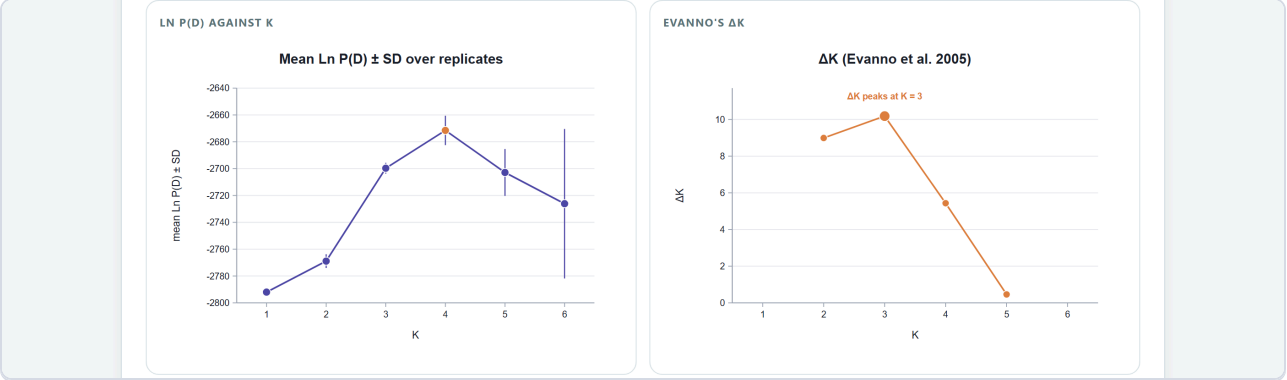



**Figure 8.3.** Mean Ln P(D) ± standard deviation (left; the orange dot is the maximum) and ΔK (right). The bar at K = 6 shows how the replicates spread out when K is excessive.

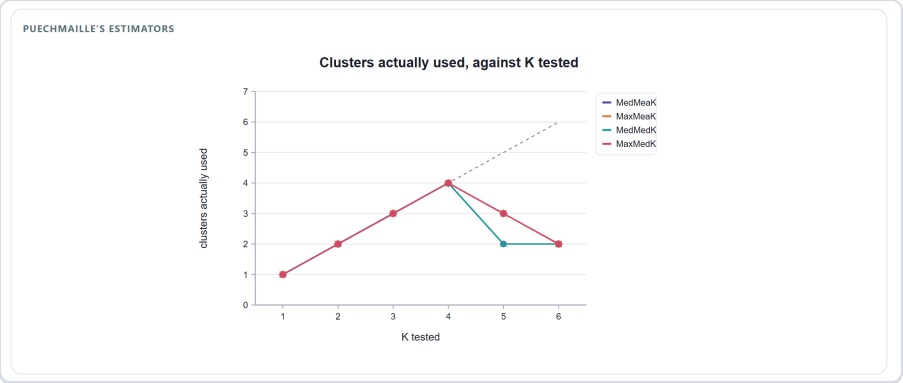
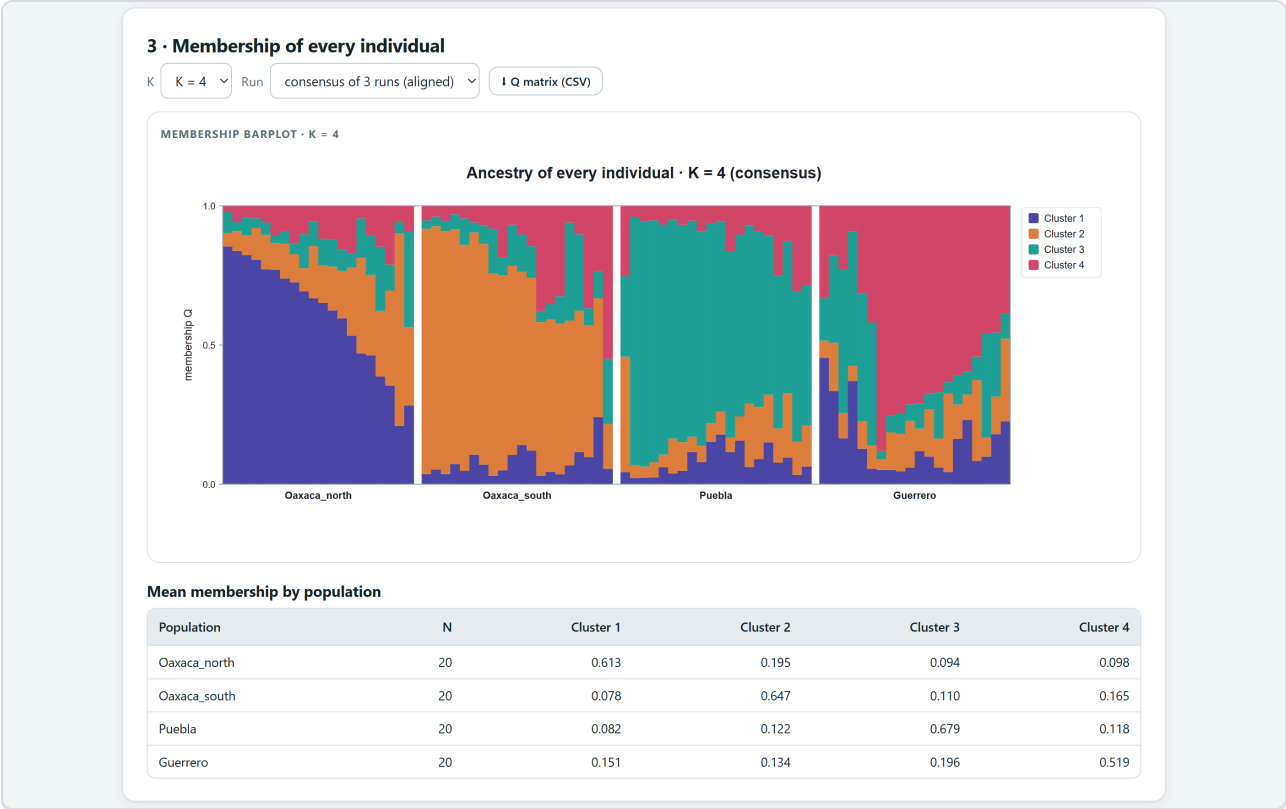



**Figure 8.4.** Puechmaille estimators against the K tested. The dotted line is K = K: as long as the estimators follow it, every new cluster is used by some population. They depart from it after K = 4.




**Mean membership by population**

| Population | N | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 |
|---|---|---|---|---|---|
| Oaxaca_north | 20 | 0.613 | 0.195 | 0.094 | 0.098 |
| Oaxaca_south | 20 | 0.078 | 0.647 | 0.110 | 0.165 |
| Puebla | 20 | 0.082 | 0.122 | 0.679 | 0.118 |
| Guerrero | 20 | 0.151 | 0.134 | 0.196 | 0.519 |

**Figure 8.5.** Membership of every individual with K = 4 (consensus of 3 replicates). Each population has its own cluster, with a mean membership of 0.61 (Oaxaca north), 0.65 (Oaxaca south), 0.68 (Puebla) and 0.52 (Guerrero), the most admixed. The four sampled populations are the four genetic clusters.

## When there are no clear clusters: maize with microsatellites

The three maize landraces differ, but only slightly: $\theta = 0.049$ in chapter 6. With 8 microsatellites and the default settings, the admixture model does not separate them, and the example shows why reading $\Delta K$ is not enough.

### 2 · How many clusters?

15 MCMC runs (admixture model, independent allele frequencies, 5,000 burn-in + 10,000 sampled iterations each) for K = 1 to 5, 3 replicates per K.

**Evanno's ΔK peaks at K = 2** (ΔK = 13.0), while the mean Ln P(D) is highest at K = 1. ΔK measures how sharply Ln P(D) changes and finds the uppermost level of structure; by construction it cannot return K = 1 and tends to favour K = 2. Ln P(D) usually rises up to the best-supported K and then levels off or falls while the runs spread apart. Here it is highest at K = 1, so the ΔK peak is not evidence of clusters by itself: accept K = 2 only if its barplot shows groups of individuals assigned mostly (Q above 0.8) to different clusters and the replicates agree.

Puechmaille's (2016) estimators, designed for uneven sampling, count at each K the clusters to which at least one population is assigned with mean (MeaK) or median (MedK) membership ≥ 0.5, as the median (Med) or the maximum (Max) over runs. Their largest values estimate K: **MedMeaK = 2**, MaxMeaK = 2, MedMedK = 2, MaxMedK = 2 (MedMeaK first reaches 2 at K = 2). At K = 2, though, the weakest counted cluster reaches a mean membership of only 0.52 in its best population, barely above the 0.5 threshold, so the count is fragile. At K = 5 no cluster reaches 0.5 in any population: the membership is spread thinly across clusters, the signature of a K well above the real structure (or of populations that are not genetically distinct at all).

Replicate similarity H′ (Jakobsson & Rosenberg 2007) stays above 0.90 for every K: the runs agree with each other, so the consensus barplots represent them faithfully (agreement between runs is not, by itself, evidence that the clusters are real).

The admixture parameter α at K = 2 is 1.32 (mean of the runs): large, so ancestry is spread evenly across clusters — either there is little structure, or K is too high.

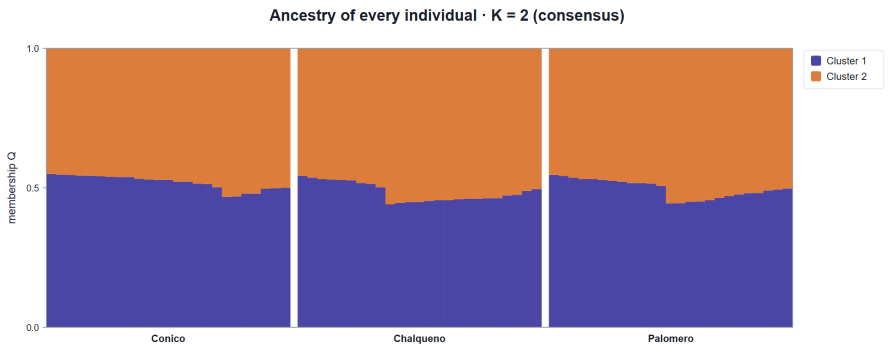
| K | Runs | mean Ln P(D) | SD | L′(K) | \|L″(K)\| | ΔK | H′ | Q ≥ 0.8 | MedMeaK | MaxMeaK | MedMedK | MaxMedK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | -1783.4 | 0.02 | — | — | — | 1.000 | 75 | 1 | 1 | 1 | 1 |
| 2 | 3 | -1798.6 | 14.04 | -15.2 | 183.1 | 13.04 | 0.959 | 0 | 2 | 2 | 2 | 2 |
| 3 | 3 | -1997.0 | 138.21 | -198.4 | 315.6 | 2.28 | 0.962 | 0 | 0 | 0 | 0 | 0 |
| 4 | 3 | -1879.8 | 73.34 | 117.3 | 179.7 | 2.45 | 0.922 | 0 | 0 | 0 | 0 | 0 |
| 5 | 3 | -1942.2 | 46.63 | -62.5 | — | — | 0.937 | 0 | 0 | 0 | 0 | 0 |

**Figure 8.6.** Maize with microsatellites. ΔK has a clear peak at K = 2 (13.04), but Ln P(D) is highest at K = 1 (–1783.4) and no individual reaches Q ≥ 0.8 with K from 2 to 5. The Puechmaille estimators count 2 clusters at K = 2, but the app warns that the mean membership barely exceeds 0.5.

### 3 · Membership of every individual

K = 2 Run consensus of 3 runs (aligned) Q matrix (CSV)

MEMBERSHIP BARPLOT · K = 2




**Mean membership by population**

| Population | N | Cluster 1 | Cluster 2 |
|---|---|---|---|
| Conico | 25 | 0.519 | 0.481 |
| Chalqueno | 25 | 0.485 | 0.515 |
| Palomero | 25 | 0.497 | 0.503 |

**Figure 8.7.** The bar plot with K = 2: all plants lie between 0.45 and 0.56 in each cluster, and the three landraces have mean memberships of 0.49 to 0.52. There are no clusters. Compare it with figure 8.5.

**FAILING TO DETECT CLUSTERS IS NOT THE SAME AS HAVING NO DIFFERENTIATION**

The AMOVA in chapter 6 found significant differences between these landraces, and the DAPC in section 8.3 distinguishes them in 72% of cases. The Bayesian model needs moderate differentiation, or many loci, to form clusters of individuals; with small differences and few loci, the honest answer is "one cluster". Moreover, the simulated data in this example have a heterozygote deficit within the landraces (F = 0.16), which the model does not expect.

## 8.3 DAPC

Discriminant analysis of principal components (Jombart, Devillard and Balloux, 2010) does not use a genetic model: it assumes neither Hardy–Weinberg equilibrium nor independent loci. It first summarises the genotypes with a principal component analysis (PCA) and then looks for the axes that best separate the groups, whether they are the populations you declared or groups the method itself finds.

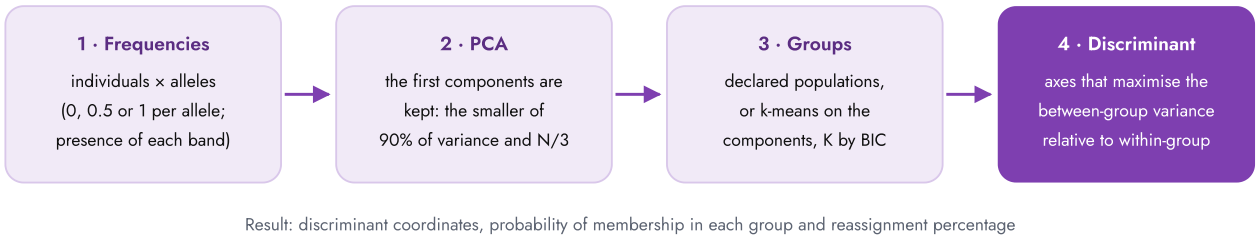



**Figure 8.8.** The four steps of DAPC. Keeping too few components loses information; keeping too many makes the discriminant axes fit the particular individuals in the sample.

**REASSIGNMENT AND LEAVE-ONE-OUT VALIDATION**

DAPC computes, for each individual, the probability of belonging to each group, and **reassigns** it to the most probable one. That success rate is optimistic, because the same individuals defined the axes. The app also computes the **leave-one-out** success rate: each individual is classified with a discriminant analysis built without it. That is the figure to report and to compare with chance (1/number of groups).

**Table 8.3.** DAPC success with the maize according to the number of components retained. Reassignment keeps rising; leave-one-out success levels off.

| Components retained | 2 | 6 | 10 | 12 | 15 | 18 | 22 (automatic) | 25 |
|---|---|---|---|---|---|---|---|---|
| Variance retained (%) | 17.4 | 42.7 | 60.9 | 68.1 | 77.2 | 84.5 | 91.0 | 94.4 |
| Reassignment (%) | 56.0 | 68.0 | 77.3 | 80.0 | 88.0 | 92.0 | 94.7 | 92.0 |
| **Leave-one-out (%)** | 54.7 | 64.0 | 68.0 | 70.7 | 70.7 | 72.0 | 72.0 | 76.0 |

### 5 · DAPC results

80 individuals × 60 variable bands. **26** principal components were retained (81.2% of the variance; the default keeps the smaller of "90% explained" and N/3, since retaining too many PCs over-fits the groups).

The discriminant analysis kept 3 axes; the first carries 45.3% of the between-group variance, the second 30%. **98.8%** of the individuals are re-assigned to their own group by their posterior membership, and **82.5%** when each individual is classified by a discriminant analysis built without it (leave-one-out). The first figure is optimistic, because the same individuals define the groups; report the leave-one-out figure and compare it with the 25% expected by chance. The gap of 16 points is the part of the first figure fitted to these very individuals. To choose the number of principal components, run DAPC with several values and keep the one beyond which the leave-one-out success stops improving.

| Group | N | Re-assigned to it | Success | Leave-one-out success |
|---|---|---|---|---|
| Oaxaca_north | 20 | 20 | 100% | 70% |
| Oaxaca_south | 20 | 19 | 95% | 90% |
| Puebla | 20 | 20 | 100% | 95% |
| Guerrero | 20 | 20 | 100% | 75% |

Coordinates and memberships (CSV)

**Figure 8.9.** DAPC results with the agave: 26 components (81.2% of the variance) and 3 discriminant axes. Reassignment is 98.8% and leave-one-out success 82.5%, against 25% expected by chance. Oaxaca north (70%) and Guerrero (75%) are the least distinct groups.

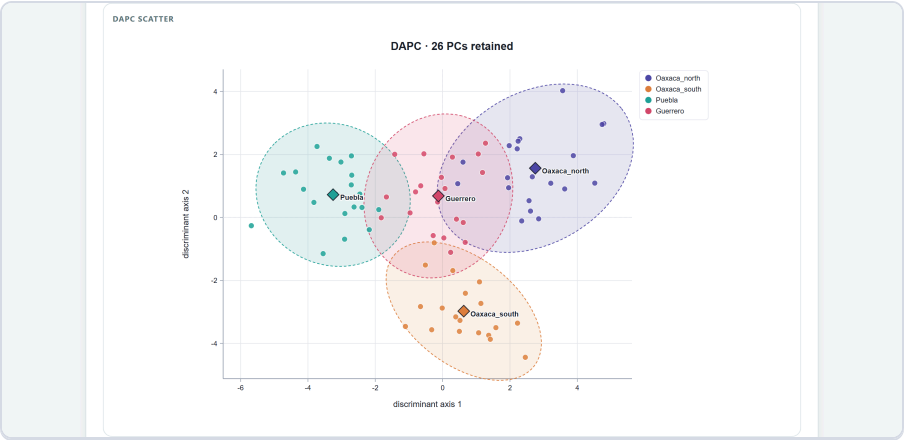



**Figure 8.10.** The first two discriminant axes (45.3% and 30% of the between-group variance), with 95% ellipses and centroids. Puebla lies on the left, Oaxaca north on the right and Oaxaca south at the bottom; Guerrero, in the centre, overlaps with Puebla and with Oaxaca north, and it is also the population that receives the most plants in the assignment of section 8.4.

**Figure 8.11.** DAPC membership probabilities (*compoplot*). Almost all plants have a probability close to 1 of belonging to their own population: this is the optimistic reassignment of figure 8.9. These are not ancestry proportions like the Q values of figure 8.5, but the certainty of the classification.

**5 · DAPC results**

15 landraces × 25 variable bands. **5** principal components were retained (71.2% of the variance; the default keeps the smaller of "90% explained" and N/3, since retaining too many PCs over-fits the groups).

**k-means clustering** ran on those PCs for K = 1 to 7; the BIC is lowest at **K = 7**, which was used as the grouping. A shallow BIC curve means the groups are not sharply separated.

**The BIC is still falling at the largest K tried.**
The curve has no minimum in this range, so K = 7 is the edge of the search, not an estimate. With few landraces the BIC of k-means keeps dropping as K grows: prefer the K at which the curve bends, and read the groups as a convenient partition rather than as genetic populations.

The discriminant analysis kept 5 axes; the first carries 75.1% of the between-group variance, the second 16%. **100%** of the landraces are re-assigned to their own group by their posterior membership, and **78.6%** when each landrace is classified by a discriminant analysis built without it (leave-one-out; 1 alone in its group cannot be tested). Both are high by construction when the groups came from the data themselves.

The least distinct groups are **Cluster 3** (50% re-assigned, leave-one-out), **Cluster 6** (50% re-assigned, leave-one-out), **Cluster 7** (67% re-assigned, leave-one-out): their individuals overlap with other groups in discriminant space.

| Group | N | Re-assigned to it | Success | Leave-one-out success |
|---|---|---|---|---|
| Cluster 1 | 1 | 1 | 100% | — |
| Cluster 2 | 2 | 2 | 100% | 100% |
| Cluster 3 | 2 | 2 | 100% | 50% |
| Cluster 4 | 2 | 2 | 100% | 100% |
| Cluster 5 | 3 | 3 | 100% | 100% |
| Cluster 6 | 2 | 2 | 100% | 50% |
| Cluster 7 | 3 | 3 | 100% | 66.7% |

**Figure 8.12.** DAPC with k-means groups for the 15 ISSR landraces. The app warns that the BIC is still falling at the largest K tested (7): this is the edge of the search, not an estimate. The groups contain 1 to 3 landraces; the one left on its own cannot be evaluated by leaving it out.

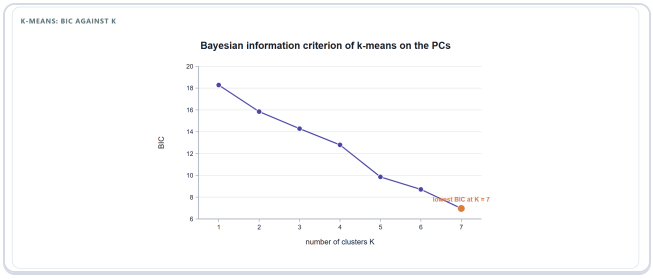



**Figure 8.13.** The BIC for the landraces falls from 18.3 (K = 1) to 7.0 (K = 7) with no minimum or elbow. The PCoA and the tree of Block 6 are a better choice.

**DECISION RULE · READING A DAPC**

| If… | then… |
|---|---|
| **Leave-one-out success clearly exceeds chance** | The groups are genetically distinguishable. Report that figure, not the reassignment. |
| **Reassignment far exceeds leave-one-out success** | Part of the success is fit to the sample. Try several numbers of components and keep the one beyond which leave-one-out success stops improving (table 8.3). |
| **One group has low success** (below 70%) | Its individuals are confused with other groups; see which ones in the scatter plot. |
| **Groups found by k-means** | Success is high by construction. Look for a minimum or an elbow in the BIC curve; if it is still falling at the largest K tested, there is no optimum and the groups are merely a convenient partition. |

# 8.4 Assignment and migrants

The assignment test answers a different question: does each individual resemble the population where you collected it more than any other? It computes the probability of its multilocus genotype in each population, assuming Hardy–Weinberg equilibrium ($p^2$ for a homozygote, $2pq$ for a heterozygote, multiplying across loci), and assigns it to the population where that probability is highest (Paetkau et al., 1995; Rannala and Mountain, 1997).

**Table 8.4.** The two ways of estimating the allele frequencies of each population. $n_a$ is the number of copies of the allele, n the total number of copies in the population and k the number of alleles at the locus across all the data.

| Method | Frequency | Comment |
|---|---|---|
| **Rannala and Mountain (1997)** Bayesian, default | $(n_a + 1/k) / (n + 1)$ | An allele absent from the population receives a small but reasonable frequency. This is the recommended method. |
| **Paetkau et al. (1995)** | $n_a / n$, or 0.01 if the allele is missing | The value 0.01 is an arbitrary convention; it is included for comparison with older studies. |

### WORKED EXAMPLE · AN AA PLANT FROM POPULATION 1

One locus with two alleles. Population 1 has 30 copies of A and 10 of B (20 plants); population 2, 12 of A and 28 of B. The plant is AA and was collected in population 1, so before the calculation its two copies of A are removed from that population (**leave-one-out**): 28 out of 38 remain.

| Population | Frequency of A (Rannala and Mountain) | $L = p^2$ | ln L |
|---|---|---|---|
| 1 (its population) | (28 + 1/2) / (38 + 1) = 0.731 | 0.534 | −0.63 |
| 2 | (12 + 1/2) / (40 + 1) = 0.305 | 0.093 | −2.38 |

The genotype is 5.7 times more likely in population 1: it is assigned to its own population. With several loci, the ln L values of each locus are added. Without leaving the plant out, the frequency of A in its population would be 0.744 and correct assignment would look slightly easier than it really is.

### THE FIRST-GENERATION MIGRANT TEST

A misassigned individual is not necessarily a migrant: if the populations are similar, many residents fit almost as well in another one. To separate the two cases, the app computes $\lambda = \ln(L_{home}/L_{max})$, where $L_{home}$ is the likelihood in the population where the individual was collected and $L_{max}$ the highest among all of them (Paetkau et al., 2004). $\lambda$ equals 0 if the individual fits best in its own population and is negative if it fits better in another. The app then simulates genotypes of residents of that population (500 by default) and computes $\lambda$ for each one; each simulated genotype is evaluated against a resample with replacement of the population, so that it has the same sampling error as a real resident evaluated with leave-one-out. The P value is the fraction of simulated residents with a $\lambda$ as low or lower. With $P < \alpha$ (0.01 by default), the individual is flagged as a possible migrant.

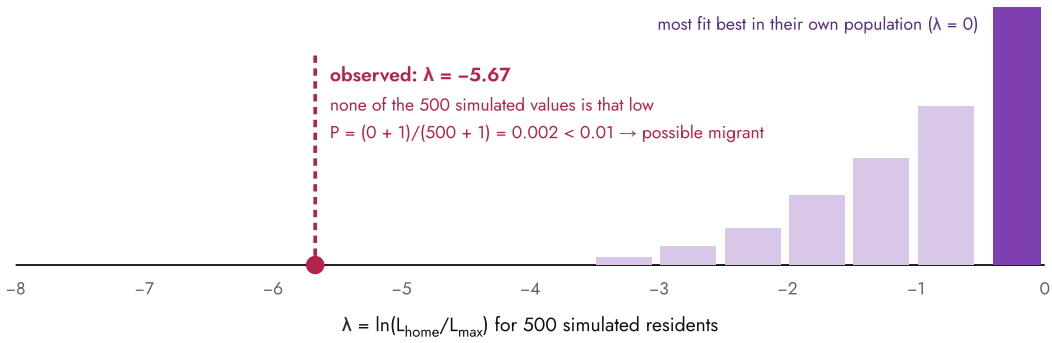
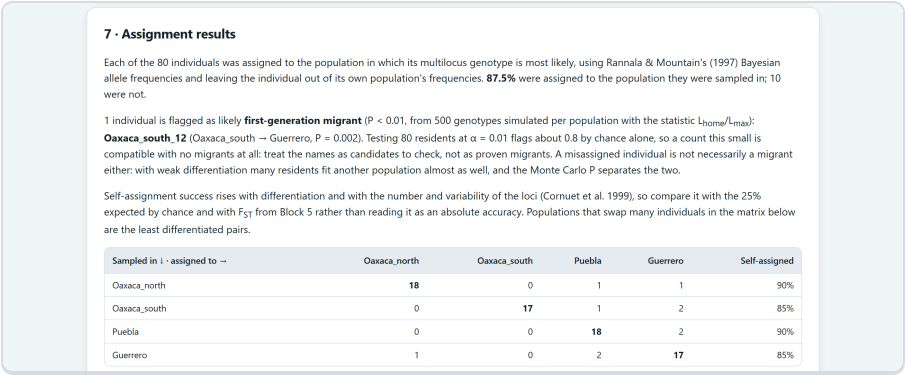



**Figure 8.14.** The idea behind the migrant test, with the values of the agave plant Oaxaca_south_12. The distribution is schematic: the one in the app comes from the simulated genotypes of the population where the plant was collected.

**DECISION RULE · ASSIGNMENT AND MIGRANTS**

| If… | then… |
|---|---|
| **Self-assignment is high** | The populations are distinguishable. Compare it with chance (1/number of populations) and with the $F_{ST}$ of chapter 6: it increases with differentiation and with the number and variability of the loci (Cornuet et al., 1999). |
| **Two populations exchange many individuals** in the matrix | They are the least differentiated pair. This is not proof of current gene flow. |
| **Few individuals flagged as migrants** | With N individuals and α = 0.01, about N × 0.01 are expected by chance. If the number flagged does not greatly exceed that value, treat them as candidates for review, not as proven migrants. |
| **A misassigned individual with a high P** | It is a resident that resembles another population, not a migrant. |
| **Populations with fewer than 5 individuals** | They are excluded from the test: their frequencies are not reliable. |

The test assumes that all possible source populations are in the data. A migrant from an unsampled population appears as an unusual individual in all of them.



**7 · Assignment results**

Each of the 80 individuals was assigned to the population in which its multilocus genotype is most likely, using Rannala & Mountain's (1997) Bayesian allele frequencies and leaving the individual out of its own population's frequencies. **87.5%** were assigned to the population they were sampled in; 10 were not.

1 individual is flagged as likely **first-generation migrant** (P < 0.01, from 500 genotypes simulated per population with the statistic $L_{home}/L_{max}$): **Oaxaca_south_12** (Oaxaca_south → Guerrero, P = 0.002). Testing 80 residents at α = 0.01 flags about 0.8 by chance alone, so a count this small is compatible with no migrants at all: treat the names as candidates to check, not as proven migrants. A misassigned individual is not necessarily a migrant either: with weak differentiation many residents fit another population almost as well, and the Monte Carlo P separates the two.

Self-assignment success rises with differentiation and with the number and variability of the loci (Cornuet et al. 1999), so compare it with the 25% expected by chance and with $F_{ST}$ from Block 5 rather than reading it as an absolute accuracy. Populations that swap many individuals in the matrix below are the least differentiated pairs.

| Sampled in ↓ · assigned to → | Oaxaca_north | Oaxaca_south | Puebla | Guerrero | Self-assigned |
|---|---|---|---|---|---|
| Oaxaca_north | **18** | 0 | 1 | 1 | 90% |
| Oaxaca_south | 0 | **17** | 1 | 2 | 85% |
| Puebla | 0 | 0 | **18** | 2 | 90% |
| Guerrero | 1 | 0 | 2 | **17** | 85% |

**Figure 8.15.** Assignment results with the agave: 87.5% of the plants are assigned to their own population (70 of 80). One plant, Oaxaca_south_12, is flagged as a possible migrant, assigned to Guerrero (P = 0.002), but with 80 plants 0.8 are expected by chance, and the app says so.

**Figure 8.16.** On the left, the assignment matrix in percentages: the diagonal ranges from 85 to 90%. On the right, the likelihood of each genotype in its own population (horizontal axis) against the best of the others (vertical axis). Plants above the diagonal fit better in another population and are labelled; the circle with a thick outline is the possible migrant.

## 8.5 Block settings, tables and figures

Block 7 opens with six or more individuals and three polymorphic markers; it does not apply to morphological characters or to a single sequence alignment. The three parts (Bayesian clustering, DAPC and assignment) are run separately, each with its own button.

**Table 8.5.** Block 7 settings.

| Setting | Options | Recommendation |
|---|---|---|
| Model | *Admixture* or *No admixture* (each individual from a single cluster). | Admixture, unless you know there are no hybrids. |
| K from · K to | 1 to 20. On entering, K runs from 1 to the number of populations plus 2. | Include K = 1 and at least two values above the expected one. |
| Replicates per K | 1 to 20 (default 3). ΔK needs 2 or more. | 10 to 20 for publication. |
| Burn-in iterations · Sampled iterations | Default 5000 + 10,000. | 1000 + 2000 to explore; 50,000 + 100,000 for publication. |
| Random seed | Integer; DAPC and assignment use it too. | Report it in the methods. |
| Groups (DAPC) | The populations from Block 2, or groups found by k-means with K chosen by BIC (Largest K to try, default 10). | Populations if there are any; k-means to explore. |
| Principal components retained | Empty: the smaller of 90% of the variance and N/3. | Compare several values using leave-one-out success. |
| Allele frequencies (assignment) | Rannala and Mountain (1997) or Paetkau et al. (1995). | Rannala and Mountain. |

**Table 8.5.** Block 7 settings. *(continued)*

| Setting | Options | Recommendation |
|---|---|---|
| Simulated genotypes per population | 0 to 10,000 (default 500); 0 skips the migrant test. | 500; 1000 or more for publication. |
| Migrant threshold α | 0.001 to 0.1 (default 0.01). | 0.01. |

**Table 8.6.** Block 7 result cards.

| Card | Contents | Download |
|---|---|---|
| 2 · How many clusters? | Interpretation, K table (Ln P(D), ΔK, H′, individuals with Q ≥ 0.8 and Puechmaille estimators), plots of Ln P(D), ΔK and Puechmaille. | — |
| 3 · Membership of every individual | Bar plot of the chosen K and run (consensus or each replicate) and mean membership by population. | Q matrix (CSV) |
| 5 · DAPC results | Interpretation, success by group (reassignment and leave-one-out), BIC curve if k-means was used, scatter plot and membership probabilities. | Coordinates and memberships (CSV) |
| 7 · Assignment results | Interpretation, assignment matrix, heat map, likelihood plot and a table of every individual sorted by P value. | Assignment of every individual (CSV) |

## 8.6 Step by step: agave with AFLP

The four agave populations, with 20 plants each and 60 bands, take us through the whole block. The values in this chapter are those produced by the app with the default settings and seed 1.

STEP BY STEP

1. In Block 2, load Agave · AFLP, build the dataset and open Block 7 (figure 8.17).
2. Click Run the MCMC. The 18 runs take about 3 minutes; the progress bar shows the Ln P(D) of each one.
3. In How many clusters?, compare Ln P(D), ΔK, the Puechmaille estimators and H' (figures 8.2 to 8.4).
4. In Membership of every individual, change K to 3, 4 and 5 and see at which value every cluster has plants of its own (figure 8.5).
5. Click Run DAPC with the populations as groups (figures 8.9 to 8.11).
6. Click Run the assignment and check the matrix, the likelihoods and the table of individuals (figures 8.15 and 8.16).

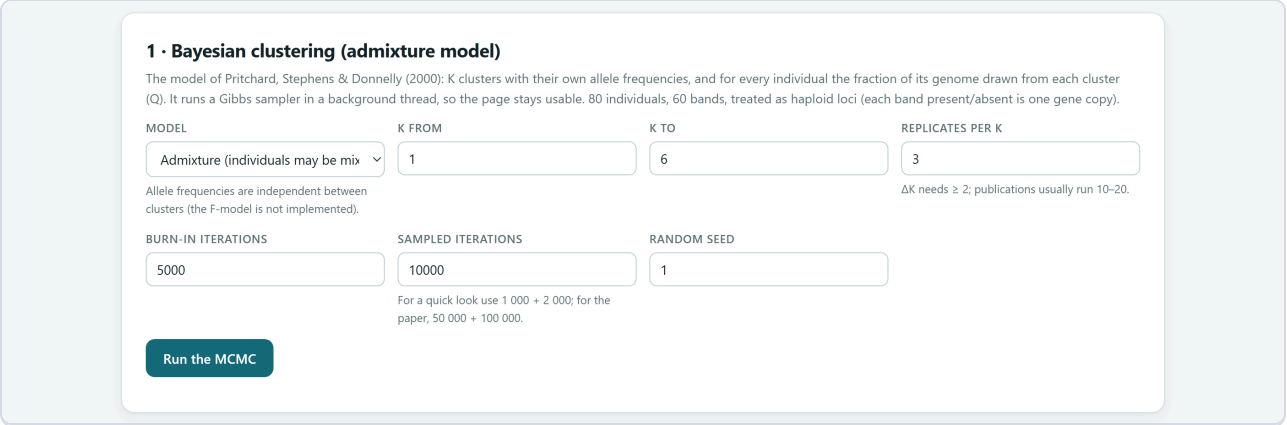



**Figure 8.17.** Bayesian clustering settings with the agave: 80 individuals and 60 bands treated as haploid loci, K from 1 to 6 and 3 replicates.

### HOW TO READ THESE RESULTS

- **Four clusters, not three.** ΔK prefers K = 3, but Ln P(D) and the four Puechmaille estimators point to K = 4, the replicates agree (H' = 0.974) and the bar plot gives each population its own cluster. The Guerrero cluster is the weakest (0.52), so it should be confirmed with long chains and more replicates before publication.
- **Differentiated populations, with admixture.** Mean membership in their own cluster ranges from 0.52 to 0.68 and α is around 0.5 with K = 3: many plants carry ancestry from other clusters, consistent with the $\theta_B$ of 0.125 in chapter 6.
- **The three methods agree on the ranking.** Guerrero and Oaxaca north are the least distinct in the DAPC (75 and 70% with leave-one-out), Guerrero is the most admixed in the bar plot and it receives five of the ten misassigned plants.
- **No evidence of migrants.** One plant flagged out of 80, when 0.8 are expected by chance.

B7

### NOTE · WHAT TO REPORT IN A PAPER

For Bayesian clustering: the model, the values of K tested, the number of replicates, the *burn-in* and sampling iterations, the seed, and all the criteria for choosing K, with the bar plot of the chosen K (and, if they differ, that of the ΔK value). For DAPC: the number of components, how the groups were defined and the leave-one-out success. For assignment: the frequency method, the self-assignment rate and, for migrants, the number of simulations, α and how many are expected by chance. The methods section of the Block 10 report writes all of this up.

**NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, the app's Ln P(D) with K = 1 was compared with its analytical value under the Dirichlet distribution (–1783.4 versus –1783.3 for the maize), and the gamma random number generator reproduced the theoretical mean and variance. With simulated data from three clusters (90 individuals, 10 loci), Ln P(D) and ΔK pointed to K = 3, H' was 0.97 and the estimated Q differed from the true one by 0.05 on average. ΔK, H' and the Puechmaille estimators were recomputed separately. The PCA, discriminant axes, membership probabilities and leave-one-out success of the DAPC matched an independent discriminant analysis (differences smaller than $10^{-12}$). The assignment likelihoods matched an independent calculation ($10^{-15}$). Without migrants, the test flagged 2 of 270 residents with P < 0.01, as expected by chance, and it detected 9 of 12 planted migrants.

The next chapter moves from markers to sequences: Block 8 analyses haplotype diversity, neutrality tests, demographic history and the haplotype network of an alignment.

# B8

CHAPTER NINE · BLOCK 8 · SEQUENCES

# DNA sequences and molecular dating

With aligned sequences you learn not only whether two plants carry the same allele but also how many mutations separate them. Block 8 uses that information to measure haplotype and nucleotide diversity, test whether the population has grown or remained stable, see whether lineages are ordered in space, draw the haplotype network and date divergences with fossils or a substitution rate.




9.1 Haplotype and nucleotide diversity

9.2 Neutrality tests and mismatch distribution

9.3 $N_{ST}$ versus $G_{ST}$

9.4 Haplotype networks

9.5 Divergence times: fossils and rates

9.6 Block settings, tables and figures

9.7 Step by step: a pine and five genera

# 9.1 Haplotype and nucleotide diversity

A haplotype is a distinct sequence. The diversity of a sample of sequences has two sides: how many haplotypes there are and how different they are from one another. Two populations with the same four haplotypes can have very different nucleotide diversity if in one the haplotypes differ by a single base and in the other by twenty.

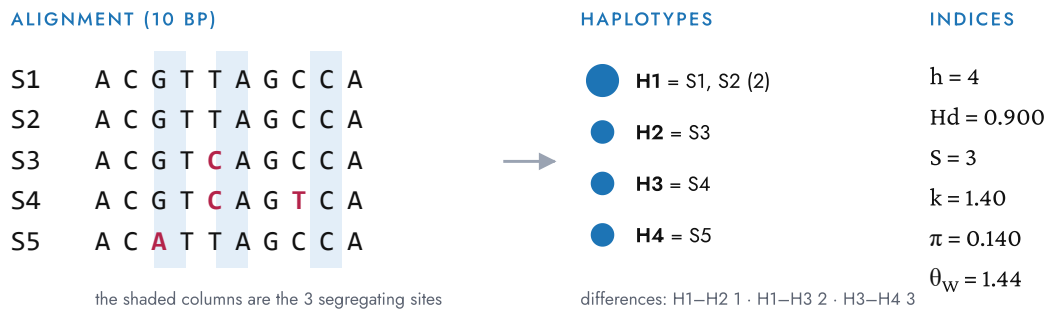



**Figure 9.1.** A practice alignment of 5 sequences and 10 base pairs, computed with the app. S1 and S2 are identical, so they form a single haplotype with two copies. The indices are defined in table 9.1.

**Table 9.1.** Sequence diversity indices. n is the number of sequences, $x_i$ the frequency of haplotype i and L the number of sites analysed.

| Index | Calculation | What it tells you |
|---|---|---|
| **h** · haplotypes | number of distinct sequences | Depends strongly on sample size. |
| **Hd** · haplotype diversity Nei (1987) | $n/(n-1) \cdot (1 - \Sigma\, x_i^2)$ | Probability that two sequences drawn at random are different haplotypes. Ranges from 0 to 1. |
| **k** · mean differences | average number of differences over all pairs of sequences | How many mutations separate two sequences on average. |
| **π** · nucleotide diversity Nei (1987) | $k / L$ | The same idea per site; it allows fragments of different lengths to be compared. |
| **S** · segregating sites | sites with more than one base | The basis of $\theta_W$ and Tajima's D. |
| **η** · minimum number of mutations | $\Sigma$ (bases at the site − 1) | Equal to S if no site has three or four bases. |
| **$\theta_W$** Watterson (1975) | $S / \Sigma_{i=1}^{n-1} 1/i$ | Another estimate of $\theta = 4N_e\mu$, based on how many sites vary. |

**NOTE · WHICH SITES ARE ANALYSED**

The app uses **complete deletion**: a site with a gap (-) or an ambiguous base (N, R, Y…) in any sequence is excluded from all the analyses in cards 2 to 7. In this way all sequences are compared over the same sites. The dating in section 9.5, by contrast, uses all sites, because its likelihood model handles missing data. Check in Sites how many sites remained.

**Table 9.2.** Joint reading of Hd and π, the one used by the automatic interpretation (Grant and Bowen, 1998).

| Hd | π | Usual reading |
|---|---|---|
| High (> 0.5) | Low (< 0.005) | Recent expansion from a small population: many haplotypes, all very similar. |
| Low | Low | Recent, severe bottleneck or founder effect. |
| High | High | Large, stable population, or a mixture of lineages that were once separated. |
| Low | High | A few highly divergent haplotypes: the intermediate ones have been lost. |

## 9.2 Neutrality tests and mismatch distribution

The history of a population leaves its mark on the shape of its genealogy. In a population of constant size, sequences descend from common ancestors spread throughout time. After an expansion, almost all of them coalesce at about the same time, close to the onset of growth, and later mutations are found in a single sequence (rare variants). If the sample mixes old lineages, mutations at intermediate frequencies abound. Neutrality tests compare these signatures with what is expected in a neutral population of constant size.

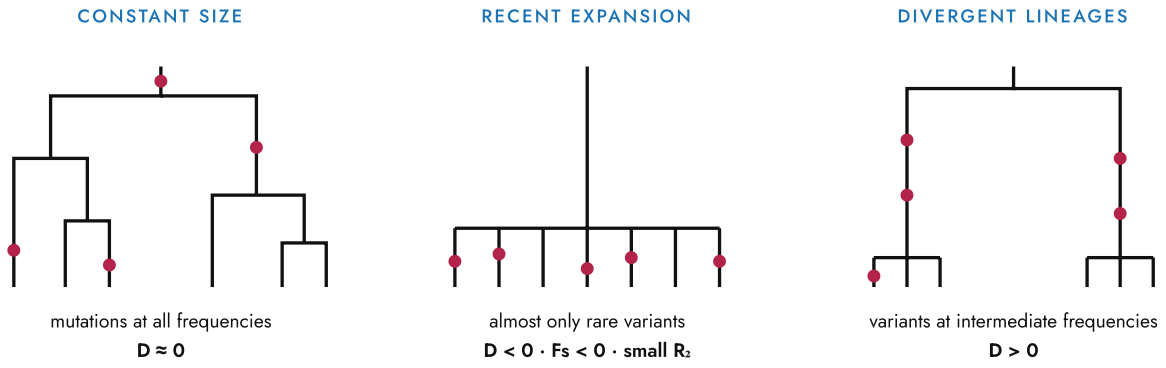



**Figure 9.2.** Three genealogies of seven sequences and their mutations (dots). A mutation on a terminal branch appears in a single sequence; one on a long internal branch appears in many. The tests read that distribution.

**Table 9.3.** The tests in the block. All P values come from coalescent simulations of a neutral population of constant size with $\theta = \theta_W$ (Hudson, 1990).

| Test | What it compares | How to read it in the app |
|---|---|---|
| **Tajima's D** (1989) | k (pairs) against $\theta_W$ (sites): they agree under neutrality and constant size. | Two-tailed, $\alpha = 0.05$. Negative: excess of rare variants; positive: excess of intermediate ones. |
| **Fu's Fs** (1997) | Are there more haplotypes than expected given k? It uses the Ewens sampling formula. | One-tailed (negative values), $\alpha = 0.02$, as Fu recommends. The most sensitive to growth. |
| **$R_2$** (Ramos-Onsins and Rozas, 2002) | Difference between the number of unique mutations in each sequence and k/2. | One-tailed (small values), $\alpha = 0.05$. The most powerful with small samples. |
| **D* and F*** (Fu and Li, 1993) | Unique mutations (singletons) against the total number of mutations $\eta$, or against k. | Shown without a test; useful to see whether the signal comes from the singletons. |
| **Raggedness index r** (Harpending, 1994) | How ragged the distribution of pairwise differences (mismatch) is. | One-tailed (small values), $\alpha = 0.05$: a smooth distribution suggests expansion. |

### THE MISMATCH DISTRIBUTION AND THE EXPANSION MODEL

If the differences between all pairs of sequences are counted, a population that grew suddenly produces a curve with a single peak, and a stable population produces a ragged curve with several peaks. The app fits by least squares the sudden expansion model of Rogers and Harpending (1992), with three parameters: $\theta_0$ and $\theta_1$ before and after the expansion, and $\boldsymbol{\tau = 2ut}$, the time since the expansion in mutational units. If you supply the mutation rate per site per year, the app converts $\tau$ into years: $t = \tau / (2\,\mu\,L)$, with L the number of sites. That date only makes sense if there is evidence of expansion (a significant r, or significant Fs and $R_2$).

### IMPORTANT · SITES WITH THREE OR FOUR BASES

The tests assume the **infinite-sites** model: each site has mutated at most once. A site with three bases needed at least two mutations. S counts sites, so with many such sites it underestimates the number of mutations, and $\theta_W$, Tajima's D and $R_2$ are distorted (D is inflated towards positive values). The app counts them in Sites, computes $\eta$, uses $\eta$ in D* and F*, and warns you when such sites are present. Within a species they are rare: if many appear, check the alignment.

**DECISION RULE · NEUTRALITY AND MISMATCH**

| If… | then… |
|---|---|
| **D, Fs and $R_2$ are not significant** | Nothing compels you to propose changes in size: the data are compatible with a stable population. |
| **Fs and $R_2$ are significant** (with or without a negative D) | Evidence of expansion. Look at the mismatch: if r is also small, read τ as the date of the expansion. |
| **D is negative and significant, but Fs and $R_2$ are not** | An excess of rare variants that may come from a selective sweep just as well as from growth. |
| **D is positive and significant** | Contraction, balancing selection or, most commonly in plants, a sample that mixes lineages or differentiated populations. Repeat the tests population by population. |
| **The mismatch is ragged** (r not significant) | There is no evidence of expansion; τ is not a date. |

Selection and demography leave similar signatures at a single locus. Only several independent loci showing the same signal allow it to be attributed to demography.

## 9.3 $N_{ST}$ versus $G_{ST}$

$G_{ST}$ measures differentiation from haplotype frequencies, as if all haplotypes were equally different. $N_{ST}$ (Pons and Petit, 1996) weights each pair of haplotypes by the number of mutations that separate them. If related haplotypes tend to occur in the same population, $N_{ST}$ exceeds $G_{ST}$: there is **phylogeographic structure**, that is, lineages are ordered in space.

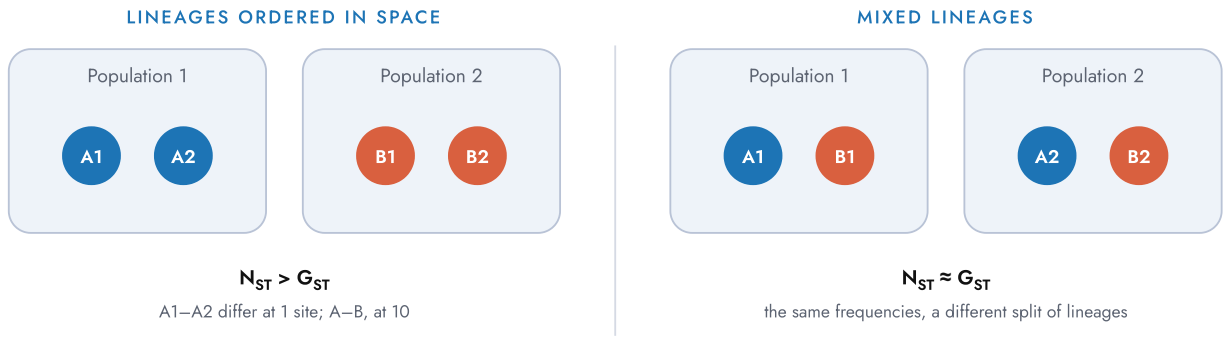



**Figure 9.3.** In both cases each population has two private haplotypes, so $G_{ST}$ is the same. On the left, the two haplotypes of lineage A (blue) are in population 1 and those of lineage B (orange) in population 2: the differences between populations are much larger than within them, and $N_{ST}$ rises.

### HOW THEY ARE COMPUTED AND TESTED

With $\pi_{ij}$ the differences between haplotypes i and j and $x_{ki}$ the frequency of haplotype i in population k, the diversity within populations is $v_S = (1/n)\ \Sigma_k\ [n_k/(n_k - 1)]\ \Sigma_{ij}\ \pi_{ij}\ x_{ki}\ x_{kj}$, the total diversity $v_T$ averages over pairs of haplotypes from different populations, and $N_{ST} = 1 - v_S/v_T$ (Pons and Petit, 1996, equations 6 to 8). $G_{ST}$ is the same with $\pi_{ij} = 1$ for every distinct pair. To test $N_{ST} > G_{ST}$, the app shuffles the haplotype identities in the distance matrix 999 times: this breaks the link between relatedness and population without altering the frequencies. It also computes $\Phi_{ST}$ with an AMOVA on the number of differences; with equal sample sizes it coincides with $N_{ST}$.

### DECISION RULE · PHYLOGEOGRAPHIC STRUCTURE

| If… | then… |
|---|---|
| **$N_{ST}$ significantly greater than $G_{ST}$** | Phylogeographic structure: related lineages share a region. Look in the haplotype network to see which lineages they are. |
| **$N_{ST} \approx G_{ST}$**, both high | Populations differentiated in frequencies, with no geographic ordering of lineages: recent isolation or weakly divergent haplotypes. |
| **Six haplotypes or fewer** | Few ways to shuffle them: the P value cannot fall very low. The app warns you; read the difference $N_{ST} - G_{ST}$. |
| **Organelle DNA** | Compare it with the nuclear $F_{ST}$ of chapter 6. DNA dispersed by seed is usually far more structured than nuclear DNA; in most conifers the chloroplast is dispersed by pollen. |

## 9.4 Haplotype networks

A tree forces each haplotype to have a single path to the others. Within a species, where ancestors are still alive in the population and there is homoplasy, a **network** represents relationships better: the observed haplotypes are nodes, lines join closely related haplotypes and loops show alternative paths. The app builds the **median-joining** network of Bandelt, Forster and Röhl (1999) with $\varepsilon = 0$.

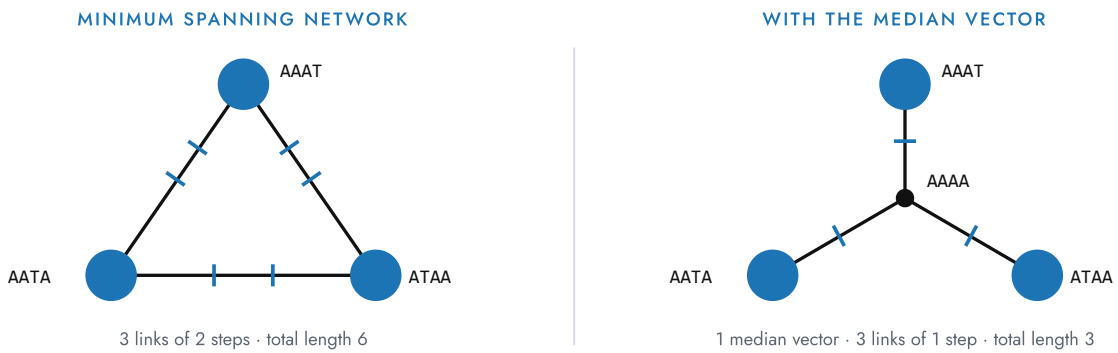



**Figure 9.4.** Three haplotypes that each differ from the others at two sites. The minimum spanning network joins them in pairs, with two mutations each time (blue ticks); the algorithm proposes their median, AAAA, an unsampled haplotype that joins them with one mutation to each. This is the simplest case the app resolves correctly, as are those of a chain, a square and a cube.

**Table 9.4.** The steps of the median-joining algorithm with ε = 0.

| Step | What it does |
|---|---|
| 1 | Builds the minimum spanning network: the union of all minimum spanning trees, so it keeps ties and may contain loops. |
| 2 | Removes obsolete median vectors, those with at most two neighbours, and repeats step 1. |
| 3 | For each triplet of haplotypes joined by at least two links, computes their median and the cost of connecting it. Adds the new medians with the lowest cost. |
| 4 | Returns to step 1 until there are no new medians. The final network is the minimum spanning network of the remaining set. |

✓ DECISION RULE · READING A HAPLOTYPE NETWORK

| If the network shows... | then... |
|---|---|
| **A star**: a frequent central haplotype and many rare derived haplotypes one step away | The picture of a recent expansion from that haplotype. |
| **Groups of haplotypes separated by many mutations** | Old lineages; if each group occupies a region, phylogeographic structure (compare with $N_{ST}$). |
| **Many loops** | Homoplasy (the same change in two lineages), recombination or intermediate haplotypes that were not sampled. |
| **Median vectors** (small black nodes) | Inferred haplotypes: extinct or unsampled. They are not observations. |

The area of each circle is proportional to the number of sequences, the sectors show the populations and each tick on a line is a mutation.

## 9.5 Divergence times: fossils and rates

A molecular clock converts substitutions into time: if two lineages accumulate differences at a rate r per site per million years, a distance d between them corresponds to t = d / (2r). But something external has to set the clock: **fossils** or published ages for nodes of a tree of species or genera, or a known **substitution rate** for haplotypes of a single species, where there are almost never any fossils.

▸ WORKED EXAMPLE · THE SIMPLEST CLOCK

Two species differ by d = 0.03 substitutions per site, and the group evolves at r = 0.0015 substitutions per site per million years. Each lineage accumulated half the distance, so they split t = 0.03 / (2 × 0.0015) = **10 million years** ago. If the true rate were 0.001, the date would be 15 Ma: the rate or the calibrations determine the time scale of the whole tree.

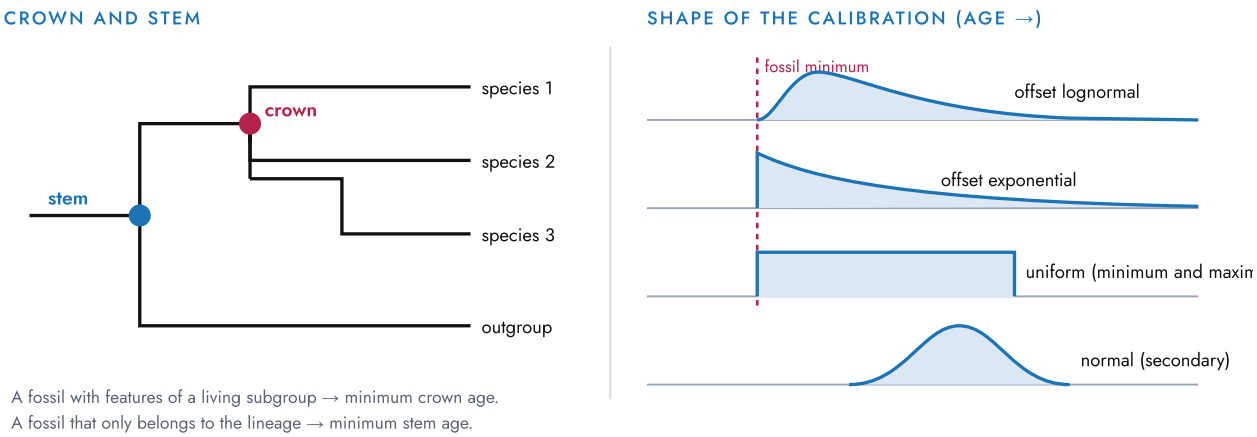



**Figure 9.5.** On the left, where a calibration is placed. On the right, the four shapes the app offers for the age of a calibrated node. The first three start at the minimum age of the fossil and leave a tail towards older ages; the normal centres the age of a secondary calibration on the published value.

**Table 9.5.** The two dating methods in the app.

| | Least squares (LSD; To et al., 2016) | Bayesian with a relaxed clock (MCMC; Drummond et al., 2006) |
|---|---|---|
| **Clock** | Strict: one rate for all branches. | Uncorrelated lognormal relaxed (each branch with its own rate) or strict. |
| **Calibrations** | Hard bounds. | Prior distributions (figure 9.5), combined with a tree model: Yule between species or coalescent within a species. |
| **Uncertainty** | Confidence interval by resampling the branch lengths. | 95% highest posterior density (HPD) interval for each node. |
| **Time** | Seconds. | Minutes; check the effective sample size (ESS ≥ 200). |
| **Use it to** | Explore and check calibrations. | Report. |

**NOTE · THE TREE THAT IS DATED**

Before dating, the app builds the tree: one tip per haplotype or per sequence, topology by neighbour-joining with K2P distances or imported in Newick or NEXUS, root on an outgroup or at the midpoint, and branch lengths by maximum likelihood. By default it chooses the substitution model with the lowest AICc among JC, K80, HKY and GTR, with or without Γ rate variation among sites. The topology is fixed: the app does not incorporate tree uncertainty, tip dates or birth–death models.

**Table 9.6.** Typical rates offered by the app for rate-based dating (substitutions per site per million years; per year in parentheses).

| Marker | Mean ± SD filled in by the app | Published range |
|---|---|---|
| Non-coding chloroplast DNA | 0.002 ± 0.0006 | 1.1–2.9 × $10^{-9}$ (Wolfe et al., 1987) |
| Plant mitochondrial DNA | 0.0006 ± 0.0002 | 0.2–1.0 × $10^{-9}$ (Wolfe et al., 1987) |

**Table 9.6.** Typical rates offered by the app for rate-based dating (substitutions per site per million years; per year in parentheses). *(continued)*

| Marker | Mean ± SD filled in by the app | Published range |
|---|---|---|
| Nuclear genes, synonymous sites | 0.006 ± 0.0005 | $5.1–7.1 \times 10^{-9}$ (Wolfe et al., 1987) |
| Nuclear ribosomal ITS | 0.0024 ± 0.0015 | $0.38–8.34 \times 10^{-9}$ (Kay et al., 2006) |

✓ DECISION RULE · CALIBRATING AND READING A DATED TREE

| If… | then… |
|---|---|
| **There are only minimum ages** | The dates may drift towards older ages: give a maximum to some calibration, a secondary calibration or a maximum age for the root. The app warns you. |
| **Any ESS is below 200** | The chain has not explored the posterior distribution well: lengthen it or rerun it with another seed before reporting. |
| **The interval for σ comes close to zero** | The data are compatible with a strict clock. |
| **The posterior of a calibration resembles its prior** | The sequences say little about that age. Compare it with the Sample from the prior only option. |
| **Data from a single species** | Use a rate and the coalescent model. The ages are coalescence times of the haplotypes, older than the split of the populations that carry them. |

## 9.6 Block settings, tables and figures

Block 8 opens only with aligned sequences (FASTA or a sheet with sequences). On entering, the sequences are analysed with the default settings; the dating in card 8 is run step by step.

**Table 9.7.** Block 8 settings.

| Setting | Options | Recommendation |
|---|---|---|
| Coalescent simulations | 0 to 20,000 (default 1000). | 1000; 10,000 for publication. |
| Permutations | 0 to 9999 (default 999), for $N_{ST}$ and $\Phi_{ST}$. | 999. |
| Mutation rate (per site per year, optional) | Optional; converts τ into years. | The published rate for your marker and group. |
| Generation time (years) | Default 1; converts years into generations. | The age at reproduction of your species. |
| Random seed | Integer. | Report it. |
| **Card 8 · dating** | | |
| Tips of the tree · Topology · Root | Haplotypes or sequences; neighbour-joining or imported tree; outgroup or midpoint. | Outgroup whenever one exists. |
| Substitution model | The best by AICc or a fixed one, with or without Γ. | The best by AICc. |

**Table 9.7.** Block 8 settings. *(continued)*

| Setting | Options | Recommendation |
|---|---|---|
| What sets the clock | Calibrations (fossil or secondary) or a rate with its standard deviation. | Calibrations between species; a rate within a species. |
| Clock · Tree prior | Relaxed or strict; Yule or coalescent. | Relaxed; Yule between species. |
| Chain length · Burn-in (%) | Default 200,000 and 25%. | Whatever is needed for ESS ≥ 200. |

**Table 9.8.** Block 8 results cards.

| Card | Content | Download |
|---|---|---|
| 2 · What the sequences say | Key numbers and interpretation of diversity, tests, mismatch, structure and network. | — |
| 3 · Sites | Sites analysed, segregating sites, sites with three or four bases, η, informative sites, transitions and transversions; map of variable sites. | — |
| 4 · Haplotypes | Frequency of each haplotype per population, its variable sites and private haplotypes. | FASTA · CSV |
| 5 · Diversity and neutrality tests | Indices and tests per population and overall, chart by population and mismatch distribution. | — |
| 6 · Structure among populations | $h_S$, $h_T$, $G_{ST}$, $v_S$, $v_T$, $N_{ST}$ and $\Phi_{ST}$ with their tests. | — |
| 7 · Haplotype network | Median-joining network with sectors by population. | Network links (CSV) |
| 8 · Divergence times | Models compared, calibrations, dated tree, age of each node and chain diagnostics. | NEXUS · Newick · CSV · MCMC log |

## 9.7 Step by step: a pine and five genera

Two examples walk through the block: 48 chloroplast sequences of a pine from four populations (12 per population, 480 base pairs) and 16 species from five invented genera plus an outgroup, simulated on a tree of known ages. The values are those produced by the app with the default settings and seed 1.

# The pine: diversity, tests, structure and network

**STEP BY STEP**

1. In Block 2 load **Pine · chloroplast DNA** and open Block 8: the analysis runs by itself with the settings in table 9.7.
2. Read the key numbers and the interpretation (figure 9.6) and check the sites (figure 9.7) and the haplotypes (figure 9.8).
3. Compare the tests by population (figure 9.9) and look at the mismatch distribution (figure 9.10).
4. Check the structure (figure 9.11) and the haplotype network (figure 9.12).

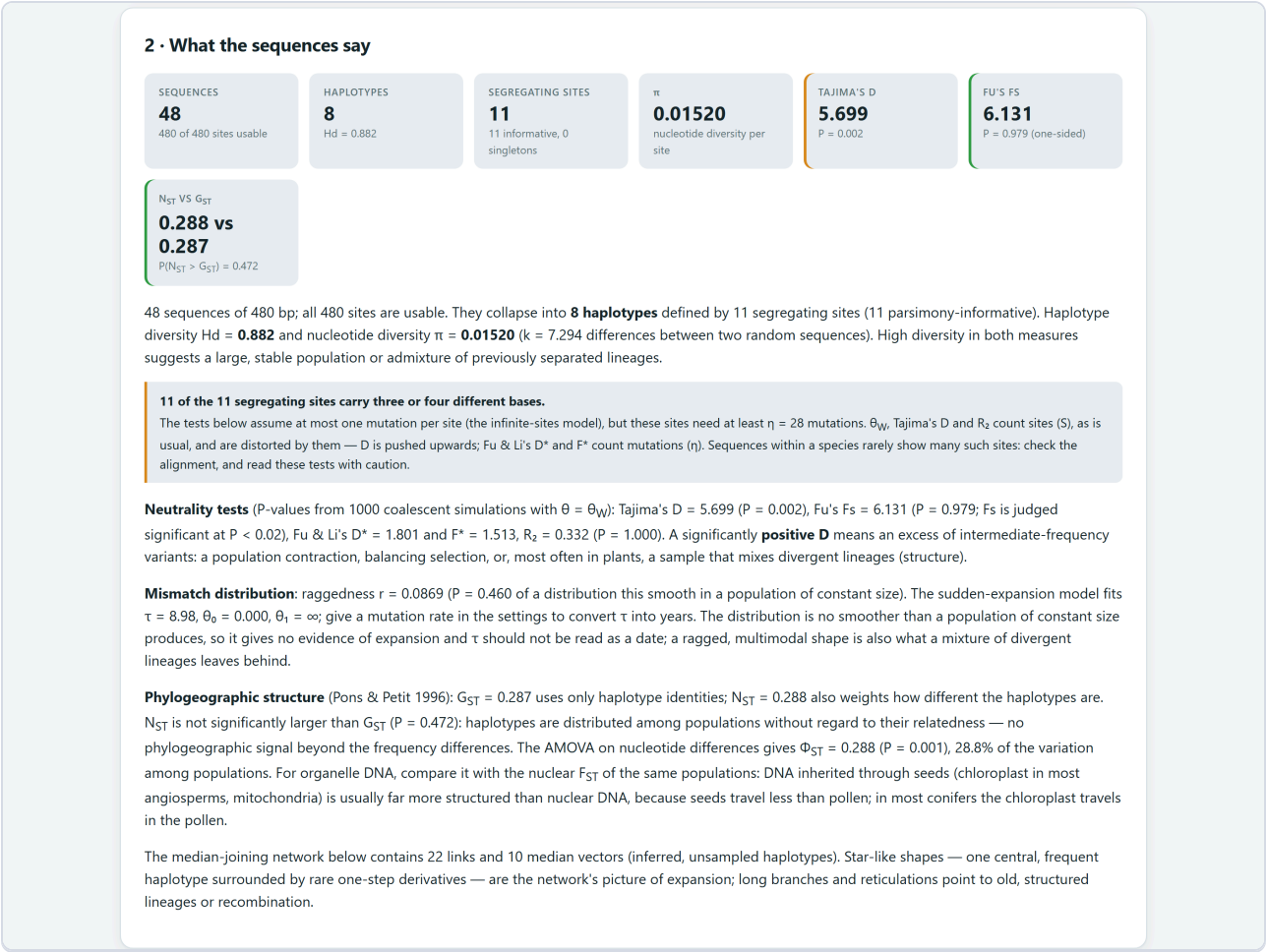



**Figure 9.6.** **What the sequences say**. The 48 sequences form 8 haplotypes (Hd = 0.882, π = 0.0152). The app warns that the 11 variable sites have three or four bases (η = 28), so Tajima's D of 5.70 is inflated.

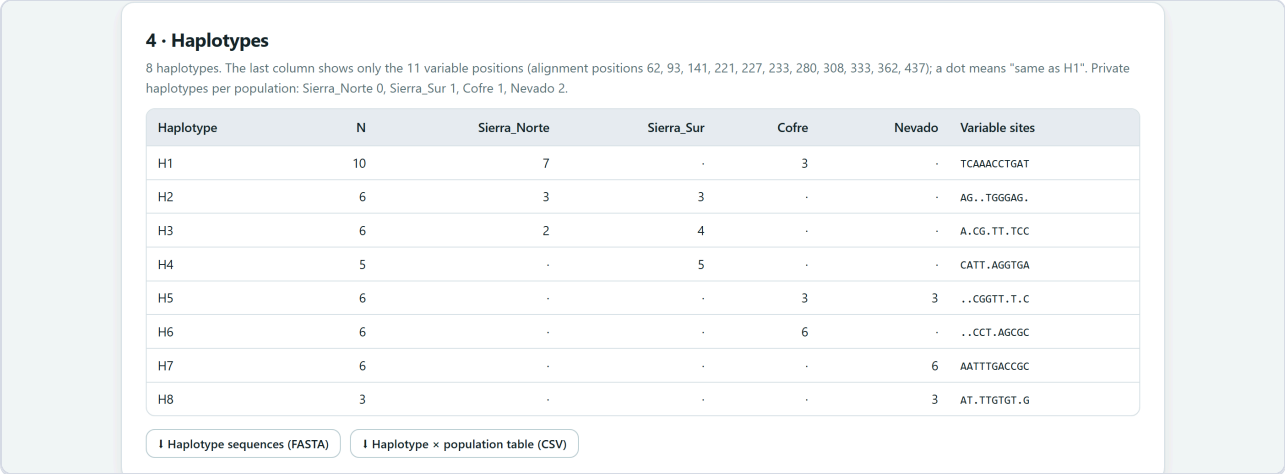
### 3 · Sites

| | |
|---|---|
| Alignment length (bp) | 480 |
| Sites analysed (no gaps or ambiguities in any sequence) | 480 |
| Sites excluded | 0 |
| Segregating sites (S) | 11 |
| Sites with three or four bases | 11 |
| Minimum number of mutations (η) | 28 |
| Parsimony-informative sites | 11 |
| Singleton sites | 0 |
| Transitions / transversions (biallelic sites) | 0 / 0 |

VARIABLE SITES ALONG THE ALIGNMENT



**Figure 9.7.** Sites: 480 sites analysed, 11 segregating, all of them informative and all with three or four bases. The map shows where they lie along the fragment.



### 4 · Haplotypes

8 haplotypes. The last column shows only the 11 variable positions (alignment positions 62, 93, 141, 221, 227, 233, 280, 308, 333, 362, 437); a dot means "same as H1". Private haplotypes per population: Sierra_Norte 0, Sierra_Sur 1, Cofre 1, Nevado 2.

| Haplotype | N | Sierra_Norte | Sierra_Sur | Cofre | Nevado | Variable sites |
|---|---|---|---|---|---|---|
| H1 | 10 | 7 | · | 3 | · | TCAAACCTGAT |
| H2 | 6 | 3 | 3 | · | · | AG..TGGGAG. |
| H3 | 6 | 2 | 4 | · | · | A.CG.TT.TCC |
| H4 | 5 | · | 5 | · | · | CATT.AGGTGA |
| H5 | 6 | · | · | 3 | 3 | ..CGGTT.T.C |
| H6 | 6 | · | · | 6 | · | ..CCT.AGCGC |
| H7 | 6 | · | · | · | 6 | AATTTGACCGC |
| H8 | 3 | · | · | · | 3 | AT.TTGTGT.G |

Haplotype sequences (FASTA) Haplotype × population table (CSV)

**Figure 9.8.** Haplotypes: H1 is the most frequent (10 sequences, in Sierra Norte and Cofre). Sierra Sur and Cofre have one private haplotype each and Nevado two; Sierra Norte has none.

### 5 · Diversity and neutrality tests

Tajima's D compares two estimates of θ — one from the number of segregating sites, one from pairwise differences — that agree only under neutrality and constant size. Fu's Fs asks whether there are more haplotypes than expected given π; it is the most sensitive to population growth. Fu & Li's D* and F* weigh the singletons; $R_2$ (Ramos-Onsins & Rozas 2002) is the most powerful for small samples. All P-values come from coalescent simulations of a neutral population of constant size, not from tables; they assume at most one mutation per site.

| Population | n | h | Hd | S | k | π | θ_W (site) | Tajima's D | Fu's Fs | D* | F* | $R_2$ | r |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sierra_Norte | 12 | 3 | 0.621 | 11 | 5.152 | 0.01073 | 0.00759 | 1.724 | 5.737 | 1.508 | 1.117 | 0.234 | 0.7225 |
| Sierra_Sur | 12 | 3 | 0.712 | 11 | 6.364 | 0.01326 | 0.00759 | **3.108** | 6.769 | 1.519 | 1.309 | 0.289 | 0.1880 |
| Cofre | 12 | 3 | 0.682 | 9 | 5.045 | 0.01051 | 0.00621 | **2.813** | 5.641 | 1.484 | 1.229 | 0.280 | 0.3616 |
| Nevado | 12 | 3 | 0.682 | 11 | 5.727 | 0.01193 | 0.00759 | **2.381** | 6.241 | 1.497 | 1.314 | 0.260 | 0.3616 |
| **All sequences** | **48** | **8** | **0.882** | **11** | **7.294** | **0.01520** | **0.00516** | **5.699** | **6.131** | **1.801** | **1.513** | **0.332** | **0.0869** |

Bold: significant by coalescent simulation of a neutral population of constant size — Tajima's D two-sided at 0.05; $R_2$ and the raggedness r one-sided for small values (growth) at 0.05; Fs one-sided at 0.02, as Fu recommends. D* and F* are shown without a test. h haplotypes · Hd haplotype diversity · S segregating sites · k mean pairwise differences · π and $\theta_W$ per site · r raggedness of the mismatch distribution (Harpending 1994). Populations with fewer than four sequences are not tested.

**Figure 9.9.** Diversity and tests by population. Each population has 3 haplotypes; π ranges from 0.0105 (Cofre) to 0.0133 (Sierra Sur). Tajima's D is positive in all of them and significant in three, while Fs, $R_2$ and r are far from indicating expansion.

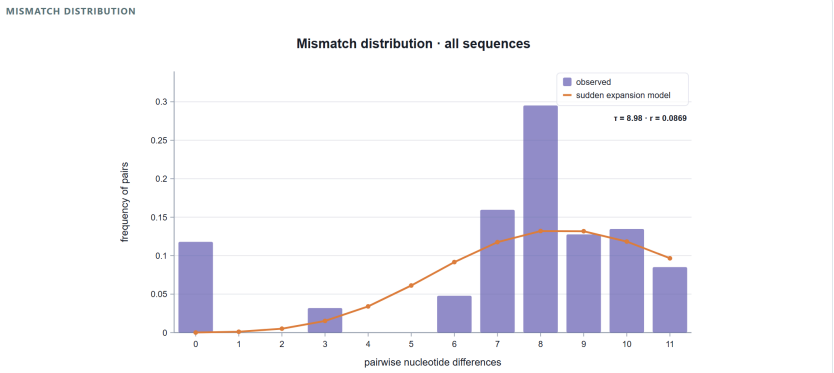



**Figure 9.10.** Mismatch distribution of the 48 sequences. It has two peaks, at 0 differences (pairs of the same haplotype) and at 8, and r = 0.087 (P = 0.46): nothing that a stable population would not produce. The fitted expansion model gives τ = 8.98, but it should not be read as a date.

### 6 · Structure among populations

Pons & Petit (1996): if $N_{ST}$, which weights haplotypes by their divergence, is significantly larger than $G_{ST}$, which treats every haplotype as equally different, then related haplotypes share populations — the lineages are geographically sorted.

| Statistic | Value | P |
|---|---|---|
| h_S (within-population haplotype diversity) | 0.674 | — |
| h_T (total) | 0.946 | — |
| G_ST (haplotype identity) | 0.287 | — |
| v_S (within, distance-weighted) | 5.572 | — |
| v_T (total, distance-weighted) | 7.821 | — |
| N_ST (distance-weighted) | 0.288 | P(N_ST > G_ST) = 0.472 (999 permutations of haplotypes) |
| Φ_ST (AMOVA on nucleotide differences) | 0.288 | P = 0.001 (999 permutations of sequences) |

**Figure 9.11.** Structure among populations: $G_{ST}$ = 0.287 and $N_{ST}$ = 0.288, with no significant difference (P = 0.47). $\Phi_{ST}$ = 0.288 is significant (P = 0.001) and equal to $N_{ST}$, because all four populations have 12 sequences.

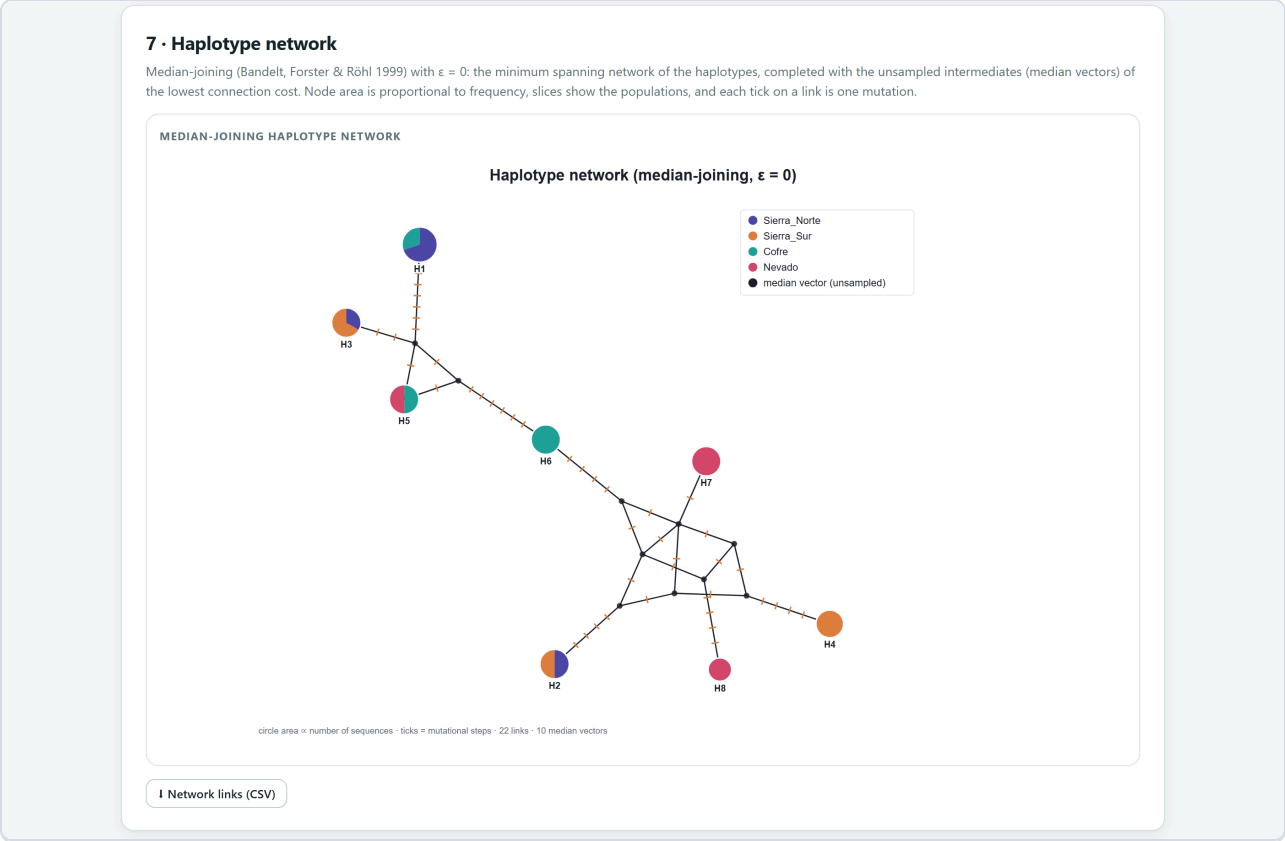



**Figure 9.12.** Median-joining network: 8 observed haplotypes, 10 median vectors and 22 links with loops. The loops reflect the sites with several mutations; the sectors show that H2 and H3 are shared by Sierra Norte and Sierra Sur, and H5 by Cofre and Nevado.

### HOW TO READ THE PINE RESULTS

- **High diversity, no expansion.** Hd and π are high, Fs is positive, $R_2$ is large (both far from significance) and the mismatch is ragged: nothing points to growth.
- **The positive D is not a reliable demographic signal here.** It comes partly from the sites with several bases (the practice data are synthetic) and partly from mixing differentiated populations.
- **Differentiated populations, unordered lineages.** Almost 29% of the variation lies among populations, the same value as the θ of 0.287 in chapter 6, but $N_{ST}$ does not exceed $G_{ST}$: related haplotypes are not concentrated in one region.
- **A rate-based date, with caution.** With the chloroplast rate offered by the app (0.002 ± 0.0006 per site per million years), LSD on the 8 haplotypes places their common ancestor at 8.5 Ma (4.4–17.5); the app warns that the α of the Γ rate variation ended up at its bound, because only 11 of the 480 sites vary.

## Five genera: dating with fossils

The sequences were simulated under an HKY + Γ model and a relaxed clock of 0.0015 substitutions per site per million years on a tree rooted at 62 Ma. The genera and the fossils are invented, so each date can be compared with the true one.

**STEP BY STEP**

1. Load **Five genera · dating with fossils**, open Block 8 and scroll down to card 8. Click **Fill in the example setup**: it sets the outgroup (*Exterocarpus*) and four calibrations.
2. Click **Build and fit the tree**. Within a few seconds it compares eight models and chooses HKY + Γ (figure 9.13). Check the calibrations in step 2 (figure 9.14 and table 9.9).
3. Click **Run LSD** and check the node table (figure 9.15).
4. Raise **Chain length** to 1,000,000 and click **Run the MCMC** (about 4 minutes). Check the summary, the dated tree, the ages and the convergence (figures 9.16 to 9.19).

**Table 9.9.** The four calibrations in the example.

| Name | Node | Evidence | Ages (Ma) | Shape |
|---|---|---|---|---|
| F1 | Crown of *Fictaria* + *Modelanthus* | Rupelian fossil (early Oligocene) | minimum 27.82 · soft maximum 45 | lognormal |
| F2 | Crown of *Demoxylon* | Aquitanian fossil (early Miocene) | minimum 20.44 | lognormal |
| F3 | Stem of *Probocarpa* | Chattian fossil (late Oligocene) | minimum 23.03 | exponential |
| S1 | Crown of the five genera | Secondary, from a published tree | 50 ± 4 | normal |

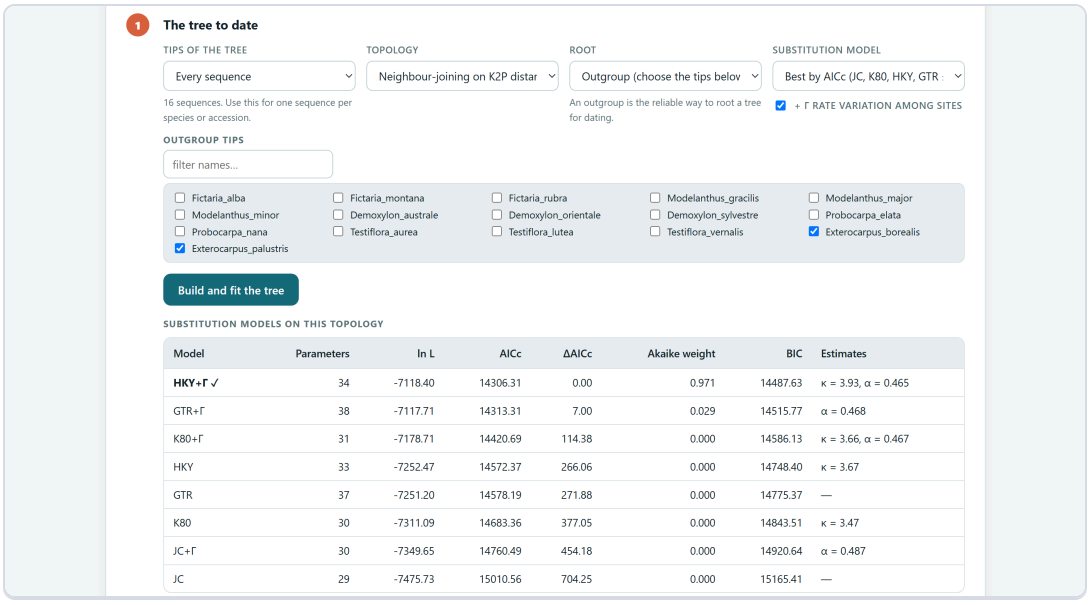



**Figure 9.13.** Step 1 with the example: all sequences as tips, neighbour-joining, root on the outgroup and the best model by AICc. HKY + Γ has an Akaike weight of 0.971, with κ = 3.93 and α = 0.465, close to the values used to simulate the data (4 and 0.5).

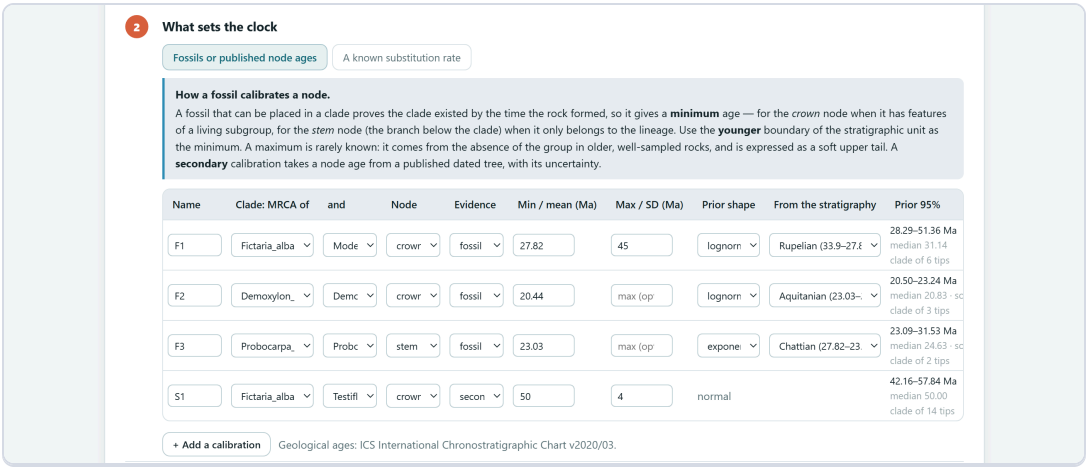



**Figure 9.14.** Step 2: the four calibrations with their node, type of evidence, ages, shape, stratigraphic unit and the resulting 95% prior interval (for example, 28.29–51.36 Ma for F1). The note under each interval states how many tips the calibrated clade has.

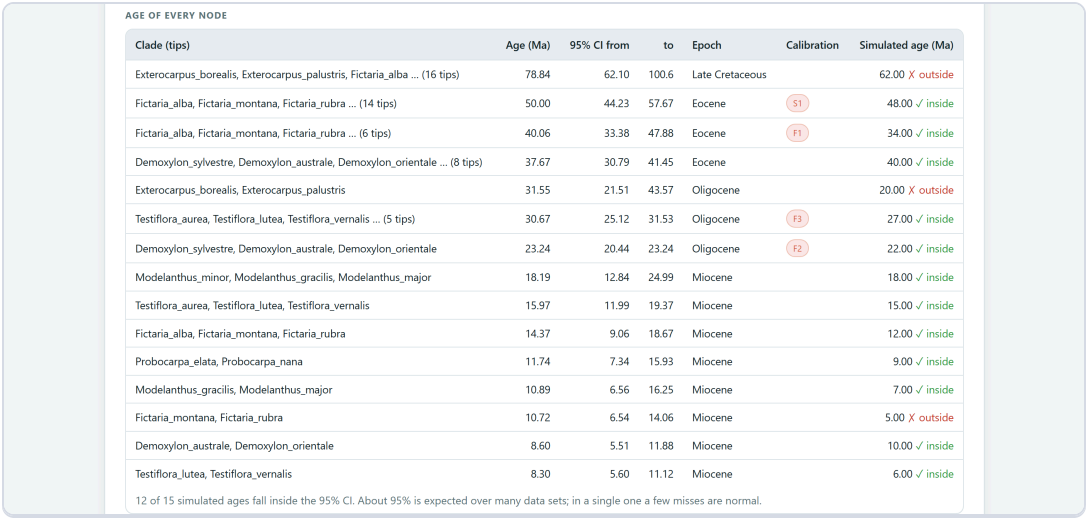



**Figure 9.15.** Ages with LSD and a strict clock: root at 78.8 Ma (62.1–100.6) and 12 of 15 simulated ages within the interval. The root and the outgroup come out too old because in this simulation the outgroup evolved faster, something a strict clock cannot absorb; the third miss is the pair *F. montana–F. rubra*.

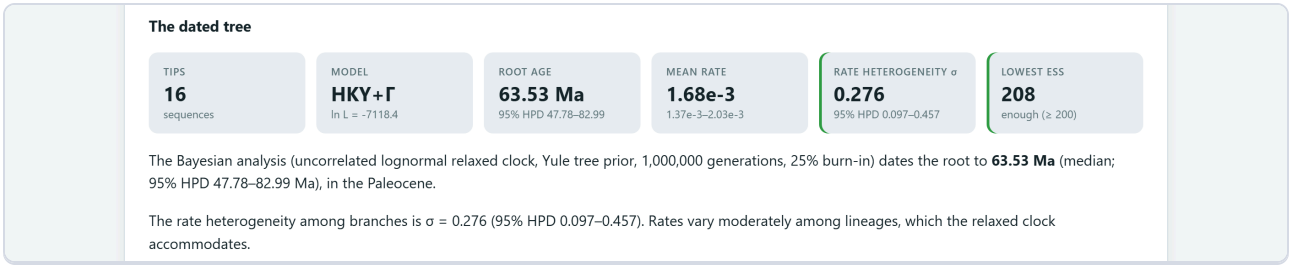



**Figure 9.16.** Summary of the MCMC with a relaxed clock and 1,000,000 generations: root at 63.5 Ma (47.8–83.0), mean rate $1.68 \times 10^{-3}$, σ = 0.276 and a lowest ESS of 208.

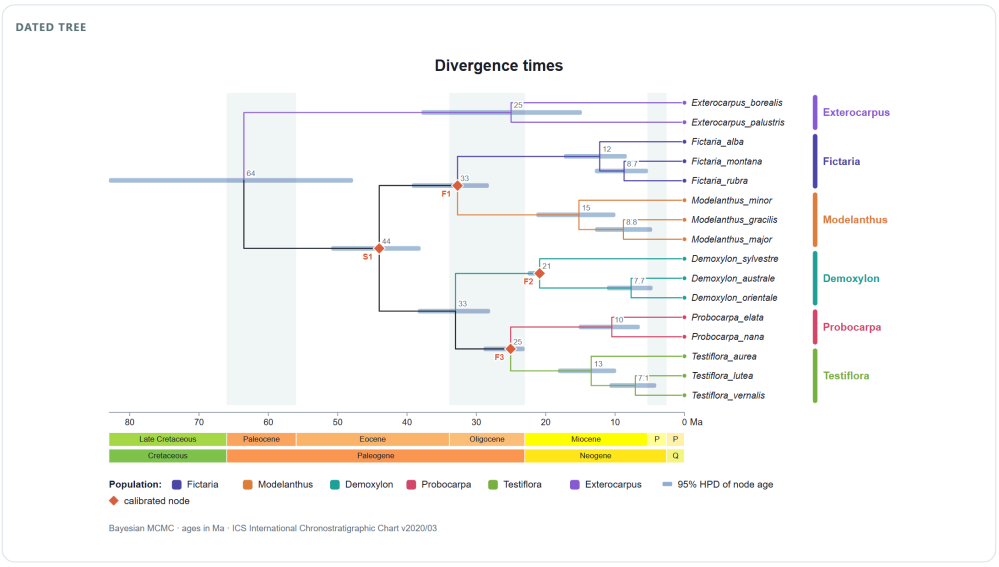



**Figure 9.17.** The dated tree: 95% HPD bars at each node, the four calibrated nodes marked in red and the geological time scale. The root bar is the widest: no calibration constrains it directly.

AGE OF EVERY NODE

| Clade (tips) | Median age (Ma) | 95% HPD from | to | Epoch | ESS | Calibration | Simulated age (Ma) |
|---|---|---|---|---|---|---|---|
| Exterocarpus_borealis, Exterocarpus_palustris, Fictaria_alba … (16 tips) | 63.53 | 47.78 | 82.99 | Paleocene | 208 | | 62.00 ✓ inside |
| Fictaria_alba, Fictaria_montana, Fictaria_rubra … (14 tips) | 44.01 | 38.04 | 50.93 | Eocene | 376 | S1 | 48.00 ✓ inside |
| Demoxylon_sylvestre, Demoxylon_australe, Demoxylon_orientale … (8 tips) | 33.01 | 28.02 | 38.45 | Oligocene | 446 | | 40.00 ✗ outside |
| Fictaria_alba, Fictaria_montana, Fictaria_rubra … (6 tips) | 32.72 | 28.20 | 39.31 | Oligocene | 598 | F1 | 34.00 ✓ inside |
| Testiflora_aurea, Testiflora_lutea, Testiflora_vernalis … (5 tips) | 25.06 | 23.04 | 28.98 | Oligocene | 590 | F3 | 27.00 ✓ inside |
| Exterocarpus_borealis, Exterocarpus_palustris | 25.00 | 14.79 | 37.93 | Oligocene | 443 | | 20.00 ✓ inside |
| Demoxylon_sylvestre, Demoxylon_australe, Demoxylon_orientale | 20.88 | 20.46 | 22.60 | Miocene | 976 | F2 | 22.00 ✓ inside |
| Modelanthus_minor, Modelanthus_gracilis, Modelanthus_major | 15.21 | 9.95 | 21.36 | Miocene | 451 | | 18.00 ✓ inside |
| Testiflora_aurea, Testiflora_lutea, Testiflora_vernalis | 13.44 | 9.85 | 18.23 | Miocene | 703 | | 15.00 ✓ inside |
| Fictaria_alba, Fictaria_montana, Fictaria_rubra | 12.22 | 8.32 | 17.39 | Miocene | 431 | | 12.00 ✓ inside |
| Probocarpa_elata, Probocarpa_nana | 10.48 | 6.44 | 15.24 | Miocene | 846 | | 9.00 ✓ inside |
| Modelanthus_gracilis, Modelanthus_major | 8.79 | 4.66 | 12.89 | Miocene | 572 | | 7.00 ✓ inside |
| Fictaria_montana, Fictaria_rubra | 8.70 | 5.26 | 12.92 | Miocene | 569 | | 5.00 ✗ outside |
| Demoxylon_australe, Demoxylon_orientale | 7.68 | 4.59 | 11.18 | Miocene | 1097 | | 10.00 ✓ inside |
| Testiflora_lutea, Testiflora_vernalis | 7.07 | 4.07 | 10.81 | Miocene | 855 | | 6.00 ✓ inside |

13 of 15 simulated ages fall inside the 95% HPD. About 95% is expected over many data sets; in a single one a few misses are normal.

**Figure 9.18.** Age of every node: the root lies 1.5 Ma from the true value (62 Ma) and 13 of 15 simulated ages fall within the HPD. The misses are the clade of *Demoxylon*, *Probocarpa* and *Testiflora* (33.0 versus 40 Ma) and the pair *F. montana–F. rubra*.

DID THE CHAIN CONVERGE?

| Parameter | Mean | 95% HPD from | to | ESS |
|---|---|---|---|---|
| log likelihood | -7134.679 | — | — | 609 |
| log posterior | -7206.354 | — | — | 489 |
| mean clock rate | 1.677e-3 | 1.368e-3 | 2.029e-3 | 240 |
| root age (Ma) | 64.387 | 47.780 | 82.989 | 208 |
| rate heterogeneity σ | 0.276 | 0.097 | 0.457 | 276 |
| κ (ts/tv) | 3.926 | 3.425 | 4.508 | 1024 |
| Γ shape α | 0.473 | 0.384 | 0.580 | 1044 |
| Yule birth rate λ | 0.035 | 6.309e-4 | 0.066 | 1103 |

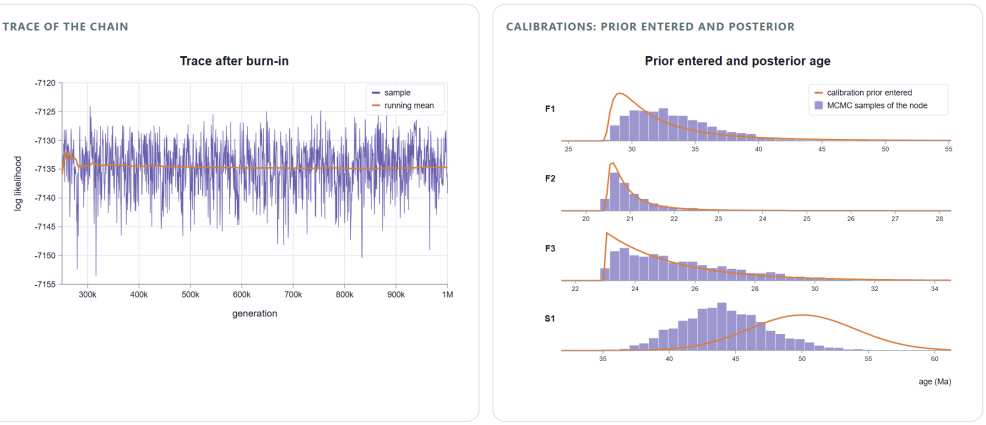



**Figure 9.19.** Did the chain converge?: parameters with their ESS (all ≥ 200), the trace of the log-likelihood and, for each calibration, the prior entered against the posterior. For S1 the posterior (44 Ma) is younger than the prior of 50 ± 4 Ma: the sequences and the other calibrations correct it. The estimated rate and σ are close to the simulated values ($1.5 \times 10^{-3}$ and 0.25).

### HOW TO READ THE GENERA RESULTS

- **The relaxed clock corrects the strict one.** LSD placed the root 17 Ma too old; the MCMC recovers it because it allows the outgroup to have its own rate.
- **Chain length matters.** With the default 200,000 generations the root comes out at 66.3 Ma, but its ESS is 29 and 12 nodes remain below 200; with 1,000,000, all exceed 200.
- **Two misses out of 15 are to be expected.** With 95% intervals you expect to miss, on average, one node in twenty; in a single simulation, two misses are normal.

### NOTE · WHAT TO REPORT IN A PAPER

For diversity: n, h, Hd, π and S per population, with complete deletion and the number of sites analysed. For the tests: the statistic, its P value, the number of simulations and the null model (neutral population of constant size). For structure: $G_{ST}$, $N_{ST}$ and the number of permutations. For the network: the method and ε. For dating: substitution model, tree and root, each calibration with its evidence, shape and ages (or the rate and its source), clock, tree model, chain length, burn-in, minimum ESS and the seed. The methods section of the Block 10 report writes all of this up.

**NOTE · VERIFICATION OF THE CALCULATIONS**

For this manual, Hd, k, $\pi$, S, $\eta$ and $\theta_W$ for the pine were recomputed independently with no differences; $G_{ST}$ and $N_{ST}$ were recomputed with equations 6 and 7 of Pons and Petit ($10^{-16}$), the r index with Harpending's formula and the Stirling numbers of Fs with exact integer arithmetic. With 4000 samples from an independent coalescent simulator, D, D* and F* had means close to 0. With 250 neutral samples, the tests rejected 4.8% (D), 3.6% (Fs, $\alpha = 0.02$), 4.0% ($R_2$) and 6.4% (r). The expansion model agreed with an independent implementation, and the median-joining network reproduced the star, chain, square and cube cases. The dating engine passes its tests of transition matrices, the $\Gamma$ distribution, exact likelihood and root placement, and in the example it recovers 13 of 15 simulated ages.

The next chapter returns to the markers to study demography and space: Block 9 looks for bottlenecks, estimates the effective population size, measures spatial genetic structure and selfing, and compares trait differentiation with genetic differentiation.

# B9

CHAPTER TEN · BLOCK 9 · DEMOGRAPHY

# Demography and spatial structure

Allele frequencies keep a memory of what happened to a population: whether it shrank recently, how many plants really contribute to the next generation, how far pollen and seed travel, how much it self-fertilises and whether its traits have diverged more than drift allows. Block 9 brings together five analyses that appear depending on the data you load.


10.1 Recent bottlenecks

10.2 Effective population size

10.3 Spatial genetic structure

10.4 Mating system

10.5 $P_{ST}$ versus $F_{ST}$

10.6 Block settings, tables and figures

10.7 Step by step: three populations, a stand and maize morphology




Not every analysis suits every dataset. Bottlenecks and effective size require codominant genotypes from diploid plants; spatial structure, a coordinate for each plant; $P_{ST}$, measured traits. The `1 · Settings` card tells you which ones apply to your data, and the block runs by itself when you open it.

**Table 10.1.** Which analysis appears with which data.

| Analysis | Data required | Example in this chapter |
|---|---|---|
| Bottlenecks | Codominant, diploid, with declared populations of 10 plants or more. | Three simulated populations |
| Effective size ($N_e$) | The same. | Three simulated populations |
| Spatial structure | At least 10 plants, each with its own latitude and longitude; any marker. | Simulated stand · agave |
| Mating system | Codominant, diploid, with declared populations. | Three simulated populations |
| $P_{ST}$ versus $F_{ST}$ | Quantitative traits in two or more populations. | Maize morphology |

## 10.1 Recent bottlenecks

When a population shrinks, it loses its rare alleles first: an allele carried by only a few plants easily disappears if those plants leave no offspring. Expected heterozygosity, by contrast, barely changes, because it depends mainly on the common alleles. For a while the population is left with **more heterozygosity than its number of alleles warrants**. Cornuet and Luikart (1996) called this heterozygosity excess and turned it into a test.

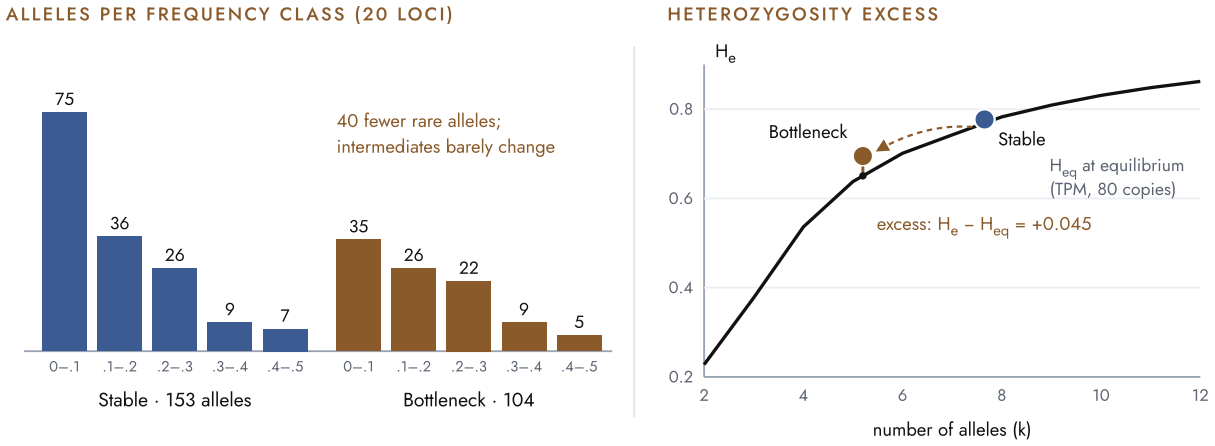



**Figure 10.1.** What a bottleneck leaves behind, using the practice data of section 10.7. Left: the population that spent 8 generations with 40 plants has 35 alleles with a frequency below 0.1, against 75 in the stable one, and almost the same number of intermediate alleles. Right: the curve is the $H_{eq}$ that the app simulates for 80 copies under the TPM model, and the points are the means over the 20 loci. With 5.2 alleles, the reduced population retains an $H_e$ of 0.695, more than that number of alleles warrants.

B9

**! IMPORTANT · TWO DIFFERENT EXCESSES**

The **heterozygosity excess** of this test compares the $H_e$ calculated from allele frequencies with the value expected for the observed number of alleles. It has nothing to do with the **heterozygote excess** ($H_o > H_e$) of Block 4, which compares observed genotypes with Hardy–Weinberg proportions.

To know how much $H_e$ "corresponds" to a given number of alleles, you have to assume how the alleles mutate. The app offers the three classic models (table 10.2) and, locus by locus, simulates samples from a stable population under the chosen model.

**Table 10.2.** Mutation models of the test.

| Model | How an allele changes | When to use it |
|---|---|---|
| **IAM** · infinite alleles | Each mutation creates an allele that did not exist before. | Allozymes and markers without an ordering of sizes. With microsatellites it finds bottlenecks too easily. |
| **SMM** · stepwise | Each mutation adds or removes one repeat. | Very regular microsatellites. It is the most conservative: it struggles to detect reductions. |
| **TPM** · two-phase<br>Di Rienzo et al. (1994) | Most mutations are single-step (95% by default) and the rest multi-step, with variance 12. | Microsatellites in general; it is the recommended model (Piry, Luikart and Cornuet, 1999). The app uses it together with the SMM. |

### HOW THE EQUILIBRIUM HETEROZYGOSITY IS OBTAINED

For each locus with n plants and k alleles, the app simulates coalescent genealogies of 2n copies under the chosen mutation model. It tunes $\theta = 4N_e\mu$ until the samples have about k alleles on average and keeps only those with exactly k (300 by default). The mean of their $H_e$ is $\mathbf{H_{eq}}$, and the fraction of simulations with an $H_e$ equal to or greater than the observed one is the excess probability of that locus.

If the plants self-fertilise, their two copies of a locus often descend from a single ancestral copy: n plants with inbreeding F carry on average $n(2 - F)$ independent copies, not 2n. Fewer independent copies show fewer alleles, and the population would appear to have an excess without having shrunk. For this reason the app estimates $F_{IS}$ in each population and, if it is positive, simulates the equilibrium with $n(2 - F)$ copies.

Each locus gives a noisy comparison; the conclusion comes from combining the loci. The app applies three statistical tests and two shape indicators (table 10.3).

**Table 10.3.** Tests and indicators of a bottleneck.

| Test | What it does | How to read it |
|---|---|---|
| **Sign test**<br>Cornuet and Luikart (1996) | Counts the loci with $H_e > H_{eq}$ and compares them with the number expected at equilibrium, which is slightly more than half because the distribution of $H_e$ is skewed. | $P < 0.05$: too many loci with an excess. Low power. |
| **Standardised differences** ($T_2$) | Sums $(H_e - H_{eq})/SD$ over all loci and divides by $\sqrt{L}$. | $P < 0.05$: average excess. Requires 20 loci or more. |
| **Wilcoxon** signed-rank test<br>Luikart et al. (1998) | Ranks the loci by the position of their $H_e$ within their own equilibrium distribution and tests whether high positions predominate. | The most powerful with few loci (from 4). It is the one used by the interpretation and the key figures. |

**Table 10.3.** Tests and indicators of a bottleneck. *(continued)*

| Test | What it does | How to read it |
|---|---|---|
| **Mode shift**<br>Luikart et al. (1998) | Distributes the alleles of all loci into frequency classes (0–0.1, 0.1–0.2...). | L-shaped: stable. If the 0–0.1 class is no longer the largest, the distribution has shifted. Qualitative; requires large samples. |
| **M ratio**<br>Garza and Williamson (2001) | M = k / (range of sizes in repeats + 1): after a reduction, gaps remain in the ladder of sizes. | M < 0.68 suggests a reduction older or longer than the one the excess detects. Requires sizes in bp and the motif. |

**NOTE · WHY WILCOXON DOES NOT USE THE RAW DIFFERENCES**

The Wilcoxon test assumes that, without a bottleneck, the differences $H_e - H_{eq}$ are distributed symmetrically around zero. They are not: given k, $H_e$ usually lies slightly above its mean and occasionally far below it. With 200 stable populations of 40 plants and 20 microsatellites, simulated for this manual, Wilcoxon on the raw differences "found" a bottleneck in 47 (23.5%), instead of the expected 5%. Applied to the position of each locus within its equilibrium distribution, converted to a normal score, it rejected 17 (8.5%); the sign test, 10 (5%). That is why the app uses the second form. That 8.5% is still somewhat above the nominal level, because θ is estimated from the same data: one more reason to require TPM and SMM to agree.

**DECISION RULE · WAS THERE A BOTTLENECK?**

| If... | then... |
|---|---|
| **Wilcoxon gives P < 0.05 for excess under both TPM and SMM** | There is a signal of a recent reduction. Confirm it with another sample or with the history of the population. |
| **Only the IAM gives an excess** | With microsatellites this is not enough: the IAM exaggerates the excess. |
| **The excess comes from one or two loci** | Check those loci: a null allele or selection affects a single locus, demography affects them all. |
| **There is a significant deficit** | The population may have grown or received immigrants carrying rare alleles. |
| **M < 0.68 without an excess** | An older or longer reduction is compatible; the excess has already faded. |
| **There are fewer than 10 polymorphic loci or fewer than 20 plants** | The test has low power: "no signal" does not prove stability (Luikart and Cornuet, 1998). |

The signal is transient: once the population reaches a new mutation–drift equilibrium, the excess disappears even if the population remains small.

# 10.2 Effective population size

The effective size, $N_e$, is the number of plants in an ideal population that would lose diversity through drift at the same rate as the real one. It is almost always smaller than the census size: not all plants reproduce, some leave far more offspring than others and population size changes from year to year. On average, $N_e$ is about one tenth of the number of adults (Frankham, 1995). It is the figure that matters for conservation, because it sets the rate at which diversity is lost and inbreeding increases.

The app estimates it with the **linkage disequilibrium method** (Hill, 1981; Waples, 2006). In a small population, drift creates random associations between alleles at unlinked loci. The association is measured with $r^2$, which has two sources: the sampling of S plants (about 1/S) and drift (about $1/(3N_e)$). The app subtracts what is expected from sampling and converts the remainder into $N_e$ with the corrections of Waples (2006). It estimates the $N_e$ of the parental generation of the sampled plants.

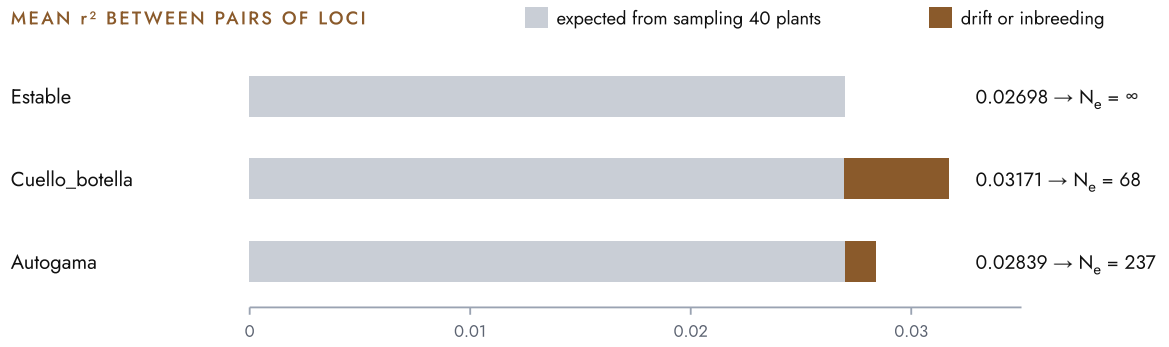



**Figure 10.2.** Mean $r^2$ over the 190 pairs of loci in the three practice populations. With 40 plants, sampling alone gives 0.02699 (grey). In the stable population nothing else remains and $N_e = \infty$. In the one that went through 40 plants, 0.00472 is left over, which gives $N_e = 68$. In the selfing population 0.00140 is left over, which comes not from drift but from inbreeding correlated across loci.

### WORKED EXAMPLE · FROM $R^2$ TO $N_e$

In the *Cuello_botella* population, S = 40 plants were analysed. Sampling contributes $E(r^2) = 1/S + 3.19/S^2 = 0.025 + 0.00199 = 0.02699$ (Waples, 2006, for $S \geq 30$). The observed $r^2$ is 0.03171, so the share due to drift is $r^{2\prime} = 0.03171 - 0.02699 = 0.00472$. Then $N_e = [1/3 + \sqrt{(1/9 - 2.76\, r^{2\prime})}] / (2\, r^{2\prime}) = (0.3333 + 0.3132) / 0.00944 =$ **68.5**. The population was simulated with 40 plants: in 150 identical simulations the median estimate was 42, and this sample fell above the 90th percentile.

**Table 10.4.** The two $N_e$ estimators and the reference figures for conservation.

| Estimator | What it is based on | When it is useful |
|---|---|---|
| **Linkage disequilibrium** Waples (2006); Waples and Do (2008) | Mean $r^2$ between pairs of loci minus the value expected from sampling. It excludes alleles with a frequency below $P_{crit}$ (0.02) because they inflate $r^2$. Interval by *jackknife* over loci. | The most precise single-sample estimator. It works well with small or moderate $N_e$; with large $N_e$ the upper limit is ∞. |
| **Heterozygote excess** Pudovkin et al. (1996) | With few parents, male and female allele frequencies differ by chance and the offspring carry more heterozygotes: $D = (H_o - H_e)/H_e$ and $N_e = 1/(2D) + 1/(2(D + 1))$. | Only with very small $N_e$ (tens). With D close to 0 its upper limit is ∞. |
| **Reference figures** · $N_e \geq 50$ avoids inbreeding depression in the short term and $N_e \geq 500$ retains evolutionary potential (Franklin, 1980); Frankham et al. (2014) propose raising them to 100 and 1000. | | |

✓ DECISION RULE · READING AN $N_e$

| If... | then... |
|---|---|
| **The estimate or its upper limit is ∞** | The $r^2$ cannot be distinguished from sampling noise: $N_e$ is large relative to the sample, not unknown. Report the lower limit. |
| **The limits fall below 50 or below 100** | The population is losing diversity rapidly; this is a conservation warning. |
| **The population self-fertilises or has a high $F_{IS}$** | Inbreeding correlated across loci inflates $r^2$ and lowers $N_e$; interpret it as a minimum. |
| **The sample mixes subpopulations or immigrants** | Admixture also creates disequilibrium: check the structure (Block 7) before reporting. |
| **There are fewer than 25 plants or few loci** | The interval will be very wide; more loci help as much as more plants. |

The method assumes random mating, unlinked loci and a closed population, and estimates $N_e$ per generation.

## 10.3 Spatial genetic structure

Within a continuous population, if seed falls close to the mother and pollen travels only short distances, neighbouring plants are relatives: full sibs, half sibs or cousins. Relatedness declines with distance and forms patches of families. Measuring how relatedness falls with distance tells you how far genes move in each generation, without tracking a single seed (Vekemans and Hardy, 2004).

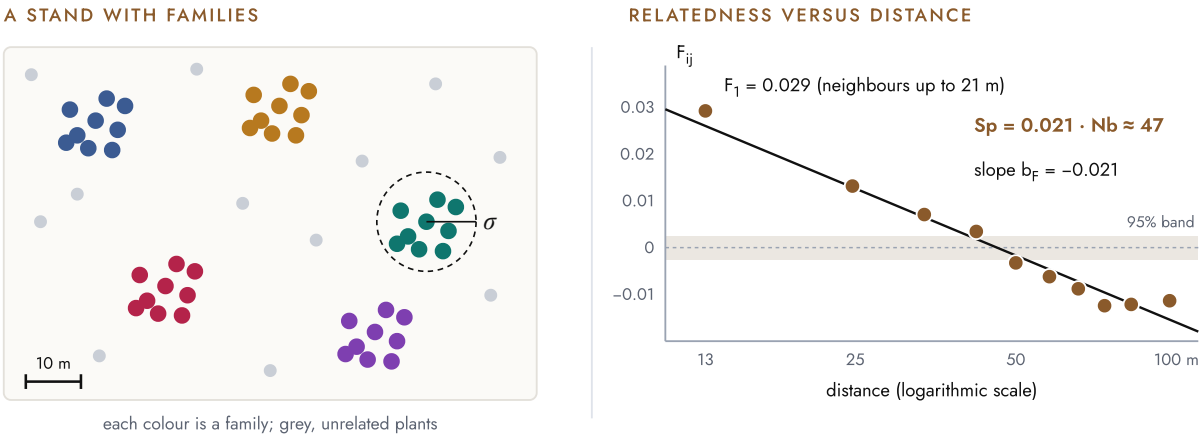



**Figure 10.3.** Left: diagram of a stand where seed falls close to the mother. Plants of the same family stay together, and σ measures how far genes move per generation. Right: mean relatedness in the 10 distance classes of the simulated stand of section 10.7, plotted against the logarithm of distance, with the regression line fitted to the 19,900 pairs. Neighbours have $F_1 = 0.029$, and the slope $b_F = -0.021$ gives Sp = 0.021.

**Table 10.5.** Spatial genetic structure statistics.

| Statistic | Definition | What it tells you |
|---|---|---|
| **$F_{ij}$** · relatedness<br>Loiselle et al. (1995) | $\Sigma_a (p_{ia} - p_a)(p_{ja} - p_a) / \Sigma_a p_a(1 - p_a) + 1/(2n - 1)$ | For codominant markers. 0 between plants taken at random; 0.25 between full sibs and 0.125 between half sibs. |
| **r** · autocorrelation<br>Smouse and Peakall (1999) | Calculated from a centred squared genetic distance between plants. | For any marker. Without structure its expected value is $-1/(n - 1)$, not 0: the permutation band takes this into account. |
| **$F_1$** | Mean relatedness in the first distance class. | How closely related neighbours are. |
| **$b_F$** | Slope of $F_{ij}$ on ln(distance), using all pairs. | How fast relatedness declines. |
| **Sp**<br>Vekemans and Hardy (2004) | $Sp = -b_F / (1 - F_1)$ | The strength of the structure, comparable across studies. |
| **Nb** · neighbourhood | $Nb \approx 1/Sp = 4\pi D\sigma^2$ | Together with the plant density D, it gives the gene dispersal variance $\sigma^2$ per generation. |

The app groups the pairs of plants into distance classes containing the same number of pairs, calculates the mean relatedness (or r) of each class and builds a 95% band by permuting the plants among the locations 999 times. A class above the band indicates that those plants are more closely related than chance can explain.

**IMPORTANT · PAIRS WITHIN POPULATIONS**

If you load several populations kilometres apart, the pairs of plants from different populations fall into the distant classes and drag the differentiation between populations, which Block 5 already measures, into the correlogram. That is why `pairs` defaults to `within populations (fine scale)`: only pairs from the same population, with relatedness referred to the allele frequencies of each population and permutations within each population, all combined into a single correlogram. `all pairs` mixes both scales. If all the plants of a population share the coordinate of the site, there is no fine-scale structure to measure: use the isolation by distance analysis of Block 6.

**WORKED EXAMPLE · FROM SP TO THE DISPERSAL DISTANCE**

In the simulated stand, $b_F = -0.0207$ and $F_1 = 0.0292$, so $Sp = 0.0207 / (1 - 0.0292) = \mathbf{0.0213}$ and $Nb \approx 1/Sp = 47$ plants. With a density of 400 plants per hectare ($D = 0.04$ plants/m²), $\sigma^2 = Nb/(4\pi D) = 47 / (4\pi \times 0.04) = 93$ m², that is, $\sigma \approx 9.7$ m per generation. The simulation used $\sigma^2 = 78$ m² ($\sigma = 8.8$ m), which corresponds to $Nb = 39$: Sp gives the correct order of magnitude, with the usual error of this approximation.

**DECISION RULE · SPATIAL GENETIC STRUCTURE**

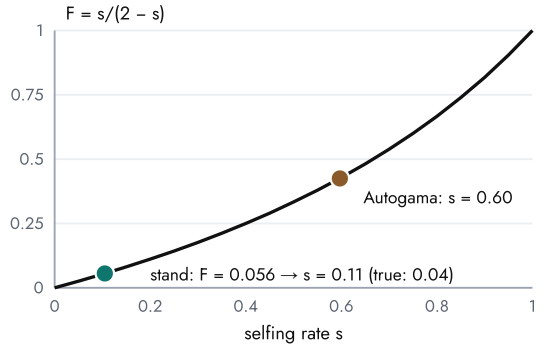
| If... | then... |
|---|---|
| **The first class lies above the band and $b_F$ is significant** | Neighbours are relatives. Report Sp and the distance up to which the signal persists. |
| **Sp is around 0.01** | Weak structure, typical of outcrossing species (mean Sp 0.013; Vekemans and Hardy, 2004). |
| **Sp is around 0.04 or more** | Strong structure, as in species with mixed mating (0.037) or selfing species, or with seed that falls nearby. |
| **No class falls outside the band** | Pollen and seed move far at this scale, or the sample lacks nearby plants. |
| **You sampled few plants or only distant plants** | About 100 plants or more are recommended, with many pairs of close neighbours. |
| **There are clones** | Ramets of the same genet inflate $F_1$: keep one ramet per genet. |



## 10.4 Mating system

A plant that self-fertilises produces more homozygous offspring. If the selfing rate s remains constant over several generations, inbreeding stabilises at $F = s/(2 - s)$, from which $s = 2F/(1 + F)$. The app uses the Weir and Cockerham $F_{IS}$ of each population (Block 4) and its bootstrap interval over loci to estimate s and the outcrossing rate $t = 1 - s$.

EQUILIBRIUM INBREEDING




FROM F TO THE SELFING RATE

| F | s = 2F/(1 + F) | t = 1 − s |
|---|---|---|
| 0.05 | 0.095 | 0.905 |
| 0.10 | 0.182 | 0.818 |
| 0.20 | 0.333 | 0.667 |
| 0.30 | 0.462 | 0.538 |
| 0.43 | 0.601 | 0.399 |
| 0.60 | 0.750 | 0.250 |
| 0.80 | 0.889 | 0.111 |

**Figure 10.4.** Left: the inbreeding reached by a population with a constant selfing rate. The points are two practice populations from section 10.7: the selfing population, simulated with s = 0.60, and the stand, where almost all the inbreeding comes from matings between related neighbours. Right: the inverse conversion used by the app, with the outcrossing rate t.

✓ DECISION RULE · HOW MUCH SELFING?

| If... | then... |
|---|---|
| **$F_{IS}$ is positive at almost all loci** | The deficit is genome-wide: selfing or matings between relatives. s is a reasonable estimate. |
| **The deficit comes from a few loci** | Suspect null alleles (Block 4) before selfing. |
| **There is spatial structure** | Matings between related neighbours also produce $F_{IS} > 0$; s overestimates selfing. |
| **You mixed plants from different groups** | The Wahlund effect creates a deficit without inbreeding. |
| **You need the actual rate for one season** | Genotype mothers and their seeds and use a mixed-mating model (Ritland, 2002). |

## 10.5 $P_{ST}$ versus $F_{ST}$

Drift differentiates populations equally across the whole genome: $F_{ST}$ from neutral markers measures by how much. A quantitative trait under selection may diverge more (divergent selection, local adaptation) or less (stabilising selection, the same optimum everywhere). Its $F_{ST}$ counterpart is $\mathbf{Q_{ST}} = \sigma^2_B/(\sigma^2_B + 2\sigma^2_W)$, calculated from additive genetic variances between (B) and within (W) populations, which require a common garden or crosses (Whitlock, 2008).

With field measurements only phenotypic variances are available. $\mathbf{P_{ST}}$ uses them under two explicit assumptions: c, the proportion of the between-population variance that is additive genetic, and $h^2$, the heritability within populations (Brommer, 2011). $P_{ST} = c\sigma^2_B/(c\sigma^2_B + 2h^2\sigma^2_W)$. With $c/h^2 = 1$ the comparison is at its most optimistic.

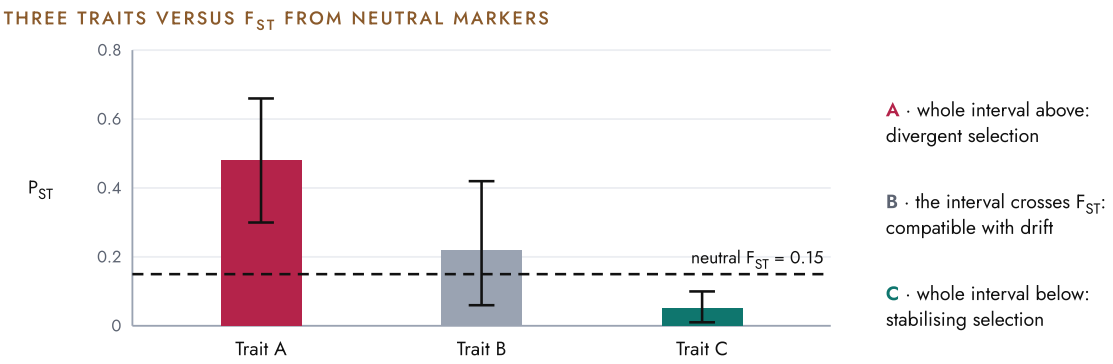



**Figure 10.5.** How to read the comparison, with three hypothetical traits. Each bar is the $P_{ST}$ of a trait and the vertical line its 95% interval; the dashed line is the $F_{ST}$ from neutral markers in the same populations.

### THE $P_{ST}$ INTERVAL

The app obtains $\sigma^2_B$ and $\sigma^2_W$ from a one-way ANOVA. In the random-effects model, the ratio $F/(1 + n_0\lambda)$ follows an F distribution with $G - 1$ and $N - G$ degrees of freedom, where $\lambda = \sigma^2_B/\sigma^2_W$ and $n_0$ is the mean population size (Searle, Casella and McCulloch, 1992). This yields exact limits for $\lambda$ and, since $P_{ST}$ increases with $\lambda$, for $P_{ST}$. With G populations, $\sigma^2_B$ rests on only $G - 1$ degrees of freedom: with three populations the interval is very wide, and a high $P_{ST}$ may be compatible with drift.

### DECISION RULE · $P_{ST}$ VERSUS $F_{ST}$

| If... | then... |
|---|---|
| **The whole $P_{ST}$ interval lies above $F_{ST}$** | The trait has diverged more than drift would cause: divergent selection, if $c/h^2$ is realistic. |
| **The whole interval lies below $F_{ST}$** | Stabilising selection: the populations are more alike than expected. |
| **The interval crosses $F_{ST}$** | There is no evidence against drift. |
| **The conclusion changes when $c/h^2$ is lowered to 0.5** | It depends on the assumption: say so, or measure the trait in a common garden. |
| **You have fewer than 10 populations** | Expect wide intervals; the comparison has low power. |

$F_{ST}$ must come from neutral markers in the same populations (θ from Block 5).

B9

## 10.6 Block settings, tables and figures

The 1 · Settings card shows only the settings of the analyses that apply to your data. After changing one, press Run the analyses.

**Table 10.6.** Block 9 settings.

| Setting | Options | Recommendation |
|---|---|---|
| IAM · TPM · SMM | Mutation models of the excess test; TPM and SMM by default. | TPM and SMM with microsatellites. |
| accepted simulations per locus | 50 to 2000 (default 300). | 300; 1000 for publication. |
| TPM single-step fraction · TPM variance | Defaults 0.95 and 12. | Those of Piry et al. (1999). |
| repeat motif (bp) for M-ratio | 1 to 6 (default 2). | The motif of your microsatellites. |
| $N_e$ (LD method): exclude alleles rarer than | 0 to 0.1 (default 0.02). | 0.02 (Waples and Do, 2010). |
| classes · permutations | 3 to 30 classes with the same number of pairs; 99 to 9999 permutations. | 10 classes and 999 permutations. |
| pairs | within populations (fine scale) or all pairs; appears with two or more populations. | Within populations. |
| c · $h^2$ · $F_{ST}$ | c and $h^2$ from 0.05 to 1; neutral reference $F_{ST}$. | Try $c/h^2$ = 1 and 0.5. |
| Random seed | Integer. | Report it. |

**Table 10.7.** Block 9 result cards.

| Card | Content | Figures |
|---|---|---|
| 2 · What the analyses say | Key figures and interpretation of all analyses; download of everything as CSV. | — |
| 3 · Recent bottlenecks | Tests by population and model, locus-by-locus table. | $H_e$ versus $H_{eq}$ · mode shift |
| 4 · Effective population size | Observed and expected $r^2$, $N_e$ with interval and $N_e$ from heterozygote excess. | $N_e$ by population |
| 5 · Fine-scale spatial genetic structure | Distance classes, value, permutation band, P, $b_F$, Sp and Nb. | Correlogram |
| 6 · Mating system | $F_{IS}$, selfing and outcrossing rates with intervals. | — |
| 7 · $P_{ST}$ versus $F_{ST}$ | $\sigma^2_B$, $\sigma^2_W$, ANOVA, $P_{ST}$ with exact interval and comparison with $F_{ST}$. | $P_{ST}$ by trait |



## 10.7 Step by step: three populations, a stand and maize morphology

Three files take you through the block. The two microsatellite files were simulated for this manual with known parameters and are in the `manual/herramientas` folder, together with the program that generates them (`generar-demografia.pl`). The third is the app's maize morphology example. The values are those produced by the app with the default settings and seed 1.

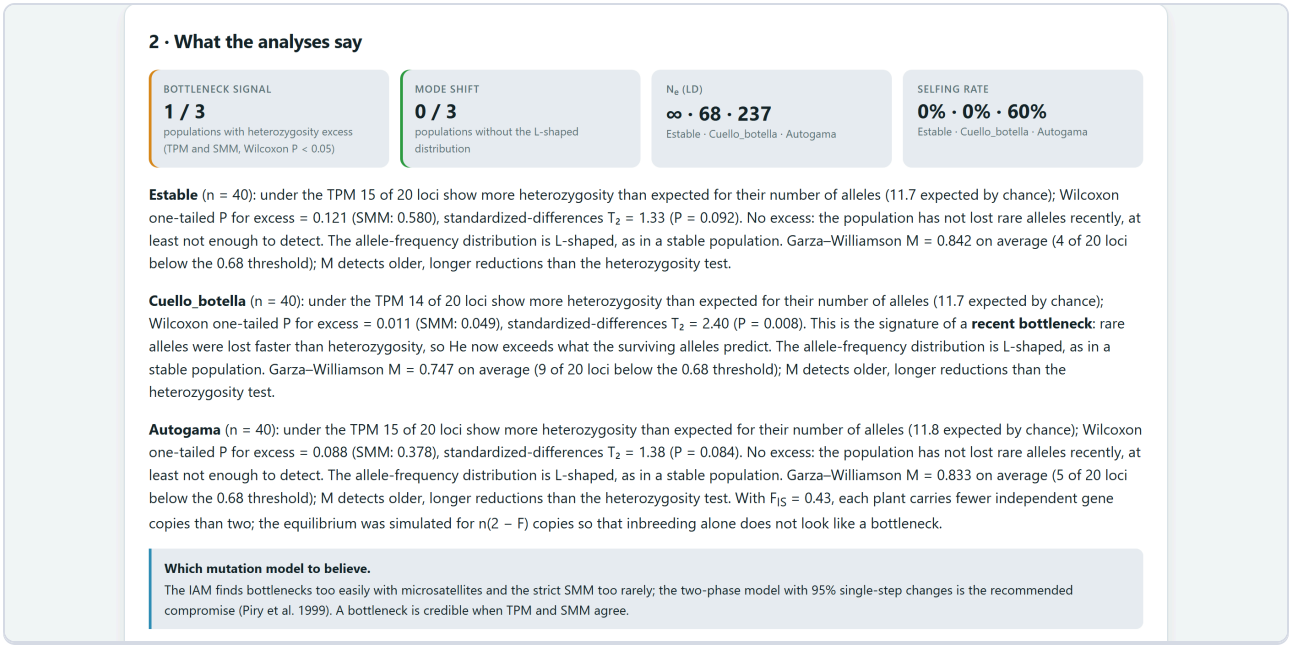
**Table 10.8.** The practice data and what was simulated in each file.

| File | Content | What was simulated |
|---|---|---|
| `ssr-demografia.csv` | 3 populations × 40 plants × 20 microsatellites (2 bp motif). | *Estable*: sample from a large population at equilibrium (TPM model). *Cuello_botella*: 40 plants for 8 generations and 400 in the last one, so $N_e \approx 40$. *Autogama*: equilibrium with 60% selfing (F = 0.43). |
| `ssr-espacial.csv` | 200 plants with latitude and longitude in a 1 ha plot, 12 microsatellites. | A continuous stand of 22,500 plants spaced 5 m apart, with 3000 generations of seed dispersal ($\sigma$ = 5 m) and pollen dispersal ($\sigma$ = 10 m from the mother): $\sigma^2 = 78$ m$^2$ and Nb = 39. |
| Maize · morphology | 3 landraces × 25 plants, 9 traits. | The mean of each landrace was fixed in the simulation; there are no common-garden data. |

## Three populations: bottlenecks, $N_e$ and selfing

STEP BY STEP

1. In Block 2, drag `ssr-demografia.csv` onto Drop your file here, check that the app sees 20 codominant loci and 3 populations, and press Build the dataset and run the checks →.
2. Open Block 9: the analyses run by themselves in about 30 seconds. Read the key figures and the interpretation (figure 10.6).
3. In 3 · Recent bottlenecks choose Cuello_botella and review the tests (figure 10.7), the locus table (figure 10.8) and the plots (figures 10.9 and 10.10).
4. Review $N_e$ (figure 10.11) and the selfing rate (figure 10.12).



**2 · What the analyses say**

| BOTTLENECK SIGNAL | MODE SHIFT | $N_e$ (LD) | SELFING RATE |
|---|---|---|---|
| **1 / 3** | **0 / 3** | **∞ · 68 · 237** | **0% · 0% · 60%** |
| populations with heterozygosity excess (TPM and SMM, Wilcoxon P < 0.05) | populations without the L-shaped distribution | Estable · Cuello_botella · Autogama | Estable · Cuello_botella · Autogama |

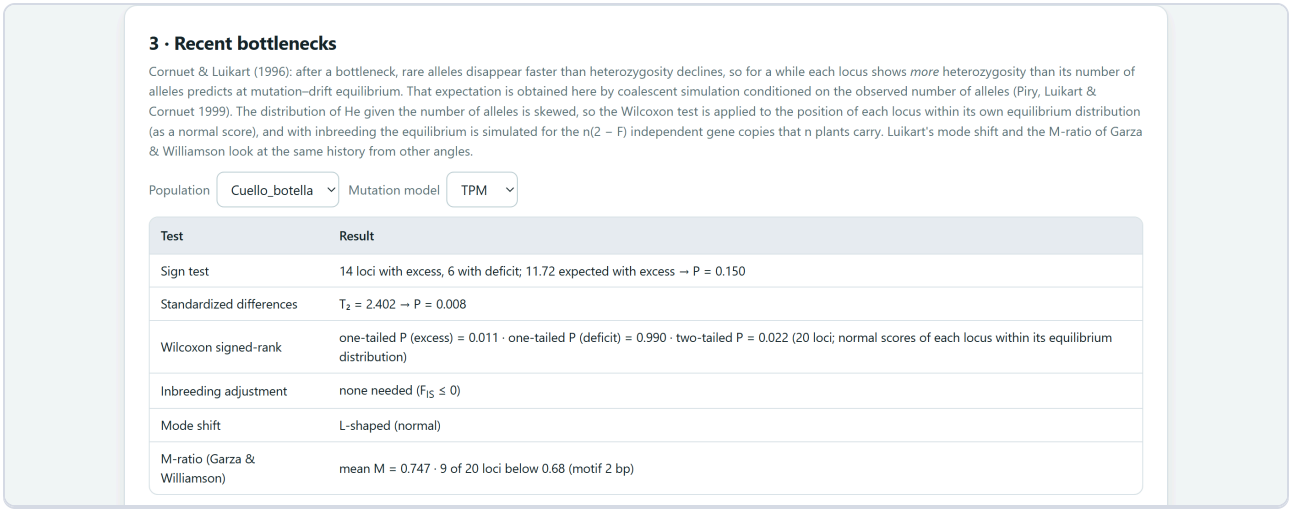
**Estable** (n = 40): under the TPM 15 of 20 loci show more heterozygosity than expected for their number of alleles (11.7 expected by chance); Wilcoxon one-tailed P for excess = 0.121 (SMM: 0.580), standardized-differences $T_2$ = 1.33 (P = 0.092). No excess: the population has not lost rare alleles recently, at least not enough to detect. The allele-frequency distribution is L-shaped, as in a stable population. Garza–Williamson M = 0.842 on average (4 of 20 loci below the 0.68 threshold); M detects older, longer reductions than the heterozygosity test.

**Cuello_botella** (n = 40): under the TPM 14 of 20 loci show more heterozygosity than expected for their number of alleles (11.7 expected by chance); Wilcoxon one-tailed P for excess = 0.011 (SMM: 0.049), standardized-differences $T_2$ = 2.40 (P = 0.008). This is the signature of a **recent bottleneck**: rare alleles were lost faster than heterozygosity, so He now exceeds what the surviving alleles predict. The allele-frequency distribution is L-shaped, as in a stable population. Garza–Williamson M = 0.747 on average (9 of 20 loci below the 0.68 threshold); M detects older, longer reductions than the heterozygosity test.

**Autogama** (n = 40): under the TPM 15 of 20 loci show more heterozygosity than expected for their number of alleles (11.8 expected by chance); Wilcoxon one-tailed P for excess = 0.088 (SMM: 0.378), standardized-differences $T_2$ = 1.38 (P = 0.084). No excess: the population has not lost rare alleles recently, at least not enough to detect. The allele-frequency distribution is L-shaped, as in a stable population. Garza–Williamson M = 0.833 on average (5 of 20 loci below the 0.68 threshold); M detects older, longer reductions than the heterozygosity test. With $F_{IS}$ = 0.43, each plant carries fewer independent gene copies than two; the equilibrium was simulated for n(2 − F) copies so that inbreeding alone does not look like a bottleneck.

**Which mutation model to believe.**
The IAM finds bottlenecks too easily with microsatellites and the strict SMM too rarely; the two-phase model with 95% single-step changes is the recommended compromise (Piry et al. 1999). A bottleneck is credible when TPM and SMM agree.

**Figure 10.6.** What the analyses say. Only *Cuello_botella* has a heterozygosity excess under both TPM and SMM (P = 0.011 and 0.049). *Estable* and *Autogama* do not (P = 0.121 and 0.088 under TPM); in *Autogama* the app simulated the equilibrium with n(2 − F) copies because $F_{IS}$ = 0.43.



**3 · Recent bottlenecks**

Cornuet & Luikart (1996): after a bottleneck, rare alleles disappear faster than heterozygosity declines, so for a while each locus shows *more* heterozygosity than its number of alleles predicts at mutation–drift equilibrium. That expectation is obtained here by coalescent simulation conditioned on the observed number of alleles (Piry, Luikart & Cornuet 1999). The distribution of He given the number of alleles is skewed, so the Wilcoxon test is applied to the position of each locus within its own equilibrium distribution (as a normal score), and with inbreeding the equilibrium is simulated for the n(2 − F) independent gene copies that n plants carry. Luikart's mode shift and the M-ratio of Garza & Williamson look at the same history from other angles.

Population: Cuello_botella · Mutation model: TPM

| Test | Result |
|---|---|
| Sign test | 14 loci with excess, 6 with deficit; 11.72 expected with excess → P = 0.150 |
| Standardized differences | $T_2$ = 2.402 → P = 0.008 |
| Wilcoxon signed-rank | one-tailed P (excess) = 0.011 · one-tailed P (deficit) = 0.990 · two-tailed P = 0.022 (20 loci; normal scores of each locus within its equilibrium distribution) |
| Inbreeding adjustment | none needed ($F_{IS} \leq 0$) |
| Mode shift | L-shaped (normal) |
| M-ratio (Garza & Williamson) | mean M = 0.747 · 9 of 20 loci below 0.68 (motif 2 bp) |

**Figure 10.7.** Tests for *Cuello_botella* under TPM: 14 of 20 loci with an excess against 11.7 expected (sign test, P = 0.150), $T_2$ = 2.40 (P = 0.008) and Wilcoxon P = 0.011. The distribution keeps its L shape and 9 of 20 loci have M < 0.68.

**Locus by locus**

| Locus | n | Alleles | He observed | Heq (equilibrium) | SD | He − Heq | Standardized | P(excess) | Simulations kept |
|---|---|---|---|---|---|---|---|---|---|
| Ssr01 | 40 | 6 | 0.741 | 0.696 | 0.076 | +0.044 | 0.586 | 0.320 | 300 |
| Ssr02 | 40 | 8 | 0.810 | 0.781 | 0.056 | +0.029 | 0.526 | 0.337 | 300 |
| Ssr03 | 40 | 5 | 0.776 | 0.639 | 0.097 | +0.137 | 1.411 | 0.023 | 300 |
| Ssr04 | 40 | 7 | 0.814 | 0.744 | 0.065 | +0.070 | 1.076 | 0.090 | 300 |
| Ssr05 | 40 | 4 | 0.730 | 0.549 | 0.128 | +0.181 | 1.414 | 0.020 | 300 |
| Ssr06 | 40 | 5 | 0.723 | 0.634 | 0.109 | +0.089 | 0.816 | 0.187 | 300 |
| Ssr07 | 40 | 4 | 0.657 | 0.531 | 0.136 | +0.127 | 0.930 | 0.190 | 300 |
| Ssr08 | 40 | 6 | 0.608 | 0.682 | 0.085 | -0.074 | -0.864 | 0.827 | 300 |
| Ssr09 | 40 | 6 | 0.615 | 0.699 | 0.075 | -0.085 | -1.131 | 0.877 | 300 |
| Ssr10 | 40 | 5 | 0.645 | 0.627 | 0.105 | +0.018 | 0.167 | 0.530 | 300 |
| Ssr11 | 40 | 5 | 0.616 | 0.639 | 0.092 | -0.023 | -0.248 | 0.663 | 300 |
| Ssr12 | 40 | 6 | 0.789 | 0.702 | 0.088 | +0.088 | 0.997 | 0.113 | 300 |
| Ssr13 | 40 | 5 | 0.682 | 0.622 | 0.098 | +0.060 | 0.611 | 0.333 | 300 |
| Ssr14 | 40 | 4 | 0.513 | 0.539 | 0.132 | -0.026 | -0.194 | 0.667 | 300 |
| Ssr15 | 40 | 5 | 0.765 | 0.627 | 0.096 | +0.138 | 1.433 | 0.027 | 300 |
| Ssr16 | 40 | 4 | 0.673 | 0.557 | 0.137 | +0.117 | 0.849 | 0.197 | 300 |
| Ssr17 | 40 | 4 | 0.757 | 0.538 | 0.132 | +0.218 | 1.656 | < 0.001 | 300 |
| Ssr18 | 40 | 6 | 0.682 | 0.692 | 0.085 | -0.011 | -0.126 | 0.663 | 300 |
| Ssr19 | 40 | 4 | 0.711 | 0.532 | 0.140 | +0.178 | 1.269 | 0.043 | 300 |
| Ssr20 | 40 | 5 | 0.586 | 0.631 | 0.103 | -0.045 | -0.439 | 0.727 | 300 |

**Figure 10.8.** Locus by locus: the excess (in red) appears at 14 loci. Five have an excess probability below 0.05, from 0.043 (Ssr19) to less than 0.001 (Ssr17), and none dominates: the signal comes from the whole genome.

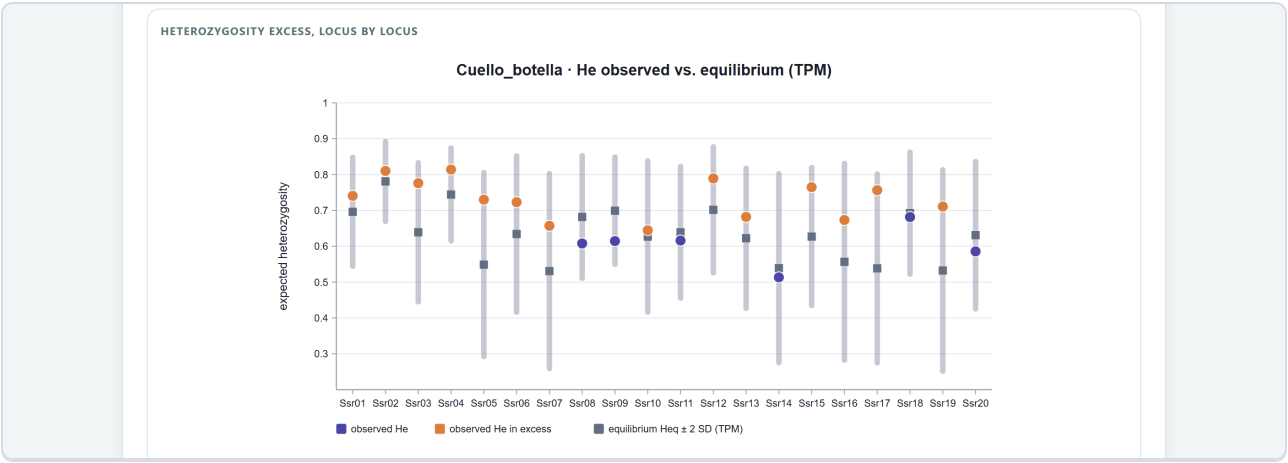



**Figure 10.9.** Observed $H_e$ (circles) against $H_{eq}$ ± 2 standard deviations (grey bars) at each locus of *Cuello_botella*. The orange circles lie above their equilibrium.

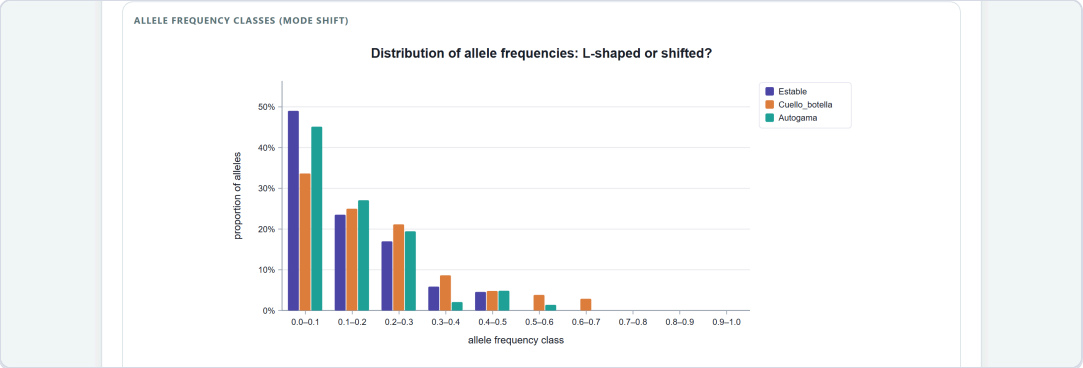



**Figure 10.10.** Mode shift: all three populations keep the L shape. *Cuello_botella* has fewer rare alleles (34% against 49% in the stable one), but the 0–0.1 class is still the largest. After 8 generations of reduction, this indicator does not yet detect it.

**4 · Effective population size**

The linkage-disequilibrium method (Hill 1981; Waples 2006; Waples & Do 2008): in a small population, drift creates associations between unlinked loci in proportion to $1/(3N_e)$. The $r^2$ expected from sampling alone is subtracted, and the remainder gives $N_e$ of the parental generation. Jackknife over loci for the confidence limits.

| Population | n | mean $r^2$ | expected (sampling) | locus pairs | $N_e$ (LD) | 95% CI (jackknife over loci) | $N_e$ (heterozygote excess) | 95% limits |
|---|---|---|---|---|---|---|---|---|
| Estable | 40 | 0.02698 | 0.02699 | 190 | ∞ | 295 – ∞ | 28 | 9 – ∞ |
| Cuello_botella | 40 | 0.03171 | 0.02699 | 190 | 68 | 46 – 127 | 27 | 11 – ∞ |
| Autogama | 40 | 0.02839 | 0.02699 | 190 | 237 | 88 – ∞ | ∞ | ∞ – ∞ |



**Figure 10.11.** $N_e$ from linkage disequilibrium: ∞ in the stable population (lower limit 295), 68 (46–127) in the reduced one and 237 (88–∞) in the selfing one. With the heterozygote excess method, the upper limit is ∞ in all three.

**6 · Mating system**

At inbreeding equilibrium, a selfing rate s produces F = s/(2 − s), so s = 2F/(1 + F). The confidence limits come from a bootstrap over loci of Weir & Cockerham's f.

| Population | n | Loci | $F_{IS}$ (W&C f) | 95% CI | Selfing rate s | 95% CI | Outcrossing t = 1 − s |
|---|---|---|---|---|---|---|---|
| Estable | 40 | 20 | -0.019 | -0.059 to 0.022 | 0% | 0% to 4.3% | 100% |
| Cuello_botella | 40 | 20 | -0.017 | -0.046 to 0.013 | 0% | 0% to 2.6% | 100% |
| Autogama | 40 | 20 | 0.425 | 0.401 to 0.451 | 59.7% | 57.2% to 62.1% | 40.3% |

**Figure 10.12.** Mating system: *Autogama* has $F_{IS}$ = 0.425 and a selfing rate of 59.7% (57.2–62.1%); the simulated rate was 60%. The other two populations show no heterozygote deficit.

### HOW TO READ THE RESULTS FOR THE THREE POPULATIONS

- **The reduction is detected and stability is not mistaken for it.** Wilcoxon gives an excess under both models only in the population that spent 8 generations with 40 plants.
- **Selfing no longer looks like a bottleneck.** Without the inbreeding adjustment, the selfing sample gave an excess with the sign test (P = 0.027) and with $T_2$ (P = 0.020). With F = 0.43, its 40 plants carry about 40 × (2 − 0.43) = 63 independent copies, not 80.
- **A single $N_e$ estimate varies.** The true $N_e$ is 40 and this sample gave 68, with an interval that does not include it. In 150 replicates of the same simulation, the median was 42 and the interval included the true value in 95% of cases: this sample is one of those that miss.
- **The selfing population shows the bias caused by inbreeding.** It comes from a huge population, but inbreeding correlated across loci raises $r^2$ and the method gives $N_e$ = 237.

## A stand: spatial genetic structure

### STEP BY STEP

1. Load `ssr-espacial.csv` in Block 2 (200 plants with `Latitude` and `Longitude`, 12 loci) and build the dataset.
2. Open Block 9 and scroll down to **5 · Fine-scale spatial genetic structure** (figures 10.13 and 10.14). With a single population, **pairs** does not appear: all pairs belong to the same population.

**5 · Fine-scale spatial genetic structure**

Loiselle et al. (1995) kinship, averaged over loci, with the allele frequencies of the 200 mapped plants as reference. Envelope from 999 random permutations of individuals among locations. Regression on ln(distance): slope $b_F$ = -2.065e-2, $r^2$ = 0.046, P = 0.002; Sp = 0.021, Nb = 47.

| Class | From (m) | To (m) | Pairs | Loiselle kinship $F_{ij}$ | 95% envelope | P |
|---|---|---|---|---|---|---|
| 1 | 4.9 | 20.5 | 1990 | 0.0292 | -0.0026 to 0.0023 | 0.002 |
| 2 | 20.5 | 29.9 | 1990 | 0.0131 | -0.0024 to 0.0025 | 0.002 |
| 3 | 29.9 | 38.1 | 1990 | 0.0071 | -0.0026 to 0.0022 | 0.002 |
| 4 | 38.1 | 46.0 | 1990 | 0.0035 | -0.0026 to 0.0022 | 0.004 |
| 5 | 46.0 | 54.0 | 1990 | -0.0033 | -0.0025 to 0.0022 | 0.012 |
| 6 | 54.0 | 61.0 | 1990 | -0.0062 | -0.0025 to 0.0024 | 0.002 |
| 7 | 61.0 | 69.6 | 1990 | -0.0088 | -0.0026 to 0.0023 | 0.002 |
| 8 | 69.6 | 77.6 | 1990 | -0.0124 | -0.0025 to 0.0024 | 0.002 |
| 9 | 77.6 | 87.3 | 1990 | -0.0122 | -0.0024 to 0.0023 | 0.002 |
| 10 | 87.3 | 130.7 | 1990 | -0.0114 | -0.0024 to 0.0019 | 0.002 |

**Figure 10.13.** Distance classes of the stand, in metres. Mean relatedness goes from 0.029 between neighbours (5 to 21 m) to negative values beyond 46 m. The note gives $b_F$ = −0.0207 (P = 0.002), Sp = 0.021 and Nb = 47.

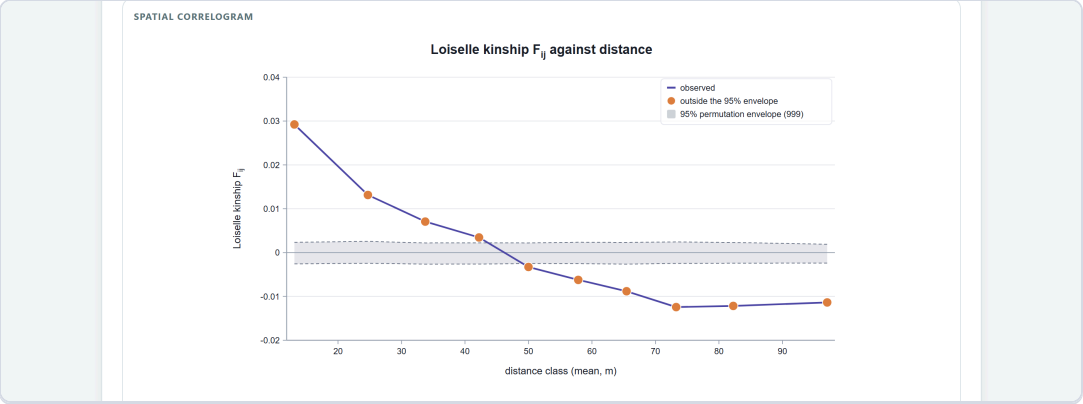



**Figure 10.14.** Correlogram of the stand. The first four classes lie above the permutation band and the distant ones below it: neighbours are relatives up to about 46 m.

The other cards also run with this file, although they are not its purpose, and they illustrate two precautions. The selfing rate comes out at 10.5%, even though in the simulation only 4% of the seeds came from the plant's own pollen: the rest of the $F_{IS}$ of 0.056 comes from matings between related neighbours, as section 10.4 warns. And the bottleneck test finds an excess only under TPM (P = 0.006; under SMM, P = 0.088): the simulation started from even allele frequencies and the stand is not at mutation–drift equilibrium, so the app reports it as a weak signal.

## Agave: no fine-scale structure within populations

The Agave · AFLP example includes coordinates for each plant in four populations 150 to 400 km apart. With within populations (fine scale), the app uses only the 760 pairs of plants from the same population and the Smouse and Peakall *r*, because the bands are dominant (figure 10.15). With all pairs, the first two classes (up to 5 km, pairs from the same population) come out with r = 0.092 and 0.083, far above the band, and the distant ones are negative: that "structure" is the differentiation between populations.

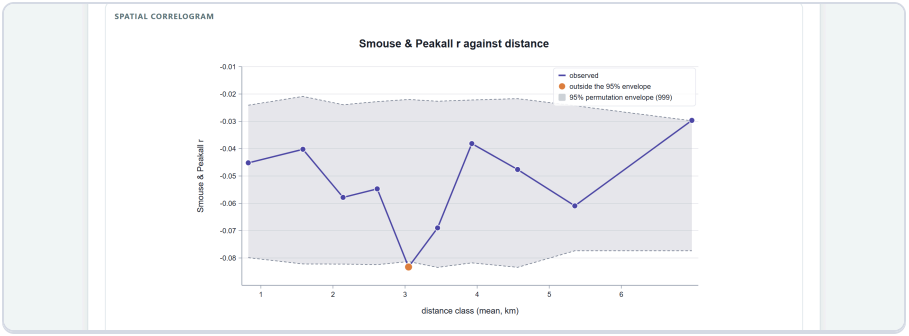



**Figure 10.15.** Agave with pairs within populations. r is around –0.05, the value expected without structure with 20 plants per population (–1/19 = –0.053), and the first class lies within the band. The 2.8 to 3.2 km class falls just below it (P = 0.042), which is to be expected when checking 10 classes. The data were simulated without structure within populations.

# Maize morphology: $P_{ST}$ versus $F_{ST}$

**STEP BY STEP**

1. Load **Maize · morphology** in Block 2, build the dataset and open Block 9.
2. In **$F_{ST}$** type 0.049, the θ of the same three landraces with microsatellites (section 6.6). Review the table and the figure (figures 10.16 and 10.17).
3. Change **c** to 0.5, press **Run the analyses** and see which trait changes its reading.

### 7 · $P_{ST}$ versus $F_{ST}$

$Q_{ST}$ is the $F_{ST}$ of a quantitative trait; if it exceeds the neutral $F_{ST}$, populations have diverged more than drift allows — divergent selection. Without a common-garden or crossing design, its phenotypic proxy $P_{ST} = c\sigma^2_B/(c\sigma^2_B + 2h^2\sigma^2_W)$ is what field measurements can give (Brommer 2011; Leinonen et al. 2013). The confidence limits are exact: they come from the F distribution of the one-way ANOVA (Searle, Casella & McCulloch 1992).

| Trait | $\sigma^2_B$ (among) | $\sigma^2_W$ (within) | ANOVA F | P | $P_{ST}$ | 95% CI (exact, F distribution) | vs $F_{ST}$ |
|---|---|---|---|---|---|---|---|
| PlantHeight_cm | 1732.9600 | 368.8228 | 118.47 | < 0.001 | 0.701 | 0.371 to 0.989 | above $F_{ST}$ · divergent selection |
| EarLength_cm | 15.8176 | 2.0953 | 189.72 | < 0.001 | 0.791 | 0.489 to 0.993 | above $F_{ST}$ · divergent selection |
| EarDiameter_mm | 134.1782 | 18.9017 | 178.47 | < 0.001 | 0.780 | 0.473 to 0.993 | above $F_{ST}$ · divergent selection |
| KernelRows | 2.3347 | 2.8056 | 21.80 | < 0.001 | 0.294 | 0.084 to 0.945 | above $F_{ST}$ · divergent selection |
| KernelsPerRow | 18.6731 | 12.4667 | 38.45 | < 0.001 | 0.428 | 0.151 to 0.968 | above $F_{ST}$ · divergent selection |
| CobWeight_g | 280.4674 | 47.5250 | 148.54 | < 0.001 | 0.747 | 0.427 to 0.992 | above $F_{ST}$ · divergent selection |
| Weight100Kernels_g | 132.2063 | 7.6809 | 431.31 | < 0.001 | 0.896 | 0.688 to 0.997 | above $F_{ST}$ · divergent selection |
| DaysToFlowering | 63.6833 | 16.4111 | 98.01 | < 0.001 | 0.660 | 0.326 to 0.987 | above $F_{ST}$ · divergent selection |
| LeafWidth_cm | 1.7827 | 0.8969 | 50.69 | < 0.001 | 0.498 | 0.194 to 0.976 | above $F_{ST}$ · divergent selection |

**Figure 10.16.** $P_{ST}$ of the nine traits with $c/h^2 = 1$. The ANOVA F values range from 21.8 to 431, so, even with only 2 degrees of freedom between landraces, the lower limit of each interval exceeds 0.049.

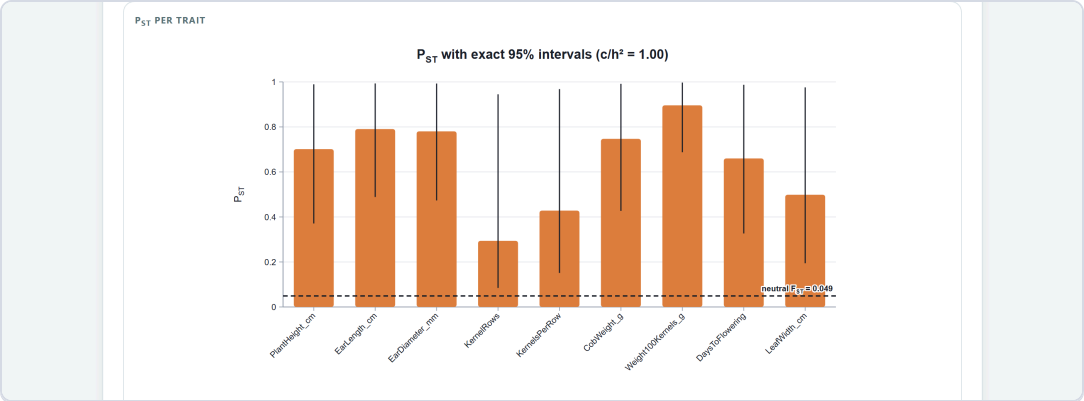



**Figure 10.17.** $P_{ST}$ with its exact intervals and the neutral $F_{ST}$. The intervals reach almost 1: with three landraces the estimate is imprecise even when the differences are enormous.

### HOW TO READ THE MAIZE RESULTS

- **The conclusion almost always holds.** With $c/h^2 = 0.5$, the lower limit for *KernelRows* drops to 0.044 and falls below $F_{ST}$: its differentiation can no longer be distinguished from drift. The other eight traits remain above it.
- **These are synthetic data.** The mean of each landrace was fixed in the simulation and there is no common garden, so the conclusion depends on the $c/h^2$ assumption.

### NOTE · WHAT TO REPORT IN A PAPER

For bottlenecks: mutation models and TPM parameters, accepted simulations per locus, tests with their P values and whether an inbreeding adjustment was applied. For $N_e$: method, $P_{crit}$, number of plants and loci, and the interval. For spatial structure: estimator, whether pairs were within populations, classes, permutations, $F_1$, $b_F$ with its P value, and Sp. For selfing: $F_{IS}$ with its interval and the caveat that it is an indirect inference. For $P_{ST}$: $c/h^2$, the reference $F_{ST}$ and how the intervals were obtained. The methods section of the Block 10 report drafts all of this, including the seed.

### NOTE · VERIFICATION OF THE CALCULATIONS

For this manual, the $H_{eq}$ simulated under the IAM matched the exact value from Ewens's formula in four cases (40 to 80 copies), within one standard error, and the unconditional expectations of the IAM and SMM matched theory. The exact Wilcoxon test matched an independent enumeration in 300 cases with ties. With an independent coalescent generator, the corrected test rejected 8.5% of 200 stable populations (the sign test, 5%), 3% of 100 selfing populations with the adjustment (15% without it) and detected 76% of 49 reduced populations. The linkage disequilibrium $N_e$ and Pudovkin's D were recalculated in another language from the genotype counts with no differences; in 150 simulations with $N_e = 40$, the median was 42 and the interval covered the true value 95% of the time. Loiselle's relatedness, $F_1$, $b_F$ and Sp of the stand, and the Smouse and Peakall r of the agave with the ordered-pairs formula, were recalculated with no differences. The exact $P_{ST}$ interval covered the true value in 94 to 96% of 2000 datasets simulated per scenario.

The next chapter concludes the analyses: Block 10 gathers figures, tables and interpretations into a report and a ZIP package ready for publication.

# B10

CHAPTER ELEVEN · BLOCK 10 · FIGURES & REPORT

# Figures and report

An analysis is finished when it can be published. Every figure in the app can be adjusted and exported in the format and resolution a journal requires. Block 10 gathers what you have run into a report: key figures, tables, interpretations, figures exactly as you left them, a methods paragraph with the settings used and a ZIP package with the data, tables, trees and figures.


**11.1** The figure editor

**11.2** Exporting for publication

**11.3** The automatic report and the ZIP package

**11.4** Step by step: the maize report


There are two ways to get results out of the app. The first is figure by figure, from the export row below each figure, in any block. The second is everything at once, from Block 10, once you have run the blocks you are interested in.

**Table 11.1.** What you can get out of the app and from where.

| Output | Where | What for |
|---|---|---|
| A figure in PNG, TIFF, SVG, JPG or WEBP | Export row below each figure | A paper, a thesis or a presentation. |
| Tables in CSV | Download buttons in each block | Rebuilding a table in the journal's format. |
| HTML report | Block 10 · **Report (HTML)** | Sharing the whole analysis in a single file that opens in any browser. |
| PDF report | Block 10 · **Print / save as PDF** | A version for printing or archiving. |
| ZIP package | Block 10 · **Full package (ZIP)** | Supplementary material and a backup of the analysis. |
| Methods paragraph | Block 10 · **Copy the methods paragraph** | A starting point for the methods section. |

## 11.1 The figure editor

All the figures in the app are vector graphics and share the same editor. Below each one, **⚙ Edit figure — titles, colours, fonts, sizes** opens two tabs: **Content & colours**, with the options specific to that figure, and **Style & typography**, with the options common to all of them. Each change redraws the figure instantly.

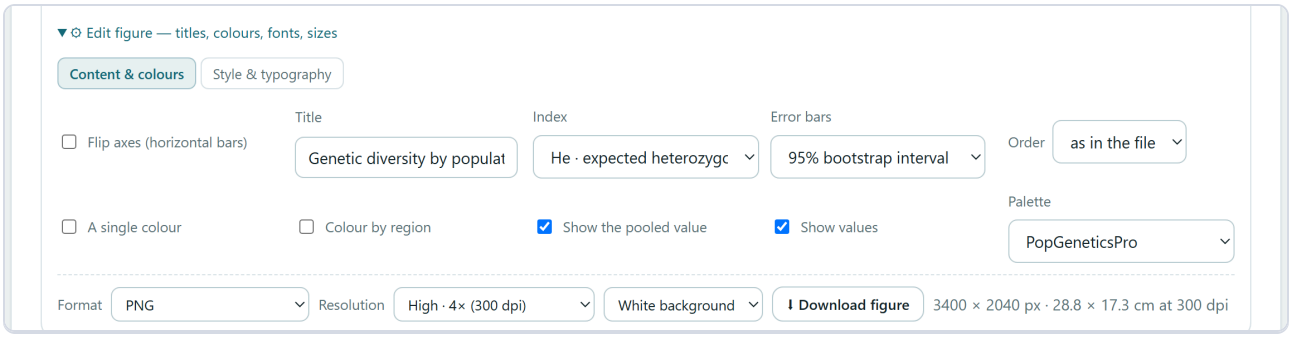



**Figure 11.1.** **Content & colours** for the diversity figure of Block 3: title, index, error bars, order, palette and value options. Below it is the export row, which reports the size in pixels and centimetres of the file to be downloaded.

The content tab changes with each figure: a tree offers root, layout and support values; a heat map, its colour scale; a bar chart also offers **Flip axes (horizontal bars)** to rotate it when the names are long. The style tab is the same for all of them (figure 11.2 and table 11.2).

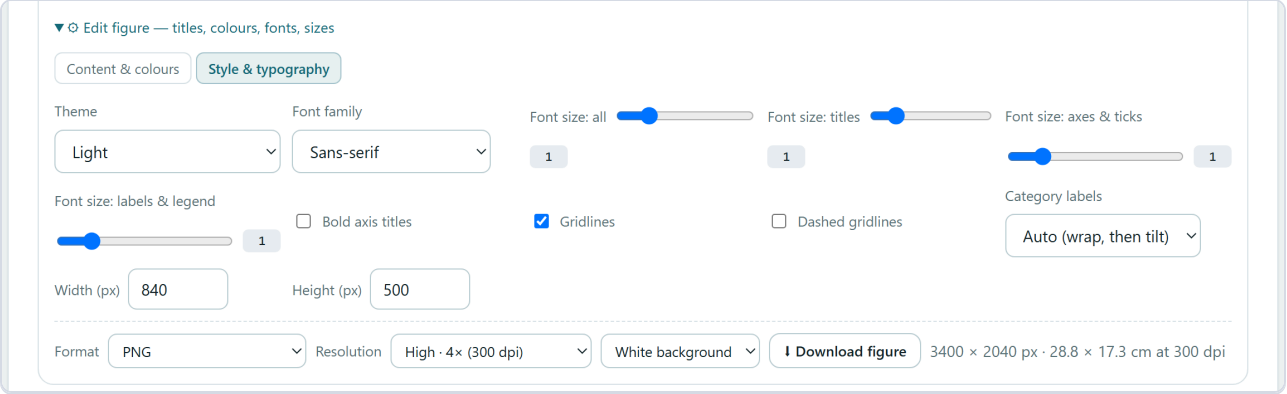



**Figure 11.2.** Style & typography: theme, font family, four font-size controls, gridlines, orientation of the category labels and figure size.

**Table 11.2.** Style & typography controls.

| Control | Options | Note |
|---|---|---|
| Theme | Light, Paper, Dark, Minimal and Journal (black axes, no grid). | Journal for papers; Dark for presentations with a dark background. |
| Font family | Sans-serif, System interface font, Serif, Monospace. | These are generic families: each computer uses whichever typeface of that kind it has installed. |
| Font size: all · titles · axes & ticks · labels & legend | Multipliers from 0.6 to 2.6 (all) and from 0.6 to 3 (each group). | If the text grows, the canvas grows with it so that nothing is cut off. |
| Bold axis titles · Gridlines · Dashed gridlines | Checkboxes. | — |
| Category labels | Auto (wrap, then tilt), Horizontal (wrap), Tilted 45° or Vertical. | Only in category charts. |
| Width (px) · Height (px) | 400 to 2400 and 300 to 2000. | Sets the size of the figure and, with it, the size of the text when printed (section 11.2). |

**NOTE · THE STYLE IS REMEMBERED**

The theme, typeface, font sizes, grid and label orientation are shared preferences: when you change them in one figure, the app saves them in this browser and applies them to the figures it draws afterwards, even in another session. The title, colours and size belong to each figure. This way all the figures in a paper share the same look.

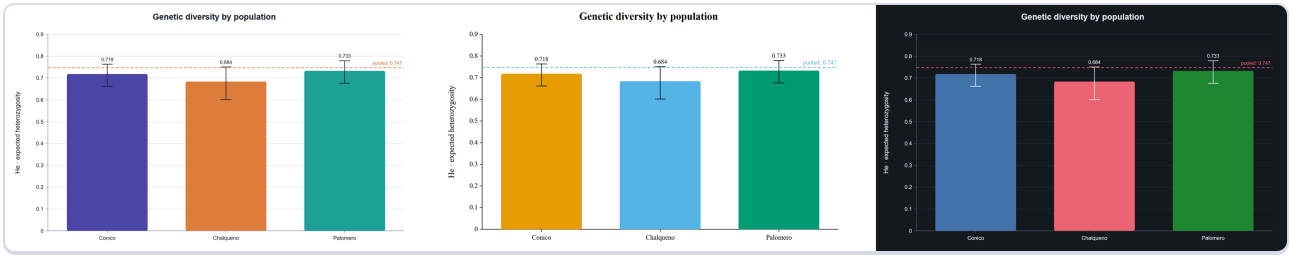



**Figure 11.3.** The diversity figure in three styles. Left: Light theme and the app's palette. Centre: Journal theme, Okabe and Ito (2008) palette, serif typeface and text 1.25 times larger. Right: Dark theme with the Tol (2021) palette.

**Table 11.3.** Palettes and colour scales.

| Type | Options |
|---|---|
| Palettes for groups | PopGeneticsPro (default), Vivid, Harvest, Soil & crop, Ocean, Sunset, Meadow, Bold, Soft, Deep, Pastel. |
| Colour-blind safe | Okabe and Ito (2008) and Tol (2021). |
| Black and white | Greyscale. |
| Continuous scales (heat maps) | Viridis, Magma, Inferno, Plasma and Cividis (colour-blind safe); diverging Red–Yellow–Blue, Red–Blue, Spectral and Blue–White–Red; sequential Greens, Yellow–Green, Browns and Heat. |

**DECISION RULE · PREPARING A FIGURE**

| If... | then... |
|---|---|
| **The figure is going into a paper** | Journal or Minimal theme, with the same typeface and sizes in every figure. |
| **You distinguish groups by colour** | Okabe and Ito or Tol: they avoid confusing red and green. In heat maps, Viridis or Cividis. |
| **It will be printed in black and white** | Greyscale, plus values or shapes that do not depend on colour. |
| **The category names are long** | Flip axes, or Category labels at 45°. |
| **The journal reduces the figure to one column** | Lower Width and raise Font size: all before exporting (section 11.2). |

## 11.2 Exporting for publication

The export row has three menus and a button: Format, Resolution, white or transparent background, and Download figure. On the right, the app reports the dimensions of the file it is about to create: in figure 11.1, *3400 × 2040 px · 28.8 × 17.3 cm at 300 dpi*.

**Table 11.4.** Export formats.

| Format | What it stores | When to use it |
|---|---|---|
| **PNG** | Lossless image with the resolution written into the file. White or transparent background. | Almost always; transparent to place it on another background. |
| **TIFF** | Uncompressed RGB image with the resolution in the file. Very large files. | Journals that require TIFF. |
| **SVG** | The vector drawing: it scales without losing sharpness and every text or colour can be edited. | Retouching in a vector drawing program, or very large figures. |
| **JPG** | Lossy compressed image (97% quality) with the resolution in the file. | Only if required; on lines and text, compression leaves halos. |
| **WEBP** | Image compressed for the web, without the resolution in the file. | Web pages and online documents. |

### PIXELS, DPI AND PRINTED SIZE

The app draws each figure at 75 pixels per inch. Resolution multiplies the pixels: 2× is 150 dpi, 4× (default) 300 dpi, 6× 450, 8× 600 and 12× 900. The printed size at that resolution does not change: an 850 px figure measures 850/75 = 11.3 inches, 28.8 cm, at any resolution. What the resolution increases is sharpness.

Text is drawn in pixels: 11 px for axis tick labels and legends, 13 px for axis titles and 17 px for the title. At natural size, 11 px is 10.6 points. If the journal reduces the figure, the text shrinks in the same proportion: **printed size (pt) ≈ 28.3 × size in px × final width (cm) / figure width (px)**.

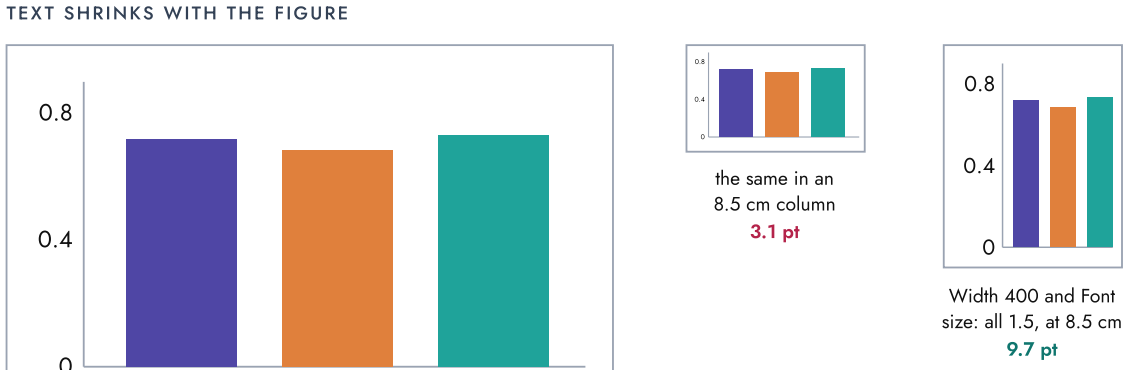



**Figure 11.4.** The diversity figure at its natural size and reduced to an 8.5 cm column. The proportions between the three cases are real; the three text sizes are drawn larger than the scale of the diagram so that they can be read. With Width 400 and Font size: all 1.5 the figure is 410 px wide and its tick labels reach 9.7 pt in the column.

**WORKED EXAMPLE · A FIGURE FOR ONE COLUMN**

The journal publishes figures in an 8.5 cm column at 600 dpi with text of at least 7 pt. The diversity figure is 850 px wide: reduced to 8.5 cm, its 11 px tick labels would end up at 28.3 × 11 × 8.5 / 850 = **3.1 pt**. With Width 400 and Font size: all 1.5, the figure is 410 px wide and the tick labels 16.5 px, which come to 28.3 × 16.5 × 8.5 / 410 = **9.7 pt**. For 600 dpi at 8.5 cm you need 8.5/2.54 × 600 = 2008 px of width; Publication · 8× (600 dpi) gives 410 × 8 = 3280 px, which is more than enough.

**IMPORTANT · VERY LARGE FIGURES**

The browser cannot create images of more than 16,000 pixels per side or more than 120 megapixels. If the chosen resolution exceeds these limits (a tree with hundreds of tips at 900 dpi, for example), the app lowers the resolution as much as necessary, keeps the proportions and shows a warning in orange next to the button. For more detail, export as SVG.

**DECISION RULE · WHICH FORMAT AND RESOLUTION**

| If… | then… |
|---|---|
| **The journal requires TIFF or PNG** | Check the final width and the resolution it demands; calculate the pixels needed and choose the factor that reaches them. |
| **The figure has thin lines and text** | PNG or TIFF at 600 dpi, or SVG; not JPG. |
| **You are going to edit it afterwards** | SVG: every text and colour can be changed. |
| **It is for a presentation** | PNG at 150 or 300 dpi, with a transparent background if the slide is coloured. |
| **The journal changes the resolution of the file** | It does not matter: what preserves quality is the number of pixels. |

## 11.3 The automatic report and the ZIP package

Block 10 does not calculate anything new: it takes what is already on screen in the other blocks. Each table and each interpretation goes in exactly as you see it, and each figure as you left it in its editor. If you change a setting or a figure, return to Block 10 and the report is rebuilt with the change.

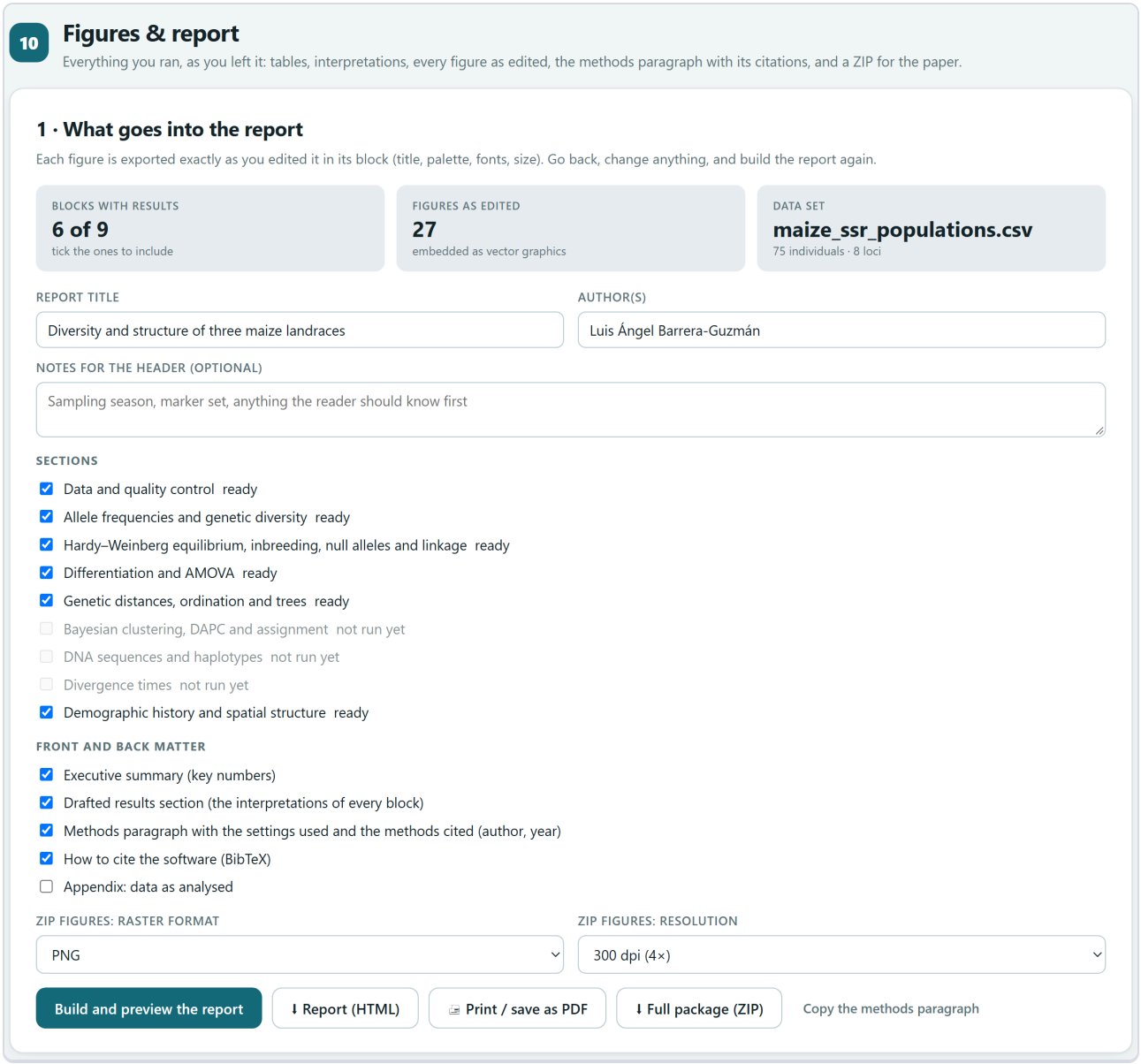



**Figure 11.5.** 1 · What goes into the report with the maize SSR data: six blocks with results and 27 figures. At the top are the title, the authors and a note; in the middle, the available sections and the optional parts; at the bottom, the format and resolution of the figures in the ZIP and the output buttons.

**Table 11.5.** Parts of the report.

| Part | Content |
|---|---|
| Header and contents | Title, authors, date, data file, number of individuals, loci and populations, optional note and table of contents with links. |
| Executive summary | Key figures: data, diversity, inbreeding, differentiation, isolation by distance, clusters, assignment, sequences, root age and spatial structure, depending on what you ran. |
| Drafted results section | The interpretations from all blocks, without the warning boxes, as a draft of the results. |

**Table 11.5.** Parts of the report. *(continued)*

| Part | Content |
|---|---|
| One section per block | Key figures, interpretation, numbered tables with titles and numbered figures. The caption of each figure adds the settings that affect it: permutations, bootstrap replicates, model or seed. |
| Methods paragraph | The methods used, with the actual settings and the authors and year of each method. |
| How to cite the software | The citation of the program, its licence and the BibTeX entry. |
| Appendix: data as analysed | The data matrix as it was analysed (optional). |

**IMPORTANT · WHAT YOU SEE IS WHAT GOES IN**

The report copies what each card shows at that moment. If in Block 9 you chose to view one population in 3 · Recent bottlenecks, that is the one that goes into its figure; if a card is hidden because it does not apply to your data, it is left out. The three output buttons rebuild the report before saving it, so a title or a checkbox you change after the preview is always included.

**NOTE · THE METHODS PARAGRAPH IS A DRAFT**

The paragraph names each method with its authors and year, and the settings it ran with (permutations, bootstrap replicates, models, seeds). It does not include the full reference list: take the references from appendix E, which gathers them with their details and marks those the paragraph may cite. Also check that the paragraph describes what you will actually report; if you removed a section from the report, its methods remain in the paragraph.

**Table 11.6.** Contents of the ZIP package. Only the files of the analyses you ran are included.

| Folder or file | Content |
|---|---|
| report.html | The report, identical to the one from Report (HTML). |
| data/ | The data as analysed (CSV) and, with sequences, the haplotypes in FASTA. |
| tables/ | Allele frequencies, diversity, Hardy–Weinberg and null alleles, F-statistics by locus, pairwise differentiation, distances, PCoA coordinates, ancestry proportions by K and choice of K, assignment, haplotypes, bottleneck tests, $N_e$, spatial autocorrelation, selfing, $P_{ST}$ and node ages. |
| trees/ | Population and individual trees (Newick) and dated trees (Newick and NEXUS). |
| dating/ | The MCMC chain log of the dating analysis. |
| figures/svg/ · figures/png/ | Each figure in SVG and in the chosen format (PNG, TIFF, JPG or WEBP) at the chosen resolution. |
| methods.txt · README.txt | The methods paragraph in plain text and the guide to the package. |

**DECISION RULE · WHICH OUTPUT TO USE**

| If… | then… |
|---|---|
| **You are writing the paper or thesis** | Copy the methods paragraph and adapt it; rebuild the tables from the CSV files; export the figures following the rule in section 11.2. |
| **You are sharing the analysis with co-authors** | The HTML report: a single file that opens in any browser, with vector figures. |
| **You are preparing the supplementary material** | The ZIP: data, tables, trees and figures. |
| **You want a fixed copy** | Print / save as PDF and the ZIP, with the seeds recorded. |

## 11.4 Step by step: the maize report

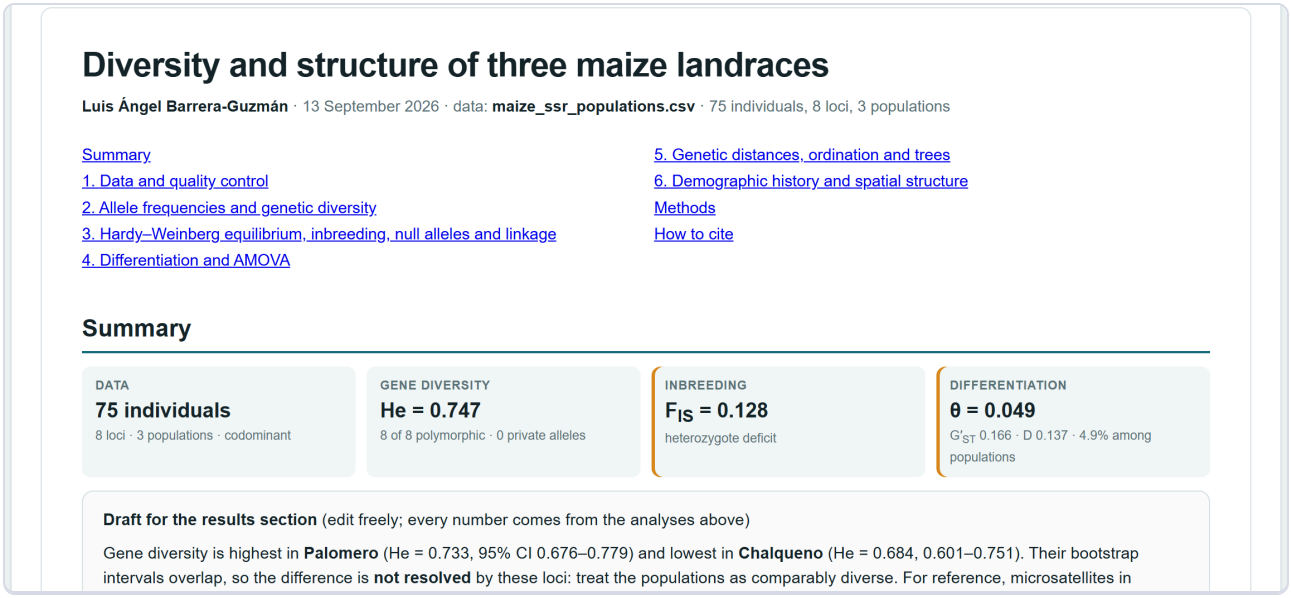
With the maize microsatellite example and five blocks run with their default settings, the report has 6 sections, 20 tables and 27 figures, weighs 346 KB and takes up 30 letter-size pages when printed.

**STEP BY STEP**

1. In Block 2, load Maize populations · SSR and press Build the dataset and run the checks →. Open Blocks 3, 4, 5, 6 and 9: each one runs when you open it.
2. In Block 3, open the editor of Diversity of every population and try out themes, palettes and sizes (figures 11.1 to 11.3). Settle on the style you will use in all of them.
3. Open Block 10, type the title and the authors and review the selected sections (figure 11.5).
4. Press Build and preview the report and look through the preview (figures 11.6 to 11.8).
5. Save the result with Report (HTML), Print / save as PDF or Full package (ZIP), and copy the methods paragraph with Copy the methods paragraph.

**WHAT TO CHECK BEFORE HANDING IN THE REPORT**

- **The summary repeats the interpretations.** The draft results section brings together the texts that later appear in each section; if the report is just for you, untick Drafted results section and it becomes shorter.
- **Every interpretation is a draft.** The sentences are generated from general rules; check that they apply to your species and your sampling design.
- **The full maize ZIP has 72 files and 7.6 MB** with PNG figures at 300 dpi: the report, the data, 12 tables, 2 trees, 27 figures in SVG and 27 in PNG, the methods and the guide. In TIFF it is several times larger.
- **When printing to PDF,** enable background graphics in the print dialog to keep the colour of the key figures. Figures are not split across pages and short tables stay with their titles.



# Diversity and structure of three maize landraces

**Luis Ángel Barrera-Guzmán** · 13 September 2026 · data: **maize_ssr_populations.csv** · 75 individuals, 8 loci, 3 populations

Summary
1. Data and quality control
2. Allele frequencies and genetic diversity
3. Hardy–Weinberg equilibrium, inbreeding, null alleles and linkage
4. Differentiation and AMOVA
5. Genetic distances, ordination and trees
6. Demographic history and spatial structure
Methods
How to cite

## Summary

DATA
**75 individuals**
8 loci · 3 populations · codominant

GENE DIVERSITY
**He = 0.747**
8 of 8 polymorphic · 0 private alleles

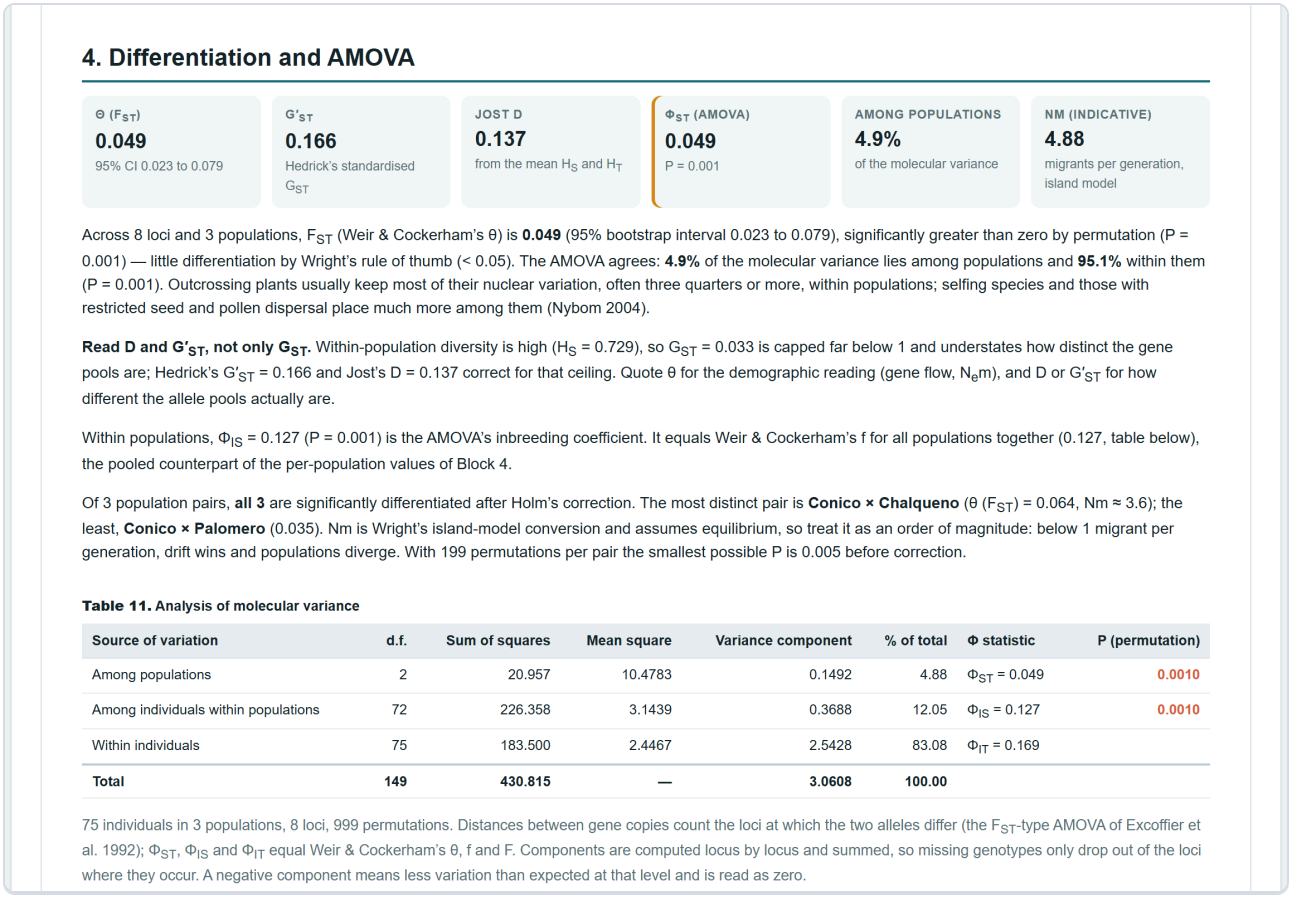
INBREEDING
**$F_{IS}$ = 0.128**
heterozygote deficit

DIFFERENTIATION
**θ = 0.049**
$G'_{ST}$ 0.166 · D 0.137 · 4.9% among populations

**Draft for the results section** (edit freely; every number comes from the analyses above)

Gene diversity is highest in **Palomero** (He = 0.733, 95% CI 0.676–0.779) and lowest in **Chalqueno** (He = 0.684, 0.601–0.751). Their bootstrap intervals overlap, so the difference is **not resolved** by these loci: treat the populations as comparably diverse. For reference, microsatellites in

**Figure 11.6.** Beginning of the report: title, authors, date and data; the table of contents with links; the key figures, with $F_{IS}$ and $G'_{ST}$ written with subscripts; and the draft results section, which brings together the interpretations from the blocks.



## 4. Differentiation and AMOVA

Θ ($F_{ST}$)
**0.049**
95% CI 0.023 to 0.079

$G'_{ST}$
**0.166**
Hedrick's standardised $G_{ST}$

JOST D
**0.137**
from the mean $H_S$ and $H_T$

$\Phi_{ST}$ (AMOVA)
**0.049**
P = 0.001

AMONG POPULATIONS
**4.9%**
of the molecular variance

NM (INDICATIVE)
**4.88**
migrants per generation, island model

Across 8 loci and 3 populations, $F_{ST}$ (Weir & Cockerham's θ) is **0.049** (95% bootstrap interval 0.023 to 0.079), significantly greater than zero by permutation (P = 0.001) — little differentiation by Wright's rule of thumb (< 0.05). The AMOVA agrees: **4.9%** of the molecular variance lies among populations and **95.1%** within them (P = 0.001). Outcrossing plants usually keep most of their nuclear variation, often three quarters or more, within populations; selfing species and those with restricted seed and pollen dispersal place much more among them (Nybom 2004).

**Read D and $G'_{ST}$, not only $G_{ST}$.** Within-population diversity is high ($H_S$ = 0.729), so $G_{ST}$ = 0.033 is capped far below 1 and understates how distinct the gene pools are; Hedrick's $G'_{ST}$ = 0.166 and Jost's D = 0.137 correct for that ceiling. Quote θ for the demographic reading (gene flow, $N_em$), and D or $G'_{ST}$ for how different the allele pools actually are.

Within populations, $\Phi_{IS}$ = 0.127 (P = 0.001) is the AMOVA's inbreeding coefficient. It equals Weir & Cockerham's f for all populations together (0.127, table below), the pooled counterpart of the per-population values of Block 4.

Of 3 population pairs, **all 3** are significantly differentiated after Holm's correction. The most distinct pair is **Conico × Chalqueno** (θ ($F_{ST}$) = 0.064, Nm ≈ 3.6); the least, **Conico × Palomero** (0.035). Nm is Wright's island-model conversion and assumes equilibrium, so treat it as an order of magnitude: below 1 migrant per generation, drift wins and populations diverge. With 199 permutations per pair the smallest possible P is 0.005 before correction.

**Table 11.** Analysis of molecular variance

| Source of variation | d.f. | Sum of squares | Mean square | Variance component | % of total | Φ statistic | P (permutation) |
|---|---|---|---|---|---|---|---|
| Among populations | 2 | 20.957 | 10.4783 | 0.1492 | 4.88 | $\Phi_{ST}$ = 0.049 | 0.0010 |
| Among individuals within populations | 72 | 226.358 | 3.1439 | 0.3688 | 12.05 | $\Phi_{IS}$ = 0.127 | 0.0010 |
| Within individuals | 75 | 183.500 | 2.4467 | 2.5428 | 83.08 | $\Phi_{IT}$ = 0.169 | |
| **Total** | **149** | **430.815** | — | **3.0608** | **100.00** | | |

75 individuals in 3 populations, 8 loci, 999 permutations. Distances between gene copies count the loci at which the two alleles differ (the $F_{ST}$-type AMOVA of Excoffier et al. 1992); $\Phi_{ST}$, $\Phi_{IS}$ and $\Phi_{IT}$ equal Weir & Cockerham's θ, f and F. Components are computed locus by locus and summed, so missing genotypes only drop out of the loci where they occur. A negative component means less variation than expected at that level and is read as zero.

**Figure 11.7.** The Block 5 section: key figures, interpretation and the AMOVA table with its note. Tables and figures are numbered automatically throughout the report; this is table 11.

## Methods

The data set (maize_ssr_populations.csv) comprised 75 individuals from 3 populations scored for 8 loci (codominant genotypes, ploidy 2). Data were screened for missing genotypes, monomorphic loci and repeated multilocus genotypes before analysis (3.8% missing data, 0 monomorphic loci, 0 repeated genotypes). Genetic diversity was described per population by the number of alleles (Na), effective number of alleles (Ne = 1/Σp²), Shannon's information index (I), observed and expected heterozygosity (Ho, He; Nei 1973) and unbiased He (Nei 1978), the fixation index F = (He − Ho)/He, the polymorphic information content (Botstein et al. 1980), the percentage of polymorphic loci (95% criterion), and allelic richness rarefied to 44 gene copies (Hurlbert 1971; El Mousadik & Petit 1996); private and locally common alleles were listed, and 95% intervals were obtained by 1000 bootstrap replicates over loci. Departures from Hardy–Weinberg equilibrium were tested per locus and population with an exact test conditional on allele counts (Guo & Thompson 1992; 1000 Monte Carlo permutations of gene copies), with one-sided versions for heterozygote deficit and excess (Rousset & Raymond 1995) and a chi-square test for comparison; P-values were corrected for multiple testing by Holm's sequential Bonferroni and combined across loci and populations with Fisher's method. Inbreeding coefficients were estimated as Weir & Cockerham's (1984) f, multilocus as ratios of summed variance components with jackknife standard errors and 1 000 bootstrap replicates over loci, and as Nei & Chesser's (1983) $F_{IS}$; the equilibrium selfing rate was derived as s = 2F/(1 + F). Null allele frequencies were estimated with the EM algorithm of Dempster et al. (1977; Chapuis & Estoup 2007), and with the estimators of Brookfield (1996) and Chakraborty et al. (1992). Multilocus linkage disequilibrium was measured with the index of association and its standardised form $\bar{r}_d$ (Agapow & Burt 2001; 999 permutations of loci among individuals), and between locus pairs by pairwise $\bar{r}_d$. Population differentiation was estimated with Weir & Cockerham's (1984) θ, f and F, Nei's (1973) $G_{ST}$ with the sample-size corrections of Nei & Chesser (1983), Hedrick's (2005) $G'_{ST}$ and Jost's (2008) D; multilocus θ used ratios of summed variance components, multilocus $G_{ST}$, $G'_{ST}$ and D were computed from the mean $H_S$ and $H_T$ over loci, and 95% intervals came from 1000 bootstrap replicates over loci. The distribution of molecular variance among populations, individuals within populations and within individuals was analysed by AMOVA (Excoffier, Smouse & Quattro 1992) on the number of allelic differences between gene copies, with 999 permutations of the appropriate units for each Φ statistic, and pairwise θ between populations was tested with 199 permutations (Holm-corrected). Nm was derived from θ under Wright's island model as an indicative value only. Genetic distances between populations were computed as the Nei (1978) unbiased genetic distance; distances between individuals used the Smouse & Peakall (1999) genotypic distance. Relationships were displayed by principal coordinates analysis (Gower 1966, on the squared distances as they are) and by neighbour-joining (Saitou & Nei 1987) and UPGMA trees of populations, with node support from 1000 bootstrap resamplings of the loci (Felsenstein 1985). Recent bottlenecks were tested by the heterozygosity-excess method of Cornuet & Luikart (1996) under the TPM and SMM mutation models (TPM with 95% single-step mutations and variance 12), obtaining the equilibrium heterozygosity by 300 coalescent simulations per locus conditioned on the observed number of alleles (for n(2 − F) independent gene copies when $F_{IS} > 0$), and evaluated with the sign and standardized-differences tests and with a Wilcoxon signed-rank test on the normal scores of each locus within its equilibrium distribution (Piry et al. 1999), together with the mode-shift indicator (Luikart et al. 1998) and the M-ratio (Garza & Williamson 2001); effective population size was estimated by the linkage-disequilibrium method (Hill 1981) with the bias corrections of Waples (2006) and the composite disequilibrium of Weir (1979), excluding alleles below 0.02, with jackknife confidence limits over loci, and by the heterozygote-excess method (Pudovkin et al. 1996). All computations were carried out in PopGeneticsPro version 1.0.1 (Barrera-Guzmán, 2026; https://doi.org/10.5281/zenodo.22740171), a browser-based platform for plant population genetics; every stochastic procedure used a fixed random seed reported in the settings of each block.

## How to cite

If this analysis is published, please cite the software and the original method papers named in the Methods section:

Barrera-Guzmán, L. Á. (2026). *PopGeneticsPro: a browser-based platform for plant population genetics with codominant, dominant, sequence and morphological data* (Version 1.0.1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.22740171

Source code: https://github.com/luisangelbg/PopGeneticsPro · online version: https://luisangelbg.github.io/PopGeneticsPro/ · license GPL-3.0-or-later.

BibTeX:

```
@software{barrera_guzman_popgeneticspro_2026,
  author  = {Barrera-Guzmán, Luis Ángel},
  title   = {PopGeneticsPro: a browser-based platform for plant population genetics with codominant, dominant, sequence and morphological data},
  year    = {2026},
  version = {1.0.1},
  doi     = {10.5281/zenodo.22740171},
  url     = {https://doi.org/10.5281/zenodo.22740171}
}
```

B10

**Figure 11.8.** The methods paragraph, with the settings that were used (1000 permutations per exact test, Holm correction, 999 AMOVA permutations, 300 simulations per locus…), and the section for citing the program with its BibTeX entry.

**NOTE · VERIFICATION OF THE OUTPUTS**

For this manual, the ZIP package was read with an independent reader of the format: its 68 test files had correct headers and directory and a CRC-32 identical to the one recalculated bit by bit, and its 27 PNG files opened without errors. The TIFF of the AMOVA figure had its tags in order, 150 dpi on both axes and the same pixels as the equivalent PNG, without a single difference. The PNG carried 5906 dots per metre (150 dpi). The editor redrew the 27 maize figures with five combinations of theme, typeface, font sizes from 0.6 to 2.2, flipped axes and widths from 520 to 1600 px: 135 figures without errors and without any text outside the canvas.

**NOTE · CORRECTIONS MADE TO THE APP WHILE PREPARING THIS CHAPTER**

The transparent PNG background came out white, because the figure paints its own background; it now stays transparent. The JPG did not store the resolution, and programs read it at 72 or 96 dpi; it now carries it. Block 10 downloads used the previous preview even if the title or the sections had been changed; the report is now rebuilt for every output. The ZIP did not include the results of Block 9 or of the dating analysis; it now contains their tables, trees and the MCMC log. The printed report numbered the table of contents twice, cut off the BibTeX entry, split the key-figure boxes and separated titles from their tables; and symbols such as $F_{ST}$ appeared with an underscore in the titles of tables, figures and key figures.

This chapter concludes the tour of the blocks. The appendices gather the file formats, the decision rules from the whole manual, a glossary, troubleshooting, the references and the third-party components.

# A–F

APPENDICES · REFERENCE MATERIAL

# Appendices

The appendices are not meant to be read from start to finish but to be consulted. Here you will find the exact rules by which the app reads and writes each file, the manual's 43 decision rules gathered in one place, a glossary of terms, abbreviations and symbols, solutions to the most common problems, the full references for the methods, and the third-party components with their licences.


**A** File formats

**B** Summary of decision rules

**C** Glossary

**D** Troubleshooting

**E** References

**F** Third-party components


## A File formats

Chapter 3 explains how to lay out your data (tables 3.1 and 3.2). This appendix collects the exact rules: how the app recognises each file, which column names it understands, what counts as missing and what each downloaded file contains.

## A.1 How the app recognises a file

**Table A.1.** File types loaded in Block 2 and how the app tells them apart.

| File | How it is recognised | What you should know |
|---|---|---|
| **Spreadsheet** `.xlsx .xls .xlsm .xlsb .ods` | By its extension. | The first sheet is read; if there are several, Sheet appears so you can choose another. Numbers are stored as numbers, so a decimal comma causes no problems. |
| **Delimited text** `.csv .txt .tsv` | Anything that is not one of the other types. | Separator: whichever of comma, semicolon and tab is most frequent in the first 20 lines (the comma if they tie). Spaces do not separate columns. UTF-8 encoding, with or without a byte-order mark. Double quotes are accepted. |
| **FASTA** `.fasta .fas .fa` | The text begins with `>`, whatever the extension. | Sequences on one or several lines per sample. Alignment is not required, but different lengths are flagged. |
| **VCF** | A `##fileformat=VCF` line at the top. | Diploid SNP genotypes; see table A.7. |
| **Project file .arp** | A `[Profile]` section with `NbSamples`, `SampleData` or `GenotypicData`. | Genotypic or haplotypic data by sample. |
| **Population file .gen** | A line that says only `POP`, and at least one comma in the text. | Diploids only. |
| **Genotype file .str** | **Only by the extension** `.str` or `.stru`. | The same file saved as `.txt` is read as a table: change its extension. |

## A.2 Tables: one row per individual

**Table A.2.** How each type of data is written in a table and when the app recognises it on its own. If it gets it wrong, correct Marker type and How genotypes are written (section 3.4).

| Data | How it is written | When it is recognised automatically |
|---|---|---|
| **Codominant,** one allele per column | Two columns per locus (three or four for polyploids). The name goes in the first; the second repeats the name, is left empty or ends in `_1/_2`, `a/b` or `A/B`. Sizes in bp or codes. | When the name pairs cover at least 80% of the data columns. |
| **Codominant,** genotype in one cell | One column per locus: `182/186`, `182\|186`, `182:186`, `182;186`, `182,186` or with a space. Also `182186` (2 or 3 digits per allele), `94-98` and `AG`. | When more than 60% of the cells look like `x/y`, `x\|y`, `x:y`, `NN-NN` or two letters. The other forms are read correctly, but you have to choose Whole genotype in one cell. |
| **Dominant** (bands) | `1` present and `0` absent; also `p/a`, `present/absent`, `+/-`, `yes/no`. `-1` is always missing. | When more than 90% of the data columns contain only 0 and 1. A column with `-1` no longer looks binary: use empty cells or choose the type by hand. |
| **Haploid** | One code per column (chloroplast haplotype, size of a cpSSR). | When more than 60% of the data columns contain text. With numeric codes, choose the type by hand. |

**Table A.2.** How each type of data is written in a table and when the app recognises it on its own. If it gets it wrong, correct Marker type and How genotypes are written (section 3.4). *(continued)*

| Data | How it is written | When it is recognised automatically |
|---|---|---|
| **Sequences in cells** | One column per gene region, with the complete sequence (20 letters or more). IUPAC letters, `-` for gaps; U is converted to T and a full stop to a gap. | When more than 60% of the cells are sequences. The regions are concatenated in column order. |
| **Morphological** | One column per character. Numbers with a decimal point or comma (`13.5` or `13,5`); class characters may be written as text. | When more than half of the columns are continuous measurements (more than 12 distinct values). A file with 12 rows or fewer must be typed by hand. |

**MISSING VALUES**

The following are always missing: an empty cell, `-9`, `-99`, `-999`, `NA`, `N/A`, `NaN`, `?`, `.`, `..`, `null`, `missing` and `x`, in upper or lower case. So is the value of Code used for missing data: `0` by default for codominant data. With haploid data and characters, 0 is a value. A genotype with only one of its alleles becomes missing and is flagged.

**TYPES OF MORPHOLOGICAL CHARACTER**

Each column is classified automatically: **binary** (only 0 and 1), **quantitative** (mostly numbers, more than 12 distinct values), **discrete** (mostly numbers, 12 values or fewer) or **nominal** (mostly text). In a numeric column, text that is not a number becomes missing and is flagged.

## A.3 Label and coordinate columns

**Table A.3.** Column names that the app assigns to each role on its own. They can be changed in the column roles card (section 3.4).

| Role | Names it recognises |
|---|---|
| **ID** | ID, Ind, Individual, Unit, Sample, Plant, Tree, Accession, Genotype, Code, OTU, Taxon, Name, Clave, Individuo, Muestra, Planta, Unidad, or the name of the chosen unit (Landrace, Accession…). |
| **Population** | Pop, Population, Población, Site, Locality, Location, Deme, Race, Raza, Landrace, Provenance, Origin, Sitio, Localidad, Procedencia. |
| **Region** | Region, Región, Zone, Zona, Group, State, Estado, Area, Área, Cluster, Subregion, Watershed, Province, Provincia, País, Country. |
| **Latitude** | Lat, Latitude, Latitud, Y_coord, Northing. |
| **Longitude** | Lon, Long, Longitude, Longitud, X_coord, Easting. |

What counts is the **whole name**, in the plural or with a unit or note: `Populations`, `Latitude (°)`, `Lat_N`. A name that merely begins with the word, such as `Population_origin`, counts as population or region if the column holds text; that is why Spanish trait names such as `Longitud_mazorca` (ear length), `Area_foliar` (leaf area) or `Estado_fenologico` (phenological stage) remain characters. Without an ID column, the app numbers the rows `Ind1, Ind2…`; without a population column, each row is its own unit.

**Table A.4.** How coordinates are read.

| If the coordinates are… | the app… | Units of the results |
|---|---|---|
| **Decimal degrees** (south and west negative) | Computes the great-circle distance (Earth radius 6,371 km). A value outside ±90° or ±180° becomes missing and is flagged. | Kilometres; in the spatial structure analysis of Block 9, metres if all pairs are less than 5 km apart. |
| **Projected** (UTM in metres) | Recognises them when most values do not fit in degrees, and uses the Euclidean distance. | Map units (metres with UTM). |
| **Per population** (all plants share the same ones) | Valid for isolation by distance between populations, but not for fine-scale spatial structure, which requires one coordinate per plant. | — |

**IMPORTANT · DETAILS THAT CAUSE CONFUSION WHEN LOADING A TABLE**

- **Leading zeros.** In a CSV, an ID `001` is read as the number 1 and no longer matches the `001` in a FASTA file or in the sample sheet. Use IDs with letters ( `P001` ) or an `.xlsx` with those cells formatted as text.
- **Decimal comma in a comma-separated CSV.** An unquoted `19,43` splits the row into two columns. Save the CSV with semicolons or use a decimal point.
- **Population names.** They are compared exactly: `Tlaxco` and `tlaxco` are two different populations.
- `Group` means region in a table, but population in the sample sheet (table A.8).

## A.4 The template row

The templates from the designer (section 3.3) carry in row 1 the cell `#PopGeneticsPro template` followed by `key=value` pairs, one per cell. Row 2 holds the column names and the data start in row 3. With this row the app guesses nothing: it reads the declared layout and compares it with what it found.

**Table A.5.** Keys of the template row. Values are case-sensitive; keys are not.

| Key | Values | Purpose |
|---|---|---|
| `v` | `1` | Template version. |
| `kind` | `codominant` · `dominant` · `haploid` · `sequence` · `morph` | Data type (SNPs are `codominant` ). |
| `ploidy` | Integer (codominant data only). | Alleles per locus. |
| `cells` | `columns` · `combined` | One allele per column, or genotype in one cell. |
| `coding` | `size` · `nucleotide` · `code` | With `size` , an allele that is not a number becomes missing. |
| `missing` | Any code; empty if missing data are empty cells. | Missing-data code. |
| `unit` | `individuals` · `landraces` · `accessions` · `varieties` · `populations` · `clones` · `genotypes` · `samples` · `OTUs` | What the rows are called throughout the app. |
| `sampling` | `pops` · `bulk` | Plants per population, or one bulk sample per unit. |

**Table A.5.** Keys of the template row. Values are case-sensitive; keys are not. *(continued)*

| Key | Values | Purpose |
|---|---|---|
| `orientation` | `rows-are-individuals` · `rows-are-loci` | Bands in rows (dominant data only). |
| `loci` · `units` · `pops` | Integers. | Only for checking what was read. |
| `sizes` · `popnames` | `8\|8\|8` · `SanJuan\|ElMirador\|LaCumbre` | Planned plants and name of each population, in order. |
| `id` · `pop` · `region` · `lat` · `lon` | Exact column name. | Which column holds each label. |

**Example.** CSV template with 3 loci and 2 populations (3 and 2 plants); row 1 is a single line.

```
#PopGeneticsPro template,v=1,kind=codominant,ploidy=2,cells=columns,coding=size,missing=0,unit=indi
viduals,sampling=pops,orientation=rows-are-individuals,loci=3,units=5,pops=2,sizes=3|2,popnames=Con
ico|Chalqueno,id=Individual,pop=Population
Individual,Population,phi024,phi024,phi053,phi053,umc1147,umc1147
Con01,Conico,122,122,0,0,0,0
Con02,Conico,126,118,160,154,96,96
Con03,Conico,124,118,152,156,102,102
Cha01,Chalqueno,126,126,164,164,96,96
Cha02,Chalqueno,126,120,160,164,96,96
```

## A.5 Layout with parameter rows

**Table A.6.** What goes in each cell of the layout with parameter rows (figure 3.7). The app recognises it when the numbers in row 1 are consistent.

| Cell | Content |
|---|---|
| **A1 · B1 · C1** | Number of loci, of individuals and of populations. |
| **D1 onwards** | Plants in each population (as many cells as populations); they must add up to B1. |
| **Next cell** (optional) | Number of regions and, to its right, the plants in each region; these also add up to B1. |
| **Row 2** | Title in A2; population names below their sizes, and region names below theirs. |
| **Row 3** | Headers: individual, population and the loci. With codominant data, two columns per locus with the name only in the first. |
| **Row 4 onwards** | Data, with the populations in consecutive blocks. `Lat` and `Long` may follow after an empty column. |

The data columns run from column C to the first empty column; divided by A1 they give the columns per locus: two means codominant; one means dominant if more than 90% are 0/1, and haploid otherwise. With dominant data, the default missing code is `-1`. This layout exists only for diploid codominant, dominant and haploid data with populations.

**Example.** 3 loci, 5 individuals in 2 populations (3 and 2 plants). For one region, write `3,5,2,3,2,1,5` in row 1 and its name in G2.

```
3,5,2,3,2
Maize SSR subset,,,Conico,Chalqueno
Sample,Pop,phi024,,phi053,,umc1147,
Con01,Conico,122,122,0,0,0,0
Con02,Conico,126,118,160,154,96,96
Con03,Conico,124,118,152,156,102,102
Cha01,Chalqueno,126,126,164,164,96,96
Cha02,Chalqueno,126,120,160,164,96,96
```

## A.6 Files with their own format

**Table A.7.** What the app reads from each special format. None of them carries regions or coordinates, and some lack populations too: these are supplied with the sample sheet (table A.8).

| Format | Structure it reads | Missing data and limits |
|---|---|---|
| **FASTA** | Header `>ID \| Population \| Region` (fields may also be separated by two or more spaces). `>ID_pop` is not split. Identical sequences form one haplotype. | `-` gap; `N` and `?` unknown. Block 8 discards sites with a gap or an ambiguous base (section 9.1); dating uses them as partial states. Repeated IDs are not flagged. |
| **VCF** | Tab-separated columns; samples from column 10 onwards. Locus: the ID or `CHROM_POS`. The GT field is read: `0` is REF and `1` the first ALT; `/` and `\|` are equivalent. | `./.` missing. An allele 2, 3... or a partial call (`0/.`) makes the genotype missing. QUAL, FILTER and INFO are not used: filter beforehand. No populations. |
| **.str** | Optional line with the marker names; column 1 ID, column 2 population; then one allele per marker. Two consecutive rows per individual = diploid; one row = haploid. | Only `-9` is missing (0 is an allele). IDs without spaces. A row with two columns per locus is not accepted. |
| **.gen** | Line 1, title; locus names up to the first `POP`; each individual: `name , codes`. Codes of 4 digits (2 per allele) or 6 (3 per allele). | `0000` or `000000` missing. Populations are named `Pop1, Pop2…` (rename them with the sample sheet). Diploids only. |
| **.arp** | `GenotypicData=1` diploid and `=0` haploid (without the key, diploid). Samples in `SampleName="…"` and `SampleData={ … }`; in genotypic data, the second allele is on the next line. | `?` missing. The frequency column is ignored: each line is one individual. Sequences enter as codes, not as an alignment. |

**Example.** VCF with two sites; in the file the columns are separated by a tab (here they are aligned with spaces for readability). Con02 at the second site (`0/2`) becomes missing because it uses the second alternative allele.

```
##fileformat=VCFv4.2
#CHROM  POS    ID        REF  ALT  QUAL  FILTER  INFO  FORMAT  Con01    Con02   Con03
1       10234  S1_10234  A    G    .     PASS    .     GT:DP   0/1:12   1|1:8   ./.:0
1       55871  .         C    T,A  .     PASS    .     GT      0/0      0/2     1/1
```

## A.7 Sample sheet

**Table A.8.** Columns of the sample sheet (Choose a sample sheet), which supplements a FASTA, a VCF or a .str, .gen or .arp file. The first sheet of an `.xlsx`, `.xls` or `.ods` workbook is read, or a CSV.

| Column | Names it recognises | Note |
|---|---|---|
| **ID** (required) | ID, Ind, Individual, Sample, Plant, Tree, Accession, Genotype, Landrace, Name, Seq, Sequence, OTU, Clave, Individuo, Muestra, Planta. | Compared ignoring case and leading or trailing spaces. |
| **Population** | Pop, Population, Población, Site, Locality, **Group**, Deme, Localidad, Sitio. | Only cells with a value replace what the file contained. Coordinates accept a decimal comma. |
| **Region** | Region, Zone, State, Estado, Area, Province, Country, País. | |
| **Latitude · Longitude** | Lat, Latitude, Latitud, Y · Lon, Long, Longitude, Longitud, X. | |

Download an empty sample sheet with these IDs writes `<file>_samples.csv` with the columns `ID, Population, Region, Latitude, Longitude` and the IDs already filled in.

## A.8 Files produced by the app

**Table A.9.** Format of each downloadable file. Figures, the report and the ZIP package are described in sections 11.2 and 11.3.

| File | Content and format |
|---|---|
| **Results tables** ( `.csv` ) | Comma-separated, with a decimal point, in UTF-8 with a byte-order mark and Windows line endings. Values that do not exist are left empty. The name is that of the data file plus a suffix: `_diversity_summary`, `_hwe_fis_null`, `_pairwise_differentiation`, `_pcoa_scores`, `_assignment`, `_haplotype_table`, `_demography` … |
| **Data sheet** | Save as: `.xlsx` or `.ods` (sheet `Data`), `.csv`, `_semicolon.csv` or `.tsv`, with the values exactly as they were entered. |
| **Templates** | `.xlsx` with the sheets `Data`, `Instructions` and `Design`; `.csv` with the template row; `_parameter_rows.csv`; `.fasta` (`>ID \| Population \| Region`, 60 bases per line); `_samples.csv`; `_instructions.txt`. Export in another layout writes the loaded data in these same layouts. |
| **Block 6 trees** ( `.nwk` ) | Newick with branch lengths to 6 decimal places and bootstrap support as a whole percentage after each closing parenthesis. Spaces and punctuation in names are replaced by `_`. |
| **Dated trees** ( `.nwk`, `.nex` ) | `_ml_tree.nwk`: lengths in substitutions per site. `_dated_lsd` and `_dated_mcmc`: chronogram with lengths in millions of years; names containing punctuation go in single quotes. The NEXUS file adds a `TRANSLATE` table and annotates each node as `[&height=…,height_95%_HPD={a,b},posterior=…,calibration="…"]`. With MCMC, `height` is the median and the interval is the 95% HPD; with LSD, the estimate and the 2.5 and 97.5 percentiles of the resampling. |

**Table A.9.** Format of each downloadable file. Figures, the report and the ZIP package are described in sections 11.2 and 11.3. *(continued)*

| File | Content and format |
|---|---|
| **Node ages** ( `.csv` ) | One row per internal node: `Clade` (tips separated by `\|`), `Tips`, `Age (Ma)` or `Median age (Ma)`, `Lower 95%`, `Upper 95%`, `ESS` (MCMC only), `Calibration` and `Epoch` (epoch or period of the geological time scale). |
| **MCMC log** ( `_mcmc.log` ) | Tab-separated text with an initial comment line. Columns: `state`, `posterior`, `log_likelihood`, `prior`, `clock_rate`; `rate_sigma` with a relaxed clock, `kappa` with K80 or HKY, `alpha` with Γ; `birth_rate` (Yule) or `theta` (coalescent); `root_age` and one `height(clade)` column per internal node. Only samples after the burn-in, about 2,000 rows. |

# B Summary of decision rules

The manual's 43 decision rules, in the order of the app. Each box gives the section where the theory behind it is explained; read them as guides to interpretation, not as laws: the biological context of your species always comes first.

Look for your question in the index first. The rule from section 2.2 on how much differentiation is a lot is repeated, in fuller form, in section 6.4, and appears here only once.

## 1 Introduction to PopGeneticsPro

| | | | |
|---|---|---|---|
| 1.8 | How many plants per population? | 1.9 | Reading a P value |

## B1 Home, theory and simulator

| | | | |
|---|---|---|---|
| 2.3 | Does selection or drift win? | 2.4 | More loci or more individuals? |

## B2 Data and quality control

| | | | |
|---|---|---|---|
| 3.3 | The design checks | 3.4 | How the app works out the marker type |
| 3.5 | Missing data | 3.5 | Monomorphic loci and extreme bands |
| 3.5 | Repeated genotypes | 3.7 | Reading the coefficient of variation of a trait |

## B3 Allele frequencies and diversity

| | | | |
|---|---|---|---|
| 4.3 | Is diversity high or low? | 4.3 | Do two populations differ in diversity? |
| 4.3 | Fixation index F | 4.3 | PIC of a marker |
| 4.3 | Private alleles and locally common alleles | 4.6 | Which estimator to use with bands? |
| 4.6 | Phenotypic H′ | | |

## B4 Hardy–Weinberg and linkage disequilibrium

| | | | |
|---|---|---|---|
| 5.1 | Which Hardy–Weinberg test to read | 5.2 | Inbreeding, Wahlund or null alleles? |
| 5.3 | Reading $\bar{r}_d$ | 5.4 | Which correction to use |

## B5 Differentiation and AMOVA

| | | | |
|---|---|---|---|
| 6.4 | How much differentiation is a lot? | 6.4 | Which estimator should you report? |
| 6.4 | Reading an AMOVA with regions | 6.4 | Population pairs and migrants |

## B6 Distances, ordination and trees

| | | | |
|---|---|---|---|
| 7.2 | Reading a PCoA | 7.3 | Support and fit of the tree |
| 7.5 | Isolation by distance | | |

## B7 Bayesian clustering and assignment

| | | | |
|---|---|---|---|
| 8.2 | Choosing K | 8.3 | Reading a DAPC |
| 8.4 | Assignment and migrants | | |

**B8 DNA sequences and molecular dating**

9.2 Neutrality and mismatch
9.3 Phylogeographic structure
9.4 Reading a haplotype network
9.5 Calibrating and reading a dated tree

**B9 Demography and spatial structure**

10.1 Was there a bottleneck?
10.2 Reading an $N_e$
10.3 Spatial genetic structure
10.4 How much selfing?
10.5 $P_{ST}$ versus $F_{ST}$

**B10 Figures and report**

11.1 Preparing a figure
11.2 Which format and resolution
11.3 Which output to use

## 1 Introduction to PopGeneticsPro

**DECISION RULE · HOW MANY PLANTS PER POPULATION?** SECTION 1.8

| Size per population | Interpretation | What to do |
|---|---|---|
| **25 to 30 or more** | Adequate | Sufficient to estimate allele frequencies reliably with microsatellites (Hale et al., 2012). All analyses are dependable. |
| **10 to 24** | Acceptable, with caution | Compare diversity using **rarefied allelic richness** rather than the raw number of alleles, and report confidence intervals. |
| **5 to 9** | Limited | Rare alleles are lost and $F_{IS}$ values are unstable. Use such samples for description, not for precise comparisons between populations. |
| **Fewer than 5** | Insufficient | The app flags this in quality control. Consider pooling neighbouring populations or restricting the analysis to distances and ordination. |

With dominant markers, more plants and more polymorphic bands are advisable, because each band carries less information than a microsatellite.

**DECISION RULE · READING A P VALUE** SECTION 1.9

| P value | Evidence against the null hypothesis | How to word it |
|---|---|---|
| **P < 0.001** | Very strong | "The departure is highly significant". |
| **0.001 ≤ P < 0.01** | Strong | "The departure is significant". |
| **0.01 ≤ P < 0.05** | Moderate | "There is evidence of a departure"; confirm it with other loci or populations. |
| **P ≥ 0.05** | Insufficient | "No departure was detected". Do not claim that the population *is* in equilibrium: the test may have lacked statistical power. |

Always report the exact P value and the effect size (for example, $F_{IS}$ with its interval), not just whether the result is significant.

## B1 Home, theory and simulator

**DECISION RULE · DOES SELECTION OR DRIFT WIN?** SECTION 2.3

| $4 N_e \lvert s \rvert$ | Which dominates | What to expect |
|---|---|---|
| **Less than 1** | Drift | The allele behaves almost as if it were neutral; it can be lost even though it is advantageous. Example: $N_e$ = 10 and $s$ = 0.01 give 0.4. |
| **1 to 10** | Both | Selection steers the change, but chance produces variable outcomes among populations. |
| **More than 10** | Selection | The change is almost deterministic. Example: $N_e$ = 100 and $s$ = 0.1 give 40. |

Practical consequence: in small populations, even adaptive variants can be lost by chance. Maintaining large effective sizes also maintains the capacity to adapt.

**DECISION RULE · MORE LOCI OR MORE INDIVIDUALS?** SECTION 2.4

| If your objective is... | ... invest in |
|---|---|
| **Estimating differentiation** ($F_{ST}$, AMOVA, distances) | **More loci.** Each locus is an independent replicate of the history of the populations, and the variance of $F_{ST}$ decreases faster when you add loci than when you add individuals. |
| **Finding rare or private alleles** | **More individuals.** An allele at a frequency of 2% has only a 64% probability of appearing in 25 diploid plants. |
| **Detecting kinship or spatial structure** | **Both**, with mapped individuals and highly polymorphic markers. |

## B2 Data and quality control

**DECISION RULE · THE DESIGN CHECKS** SECTION 3.3

| If the design has... | the designer warns that... |
|---|---|
| **Fewer than 5 plants** in a population | this is not enough for any within-population test; 20 to 30 are needed to estimate frequencies well (Hale et al., 2012). |
| **20 or more plants** per population | this is enough for allele frequencies, $H_e$, $F_{IS}$ and Hardy–Weinberg tests. |
| **Fewer than 5 polymorphic loci** | multilocus analyses (Bayesian clustering, assignment, $\bar{r}_d$) are not reliable; clustering is usually run with 10 or more. |
| **Few populations per region** | the number of possible permutations limits the smallest P value of $\Phi_{CT}$ in the AMOVA (Fitzpatrick, 2009). With 2 + 2 populations there are only 3 distinct arrangements and P never falls below 0.33. |
| **10 or more plants** per population (codominant) | $N_e$ from linkage disequilibrium and the bottleneck tests of Block 9 become available; their precision improves greatly with 30 or more plants and 10 or more polymorphic loci. |

**DECISION RULE · HOW THE APP WORKS OUT THE MARKER TYPE** SECTION 3.4

| # | If it finds... | then the data are... |
|---|---|---|
| 1 | row 1 of a PopGeneticsPro template | the type declared in it |
| 2 | a parameter row with two columns per locus | codominant (with one column per locus: dominant or haploid) |
| 3 | more than 90% of the data columns containing only 0 and 1 | dominant |
| 4 | columns with complete DNA sequences | sequences |
| 5 | columns with genotypes in one cell (182/186, A/G) | codominant, genotype in one cell |
| 6 | column names that come in pairs | codominant, one allele per column |
| 7 | most columns with continuous measurements | morphological |
| 8 | most columns with a text code | haploid |
| 9 | none of the above | codominant (check it) |

Only the data columns are examined: a latitude or ID column does not turn a matrix of 0s and 1s into measurements. The orientation is worked out first: a matrix of 0s and 1s whose first column repeats names, or with many more rows than columns, is read with the bands in rows.

**DECISION RULE · MISSING DATA** SECTION 3.5

| Missing data | Reading | What to do |
|---|---|---|
| **Less than 5%** | Very good | Carry on without changes. |
| **5 to 10%** | Acceptable | Carry on, but check whether the missing data are concentrated in one locus or one population. |
| **More than 10%** | Worrying | Remove the loci or individuals with the most missing data rather than letting the gaps drive the result. |
| **An individual with more than 25%** | Weak individual | The app names it. Genotype it again, or exclude it if its absence does not leave the population too small. |
| **A locus with far more missing data than the rest** | Problematic locus | It may have null alleles or amplify poorly. Check it in Block 4 as well. |

**DECISION RULE · MONOMORPHIC LOCI AND EXTREME BANDS** SECTION 3.5

| If a locus or band... | then... |
|---|---|
| **Is monomorphic** (a single allele, or a band present in all or in none) | It contributes nothing to differentiation or distances. The app keeps it and warns you; it counts towards the percentage of polymorphic loci (%P). |
| **Is a dominant band with a frequency below 5% or above 95%** | Its frequency is poorly estimated from a gel. It is common practice to remove such bands before analysis; the app counts them in a warning. |
| **Has an allele found in a single plant only** | It may be real (a rare allele) or a scoring error. Check the electropherogram before reporting it as a private allele. |

**DECISION RULE · REPEATED GENOTYPES** SECTION 3.5

| If you find... | the most likely explanation is... | what to do |
|---|---|---|
| **Identical pairs in a clonal species** | ramets of the same genet | keep one per genotype |
| **Identical pairs in a species without vegetative reproduction** | the same plant sampled or entered twice, or too few loci | check the labels and the field sheet |
| **Pairs that match at 95% or more without being identical** | close relatives, or the same plant with a scoring error | check the loci that differ |
| **Many identical pairs with few loci** | a lack of power to tell individuals apart | add loci before concluding that there is clonality |

With sequences, sharing a haplotype is normal, and with traits there is no multilocus genotype: in both cases this check does not apply.

**DECISION RULE · READING THE COEFFICIENT OF VARIATION OF A TRAIT** SECTION 3.7

| CV | Variation | In the example |
|---|---|---|
| **Less than 10%** | Low | Days to flowering, 9.8%: a very stable trait across plants. |
| **10 to 30%** | Moderate | Plant height (17.3%), ear length (25.9%) and ear diameter (25.8%). |
| **More than 30%** | High | Cob weight (41.6%) and 100-kernel weight (39.3%): good candidates for differentiating populations or for selection. |

An indicative guide: the expected CV depends on the trait and the environment. A very high CV may also reveal data-entry errors, such as a weight in grams entered as kilograms; check the minimum and maximum.

## B3 Allele frequencies and diversity

**DECISION RULE · IS DIVERSITY HIGH OR LOW?** SECTION 4.3

| Marker | Typical $H_e$ | Interpretation |
|---|---|---|
| **Microsatellites** in outcrossing plants | 0.45 to 0.85 | Below 0.45, low diversity (bottleneck, isolation, selfing); above 0.85, very high. |
| **Microsatellites** in selfing plants | below 0.45 | Low values are normal; compare with other species with a similar mating system. |
| **AFLP, ISSR, RAPD** | 0.15 to 0.40 | The theoretical maximum is 0.5, because each band has only two states. Never compare $H_e$ from dominant markers with $H_e$ from microsatellites. |
| Chloroplast **haplotypes** or sequences | highly variable | Depends on the marker and on history. Report it together with the number of haplotypes and the nucleotide diversity from Block 8. |

These are indicative ranges taken from the plant literature: they help you place a number, not judge a dataset. What is really informative is the comparison among your own populations with the same marker.

**DECISION RULE · DO TWO POPULATIONS DIFFER IN DIVERSITY?** SECTION 4.3

| If… | then… |
|---|---|
| **The 95% intervals of $H_e$ do not overlap** | The difference is supported by the loci sampled. |
| **The intervals overlap** | The difference is not resolved: treat the populations as equally diverse. The app says so explicitly. |
| **There are fewer than three loci** | Loci cannot be resampled and intervals cannot be calculated: the difference is not tested (the case of a single sequence alignment). |
| **Sample sizes differ by more than a factor of two** among populations | Do not compare $N_a$ or the number of private alleles; use $uH_e$ and $A_r$. The app warns you. |

The intervals are obtained by *bootstrap* over loci: loci are resampled with replacement 1000 times and the 2.5 and 97.5 percentiles of the mean are taken. They reflect how much the result would change with other loci, not with other plants.

### FIXATION INDEX F

SECTION 4.3

| F | Interpretation |
|---|---|
| **Greater than 0.1** | Clear heterozygote deficit. |
| **−0.05 to 0.1** | Close to equilibrium. |
| **Less than −0.05** | Heterozygote excess. |

F is a description. Whether there is inbreeding, null alleles or a Wahlund effect is decided in Block 4 with $F_{IS}$ and its tests.

### PIC OF A MARKER

SECTION 4.3

| PIC | Interpretation |
|---|---|
| **Greater than 0.5** | Highly informative. |
| **0.25 to 0.5** | Moderately informative. |
| **Less than 0.25** | Slightly informative. |

Categories of Botstein et al. (1980). Useful for choosing loci for variety identification or paternity testing.

### DECISION RULE · PRIVATE ALLELES AND LOCALLY COMMON ALLELES

SECTION 4.3

| If you find… | the interpretation is… |
|---|---|
| **A private allele with frequency ≥ 5%** (*frequent*) | Evidence of a distinct gene pool: restricted gene flow or an independent history. A priority for conservation. |
| **A rare private allele** (*rare*, < 5%) | It may be real or simply a sampling accident; with more plants it might turn up in other populations. |
| **A locally common allele**, ≥ 5% in 25% or fewer of the populations | Strong signal of restricted gene flow. |
| **A locally common allele**, ≥ 5% in half or fewer of the populations | Moderate signal. The app reports both thresholds when there are four or more populations. |

### DECISION RULE · WHICH ESTIMATOR TO USE WITH BANDS?

SECTION 4.6

| If your species… | use… |
|---|---|
| **Is outcrossing** and you have 20 or more plants per population | Lynch and Milligan. With large samples it coincides with the square root; with few plants it corrects the square root's bias. |
| **Is outcrossing** but some bands are absent from almost every plant | Bayesian with F = 0: it does not leave frequencies at 0 or 1 when a band is missing from the whole sample. |
| **Is selfing** or has known inbreeding | Bayesian with F between 0.8 and 0.95: without this correction the frequency of the recessive allele is underestimated. |

Whichever method you choose, report it: $H_e$ from dominant markers depends on it. Comparisons among populations hardly change, as table 4.6 shows.

**DECISION RULE · PHENOTYPIC H'** SECTION 4.6

| H' | Diversity | Interpretation |
|---|---|---|
| **0.6 or more** | High | The plants cover almost the whole range of the trait. Useful material as a source of variation. |
| **0.4 to 0.6** | Medium | Moderate variation, common within a landrace or local variety. |
| **Less than 0.4** | Low | Trait uniform in that population: fixed by selection or with a narrow genetic base. |

An indicative guide used in studies of local varieties. H' mixes genetic and environmental variation: to find out whether differences among populations exceed what drift would produce, $P_{ST}$ is compared with $F_{ST}$ in Block 9.

## B4 Hardy–Weinberg and linkage disequilibrium

**DECISION RULE · WHICH HARDY–WEINBERG TEST TO READ** SECTION 5.1

| If... | then... |
|---|---|
| **You use microsatellites or SNPs with rare alleles** | Read the exact test. The app flags unreliable $\chi^2$ values with †. |
| **You want to know the direction** of the departure | Use the one-tailed tests: one-sided: heterozygote deficit or excess (Rousset and Raymond, 1995). They are more powerful than the global test when you already suspect inbreeding. |
| **The exact P is close to α** | Raise the permutations to 10,000. With 1000, a P of 0.05 has an error of ±0.007; with 10,000, of ±0.002. |
| **You are comparing with an older paper** | Show $\chi^2$ for reference only and explain that the conclusion rests on the exact test. |

A Monte Carlo test never gives P = 0: with B permutations and b tables as unusual as the observed one, P = (b + 1)/(B + 1). The smallest possible value with 1000 permutations is 0.001.

**DECISION RULE · INBREEDING, WAHLUND OR NULL ALLELES?** SECTION 5.2

| Pattern | Likely cause | What to do |
|---|---|---|
| **Positive $F_{IS}$ at most loci** and an interval for *f* that excludes 0 | Inbreeding | Report *f* with its interval and the selfing rate *s* for reference. Keep the loci. |
| **One locus with a significant deficit in half or more of the populations** and a null frequency above 5% | Null allele | The app flags the locus as `suspected null allele`. Recompute $F_{ST}$ without it in Block 5 and compare. |
| **General deficit and linkage disequilibrium within the population** | Wahlund or selfing | Look for subgroups with Block 7. If they appear, the "population" mixes gene pools. |
| **Both patterns at once** | Inbreeding plus a null | Estimate *f* from the unflagged loci and report both results. |
| **Scattered departures**, with no repeated locus or population | Chance or scoring errors | This does not justify removing loci. Check the gels for the extreme cases. |

The automatic diagnosis in card 2 applies this rule. A significant excess of heterozygotes (*f* with an interval below 0) suggests clonality, a recent cross between different sources or, in small samples, mere chance.

**DECISION RULE · READING $\bar{r}_d$** SECTION 5.3

| If... | then... |
|---|---|
| **$\bar{r}_d$ ≈ 0 and not significant** in every population | The loci behave as independent. The Block 7 methods, which assume this, can be used without reservation. |
| **$\bar{r}_d$ significant and below 0.05** | Detectable but weak association; it does not bias analyses that assume independence. |
| **$\bar{r}_d$ significant and above 0.05** within populations | Selfing, clonality or recent admixture. Read Bayesian clustering and assignments with caution. |
| **Significant only in the pooled sample** | A sign of differentiation between populations, not of physical linkage. |
| **One particular pair stands out** in several populations | Possible physical linkage between those two loci. Consider using only one of them in analyses that require independence. |

The 0.05 threshold is a guide: in outcrossing plants with unlinked markers, $\bar{r}_d$ usually stays below 0.01; in clonal or highly selfing species, values of 0.1 to 0.5 are common.

**DECISION RULE · WHICH CORRECTION TO USE** SECTION 5.4

| If... | use... |
|---|---|
| **You have dozens of tests** and every rejection matters (HWE per locus) | Holm, the default option. |
| **You have hundreds or thousands of tests** (pairs of loci with dominant markers) | Benjamini–Hochberg: Holm would be so strict that it would detect nothing. |
| **You are interested in the population as a whole**, not each locus | Fisher's method, in addition to the per-locus correction. |
| **No individual test survives** but there are many small P values | Examine the pattern (do they recur at one locus or in one population?) before concluding that nothing is going on. |

## B5 Differentiation and AMOVA

**DECISION RULE · HOW MUCH DIFFERENTIATION IS A LOT?** SECTIONS 6.4 AND 2.2

| $F_{ST}$ or θ | Differentiation | Interpretation |
|---|---|---|
| **Below 0.05** | Low | The populations share almost all their frequencies; abundant gene flow or recent separation. |
| **0.05 to 0.15** | Moderate | Clear but shallow structure. Common in outcrossing plants with nuclear markers. |
| **0.15 to 0.25** | High | Well-differentiated populations; little exchange or a separate history. |
| **Above 0.25** | Very high | Common in selfing species, isolated populations and chloroplast markers. |

Wright's (1978) scale. It was designed for $F_{ST}$ with markers that have few alleles: with microsatellites, whose $H_S$ is high, a θ of 0.05 can conceal quite different populations (figure 6.2), so read it together with $G'_{ST}$ or D.

**DECISION RULE · WHICH ESTIMATOR SHOULD YOU REPORT?** SECTION 6.4

| If your question is... | report... |
|---|---|
| **How much structure there is** and how it relates to drift and migration | θ ($F_{ST}$) with its interval, and $\Phi_{ST}$ or $\Phi_{PT}$ from AMOVA with its P. |
| **How different the allele pools are**, for example for conservation | Jost's D or $G'_{ST}$, in addition to θ. |
| **Comparison with another study or another marker** | $G'_{ST}$ or D, which do not depend on the $H_S$ ceiling, and the same kind of locus average used in the other study. |
| **Comparison with an older study** that only gave $G_{ST}$ | $G_{ST}$ as a reference, noting its ceiling. |

The general recommendation is not to report a single estimator: θ plus D or $G'_{ST}$ covers both interpretations (Meirmans and Hedrick, 2011).

**DECISION RULE · READING AN AMOVA WITH REGIONS** SECTION 6.4

| If… | then… |
|---|---|
| **$\Phi_{CT}$ is significant** and the region component is larger than the population component | The grouping into regions is real and captures most of the structure. |
| **$\Phi_{CT}$ and $\Phi_{SC}$ are significant**, with populations weighing more | The regions differ, but within each one the populations differ even more. |
| **$\Phi_{SC}$ is significant and $\Phi_{CT}$ is not** | The populations differ, but they do not group according to the regions you defined. Try a grouping suggested by the data (Block 7). |
| **Few populations** (table 6.3) | The test of $\Phi_{CT}$ cannot reach significance: read the size of the component. |
| **None is significant** | At this scale, the species behaves almost as a single large population. |

**DECISION RULE · POPULATION PAIRS AND MIGRANTS** SECTION 6.4

| If… | then… |
|---|---|
| **The corrected P of a pair is below 0.05** | The two populations differ in frequencies; treat them as distinct units. |
| **Nm is above 4** | Gene flow is sufficient to keep them similar. |
| **Nm is between 1 and 4** | Moderate exchange; drift leaves a mark but does not separate them completely. |
| **Nm is below 1** | Drift dominates and the populations diverge; conserving each one is a priority. |
| **All corrected P values are equal** and small | This is the minimum possible with those permutations (with 199 and 3 pairs, 0.005 × 3 = 0.015). Increase the permutations to tell them apart. |

The Nm thresholds are indicative and inherit all the limitations of the island model.

## B6 Distances, ordination and trees

**DECISION RULE · READING A PCOA** SECTION 7.2

| If… | then… |
|---|---|
| **Axes 1 and 2 add up to more than 50%** | The map is a good summary: closeness on the plane reflects the matrix. |
| **They add up to less than 25%** | This is usual with individuals and highly variable markers. The map shows trends; two nearby points may differ on other axes. Check axis 3 as well. |
| **The population ellipses overlap a great deal** | Low differentiation, consistent with a small $F_{ST}$ (chapter 6). A significant AMOVA is not a contradiction. |
| **An individual falls within the cloud of another population** | Possible migrant, labelling error or admixture; check it with the assignment in Block 7. |
| **There are many large negative eigenvalues** | Turn on the Lingoes correction before reporting percentages. |

**DECISION RULE · SUPPORT AND FIT OF THE TREE** SECTION 7.3

| If... | then... |
|---|---|
| **Support of 70% or more** | The grouping is robust to the loci sampled; it can be described. |
| **Support of 50 to 70%** | Weak grouping; mention it as a trend. |
| **Support below 50%** | Do not interpret the branching order. The app hides it by default. |
| **Only three populations** | An unrooted three-tip tree has no internal branches to test: it only shows lengths. The app says so. |
| **UPGMA cophenetic correlation greater than 0.85** | The tree reproduces the distances well. If it is lower, the tree distorts them; prefer NJ and the PCoA. |

With few loci, low support is normal and does not mean the tree is wrong: it means the data are not enough to resolve it. Zero-length branches (identical units) receive no support, because the order in which they are joined is arbitrary.

**DECISION RULE · ISOLATION BY DISTANCE** SECTION 7.5

| If... | then... |
|---|---|
| **Positive slope and significant Mantel test** between populations | Isolation by distance: gene flow decreases with distance. |
| **Not significant** | Distance does not predict differentiation at this scale: extensive gene flow, structure due to another cause (habitat, altitude, history) or too few populations. |
| **Few populations** (6 or fewer) | With $n$ populations there are only $n!$ ways to reorder them; with 4, the smallest possible P is 1/24 ≈ 0.04. Read the slope and the figure, not the P value. |
| **Mantel significant only between individuals** | Pairs from the same population are close together and alike, so the correlation mainly reflects the differences between populations. Fine-scale spatial structure is studied in Block 9. |

## B7 Bayesian clustering and assignment

**DECISION RULE · CHOOSING K** — SECTION 8.2

| If... | then... |
|---|---|
| **ΔK, Ln P(D) and Puechmaille agree** | That K is well supported. Confirm it in the bar plot. |
| **ΔK gives a smaller K than Ln P(D)** | ΔK marks the uppermost level of structure; the K from Ln P(D) may be the subdivision within it. Choose the largest K at which every cluster has individuals with high Q and the replicates agree. |
| **Ln P(D) is highest at K = 1** | The ΔK peak is not evidence of clusters on its own, because ΔK never chooses K = 1. Accept K = 2 only if there are individuals with Q ≥ 0.8 in both clusters. |
| **H′ falls below 0.90** | Replicates are in different solutions: lengthen the chains, increase the replicates and do not interpret the consensus plot for that K. |
| **A Puechmaille estimator counts a cluster whose mean Q does not reach 0.6** in any population | The count is fragile and the app warns about it: confirm it with the bar plot and with the individuals with Q ≥ 0.8. |

**DECISION RULE · READING A DAPC** — SECTION 8.3

| If... | then... |
|---|---|
| **Leave-one-out success clearly exceeds chance** | The groups are genetically distinguishable. Report that figure, not the reassignment. |
| **Reassignment far exceeds leave-one-out success** | Part of the success is fit to the sample. Try several numbers of components and keep the one beyond which leave-one-out success stops improving (table 8.3). |
| **One group has low success** (below 70%) | Its individuals are confused with other groups; see which ones in the scatter plot. |
| **Groups found by k-means** | Success is high by construction. Look for a minimum or an elbow in the BIC curve; if it is still falling at the largest K tested, there is no optimum and the groups are merely a convenient partition. |

**DECISION RULE · ASSIGNMENT AND MIGRANTS** SECTION 8.4

| If... | then... |
|---|---|
| **Self-assignment is high** | The populations are distinguishable. Compare it with chance (1/number of populations) and with the $F_{ST}$ of chapter 6: it increases with differentiation and with the number and variability of the loci (Cornuet et al., 1999). |
| **Two populations exchange many individuals** in the matrix | They are the least differentiated pair. This is not proof of current gene flow. |
| **Few individuals flagged as migrants** | With N individuals and $\alpha = 0.01$, about $N \times 0.01$ are expected by chance. If the number flagged does not greatly exceed that value, treat them as candidates for review, not as proven migrants. |
| **A misassigned individual with a high P** | It is a resident that resembles another population, not a migrant. |
| **Populations with fewer than 5 individuals** | They are excluded from the test: their frequencies are not reliable. |

The test assumes that all possible source populations are in the data. A migrant from an unsampled population appears as an unusual individual in all of them.

## B8 DNA sequences and molecular dating

**DECISION RULE · NEUTRALITY AND MISMATCH** SECTION 9.2

| If... | then... |
|---|---|
| **D, Fs and $R_2$ are not significant** | Nothing compels you to propose changes in size: the data are compatible with a stable population. |
| **Fs and $R_2$ are significant** (with or without a negative D) | Evidence of expansion. Look at the mismatch: if r is also small, read τ as the date of the expansion. |
| **D is negative and significant, but Fs and $R_2$ are not** | An excess of rare variants that may come from a selective sweep just as well as from growth. |
| **D is positive and significant** | Contraction, balancing selection or, most commonly in plants, a sample that mixes lineages or differentiated populations. Repeat the tests population by population. |
| **The mismatch is ragged** (r not significant) | There is no evidence of expansion; τ is not a date. |

Selection and demography leave similar signatures at a single locus. Only several independent loci showing the same signal allow it to be attributed to demography.

**DECISION RULE · PHYLOGEOGRAPHIC STRUCTURE** SECTION 9.3

| If... | then... |
|---|---|
| **$N_{ST}$ significantly greater than $G_{ST}$** | Phylogeographic structure: related lineages share a region. Look in the haplotype network to see which lineages they are. |
| **$N_{ST} \approx G_{ST}$**, both high | Populations differentiated in frequencies, with no geographic ordering of lineages: recent isolation or weakly divergent haplotypes. |
| **Six haplotypes or fewer** | Few ways to shuffle them: the P value cannot fall very low. The app warns you; read the difference $N_{ST} - G_{ST}$. |
| **Organelle DNA** | Compare it with the nuclear $F_{ST}$ of chapter 6. DNA dispersed by seed is usually far more structured than nuclear DNA; in most conifers the chloroplast is dispersed by pollen. |

**DECISION RULE · READING A HAPLOTYPE NETWORK** SECTION 9.4

| If the network shows... | then... |
|---|---|
| **A star**: a frequent central haplotype and many rare derived haplotypes one step away | The picture of a recent expansion from that haplotype. |
| **Groups of haplotypes separated by many mutations** | Old lineages; if each group occupies a region, phylogeographic structure (compare with $N_{ST}$). |
| **Many loops** | Homoplasy (the same change in two lineages), recombination or intermediate haplotypes that were not sampled. |
| **Median vectors** (small black nodes) | Inferred haplotypes: extinct or unsampled. They are not observations. |

The area of each circle is proportional to the number of sequences, the sectors show the populations and each tick on a line is a mutation.

**DECISION RULE · CALIBRATING AND READING A DATED TREE** SECTION 9.5

| If... | then... |
|---|---|
| **There are only minimum ages** | The dates may drift towards older ages: give a maximum to some calibration, a secondary calibration or a maximum age for the root. The app warns you. |
| **Any ESS is below 200** | The chain has not explored the posterior distribution well: lengthen it or rerun it with another seed before reporting. |
| **The interval for σ comes close to zero** | The data are compatible with a strict clock. |
| **The posterior of a calibration resembles its prior** | The sequences say little about that age. Compare it with the Sample from the prior only option. |
| **Data from a single species** | Use a rate and the coalescent model. The ages are coalescence times of the haplotypes, older than the split of the populations that carry them. |

## B9 Demography and spatial structure

**DECISION RULE · WAS THERE A BOTTLENECK?** SECTION 10.1

| If... | then... |
|---|---|
| **Wilcoxon gives P < 0.05 for excess under both TPM and SMM** | There is a signal of a recent reduction. Confirm it with another sample or with the history of the population. |
| **Only the IAM gives an excess** | With microsatellites this is not enough: the IAM exaggerates the excess. |
| **The excess comes from one or two loci** | Check those loci: a null allele or selection affects a single locus, demography affects them all. |
| **There is a significant deficit** | The population may have grown or received immigrants carrying rare alleles. |
| **M < 0.68 without an excess** | An older or longer reduction is compatible; the excess has already faded. |
| **There are fewer than 10 polymorphic loci or fewer than 20 plants** | The test has low power: "no signal" does not prove stability (Luikart and Cornuet, 1998). |

The signal is transient: once the population reaches a new mutation–drift equilibrium, the excess disappears even if the population remains small.

**DECISION RULE · READING AN $N_e$** SECTION 10.2

| If... | then... |
|---|---|
| **The estimate or its upper limit is ∞** | The $r^2$ cannot be distinguished from sampling noise: $N_e$ is large relative to the sample, not unknown. Report the lower limit. |
| **The limits fall below 50 or below 100** | The population is losing diversity rapidly; this is a conservation warning. |
| **The population self-fertilises or has a high $F_{IS}$** | Inbreeding correlated across loci inflates $r^2$ and lowers $N_e$; interpret it as a minimum. |
| **The sample mixes subpopulations or immigrants** | Admixture also creates disequilibrium: check the structure (Block 7) before reporting. |
| **There are fewer than 25 plants or few loci** | The interval will be very wide; more loci help as much as more plants. |

The method assumes random mating, unlinked loci and a closed population, and estimates $N_e$ per generation.

**DECISION RULE · SPATIAL GENETIC STRUCTURE** SECTION 10.3

| If... | then... |
|---|---|
| **The first class lies above the band and $b_F$ is significant** | Neighbours are relatives. Report Sp and the distance up to which the signal persists. |
| **Sp is around 0.01** | Weak structure, typical of outcrossing species (mean Sp 0.013; Vekemans and Hardy, 2004). |
| **Sp is around 0.04 or more** | Strong structure, as in species with mixed mating (0.037) or selfing species, or with seed that falls nearby. |
| **No class falls outside the band** | Pollen and seed move far at this scale, or the sample lacks nearby plants. |
| **You sampled few plants or only distant plants** | About 100 plants or more are recommended, with many pairs of close neighbours. |
| **There are clones** | Ramets of the same genet inflate $F_1$: keep one ramet per genet. |

**DECISION RULE · HOW MUCH SELFING?** SECTION 10.4

| If... | then... |
|---|---|
| **$F_{IS}$ is positive at almost all loci** | The deficit is genome-wide: selfing or matings between relatives. s is a reasonable estimate. |
| **The deficit comes from a few loci** | Suspect null alleles (Block 4) before selfing. |
| **There is spatial structure** | Matings between related neighbours also produce $F_{IS} > 0$; s overestimates selfing. |
| **You mixed plants from different groups** | The Wahlund effect creates a deficit without inbreeding. |
| **You need the actual rate for one season** | Genotype mothers and their seeds and use a mixed-mating model (Ritland, 2002). |

**DECISION RULE · $P_{ST}$ VERSUS $F_{ST}$** SECTION 10.5

| If... | then... |
|---|---|
| **The whole $P_{ST}$ interval lies above $F_{ST}$** | The trait has diverged more than drift would cause: divergent selection, if $c/h^2$ is realistic. |
| **The whole interval lies below $F_{ST}$** | Stabilising selection: the populations are more alike than expected. |
| **The interval crosses $F_{ST}$** | There is no evidence against drift. |
| **The conclusion changes when $c/h^2$ is lowered to 0.5** | It depends on the assumption: say so, or measure the trait in a common garden. |
| **You have fewer than 10 populations** | Expect wide intervals; the comparison has low power. |

$F_{ST}$ must come from neutral markers in the same populations ($\theta$ from Block 5).

## B10 Figures and report

| ✓ DECISION RULE · PREPARING A FIGURE | SECTION 11.1 |
|---|---|
| **If…** | **then…** |
| **The figure is going into a paper** | Journal or Minimal theme, with the same typeface and sizes in every figure. |
| **You distinguish groups by colour** | Okabe and Ito or Tol: they avoid confusing red and green. In heat maps, Viridis or Cividis. |
| **It will be printed in black and white** | Greyscale, plus values or shapes that do not depend on colour. |
| **The category names are long** | `Flip axes`, or `Category labels` at 45°. |
| **The journal reduces the figure to one column** | Lower `Width` and raise `Font size: all` before exporting (section 11.2). |

| ✓ DECISION RULE · WHICH FORMAT AND RESOLUTION | SECTION 11.2 |
|---|---|
| **If…** | **then…** |
| **The journal requires TIFF or PNG** | Check the final width and the resolution it demands; calculate the pixels needed and choose the factor that reaches them. |
| **The figure has thin lines and text** | PNG or TIFF at 600 dpi, or SVG; not JPG. |
| **You are going to edit it afterwards** | SVG: every text and colour can be changed. |
| **It is for a presentation** | PNG at 150 or 300 dpi, with a transparent background if the slide is coloured. |
| **The journal changes the resolution of the file** | It does not matter: what preserves quality is the number of pixels. |

| ✓ DECISION RULE · WHICH OUTPUT TO USE | SECTION 11.3 |
|---|---|
| **If…** | **then…** |
| **You are writing the paper or thesis** | Copy the methods paragraph and adapt it; rebuild the tables from the CSV files; export the figures following the rule in section 11.2. |
| **You are sharing the analysis with co-authors** | The HTML report: a single file that opens in any browser, with vector figures. |
| **You are preparing the supplementary material** | The ZIP: data, tables, trees and figures. |
| **You want a fixed copy** | `Print / save as PDF` and the ZIP, with the seeds recorded. |

# C Glossary

The terms, abbreviations and symbols used in the manual, with a brief definition and the section where each is explained at length. Below many terms is their Spanish name, as it appears in the Spanish edition of this manual and in the Spanish-language literature.

## C.1 Abbreviations

**Table C.1.** Abbreviations used in the manual and in the app. File formats are described in appendix A.

| Abbreviation | Meaning | Abbreviation | Meaning |
|---|---|---|---|
| **AFLP** | Amplified fragment length polymorphism | **IUPAC** | Ambiguous base codes of the International Union of Pure and Applied Chemistry (R = A or G, Y = C or T, N = any) |
| **AICc** | Akaike information criterion corrected for small samples | **JC** | Jukes and Cantor substitution model |
| **AMOVA** | Analysis of molecular variance | **K80, K2P** | Kimura two-parameter substitution model |
| **BIC** | Bayesian information criterion | **LD** | Linkage disequilibrium |
| **bp** | Base pairs | **LSD** | *Least-squares dating* of a tree |
| **CI** | Confidence interval | **Ma** | Millions of years |
| **cpDNA** | Chloroplast DNA | **MCMC** | Markov chain Monte Carlo |
| **CV** | Coefficient of variation | **mtDNA** | Mitochondrial DNA |
| **DAPC** | Discriminant analysis of principal components | **NJ** | *Neighbour-joining*, a tree-building method |
| **d.f.** | Degrees of freedom | **OTU** | Operational taxonomic unit: each tip of a tree |
| **dpi** | Dots per inch | **PCA** | Principal component analysis |
| **EM** | Expectation–maximisation algorithm | **PCoA** | Principal coordinates analysis |
| **ESS** | Effective sample size of an MCMC chain | **PIC** | Polymorphism information content |
| **FDR** | False discovery rate | **RAPD** | Random amplified polymorphic DNA |
| **GTR** | General time-reversible substitution model | **SD · SE** | Standard deviation · standard error |
| **HKY** | Hasegawa, Kishino and Yano substitution model | **SMM** | Stepwise mutation model |
| **HPD** | Highest posterior density interval | **SNP** | Single nucleotide polymorphism |
| **HWE** | Hardy–Weinberg equilibrium | **SRAP** | Sequence-related amplified polymorphism |
| **IAM** | Infinite alleles model | **SSR** | Simple sequence repeats: microsatellites |
| **ISSR** | Inter-simple sequence repeats | **TPM** | Two-phase mutation model |
| **ITS** | Internal transcribed spacers of ribosomal DNA | **UPGMA** | Unweighted pair group method with arithmetic mean |

## C.2 Terms

**Table C.2.** Glossary in alphabetical order.

| Term | Definition | Section |
|---|---|---|
| **A** | | |
| Allele<br>alelo | Each variant that exists at a locus. A diploid plant carries two alleles per locus. | 1.3 |
| Allele frequency | The proportion of the gene copies in a population that belong to a given allele. It is the basic unit of population genetics. | 1.3 |
| Allelic richness ($A_r$) | The number of alleles expected with the same number of gene copies in every population (rarefaction). An exact calculation, not a random draw. | 4.1 |
| AMOVA<br>análisis de varianza molecular | Partitions the squared genetic distances among the levels of the sampling design (regions, populations, individuals). Each level yields a variance component and a Φ. | 6.2 |
| Assignment test<br>asignación | Computes the probability of each individual's genotype in each population, assuming Hardy–Weinberg, and assigns it to the most probable one. | 8.4 |
| **B** | | |
| Bootstrap | Resampling with replacement (of loci, or of sites with sequences) to obtain intervals or support values. It shows how much the result would change with other loci, not with other plants. | 4.3 |
| Bootstrap support | Percentage of replicates in which a grouping of the tree reappears. It measures how much the grouping depends on a few loci, not the probability that it is true. | 7.3 |
| Bottleneck<br>cuello de botella | A sharp, recent reduction in the size of a population. Rare alleles are lost first, leaving a transient heterozygosity excess. | 10.1 |
| Bulk sample<br>muestra compuesta | A single sample per unit (for example, per landrace). It does not allow within-population analyses. | 3.3 |
| Burn-in | Initial iterations of an MCMC chain that are discarded while it stabilises. | 8.1 |
| **C** | | |
| Calibration<br>calibración | A known age that sets the clock of a tree: a fossil (minimum age of a node) or a published age (secondary calibration). | 9.5 |
| Coalescent<br>coalescente | A model that traces the genealogies of a sample backwards in time to their common ancestor. The app uses it to simulate the P values of the neutrality and bottleneck tests. | 9.2 |
| Codominant<br>marcador codominante | A marker in which both alleles of each individual are scored (SSR, SNP, isozymes). It allows all the equilibrium and inbreeding tests. | 1.4 |
| Complete deletion<br>deleción completa | A site with a gap or an ambiguous base in any sequence is excluded from all diversity, neutrality and network analyses. | 9.1 |
| Confidence interval | The range of values compatible with the data. If it includes values with opposite readings, the data cannot distinguish between them. | 1.9 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| Cophenetic correlation | Correlation between the original distances and those reproduced by a tree. Above 0.85, the tree represents them well. | 7.3 |
| D | | |
| $D_A$, net<br>divergencia nucleotídica neta | Mean divergence between sequences from two populations minus the divergence within each. The default distance with sequences. | 7.1 |
| DAPC | Summarises the genotypes with principal components and finds the axes that best separate the groups. It does not assume Hardy–Weinberg; its probabilities measure the certainty of the classification, not ancestry. | 8.3 |
| Delta K (ΔK) | How abrupt the change in Ln P(D) is between consecutive values of K. It finds the uppermost level of structure; it cannot choose K = 1 and needs two or more replicates. | 8.2 |
| Dominant<br>marcador dominante | A marker that records only the presence or absence of a band (AFLP, ISSR, RAPD); it does not distinguish heterozygotes. Frequencies are estimated assuming Hardy–Weinberg. | 1.4 |
| E | | |
| Effective number of alleles ($N_e$)<br>alelos efectivos | $1/\Sigma p_i^2$: how many equally frequent alleles would give the same diversity. Not to be confused with effective population size. | 4.2 |
| ESS | The number of independent samples to which an MCMC chain is equivalent. Below 200, lengthen the chain or repeat it with another seed. | 9.5 |
| F | | |
| F (fixation index) | $(H_e - H_o)/H_e$: positive with a heterozygote deficit and negative with an excess. It is a description; the cause is decided in Block 4. | 4.2 |
| $F_{ij}$ (Loiselle's kinship) | Kinship coefficient between two plants with codominant markers: 0 between random plants, 0.25 between full sibs and 0.125 between half sibs. | 10.3 |
| $F_{IS}$ | Inbreeding coefficient: $1 - H_o/H_e$. It is 0 under random mating, positive when heterozygotes are lacking and negative when they are in excess. | 5.2 |
| $F_{ST}$ | The fraction of total diversity due to differences among populations: 0 if all have the same frequencies, 1 if each is fixed for a different allele. Weir and Cockerham's θ is reported. | 6.1 |
| First-generation migrant | An individual whose genotype is much less probable in its own population than in another; it is tested against simulated resident genotypes. | 8.4 |
| Fisher's method | Combines the P values of independent tests ($-2\ \Sigma \ln P$ against $\chi^2$) to ask whether a population as a whole departs from equilibrium. | 5.4 |
| Fu's Fs | Tests whether there are more haplotypes than expected for the observed diversity. Strongly negative: population growth; it is significant at $P < 0.02$. | 9.2 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| **G** | | |
| $G_{ST}$ and $G'_{ST}$ | $G_{ST}$: the fraction of diversity among populations, with a ceiling at 1 – $H_S$. $G'_{ST}$: $G_{ST}$ divided by its maximum possible value, comparable across markers. | 6.1 |
| Gene flow<br>flujo génico | Movement of genes between populations through pollen or seed. It increases diversity within populations and reduces differentiation. | 1.3 |
| Genetic distance | A number that summarises how different two individuals or two populations are. Each coefficient has its own scale: do not compare values from different coefficients. | 7.1 |
| Genetic drift<br>deriva génica | Random change in allele frequencies, stronger in small populations. It removes $1/(2N_e)$ of the heterozygosity per generation and differentiates populations. | 2.3 |
| Genotype | The alleles of an individual at a locus: homozygous if they are the same (152/152), heterozygous if they differ (152/158). | 1.3 |
| **H** | | |
| H′ (phenotypic) | Standardised Shannon index of a morphological character, from 0 (all plants in one class) to 1. | 4.6 |
| H′ (between replicates) | Similarity between the Q matrices of two clustering runs: 1 if they are identical; below 0.90, they reached different solutions. | 8.2 |
| $H_c$ (expected heterozygosity) | The probability that two gene copies drawn at random are different; also called gene diversity. $uH_e$ corrects it for sample size. | 4.1 |
| $H_o$ (observed heterozygosity) | The proportion of individuals counted as heterozygous. Codominant markers only. | 4.2 |
| $H_S$ and $H_T$ | Mean gene diversity within populations and in all of them pooled. | 6.1 |
| Haploid | Data with a single allele per individual: chloroplast or mitochondrial codes, inherited from one parent only. | 1.4 |
| Haplotype | Each distinct sequence, or each distinct combination of alleles in a haploid genome. | 9.1 |
| Haplotype diversity (Hd) | The probability that two sequences drawn at random are different haplotypes, corrected for sample size. It ranges from 0 to 1. | 9.1 |
| Hardy–Weinberg equilibrium | With random mating and no other forces, genotype proportions are $p^2$, 2pq and $q^2$. A deviation is informative: inbreeding, null alleles or admixture. | 5.1 |
| Hardy–Weinberg exact test | Keeps the allele counts fixed, distributes them at random among the plants and asks how unusual the observed table of genotypes is. | 5.1 |
| Heterozygosity excess ($H_{eq}$)<br>exceso de heterocigosis | The observed $H_e$ exceeds the value expected for its number of alleles in a population at equilibrium; a sign of a recent bottleneck. It is not the heterozygote excess ($H_o > H_e$). | 10.1 |
| Holm correction | Orders the P values and multiplies the smallest by the number of tests m, the next by m – 1, and so on. It controls false positives and never rejects fewer than Bonferroni. | 5.4 |
| HPD | The shortest interval containing 95% of the posterior distribution of an age. | 9.5 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| **I** | | |
| I | Two quantities: the Shannon index of allelic diversity (4.2) and Nei's genetic identity, whose negative logarithm is the distance (7.1). | 4.2 · 7.1 |
| $I_A$ and $\bar{r}_d$ | Index of association between loci and its standardised version. $\bar{r}_d$ does not depend on the number of loci and can be compared across studies. | 5.3 |
| Inbreeding<br>endogamia | Mating between relatives, or selfing. It raises $F_{IS}$ at almost all loci by a similar amount. | 5.2 |
| Island model | Populations of equal size that exchange migrants equally. It relates $F_{ST}$ to Nm, but no real population fulfils it. | 6.3 |
| Isolation by distance<br>aislamiento por distancia | Nearby populations exchange more migrants than distant ones and are more alike. It is tested with a Mantel test between $\theta/(1-\theta)$ and the logarithm of distance; a positive, significant slope indicates it. | 7.5 |
| **J · K** | | |
| *Jackknife* | Resampling that removes one locus at a time to estimate standard errors or intervals. | 6.1 |
| Jost's D | How distinct the sets of alleles of the populations are: 0 if identical, 1 if they share none. It says nothing about migration. | 6.1 |
| K | The number of genetic groups tested in Bayesian clustering or in k-means. | 8.1 |
| **L** | | |
| Leave-one-out<br>dejando uno fuera | Each individual is classified with an analysis built without it. This is the success percentage worth reporting from a DAPC. | 8.3 |
| Lingoes correction | Adds a constant to the squared distances to remove the negative eigenvalues of a PCoA; the percentages become exact, though somewhat smaller. | 7.2 |
| Linkage disequilibrium<br>desequilibrio de ligamiento | Non-random association between alleles at different loci. It is created by physical proximity, selfing, clonality, population admixture, drift and selection. | 5.3 |
| Ln P(D) | The likelihood of the data for a given K. It usually rises up to the best-supported K and then levels off or falls. | 8.2 |
| Locus (plural loci) | A specific place in the genome: a microsatellite, an AFLP band or a fragment of chloroplast DNA. | 1.3 |
| **M** | | |
| M ratio | The number of alleles divided by the range of sizes in repeats plus 1. Below 0.68 it suggests an old or prolonged reduction. | 10.1 |
| Mantel test | Correlation between two distance matrices (genetic and geographic), tested by permuting rows and columns. | 7.5 |
| MCMC | A method that explores a posterior distribution by taking random steps; the samples after the *burn-in* describe the parameters. | 8.1 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| Median-joining (network) | A haplotype network in which loops show alternative paths and median vectors are inferred haplotypes, extinct or unsampled. | 9.4 |
| *Mismatch* distribution | The frequencies of the number of differences between all pairs of sequences. A single peak suggests sudden expansion; several peaks, a stable size. | 9.2 |
| Mode shift<br>cambio de moda | A stable population has more rare alleles (0–0.1) than alleles in any other frequency class; if that class is no longer the largest, the distribution has been shifted by a reduction. | 10.1 |
| Molecular clock | Converts substitutions into time. Strict: one rate for all branches; relaxed: one rate per branch. | 9.5 |
| Mutation–drift equilibrium | The state in which the variation created by mutation balances the variation removed by drift. | 2.3 |
| **N · O** | | |
| $N_a$ | The number of distinct alleles observed. It increases with sample size. | 4.2 |
| Nb (neighbourhood) | $Nb \approx 1/Sp = 4\pi D\sigma^2$: the number of plants that act as a local reproductive unit. With the density D it gives the gene dispersal $\sigma^2$. | 10.3 |
| $N_c$ (effective size) | The number of plants in an ideal population that would lose diversity through drift at the same rate as the real one; it is usually around one tenth of the adults. | 10.2 |
| Neighbour-joining (NJ) | A tree-building method that allows branches of different lengths and produces an unrooted tree. | 7.3 |
| Neutrality tests | Compare the footprints in the sequences (Tajima's D, Fu's Fs, $R_2$) with what is expected in a neutral population of constant size. | 9.2 |
| Newick and NEXUS | Text formats for trees; NEXUS also accepts annotations such as ages and intervals (table A.9). | A.8 |
| Nm | $(1 - \theta)/(4\theta)$: the migrants per generation that would produce that θ under the island model. An order of magnitude only. | 6.3 |
| $N_{ST}$ | Differentiation that weights each pair of haplotypes by the mutations separating them. If it exceeds $G_{ST}$, there is phylogeographic structure. | 9.3 |
| Nucleotide diversity (π) | The mean number of differences per site between two sequences. It allows fragments of different lengths to be compared. | 9.1 |
| Null allele<br>alelo nulo | An allele that fails to amplify because of a mutation at the primer site: its heterozygotes are scored as homozygotes. It affects one locus in almost all populations. | 5.2 |
| Outcrossing and selfing<br>alógama, autógama | A plant that crosses with others, or one that fertilises itself. In outcrossing species, 80 to 95% of the variation lies within populations; in selfing species the opposite is usually true. | 2.4 |
| Outgroup<br>grupo externo | Tips known to lie outside the group of interest, which provide the root of a tree. Without one, the root is placed at the midpoint of the longest path. | 7.3 |
| **P** | | |
| P value | The probability of obtaining a result as extreme as the one observed if there were no effect. A $P \geq 0.05$ does not prove that there is no effect. | 1.9 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| PCoA | Turns a distance matrix into a map with a few axes; it works with any coefficient. | 7.2 |
| Permutation test | Shuffles the data to destroy the association and counts how many times the observed value is equalled or exceeded: P = (b + 1)/(B + 1), never 0. | 5.3 |
| $\boldsymbol{\Phi}_{ST}$, $\boldsymbol{\Phi}_{PT}$, $\boldsymbol{\Phi}_{CT}$ | Fractions of the molecular variation among populations ($\Phi_{PT}$ with bands) and among regions estimated by the AMOVA. | 6.2 |
| PIC | The ability of a marker to distinguish individuals: above 0.5, highly informative; below 0.25, slightly informative. | 4.3 |
| Polymorphic locus | A locus whose most common allele does not exceed 95% (default criterion). A monomorphic locus has a single allele and provides no information. | 4.2 |
| Private alleles<br>alelos privados | Alleles present in only one population (with dominant markers, exclusive bands). A frequent one (≥ 5%) suggests a distinct gene pool. | 4.2 |
| $P_{ST}$ and $Q_{ST}$ | Differentiation of quantitative characters. $Q_{ST}$ uses additive variances from a common garden; $P_{ST}$, phenotypic variances with assumed c and $h^2$. They are compared with $F_{ST}$. | 10.5 |
| **Q · R** | | |
| Q (membership) | The fraction of each individual's genome that comes from each group in Bayesian clustering. | 8.1 |
| $R_2$ | A neutrality test based on the unique mutations of each sequence; the most powerful with small samples. | 9.2 |
| Raggedness r<br>r de irregularidad | How jagged the *mismatch* distribution is; a smooth distribution suggests expansion. | 9.2 |
| Ramet and genet<br>rameto, geneto | In a clonal plant, each independent shoot is a ramet; all those that arose from the same original individual form a genet. | 3.5 |
| Random seed<br>semilla aleatoria | A number that fixes the randomness of permutations, resampling and chains: the same seed and the same data give the same results. | 1.6 |
| **S** | | |
| Sample sheet<br>hoja de muestras | A table with ID, population, region and coordinates that supplements a FASTA, a VCF or another file that lacks them. | 3.1 |
| Segregating sites (S) and η | S: sites with more than one base. η: the minimum number of mutations; equal to S if no site has three or four bases. | 9.1 |
| Selection | Change in frequencies because some genotypes leave more offspring. Divergent selection differentiates populations; stabilising selection keeps them alike. | 2.3 |
| Selfing (rate s)<br>autofecundación | The proportion of seeds produced by self-pollination. At equilibrium F = s/(2 − s), whence s = 2F/(1 + F); the outcrossing rate is t = 1 − s. | 10.4 |
| Sign, $T_2$ and Wilcoxon tests | Three ways of testing whether more loci show a heterozygosity excess than expected. Wilcoxon is the most powerful with few loci. | 10.1 |

**Table C.2.** Glossary in alphabetical order. *(continued)*

| Term | Definition | Section |
|---|---|---|
| Sp | $-b_F/(1 - F_1)$, where $b_F$ is the slope of kinship on the logarithm of distance and $F_1$ the kinship of the first class. The strength of spatial structure, comparable across studies. | 10.3 |
| Spatial autocorrelation r<br>autocorrelación espacial | Genetic correlation between plants in a distance class, computed from a squared genetic distance. It works with any marker; without structure it equals $-1/(n - 1)$, not 0. | 10.3 |
| Spatial genetic structure<br>estructura genética espacial | Relatedness between neighbouring plants caused by short-distance pollen and seed dispersal. It is measured as the decline of kinship with distance (correlogram). | 10.3 |
| Substitution model | Describes how bases change over time (JC, K80, HKY, GTR, with or without Γ rate variation among sites); the app chooses the one with the lowest AICc. | 9.5 |
| **T** | | |
| **τ** (tau) | Time since an expansion in mutational units: $\tau = 2ut$. | 9.2 |
| **θ** of Weir and Cockerham | An estimator of $F_{ST}$ that corrects for the size and number of samples; $\theta_B$ is its version for bands. | 6.1 |
| $\theta_W$ (Watterson) | Estimate of $\theta = 4N_e\mu$ from the number of segregating sites. | 9.1 |
| Tajima's D | Compares pairwise diversity with the diversity predicted by the number of segregating sites. Negative: excess of rare variants (expansion or purifying selection); positive: excess of intermediate variants. | 9.2 |
| Template<br>plantilla | A sheet written by the designer in Block 2 with the layout declared in row 1. | 3.3 |
| TPM, SMM and IAM | Mutation models for microsatellites: two-phase (mostly one-repeat steps), stepwise and infinite alleles. The TPM is the recommended one; the SMM, the most conservative. | 10.1 |
| **U · V · W** | | |
| UPGMA | A tree-building method that joins the closest pairs first and assumes the same rate on all branches: all tips end up at the same height. | 7.3 |
| Variance component | The part of the variation that corresponds to one level of the AMOVA. A negative one is not an error: it is interpreted as zero. | 6.2 |
| Vector format (SVG) | A figure saved as a drawing: it can be enlarged without losing sharpness, and every text or colour can be edited. | 11.2 |
| Wahlund effect | A heterozygote deficit that appears when plants from different gene pools are pooled in one sample. It raises $F_{IS}$ at many loci and creates linkage disequilibrium. | 5.2 |

## C.3 Symbols with more than one meaning

**Table C.3.** The literature uses the same letters for different quantities; the block and the section make clear which one is meant.

| Symbol | Meanings (section) |
|---|---|
| $N_c$ | Effective number of alleles (4.2) · effective population size (10.2). |
| $H'$ | Standardised phenotypic diversity (4.6) · similarity between clustering replicates (8.2). |
| $I$ | Shannon index (4.2) · Nei's genetic identity (7.1). |
| $D$ | Jost's D (6.1) · Nei's distance (7.1) · Tajima's D (9.2) · Pudovkin's heterozygote deviation (10.2) · plant density (10.3). |
| $\theta$ | Weir and Cockerham's $F_{ST}$ estimator (6.1) · population parameter $4N_e\mu$ (9.1). |
| $s$, $S$ | Selfing rate (10.4) · selection coefficient in the simulator (2.3) · segregating sites (9.1). |
| $\alpha$ | Significance level (1.9) · admixture parameter in clustering (8.1) · shape of the Γ rate variation among sites (9.5). |
| $\lambda$ | Prior on the frequencies in clustering (8.1) · likelihood ratio of a migrant (8.4) · variance ratio of $P_{ST}$ (10.5). |
| $k$ | Number of alleles (4.2) · mean pairwise differences between sequences (9.1) · number of clades of a cut tree (7.4). |
| $r$ | Spatial autocorrelation (10.3) · raggedness index (9.2) · substitution rate (9.5) · Mantel correlation (7.5). |
| $t$ | Time or generations (2.2, 9.5) · outcrossing rate (10.4). |
| $\sigma$ | Gene dispersal (10.3) · variation of rates in the relaxed clock (9.5). |
| $F$ | Fixation index (4.2) · Weir and Cockerham's $F_{IT}$ (6.1) · ratio in the analysis of variance (10.5). |

# D Troubleshooting

The most common problems, grouped by the moment at which they arise. The app's messages are quoted exactly as they appear on screen; `{n}` marks a part that varies.

## D.1 Opening the app and keeping your work

**Table D.1.** Problems when opening the app and when returning to it.

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| The page does not respond, or appears without colours or buttons. | Folders are missing (`js/`, `css/`, `vendor/`) or the browser is out of date: the app uses recent features and an old browser neither runs it nor shows an error. | Keep the complete folder and use an up-to-date browser. | 1.5 |
| `server.ps1` says it could not start on the port. | The port is in use, almost always by another open copy. | Close the other window or use the port suggested by the message (`-Port 9001`). Leave the PowerShell window open while you work. | 1.5 |
| The data sheet, the OTU images or the figure style "disappear". | They are stored in the browser, separately for each way of opening the app: what was saved after a double-click does not appear from `localhost`, nor in another browser or a private window. Clearing the site data deletes them. | Always open the app the same way. Save the sheet with Save as and keep the original image files. | 3.2 · 7.4 |
| The OTU images do not appear when the data are reloaded. | They are tied to the name of the data file; under another name, or from the built-in sheet, the app cannot find them. | Always load the same file under the same name. | 7.4 |
| The page freezes during the MCMC of Block 7 or of dating, and Cancel is slow to respond. | The browser did not allow the background thread, and the chain runs in the page. | Wait for the current run to finish; open the app with `server.ps1`. For exploration, use fewer iterations or replicates. | 8.1 |
| There is no way to stop a long calculation. | Only Block 7 clustering and the dating MCMC have Cancel. | While exploring, lower the permutations and resamples; raise them for the final result. | 1.6 |
| Seeds 0 and 1 give the same result. | An empty seed or 0 is taken as 1. | Use integers from 1 upwards and record them in your methods. | 1.6 |

## D.2 Loading the data

**Table D.2.** Problems when reading a file in Block 2.

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| `The file could not be read.` | The file is open in another program or stored in the cloud without being downloaded. | Close it or copy it to a folder on your computer. | — |
| Spreadsheet reader error when opening an `.xlsx`. | Password-protected or damaged workbook. | Remove the protection or save it again as `.xlsx` or CSV. | A.1 |
| Accented letters turned into "`�`" (`Poblaci◆n`). | CSV saved with an old encoding, not UTF-8. | Save it as CSV UTF-8, use `.xlsx` or paste the data into the built-in sheet. | A.1 |
| `This file looks like {format} but could not be parsed: …` | Incomplete structure in a FASTA, VCF, .str, .gen or .arp file. | Check it against table A.7. | A.6 |
| A genotype file with two rows per individual is read as a single-column table. | That format is recognised only with the extension `.str` or `.stru`. | Change the file extension. | A.1 |
| A row containing `19,43` is split into two columns. | Comma-separated CSV with a decimal comma. | Save it as a semicolon-separated CSV or use a decimal point. | A.3 |
| The IDs `001`, `002` in a CSV do not match those in the FASTA or in the sample sheet. | In delimited text, numbers lose their leading zeros. | Use IDs with letters (`P001`) or an `.xlsx` with those cells as text. | A.3 |
| `None of the IDs in {file} matches the names in the data file` or `{m} of {n} … matched`. | The IDs in the sample sheet are not those in the data. | Press Download an empty sample sheet with these IDs and fill in that sheet. | A.7 |
| Unexpected data were loaded from a workbook with several sheets. | The first sheet is read. | Choose the right sheet in Sheet. | 3.1 |
| Rows or columns are missing from a very large table pasted into the built-in sheet. | The sheet holds up to 20,000 rows and 5,000 columns; anything beyond that is discarded without warning. | Load the data from a file. | 3.2 |

## D.3 Interpretation, column roles and quality control

**Table D.3.** Problems when checking how the data were read.

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| The app infers a different marker type (`…because no clearer pattern was found`). | None of the detection rules was met; for example, bands with `-1` as missing, or genotypes written as `182186`. | Choose Marker type and How genotypes are written. | 3.4 |
| A character or a locus disappears from the list of data. | Its name matches that of a label and it took on that role. | In the column roles card, give it back the Data role. | A.3 |

**Table D.3.** Problems when checking how the data were read. *(continued)*

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| `no population column — each row will be treated as its own {unit}` | The population column was not detected. | Assign it the Population role. | 3.4 |
| `{n} further columns are treated as data.` | There are more than 10 columns and the hidden ones (notes, dates) count as data. | Tick Show every column and assign them Ignore this column. | 3.4 |
| `The dataset could not be built: No data columns were left after assigning roles.` | All columns ended up as labels or ignored. | Give the Data role back to the locus columns. | 3.4 |
| `coordinates out of range (latitude ±90°, longitude ±180°) were set to missing` | A typing error in coordinates in degrees, or latitude and longitude swapped. | Correct the value. With UTM, make sure the whole column is in metres. | A.3 |
| A population appears twice under almost identical names. | Names are compared exactly: `Tlaxco` and `tlaxco` are different. | Make the spelling consistent in the file. | 3.1 |
| `{n} value(s) in the sheet could not be read as intended` | Invalid cells: bands that are not 0/1, half genotypes, letters in sizes… | Correct each cell using the row and column given by Values to fix in the sheet. | 3.5 |
| Monomorphic loci remain in the analysis. | The app flags them but does not remove them. | Delete them from the file if you do not want them. | 3.5 |
| Near-identical genotype pairs do not appear. | That search is only run with 400 individuals or fewer and three or more loci. | Check the distances between individuals in Block 6. | 3.5 |

## D.4 Analyses

**Table D.4.** Messages and missing results in the analysis blocks.

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| A block appears greyed out, or `…so it is not available for these data`. | The data do not meet its requirement (marker type, replicates, number of loci). | Check the card of available analyses in Block 2. | 3.6 |
| `…the difference is not tested here` | Fewer than three loci, or resampling set to 0. | With few loci, compare without intervals and with caution. | 4.3 |
| `No population has the five or more plants needed to test linkage disequilibrium` | All populations have fewer than five plants. | Only the pooled sample is reported. | 5.5 |
| Block 4 says `Dominant or haploid data` or `{n}-ploid genotypes`. | Hardy–Weinberg, $F_{IS}$ and null alleles are defined only for diploid codominant data. | Linkage disequilibrium is still computed. | 5.5 |

**Table D.4.** Messages and missing results in the analysis blocks. *(continued)*

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| The tick box for leaving out loci with null alleles does not appear. | Block 4 has not been run or did not flag any locus. | Open Block 4 and return to Block 5. | 6.5 |
| `∞ · {n} pair(s) share no allele` | Two populations share no alleles: Nei's distance is infinite. | Choose the chord distance or Rogers' distance; with sequences, $D_A$. | 7.1 |
| There is no tree of individuals, or it has no support. | Trees of individuals up to 400 units (NJ up to 150); support for individuals between 4 and 60. | Use the population tree or a subset of individuals. | 7.3 |
| `outgroup not monophyletic — root left as it was` or `outgroup names not found` | Misspelt names or tips that do not form a single clade. | Type the exact names, separated by commas, or root at the midpoint. | 7.3 |
| `ΔK could not be computed: it needs at least three consecutive K values and two replicates per K` | Short range of K or a single replicate. | Test three or more K values with two or more replicates. | 8.2 |
| `Assignment needs at least two populations with five or more individuals.` | Populations with fewer than five individuals are excluded from assignment. | Combine samples or extend the sampling. | 8.4 |
| `…sites with gaps or ambiguous bases were excluded (complete deletion)` and few sites remain. | Ends of the alignment with gaps in some sequences. | Trim the ends before loading the alignment. | 9.1 |
| `the search stopped at {n} median vectors, so the network may be incomplete` | A very complex network: the cap of 100 median vectors was reached. | Use fewer sequences or a shorter fragment. | 9.4 |
| `{n} tips are too many to date in the browser` | More than 400 tips. | Choose One tip per haplotype or a subset. | 9.5 |
| `Tips of the tree not found among the sequences` | The names in the imported tree are not those in the FASTA (a space and an underscore count as the same). | Make the names match. | 9.5 |
| `Unexpected "{c}" at position {i} of the Newick string.` | A malformed tree: parentheses, commas or the final `;`. | Check the tree text at that position. | 9.5 |
| `The starting ages violate a calibration…` | Incompatible minimum and maximum ages between nested nodes. | Correct the calibrations. | 9.5 |
| `too low: run a longer chain` (ESS below 200) | The chain did not explore the posterior distribution well. | Lengthen it or repeat it with another seed. | 9.5 |
| The dating disappears. | Loading other data deletes the dating and stops the running chain. | Export trees, ages and log before changing data. | 9.5 |

**Table D.4.** Messages and missing results in the analysis blocks. *(continued)*

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| `{population}: too few locus pairs above the frequency threshold for the LD estimate.` | Few loci, or alleles that are almost all rare. | Use more loci or lower the rare-allele threshold. | 10.2 |
| `Spatial genetic structure: not computed – fewer than 30 pairs of plants at different locations` | Coordinates per population rather than per plant, or few plants. | Give each plant its own coordinates, or use the isolation by distance analysis of Block 6. | 10.3 |
| `{populations} left out with fewer than 5 mapped plants` | Those populations have fewer than five plants with coordinates. | This is a warning: the remaining populations are analysed. | 10.3 |

## D.5 Figures, report and package

**Table D.5.** Problems when exporting.

| Symptom or message | Cause | What to do | See |
|---|---|---|---|
| Orange warning `This figure is too large for {n} dpi in a browser…`, or `Export failed: The browser could not create an image this large.` | The image would exceed 16,000 pixels on a side or 120 megapixels. | Export as SVG, lower the resolution or reduce the size of the figure or the row height. | 11.2 |
| `Allow pop-ups to print.` | The browser blocked the report window for printing. | Allow pop-ups for the app and press the button again. | 11.3 |
| The PDF comes out without the colours of the key figures. | Background graphics are switched off in the print dialogue. | Turn on background graphics. | 11.4 |
| `Could not build the package: {message}`, or images of some figures are missing from the ZIP. | Not enough memory, almost always with TIFF at high resolutions. | Lower the resolution of the ZIP or choose SVG only. | 11.3 |
| A table in the report says `Showing {n} of {m} rows.` | The report copies the on-screen table, which shows only the first rows. | Use the complete CSV in the `tables/` folder of the ZIP. | 11.3 |
| `Select and copy the text below.` | The browser did not allow the methods paragraph to be copied to the clipboard. | Select the text in the box and copy it by hand. | 11.3 |
| A CSV opens in a single column in your spreadsheet program. | The computer uses a decimal comma and expects semicolons between columns. | Import it as text, specifying the comma as separator. | A.8 |

**NOTE · CORRECTIONS MADE TO THE APP WHILE PREPARING THE APPENDICES**

Checking the formats and messages against the code led to the following corrections in the app:

- A column written entirely with a decimal comma ( `13,5` ) was read as text; it is now numeric.
- UTM coordinates became missing because they did not fit in degrees; they are now used as map distances.
- Columns called `Longitud_mazorca` or `Area_foliar` were taken as a coordinate or a region; the whole name now counts.
- A `0/0` in a cell was flagged as a half genotype.
- The U of an RNA FASTA file was not converted to T.
- The dated tree in NEXUS stored branches in substitutions and ages in millions of years; both are now in millions of years.
- Block 4 called polyploids "dominant or haploid", and the card of available analyses treated them as valid for Hardy–Weinberg and demography.
- In Block 9, a rare-allele threshold of 0 was changed to 0.02.
- With hundreds of plants, the note on distances between individuals and the isolation by distance figure of Block 6, and the spatial structure of Block 9, could stop with an error.
- The list of references on the Home page had truncated titles.

# E References

The works cited in the manual and in the app's texts, with full bibliographic details.

The symbol ◆ marks those that the methods paragraph of Block 10 may cite: use them to complete the reference list of your paper (section 11.3). Five original titles include the name of a program; it is replaced by **[…]**. The records were checked against the publishers' and journals' databases.


Agapow, P.-M. and Burt, A. (2001). Indices of multilocus linkage disequilibrium. *Molecular Ecology Notes* 1: 101–102. doi:10.1046/j.1471-8278.2000.00014.x ◆

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Benjamini, Y. and Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics* 29: 1165–1188. doi:10.1214/aos/1013699998

Botstein, D., White, R.L., Skolnick, M. and Davis, R.W. (1980). Construction of a genetic linkage map in man using restriction fragment length polymorphisms. *American Journal of Human Genetics* 32: 314–331. ◆

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Brommer, J.E. (2011). Whither $P_{ST}$? The approximation of $Q_{ST}$ by $P_{ST}$ in evolutionary and conservation biology. *Journal of Evolutionary Biology* 24: 1160–1168. doi:10.1111/j.1420-9101.2011.02268.x ◆

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# F Third-party components

PopGeneticsPro is an original work: its code, its figure engine, its colour palettes and scales, its illustrations and its practice data were made for this program. Only one software component comes from a third party. The app and the manual also use some published data and designs, which are declared here together with their terms of use.

**Table F.1.** Third-party software component included in the app.

| Component | Version | Licence | Use and notice |
|---|---|---|---|
| **Spreadsheet reading and writing library** `vendor/xlsx.full.min.js` | 0.18.5 | Apache 2.0 | Reads `.xlsx`, `.xls` and `.ods` workbooks and writes `.xlsx` and `.ods` in Block 2: data loading, the built-in sheet and templates. It is an unmodified copy that is not part of the original work. Its copyright notice, its source and the full text of the licence are in `vendor/THIRD-PARTY-NOTICES.txt`, which must always accompany the app. |

**Table F.2.** Published data and designs used by the app.

| Resource | Authorship | Terms | Where it is used |
|---|---|---|---|
| **Colour-blind-safe palette** | Okabe and Ito (2008): eight colours; the app adds a grey and a dark brown. | Freely accessible guide; the app and the manual cite its authors. | Palette Okabe & Ito 2008 (colour-blind safe). |
| **Qualitative colour schemes** | Tol (2021): the *bright* scheme plus black and two colours from the *vibrant* scheme. | Freely accessible technical note; the author is cited. | Palette Tol 2021 (colour-blind safe). |
| **Viridis, Magma, Inferno and Plasma colour maps** | Nathaniel J. Smith and Stéfan van der Walt; Viridis also with Eric Firing (2015). | Dedicated to the public domain (CC0 1.0); their authors appreciate credit. | Continuous scales for heatmaps (five colours from each). |
| **Cividis colour map** | Núñez, Anderton and Renslow (2018). | Open-access article (CC BY 4.0); it is cited. | Scale Cividis (colour-blind safe). |
| **Geological time scale** | Boundaries and colours of the International Chronostratigraphic Chart, version 2020/03 (Cohen et al., 2013, updated). | The app cites the chart and its version. | Dated trees and the aid for reading the age of a stage (Block 8). |

**NOTE · REVIEW OF COMPONENTS**

While preparing this appendix, the origin of every colour in the app was reviewed. Seven of the heatmap scales copied published colour specifications whose licence requires an acknowledgement in the documentation. They were replaced by the original scales of figure F.1, keeping the same names in the menu, and the screenshots in the chapters that showed them were redone (the figures for null alleles, pairwise differentiation, phenotypic diversity and assignment). The values in the figures do not change, only their colours.

THE APP'S OWN HEATMAP SCALES

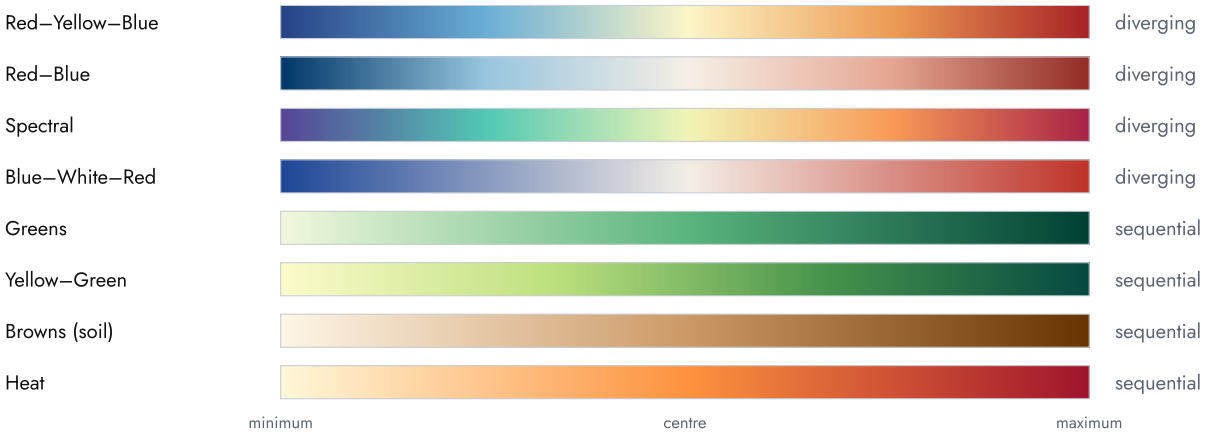


**Figure F.1.** The eight heatmap scales designed for the app, with their names as they appear in the menu. They are built from colours anchored in a perceptual colour space: in the diverging scales, lightness rises towards the centre and falls equally towards the ends; in the sequential scales, it falls continuously. The ends of the diverging scales remain distinguishable with red–green colour blindness.

**Table F.3.** Resources used in the manual.

| Resource | Authorship and licence | Use |
|---|---|---|
| **Typefaces** Cormorant, Crimson Pro and Jost | Christian Thalmann; Jacques Le Bailly; Owen Earl. SIL Open Font License (OFL), version 1.1. | Headings, text and labels. The HTML pages load them from a public font service and the PDF embeds them, as the licence permits; offline, the browser uses system fonts. |
| **Illustrations and diagrams** | Original to this manual. | Vector cover, explanatory figures and the illustrations of the invented genera in the dating example. |
| **Practice data** | By the author. The ISSR matrix of 15 maize landraces contains real data; the other examples were simulated with the programs included (`data/generate_examples.pl`, `data/generate_dating_example.pl` and `manual/herramientas/generar-demografia.pl`), with fixed seeds. | The app's examples and the files in `manual/herramientas/`. The locus names in the maize microsatellite example are public marker names; their genotypes are simulated. |
| **Screenshots** | Of the app itself. | Chapter figures. |

**! IMPORTANT · WHEN SHARING OR REDISTRIBUTING THE APP**

The PopGeneticsPro code is distributed under the GNU General Public License, version 3 or later (file `LICENSE`). If you share the folder, keep it complete: `LICENSE`, `vendor/THIRD-PARTY-NOTICES.txt` and the library in `vendor/` must travel together. If you publish a figure made with the Okabe and Ito palette, the Tol palette or the Cividis scale, cite their authors.

This appendix concludes the English edition of the manual.